



UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

Course of Robot Control

Prof. Antonella Ferrara



Third Part

Outline of the Course

Basic Terminology and Concepts

Reference Frames and Rotation Matrices

Robot Direct Kinematics: the Geometric Approach

Robot Direct Kinematics: the Denavit-Hartenberg Convention Based Approach

Robot Inverse Kinematics

Robot Differential Kinematics

Robot Dynamical Model

Robot Control



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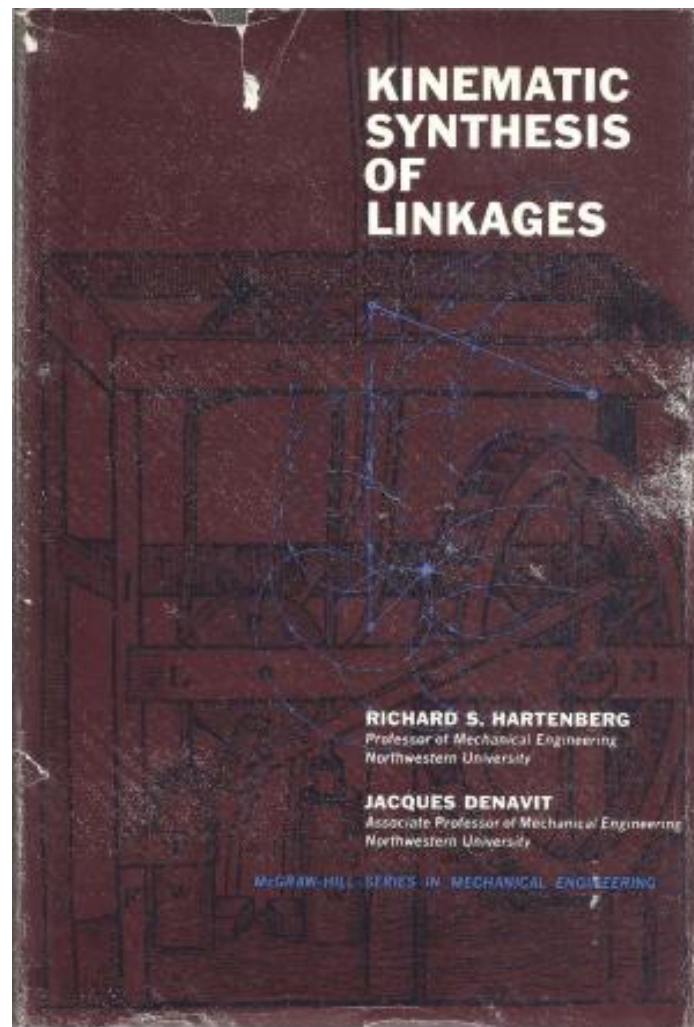
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Denavit-Hartenberg Convention Based Approach

The Denavit-Hartenberg Convention

D-H Convention



Systematic Mechanics, Design., in: Machine design

Kinematic notation for lower-pair mechanisms based on

Kinematic Synthesis of Linkages
Hartenberg, Richard S., and Denavit, Jacques
McGraw-Hill Book Company, 1964

The Denavit-Hartenberg Convention

D-H Convention



**Hartenberg, Richard
Scheunemann (1907 - 1997)**



Jacques Denavit (1930-2012)

<https://robotics.northwestern.edu/history.html>



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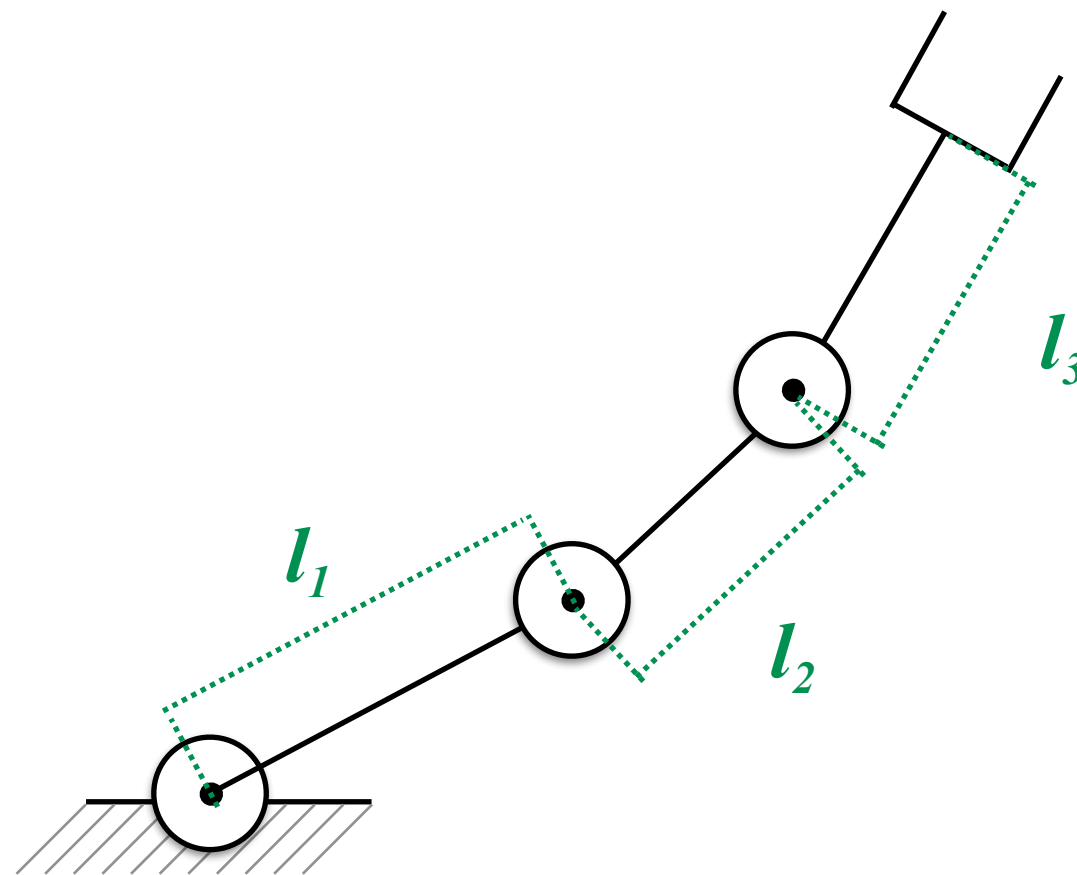
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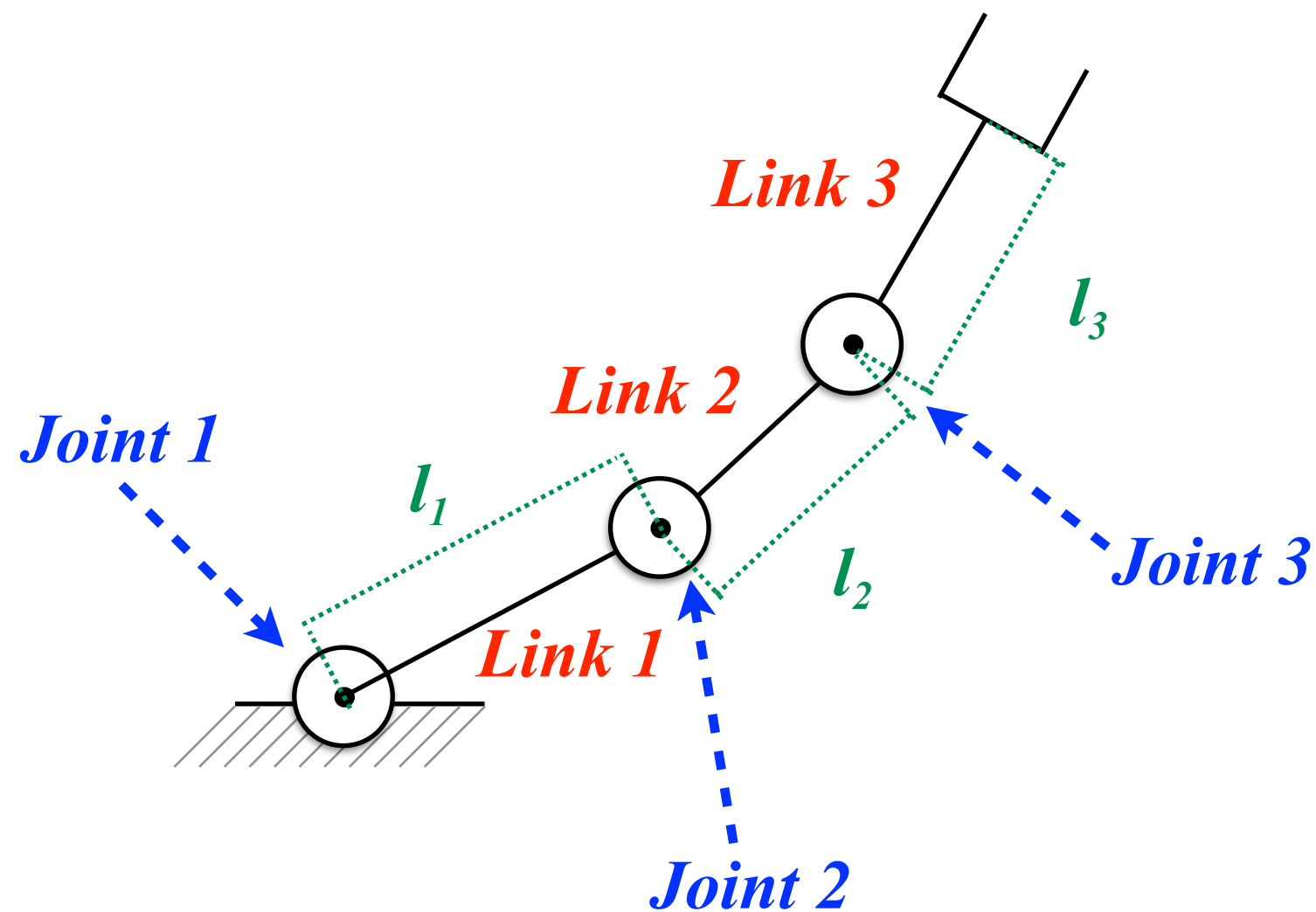


Choice of the Reference Frames

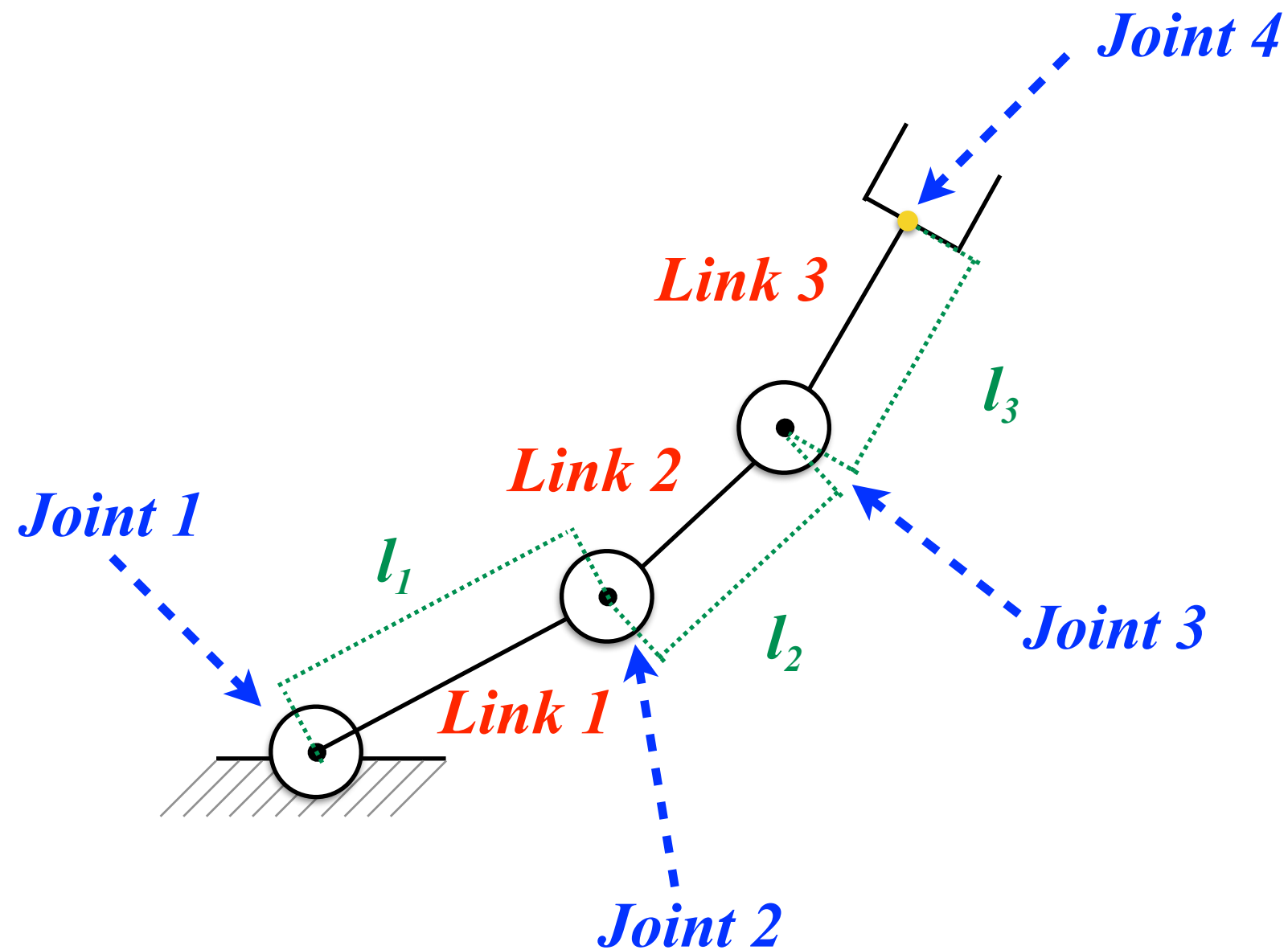
The Denavit-Hartenberg Convention



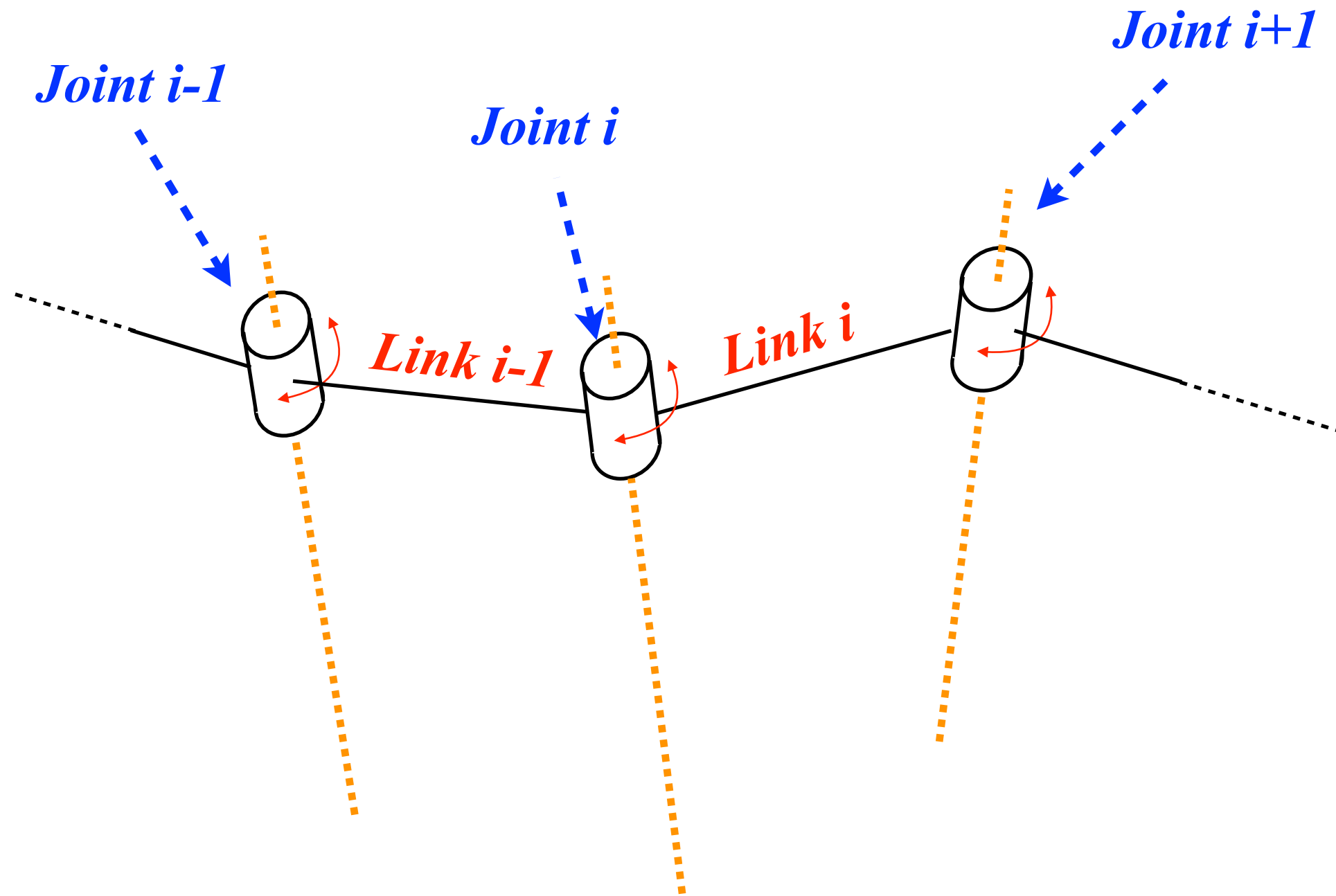
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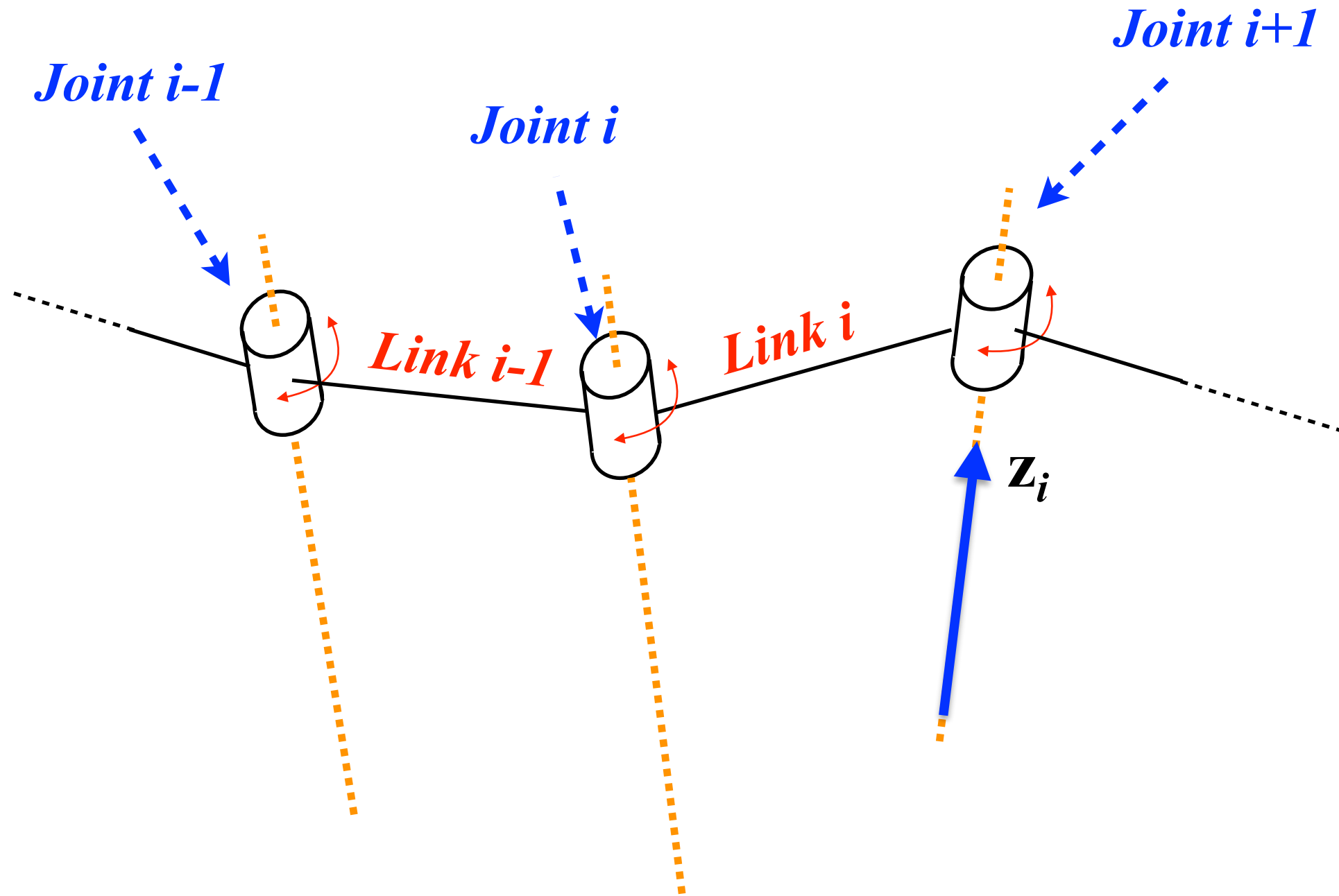
The Denavit-Hartenberg Convention



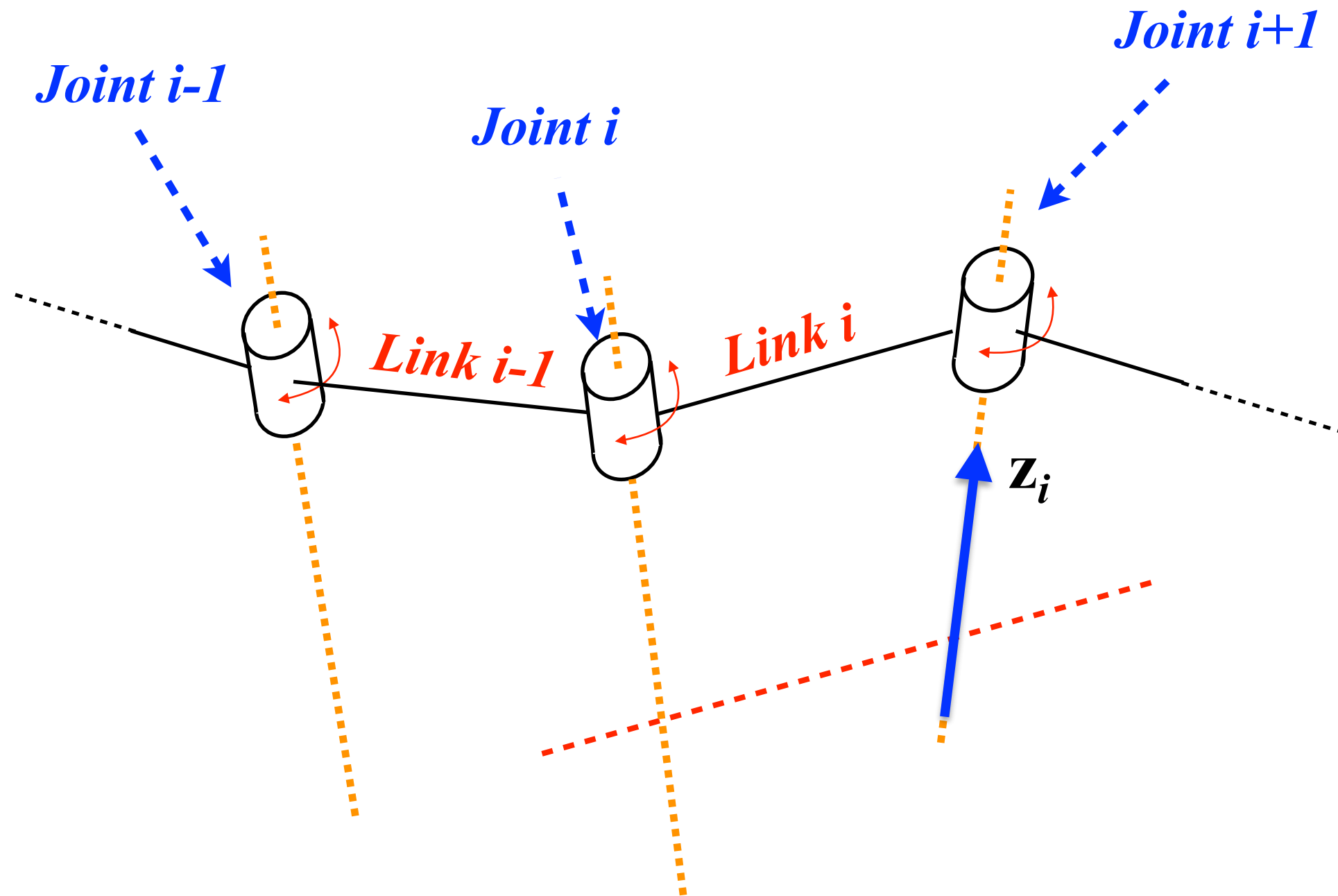
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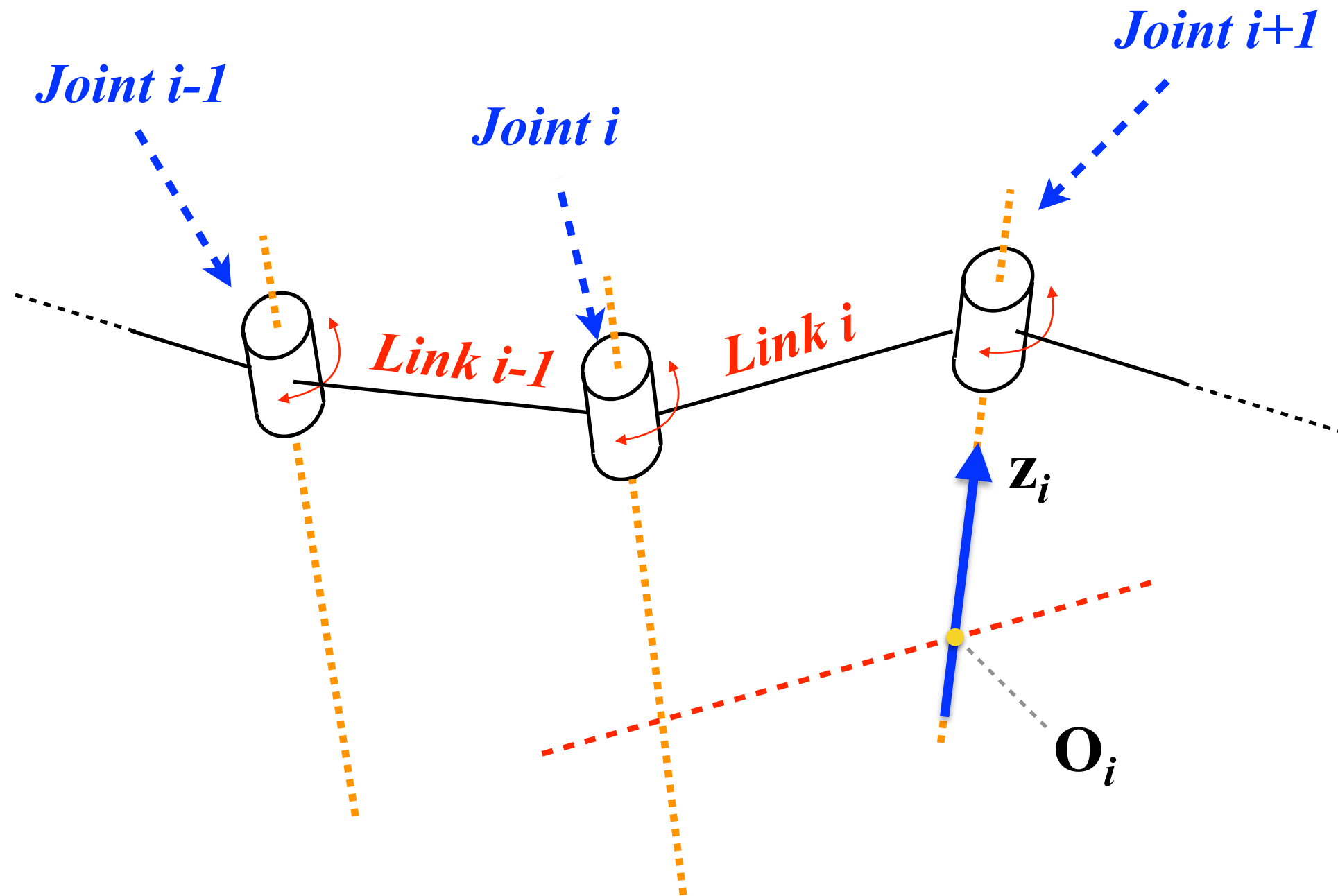
The Denavit-Hartenberg Convention



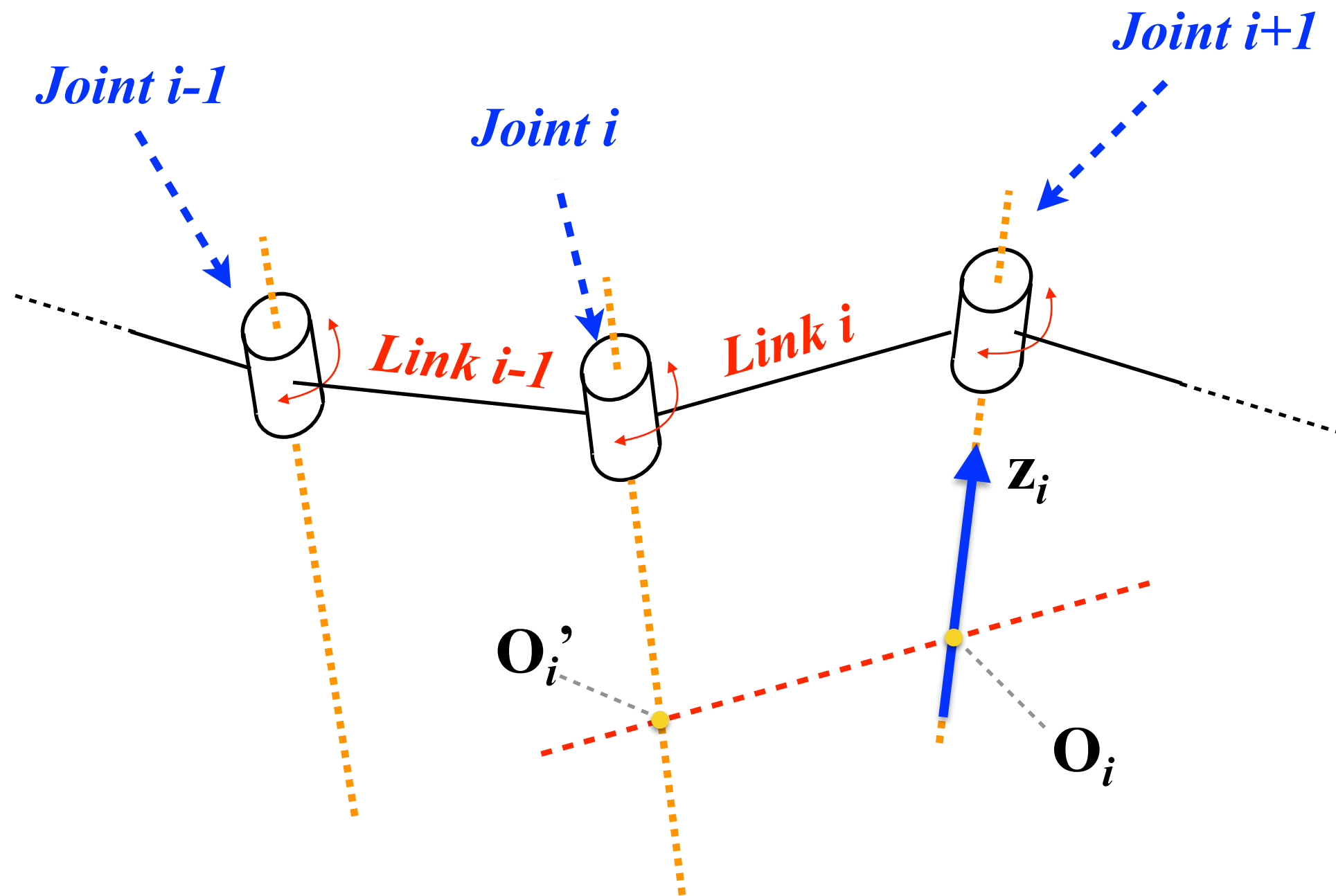
The Denavit-Hartenberg Convention



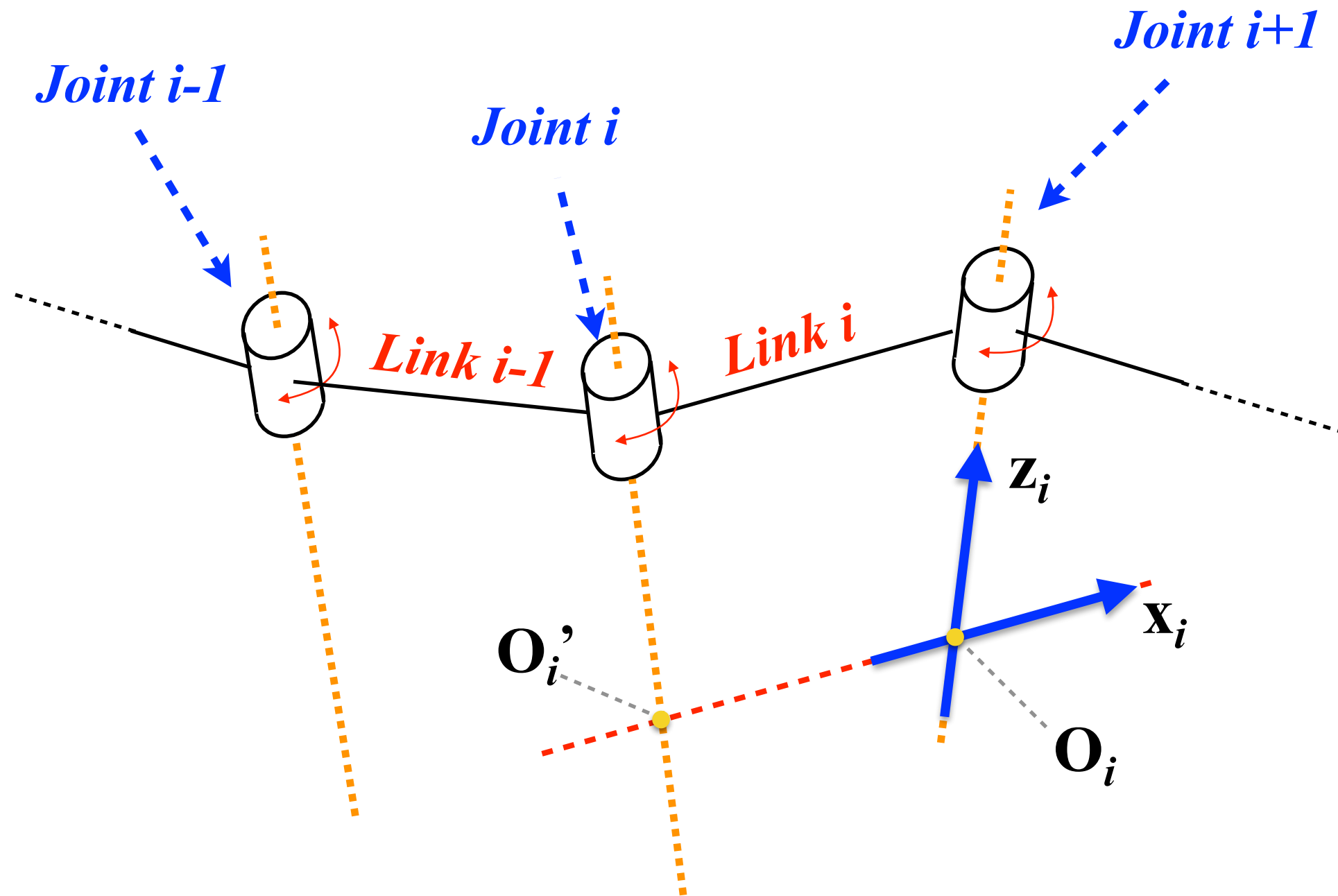
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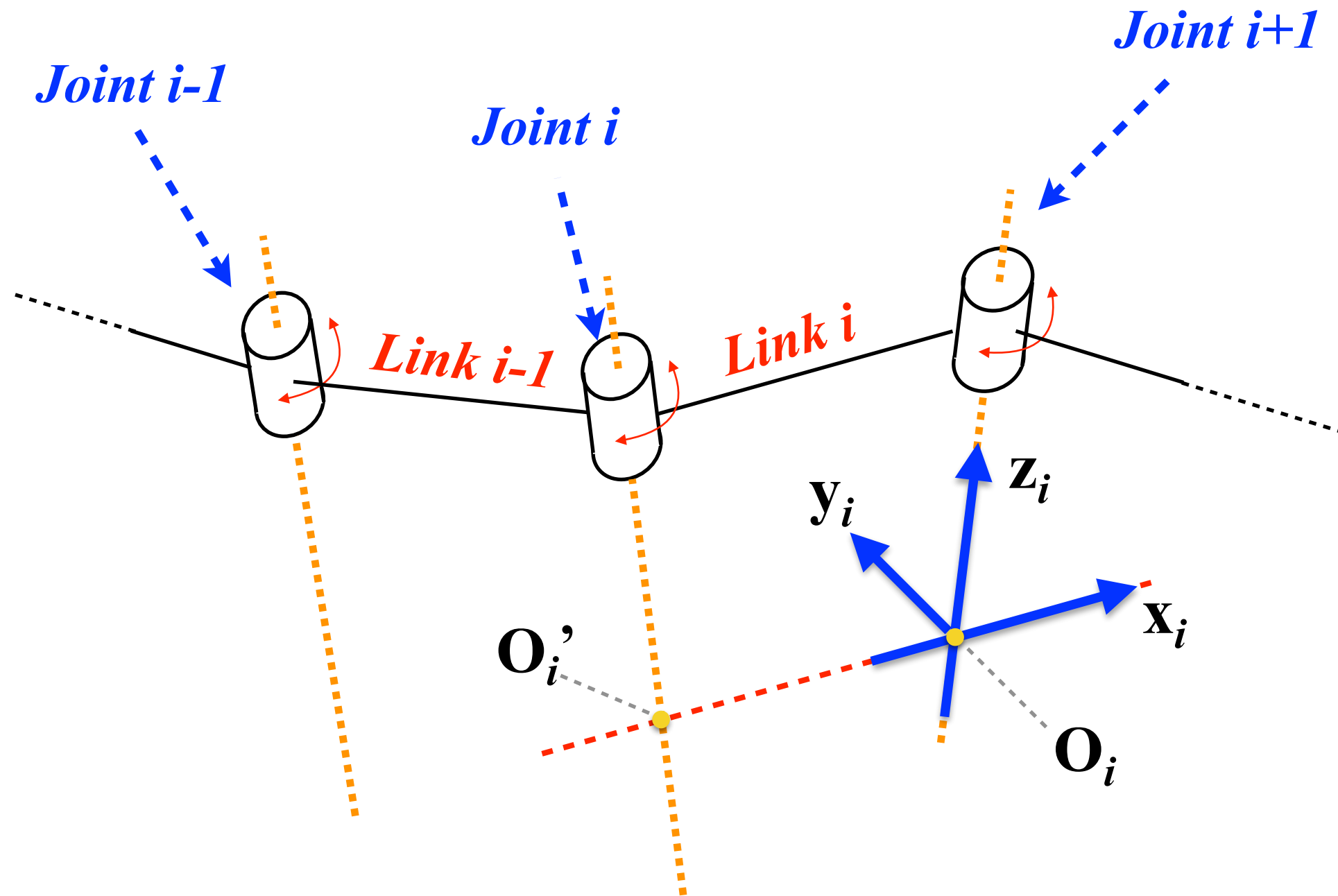
The Denavit-Hartenberg Convention



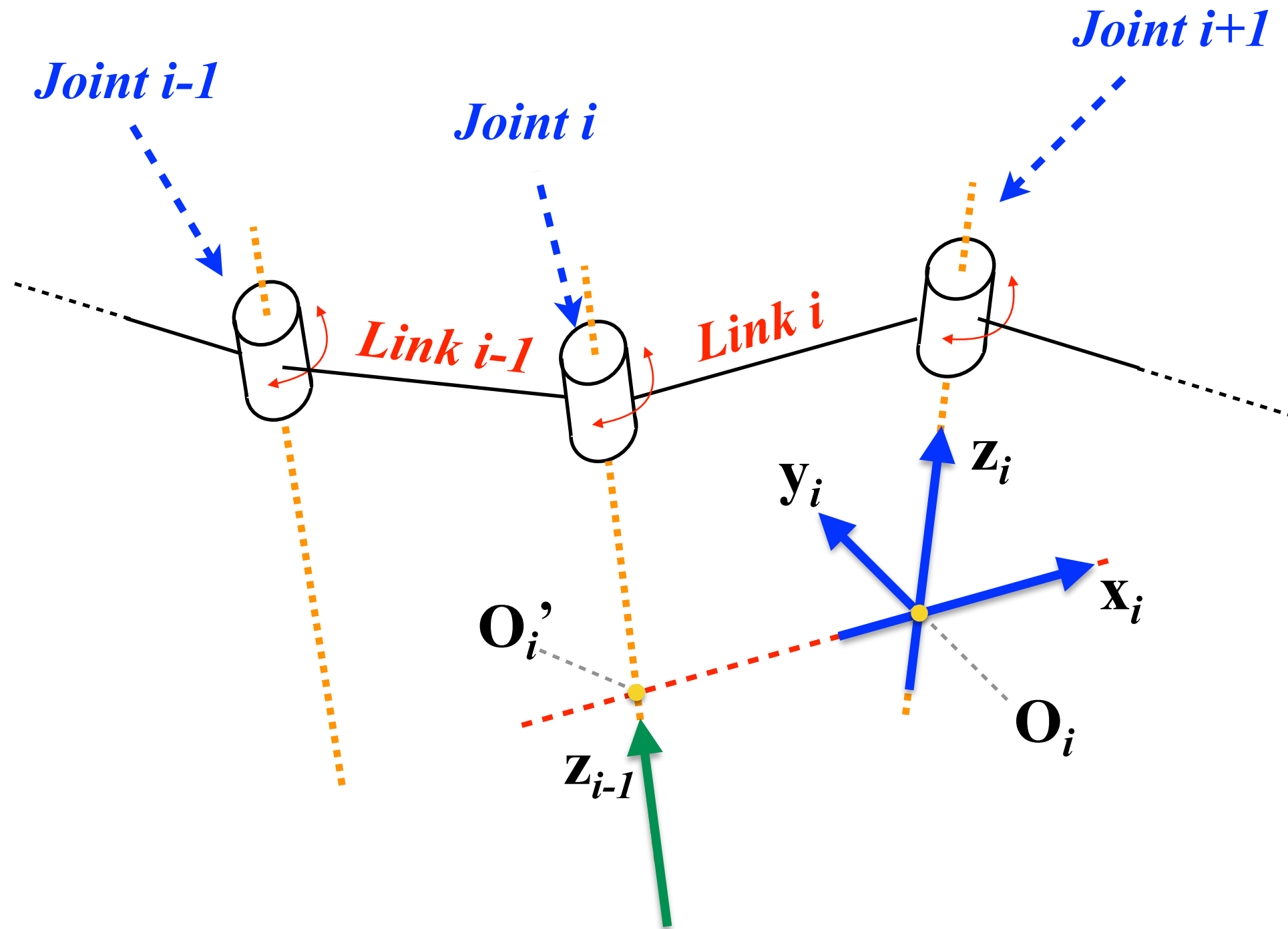
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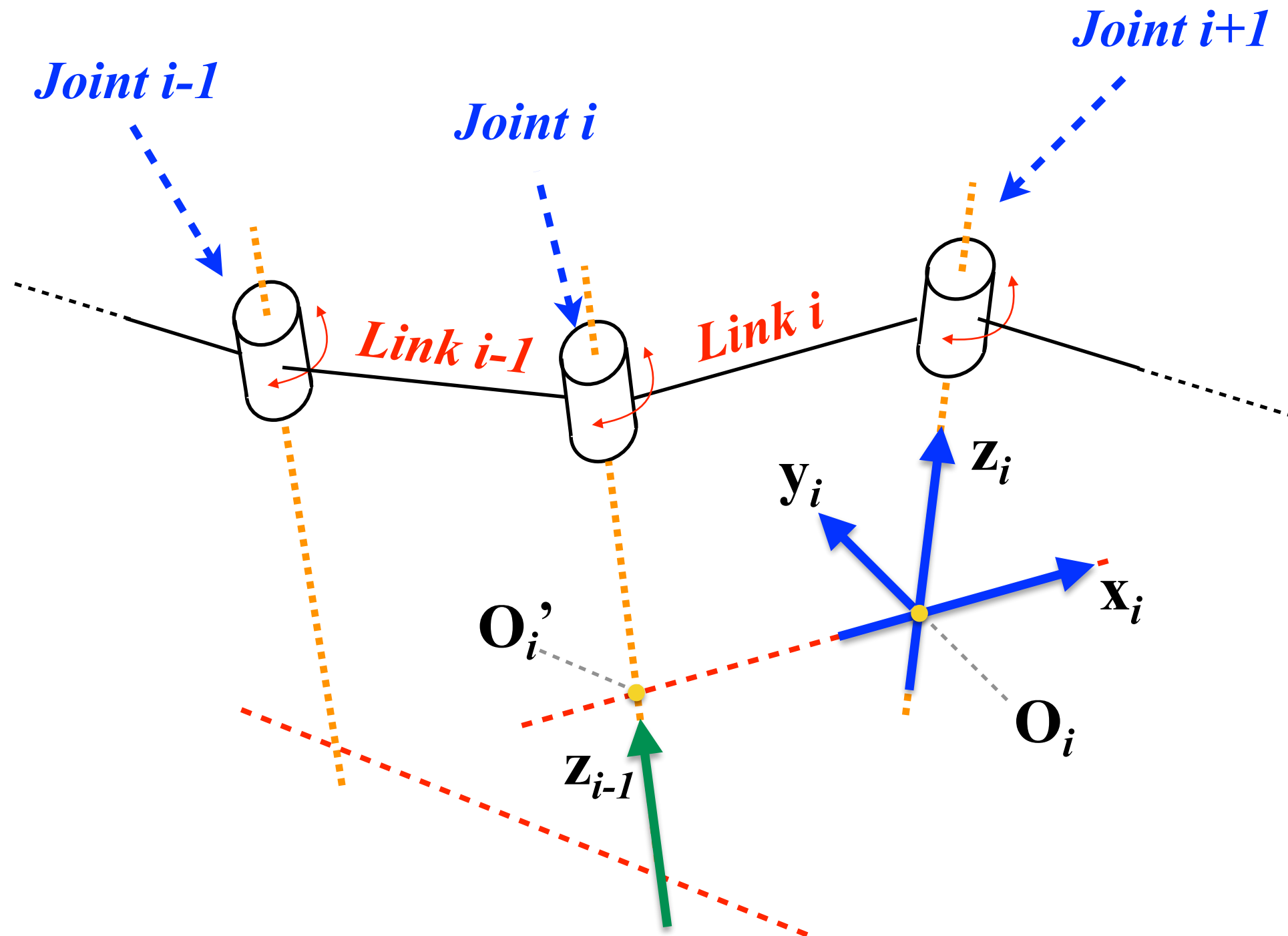
The Denavit-Hartenberg Convention



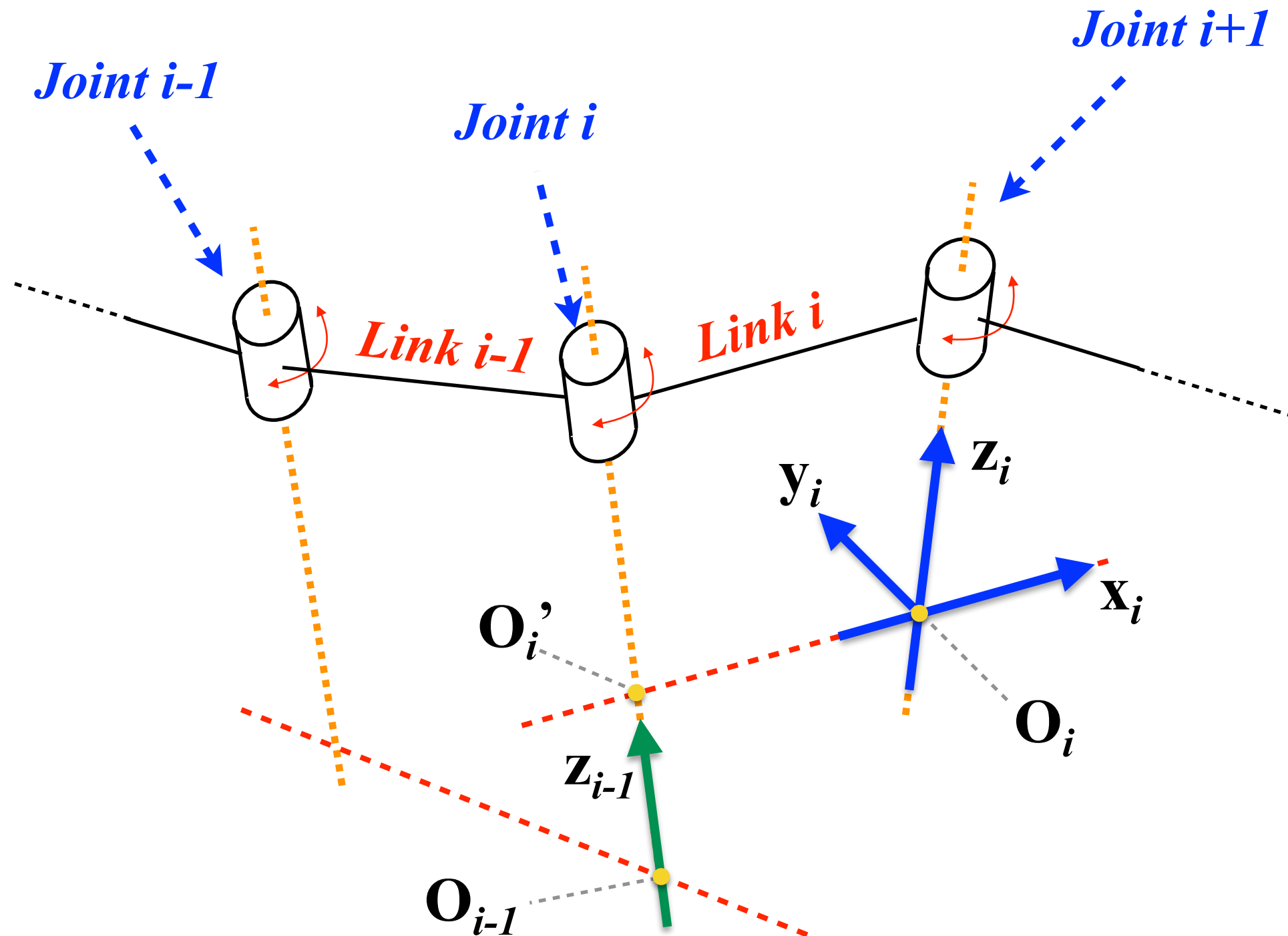
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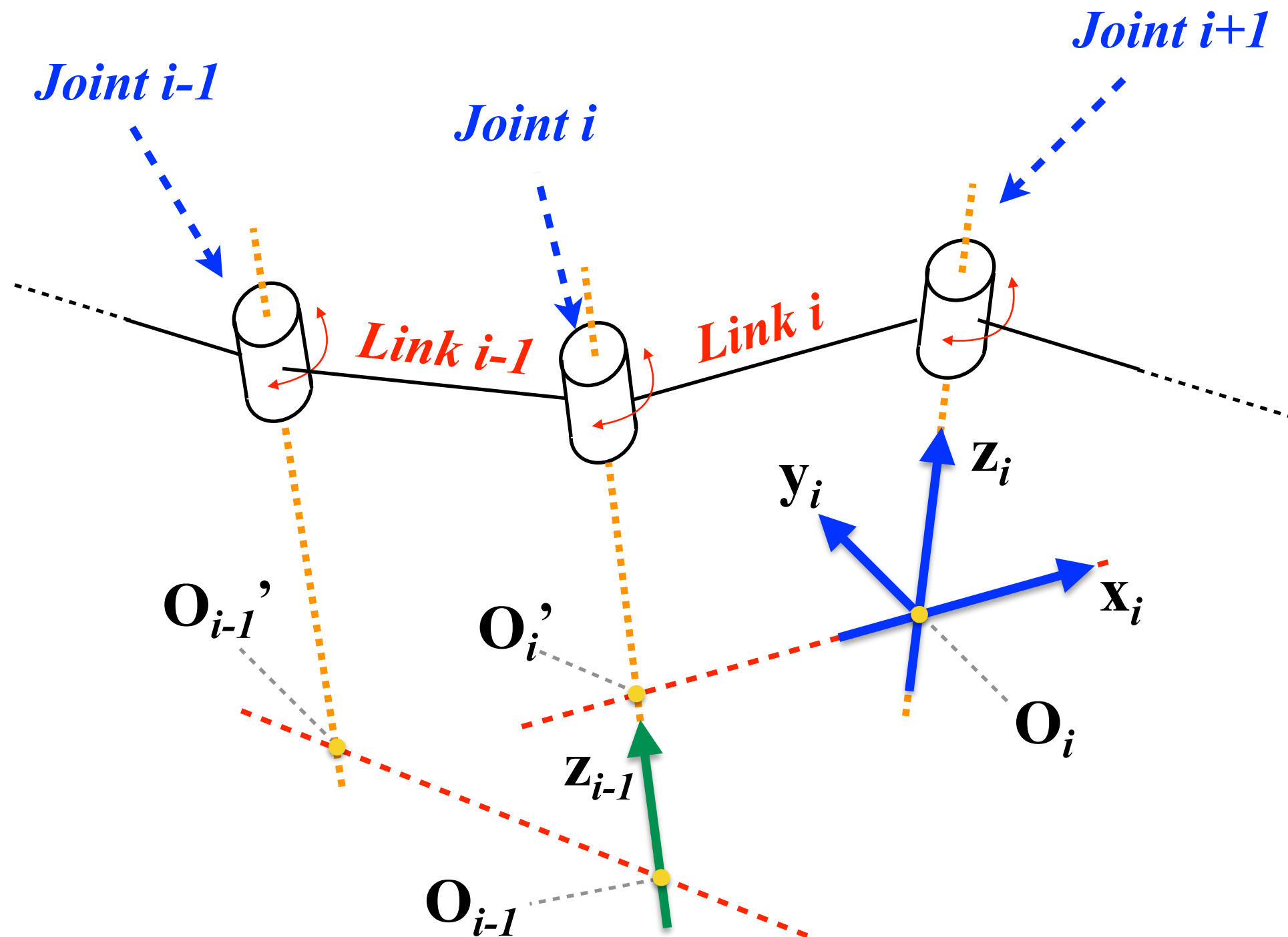
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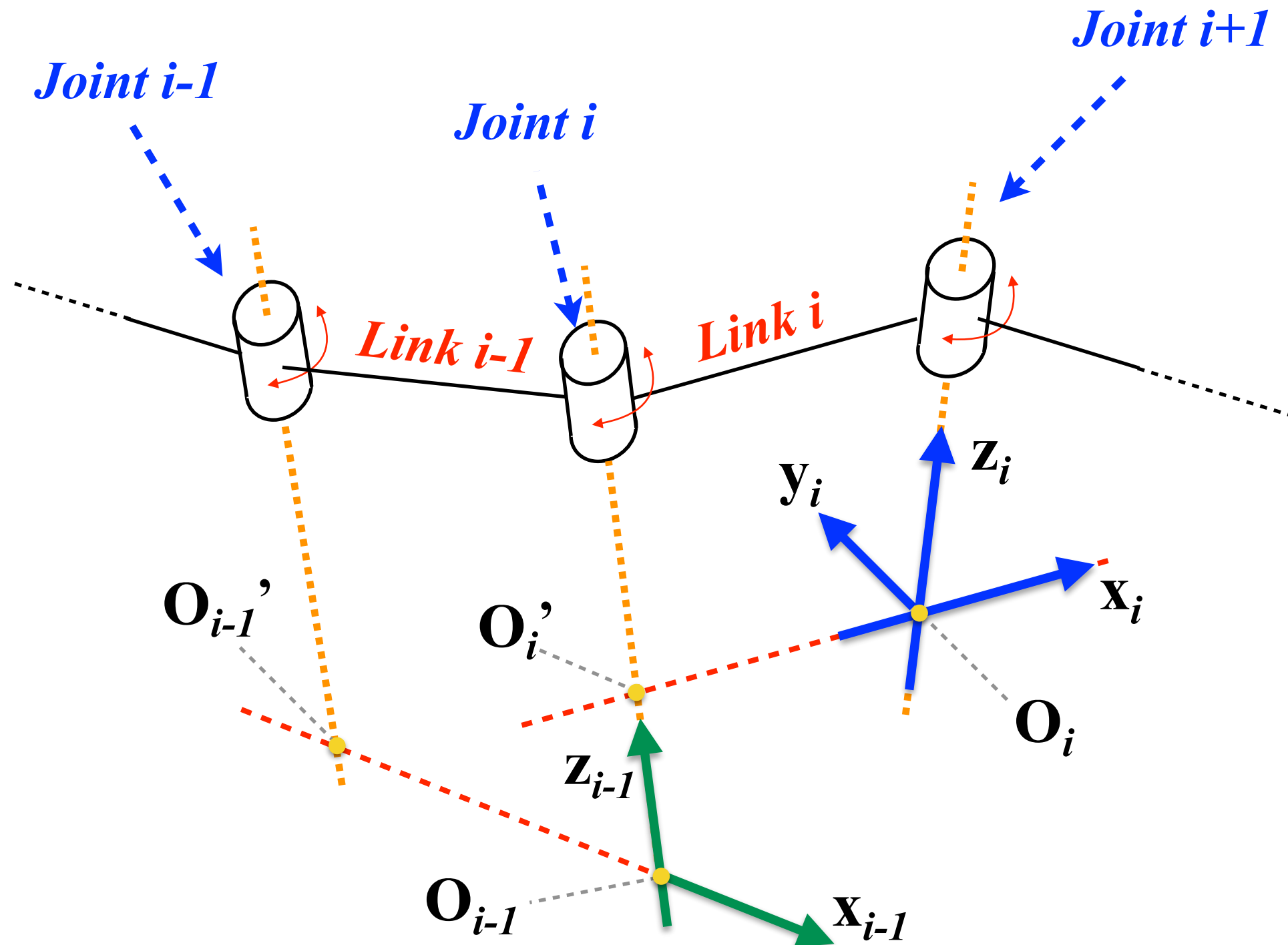
The Denavit-Hartenberg Convention



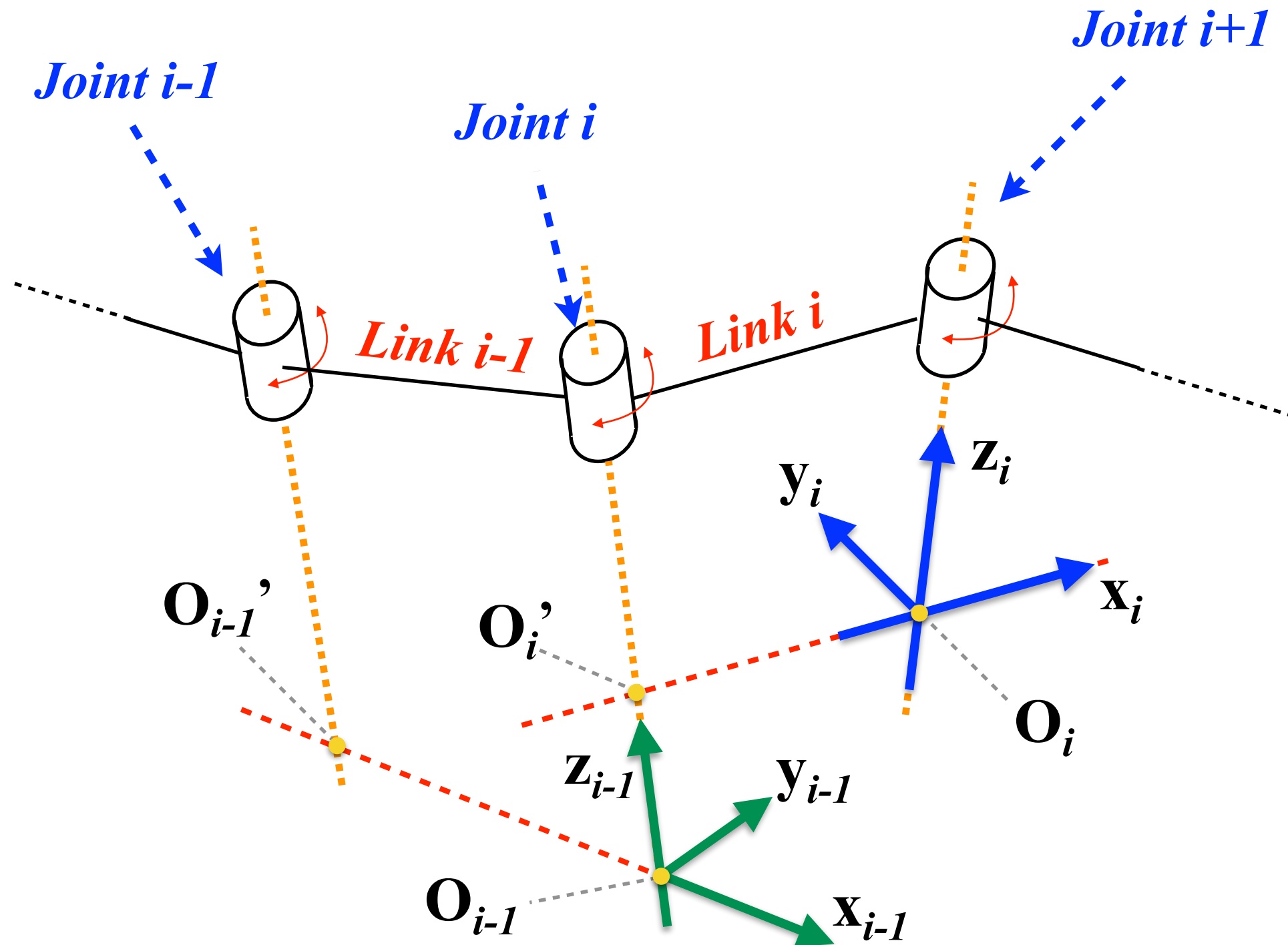
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The Denavit-Hartenberg Convention



The Denavit-Hartenberg Convention





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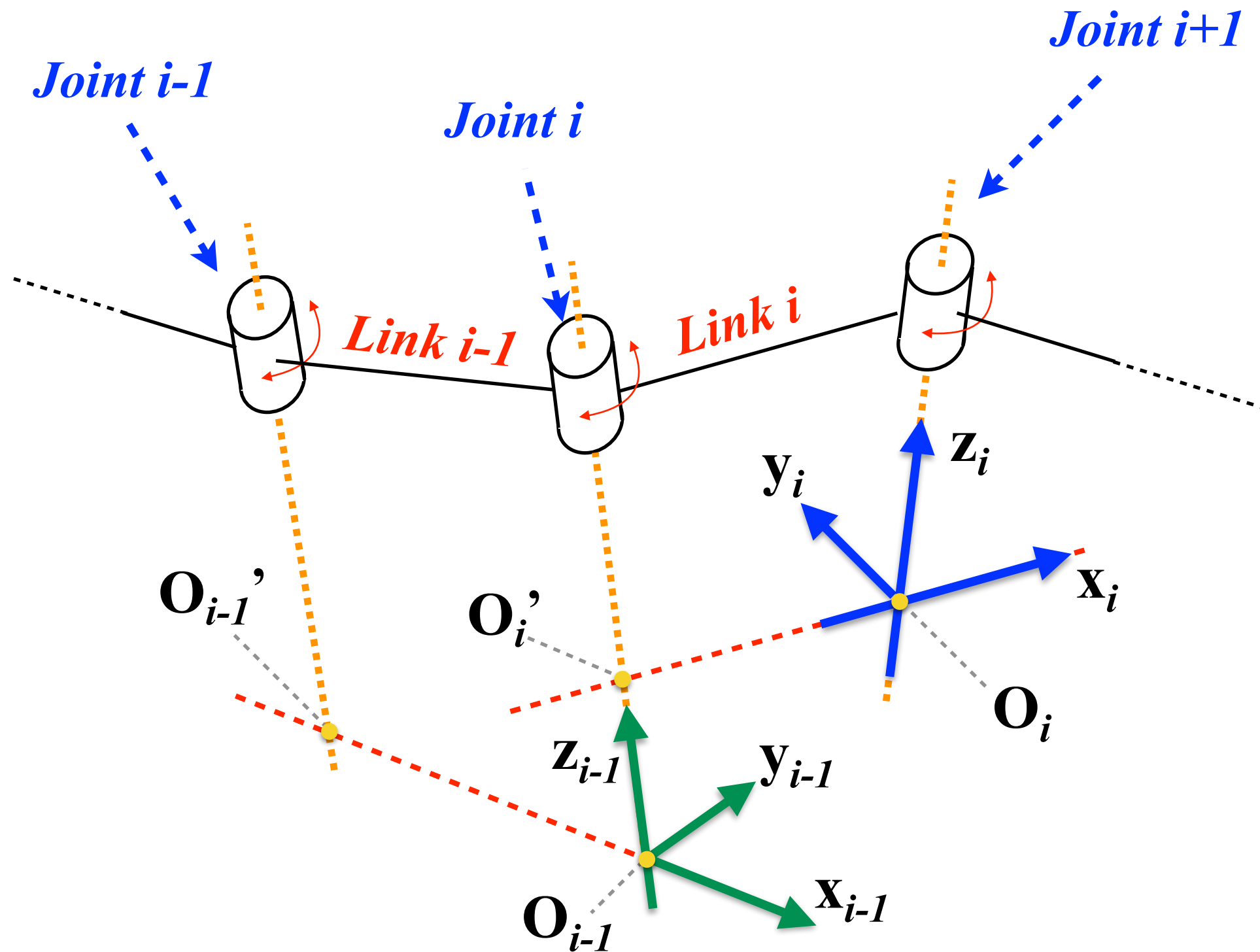
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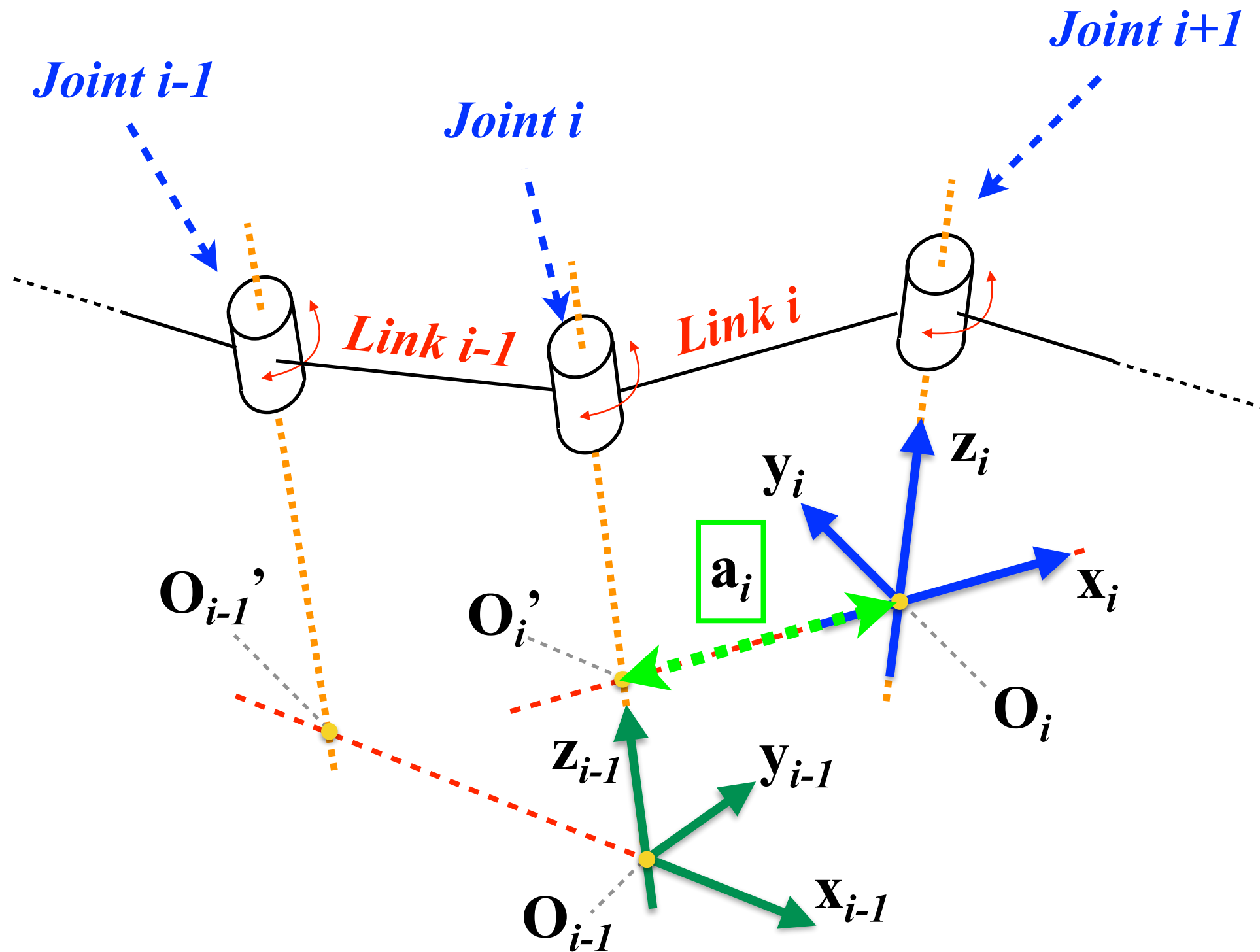


The D-H Parameters

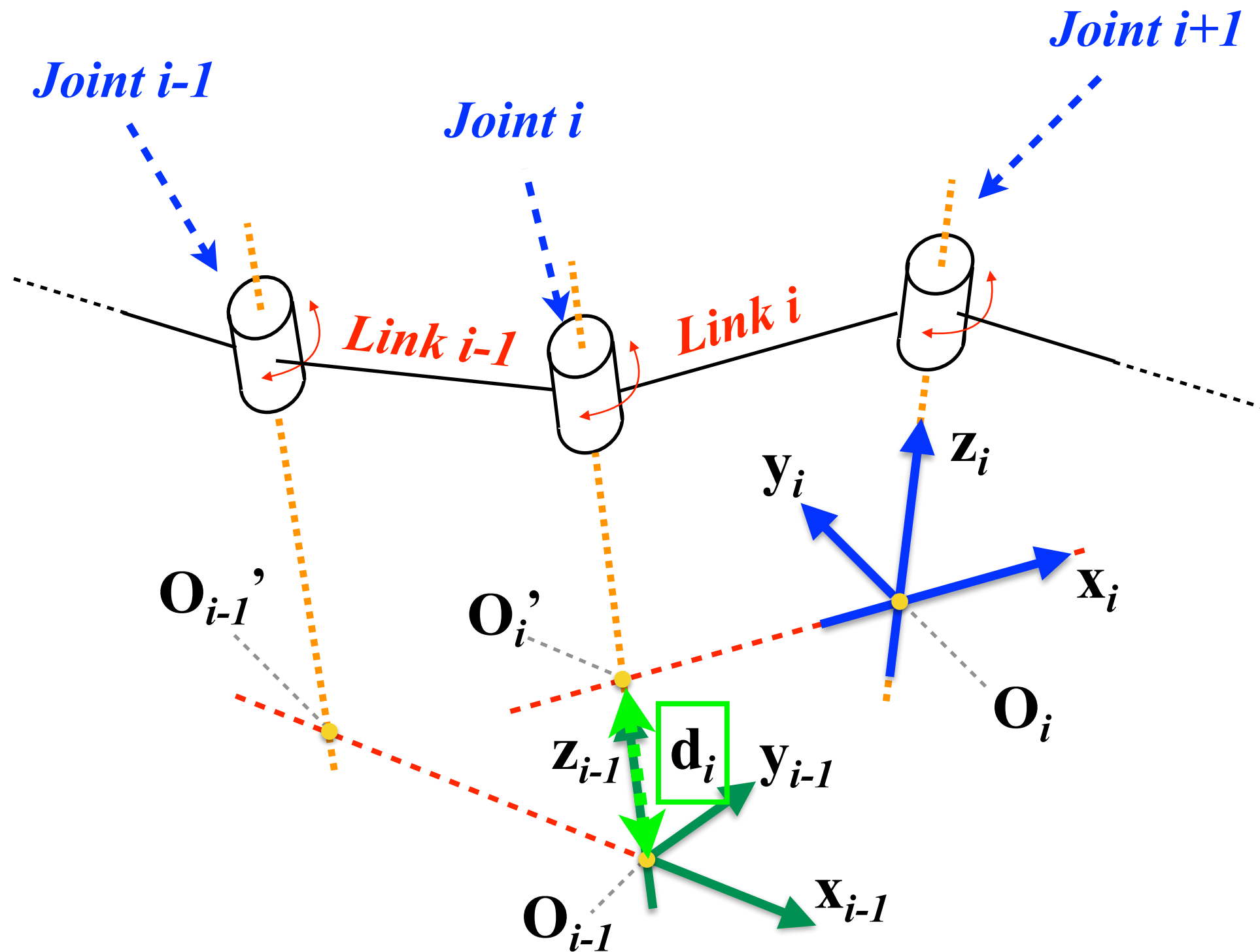
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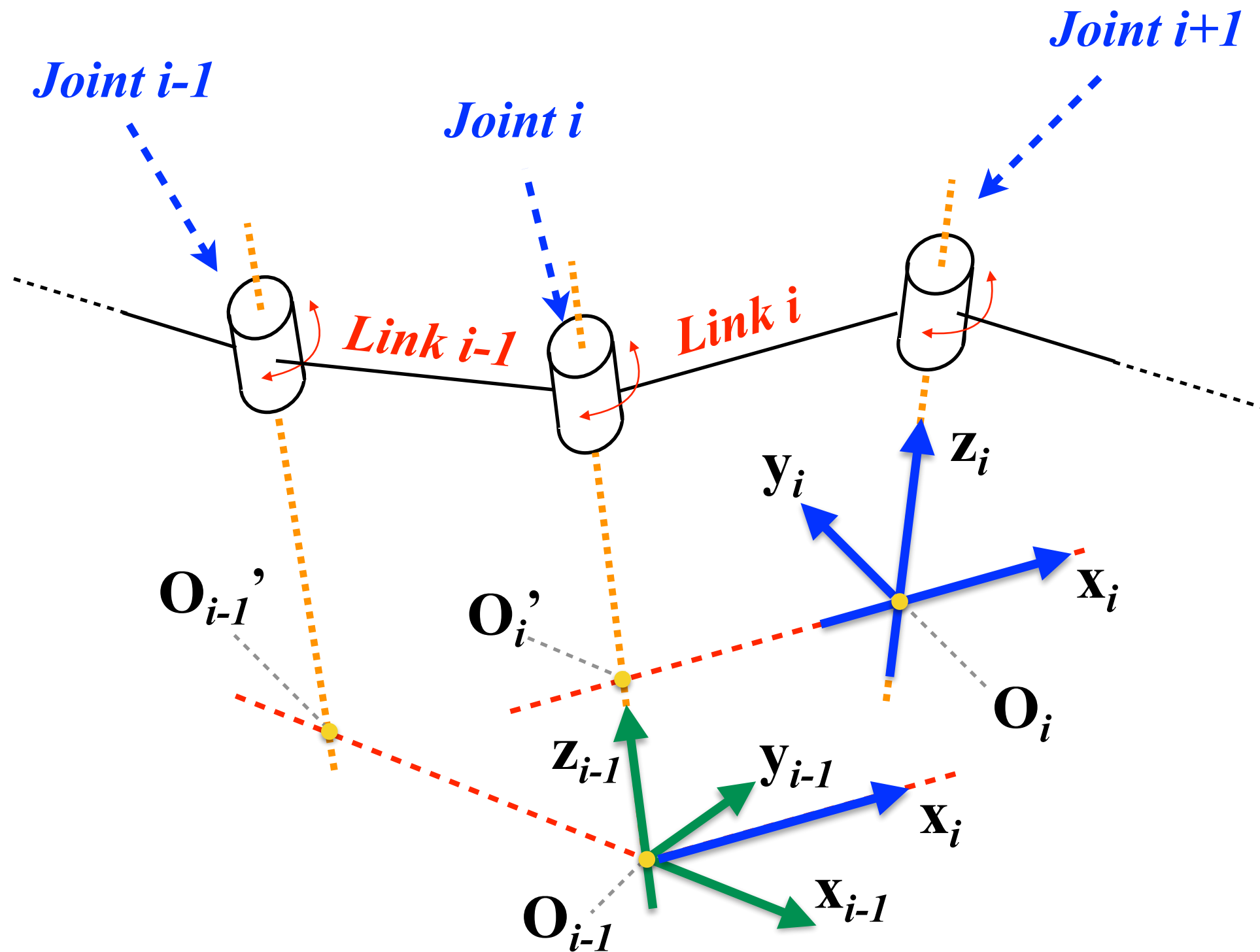
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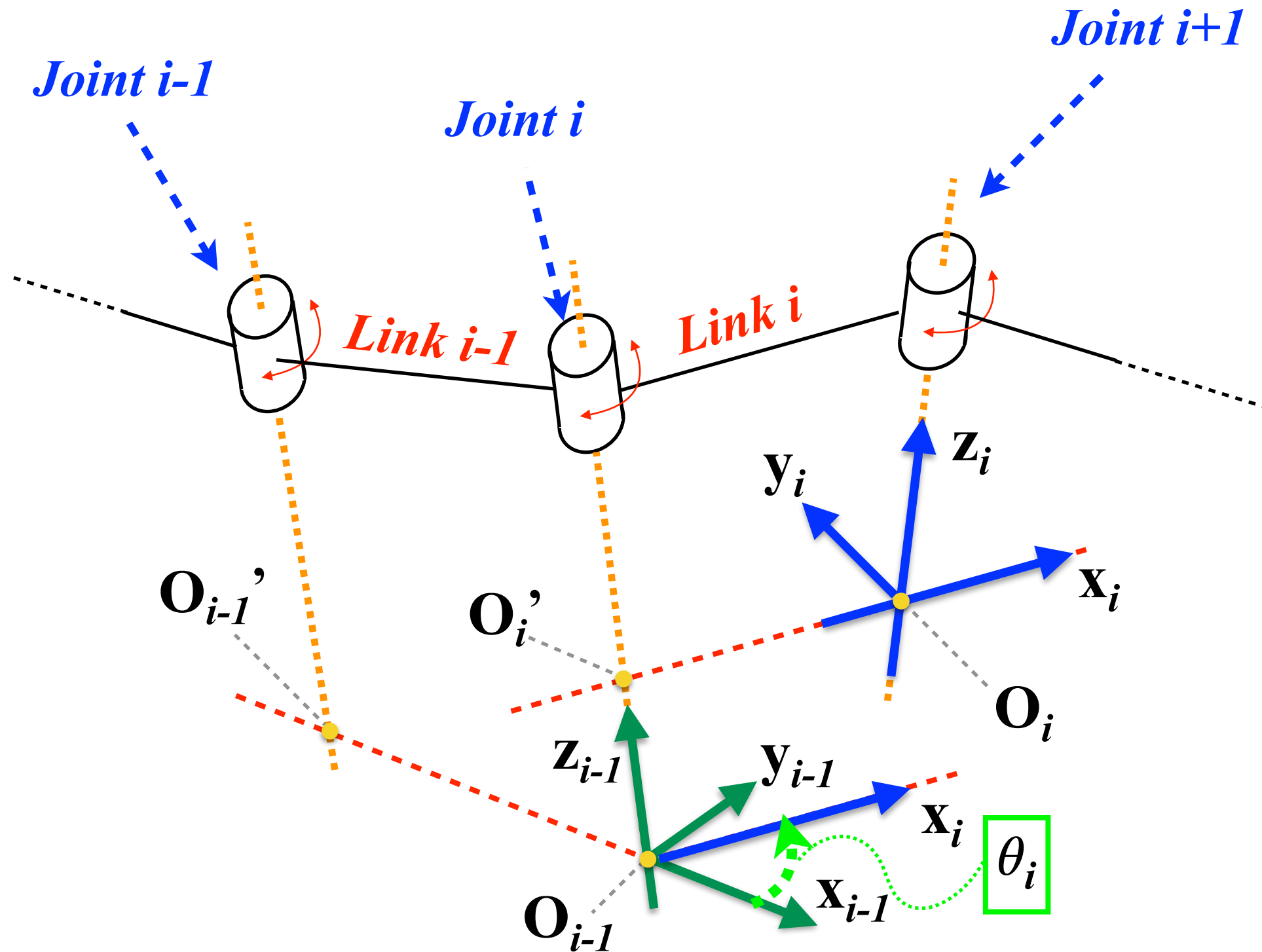
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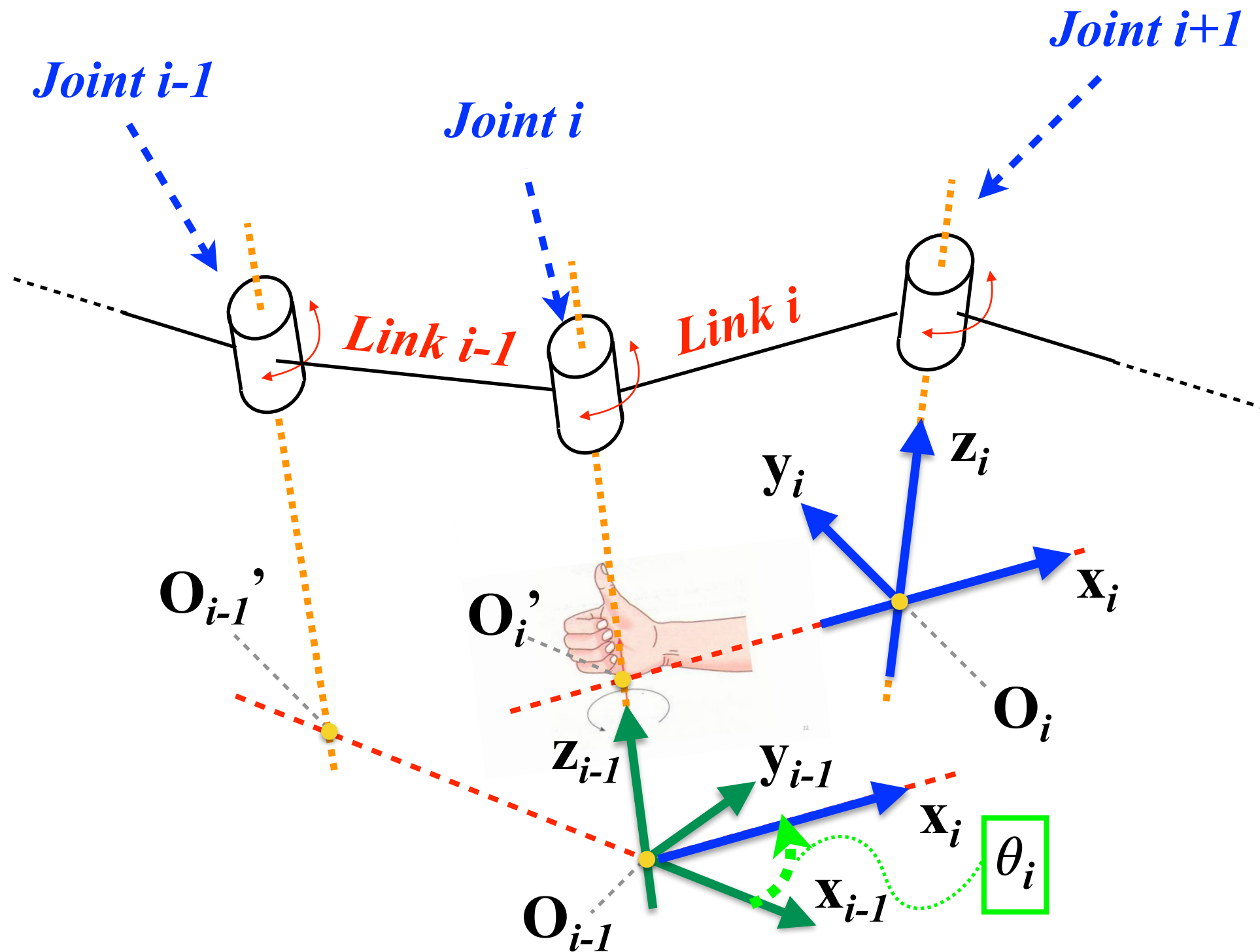
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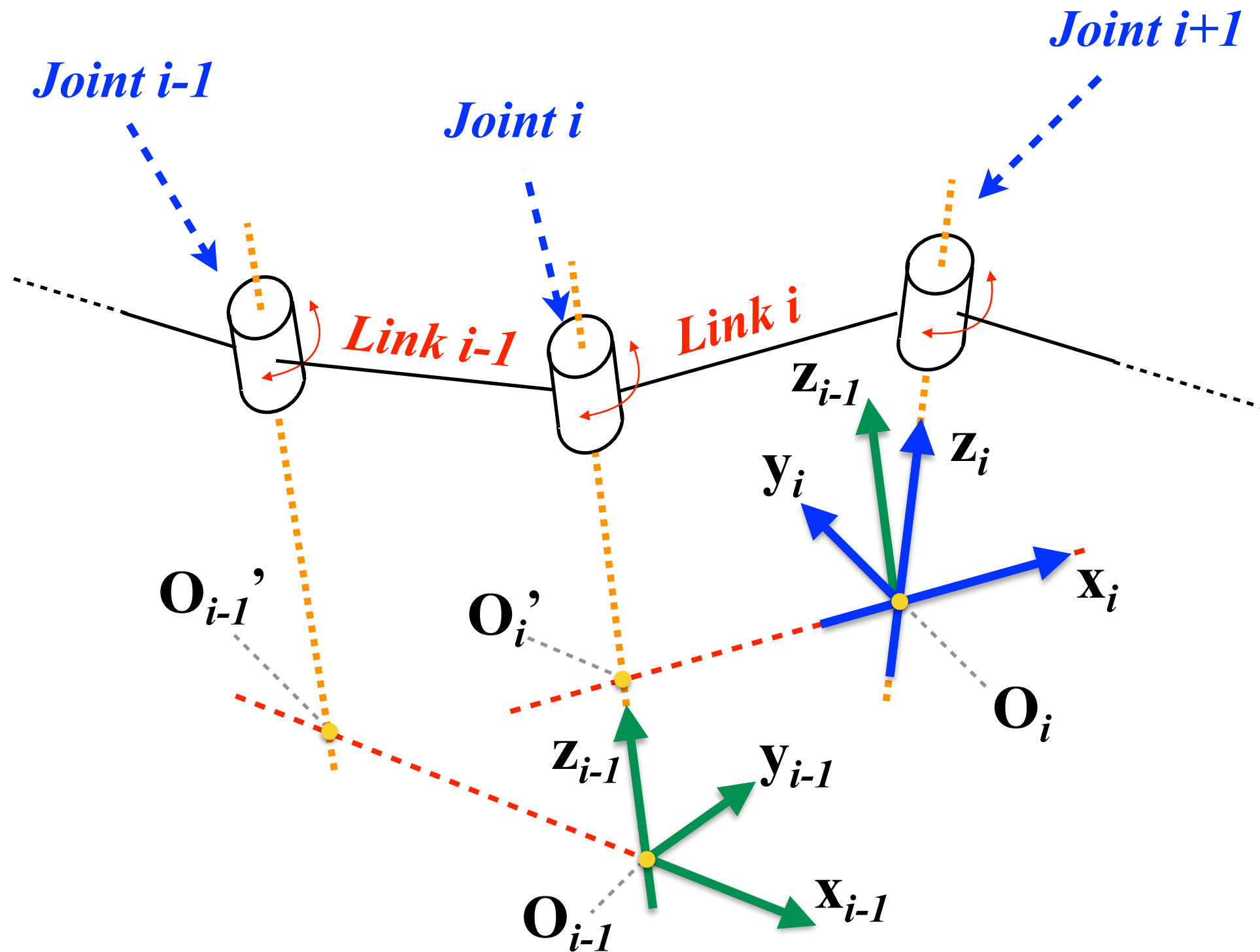
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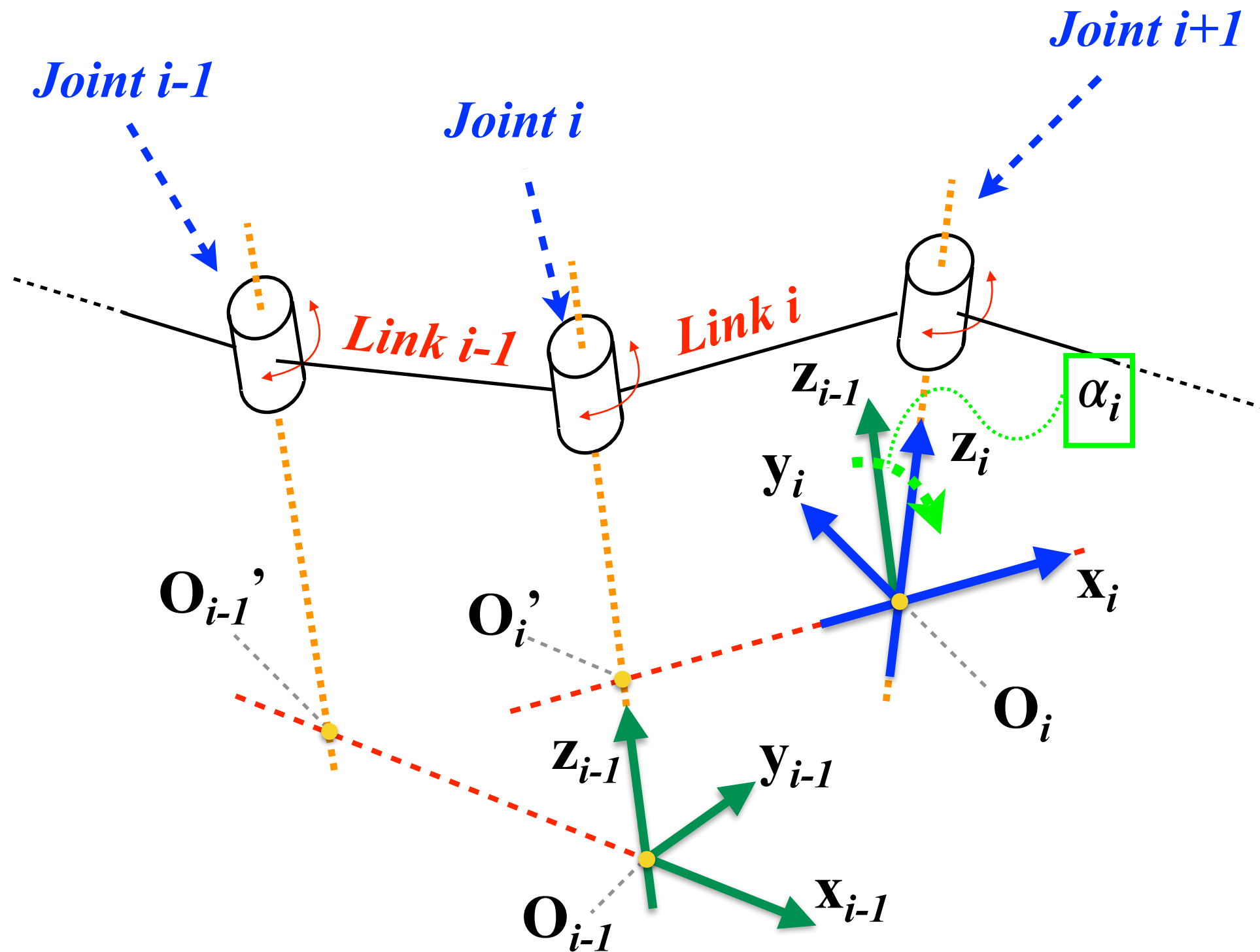
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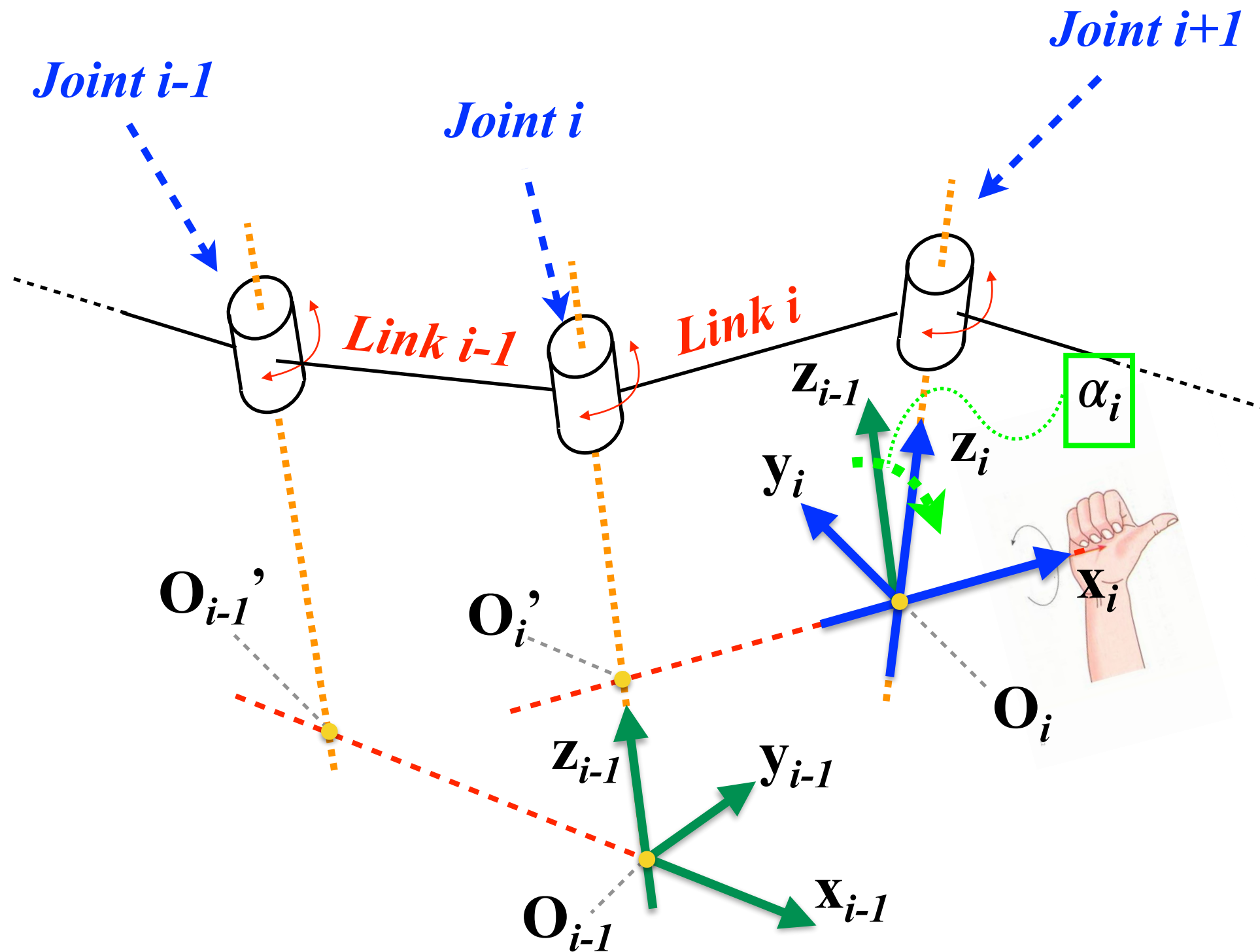
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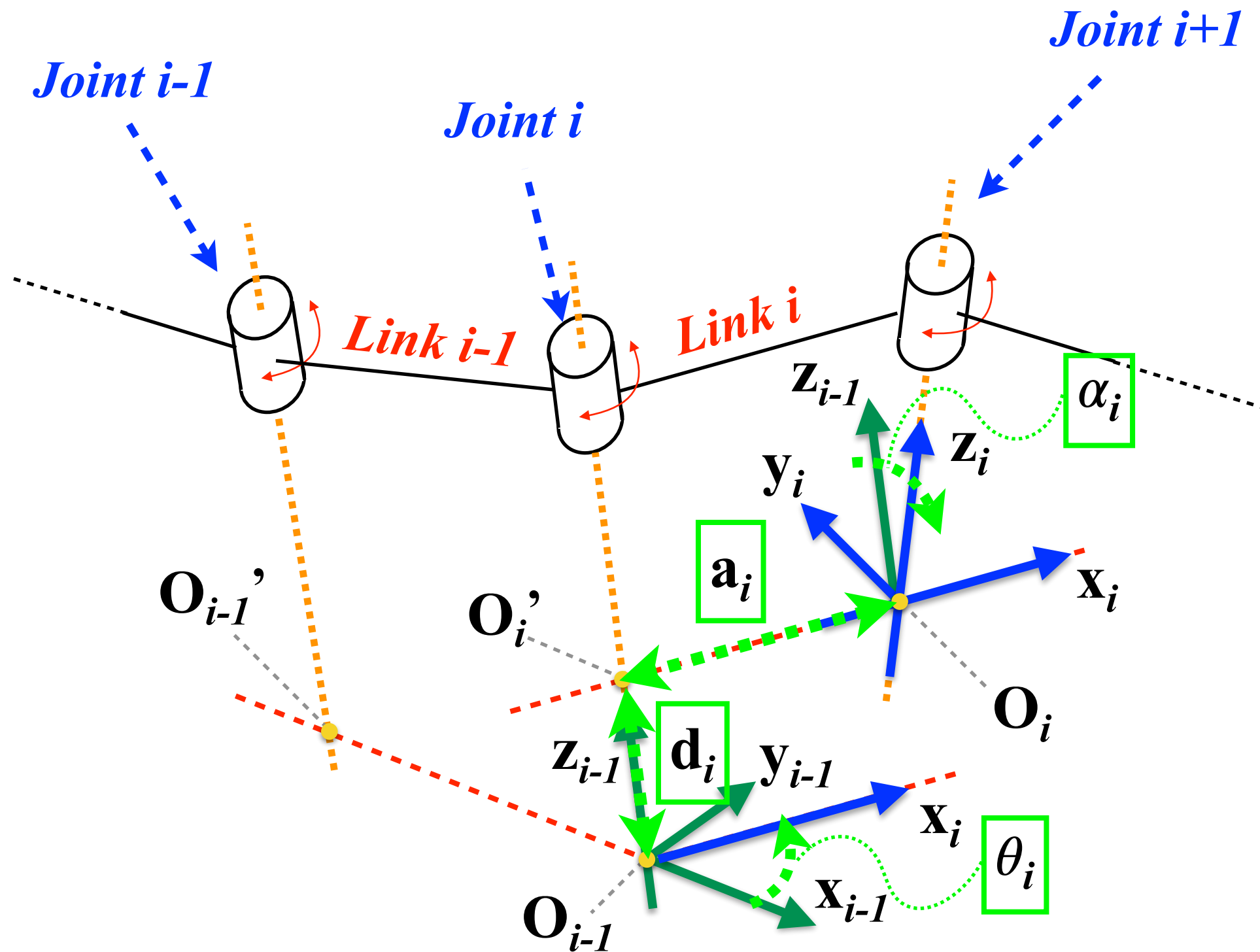
Denavit-Hartenberg Parameters



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Denavit-Hartenberg Parameters





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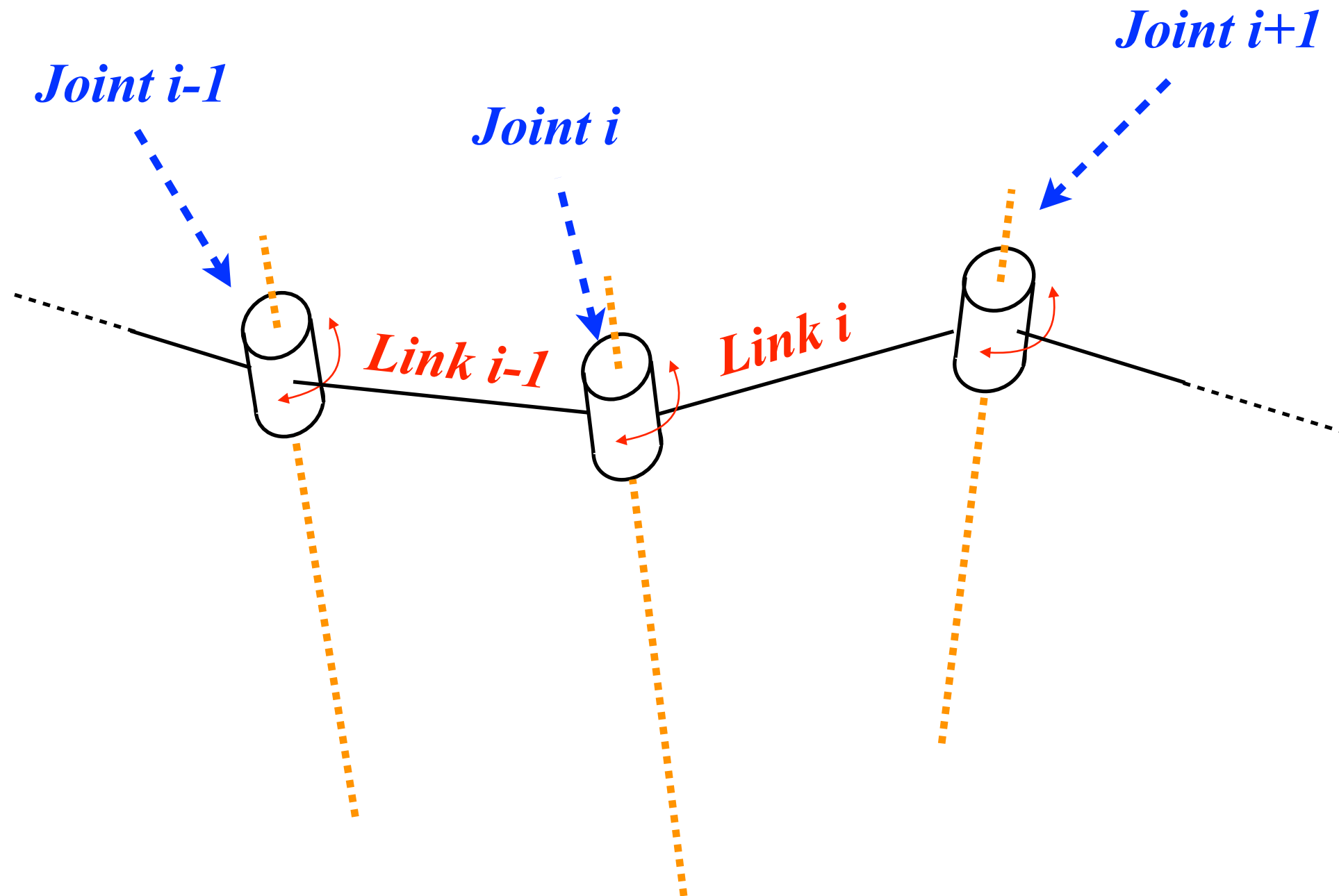
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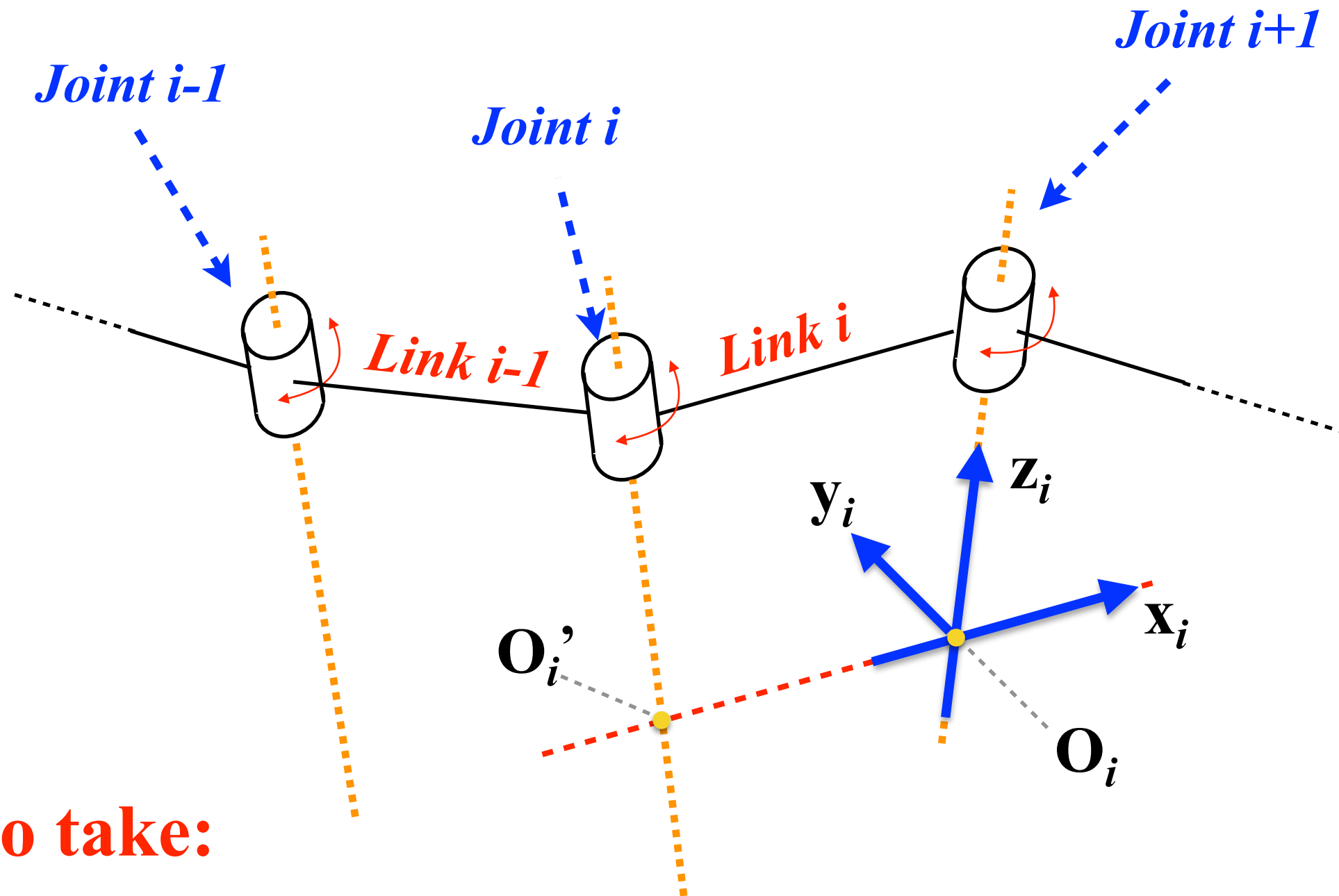


Summary

D-H Convention: Summary



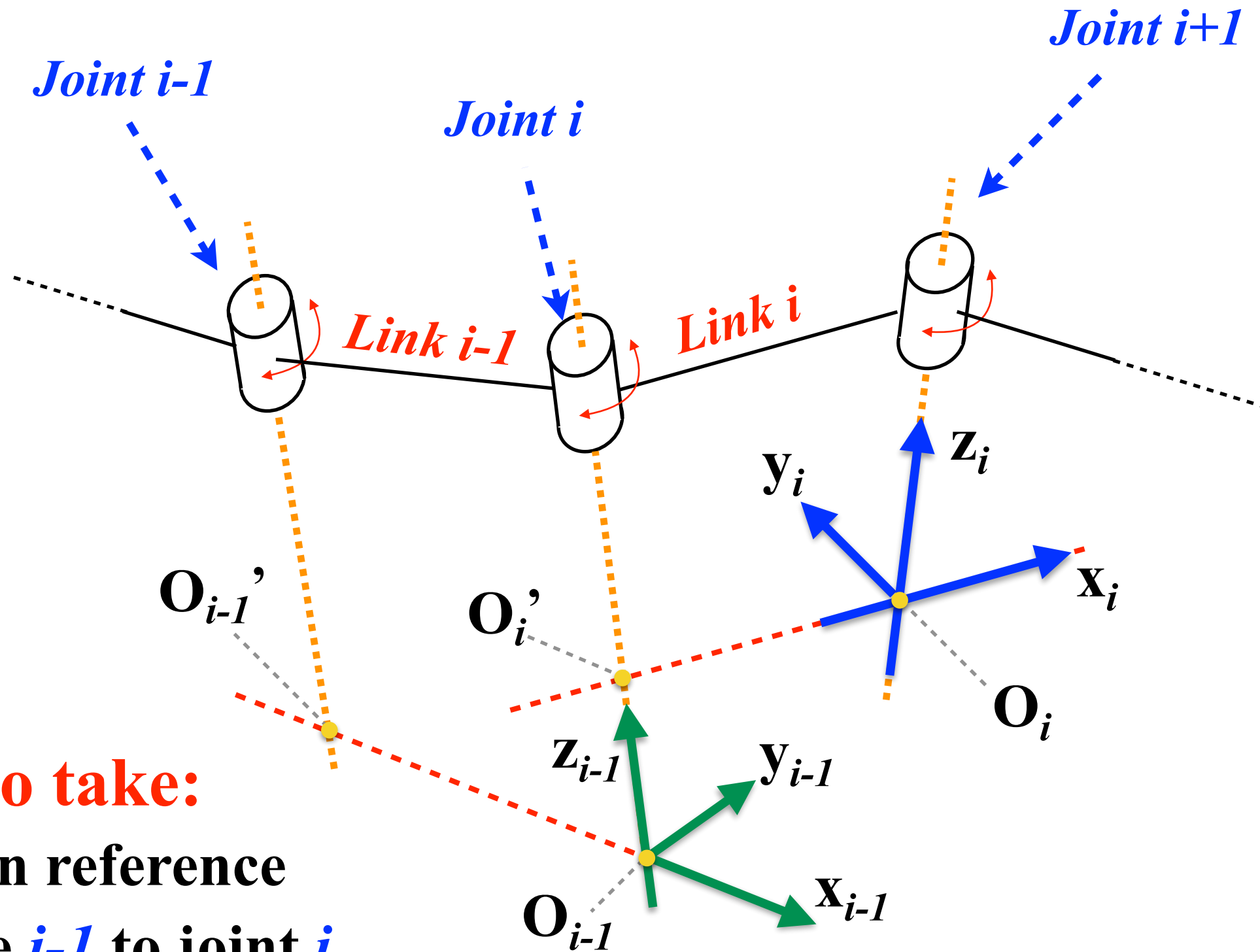
D-H Convention: Summary



Steps to take:

1. Assign reference frame *i* to joint *i+1*

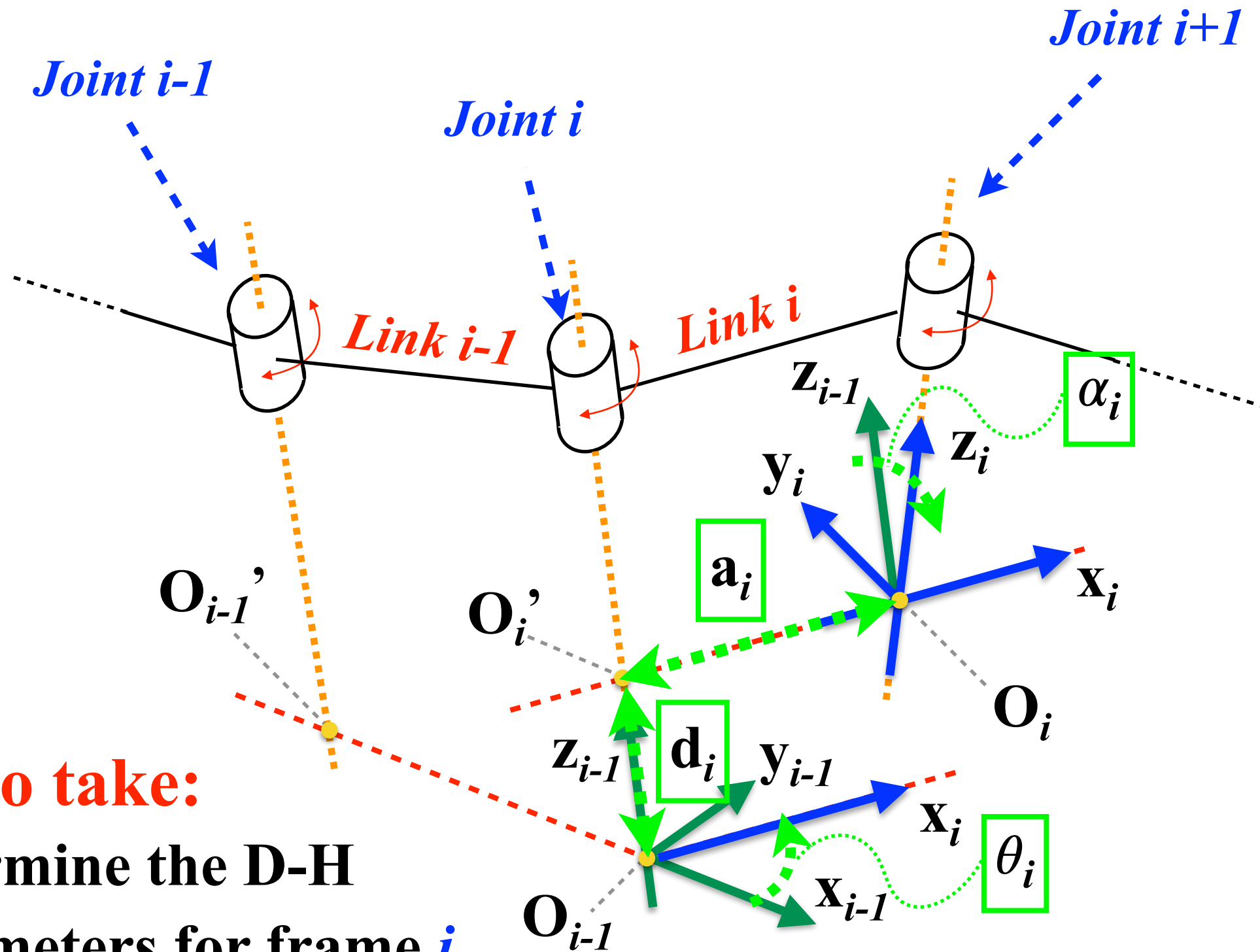
D-H Convention: Summary



Steps to take:

**2. Assign reference
frame *i-1* to joint *i***

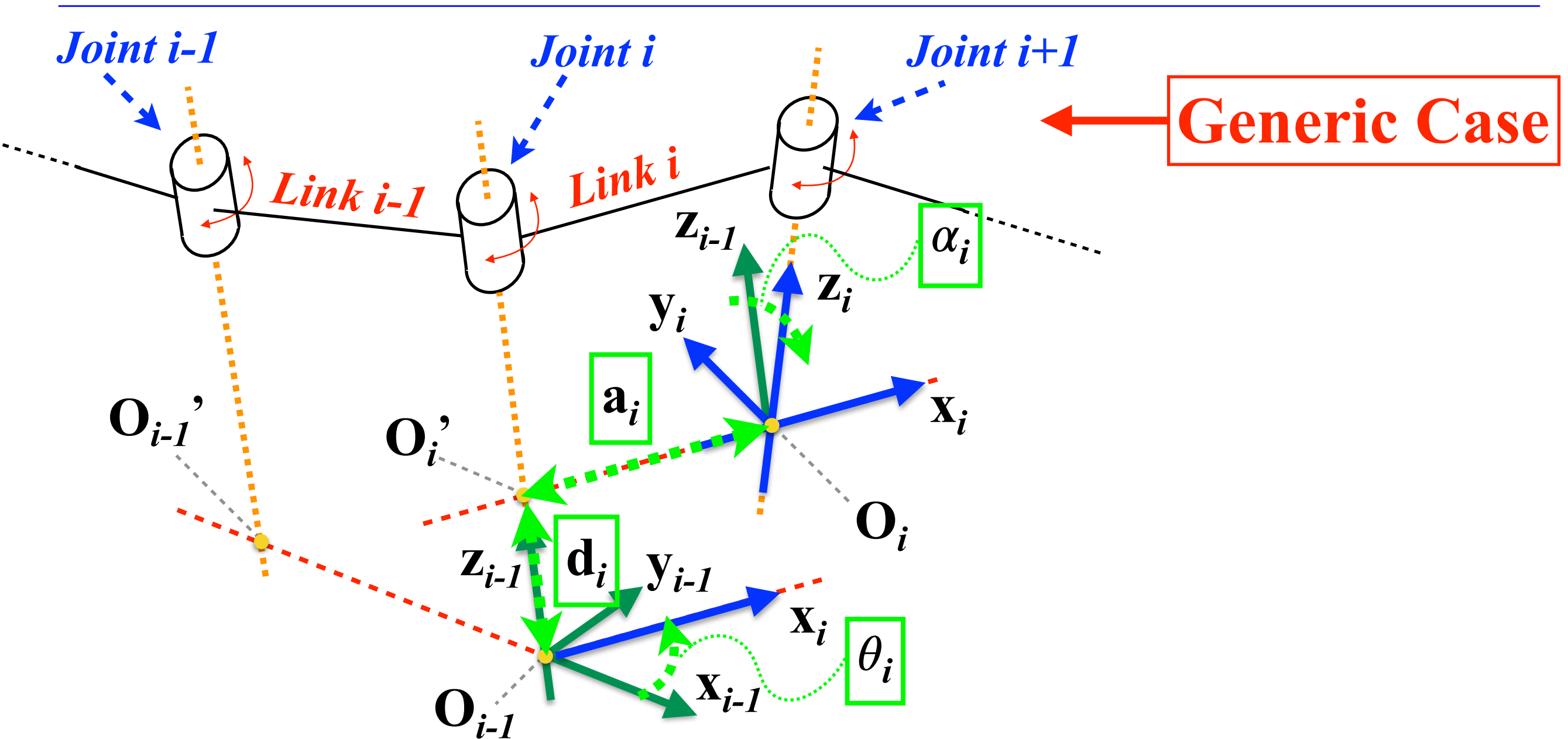
D-H Convention: Summary



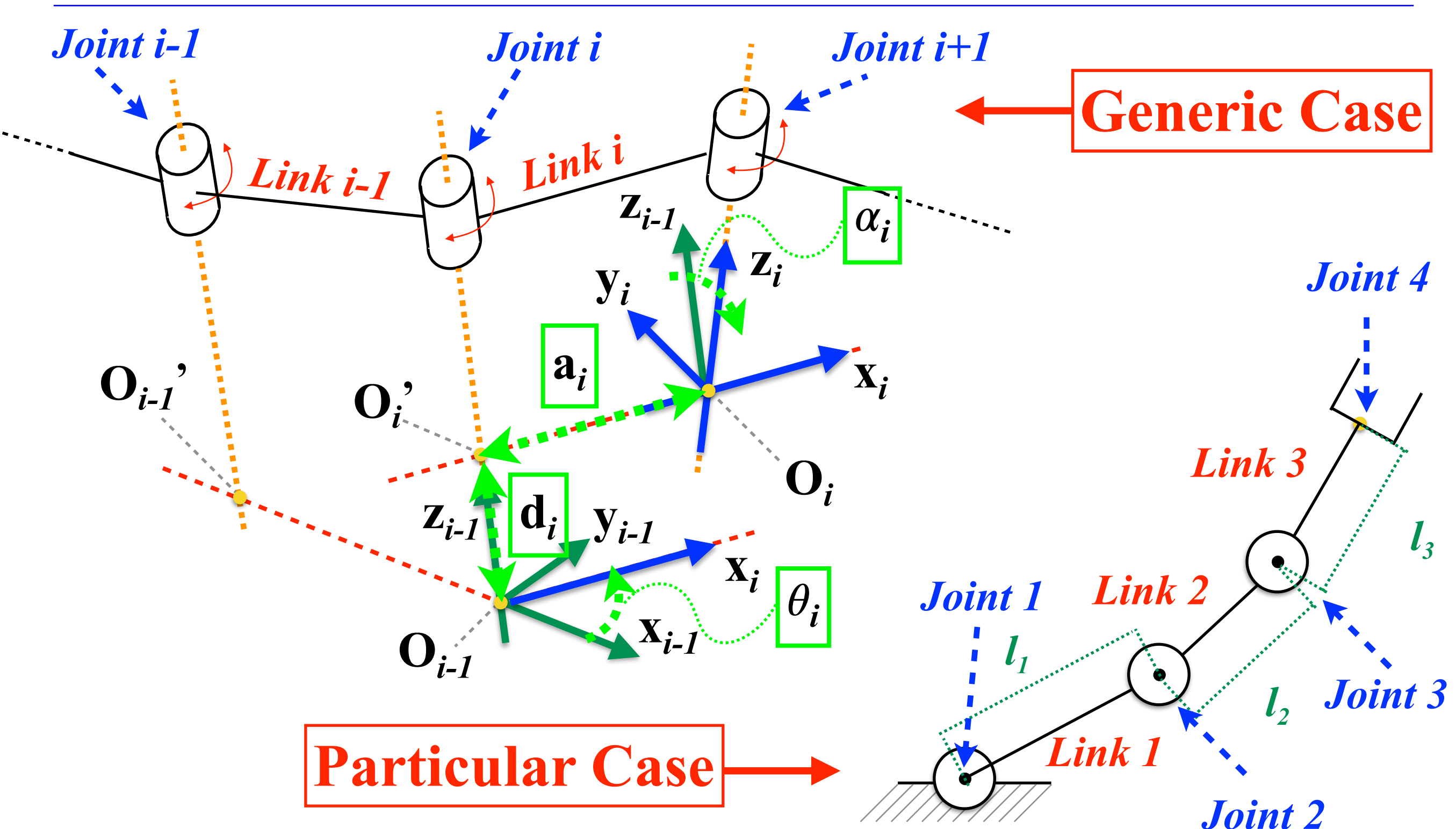
Steps to take:

3. Determine the D-H Parameters for frame i

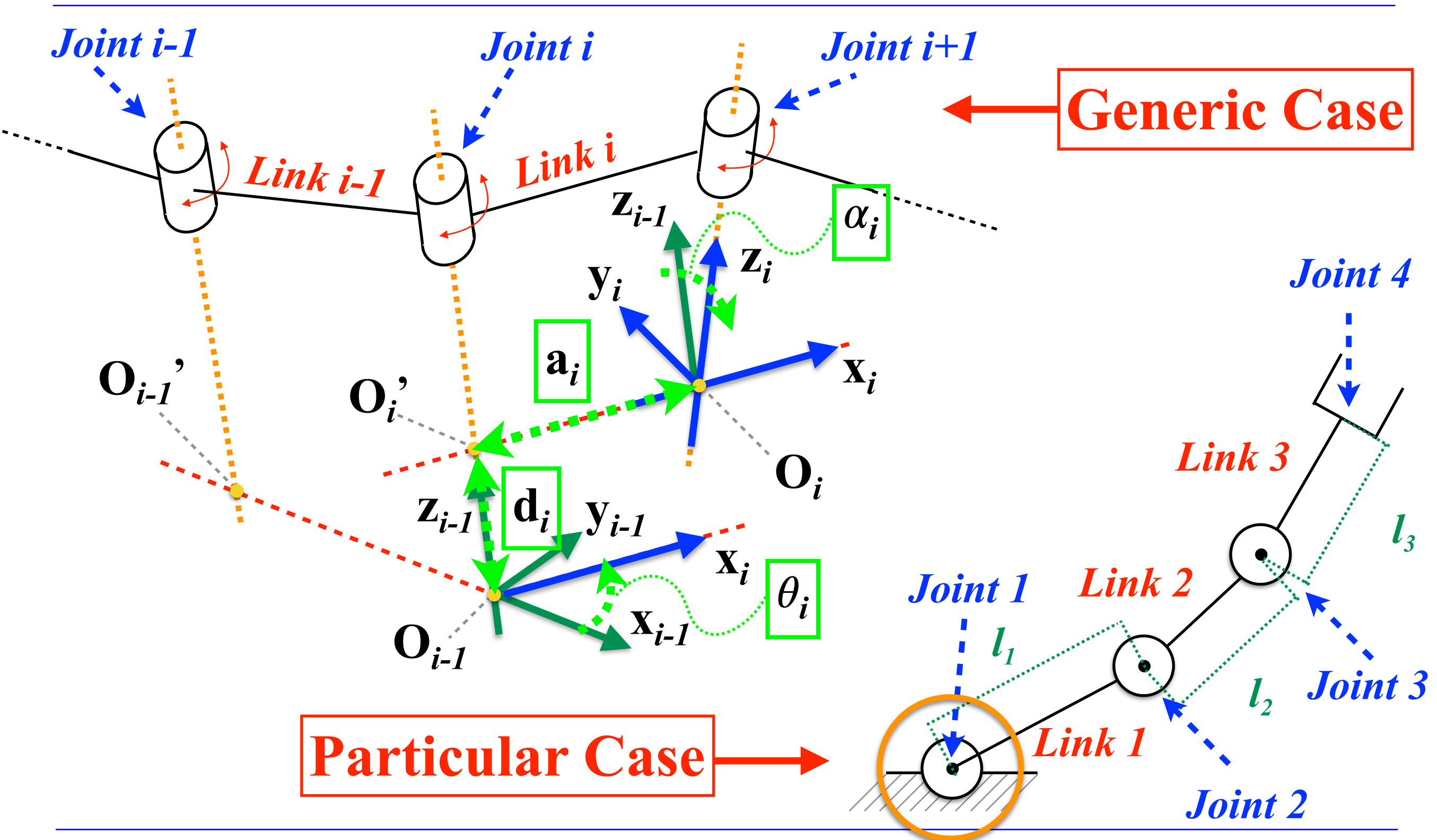
D-H Convention: Particular Cases



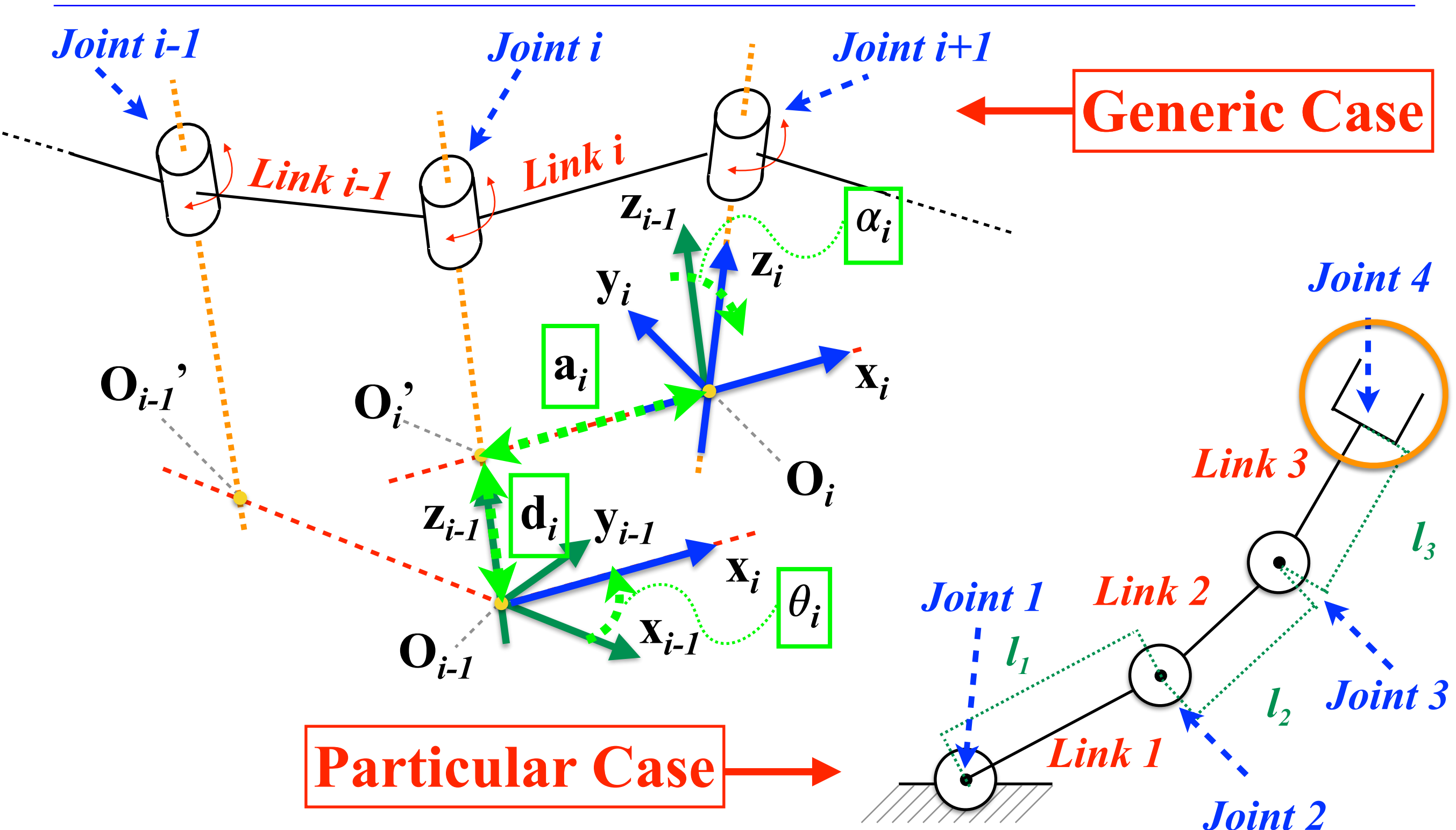
D-H Convention: Particular Cases



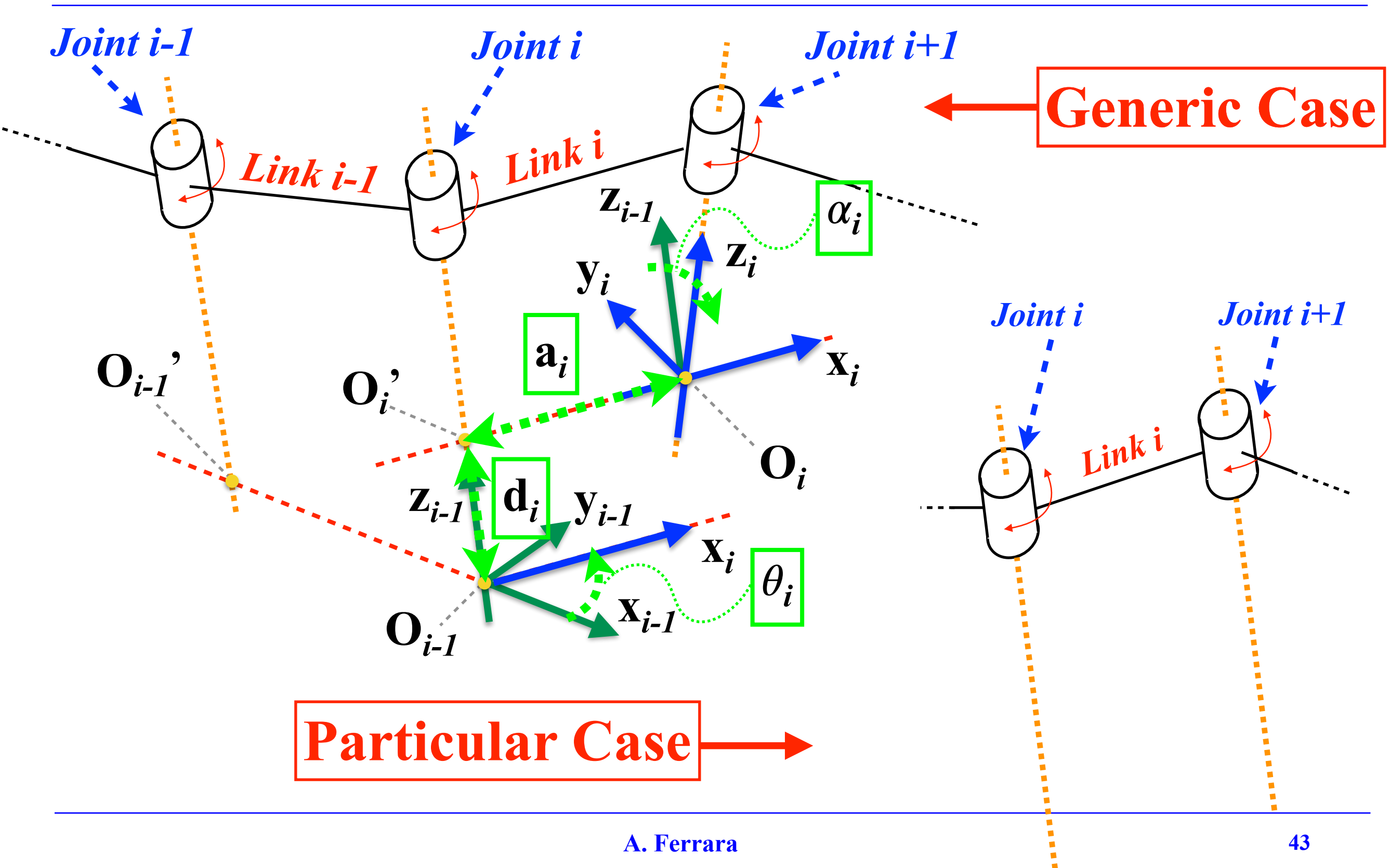
D-H Convention: Particular Cases



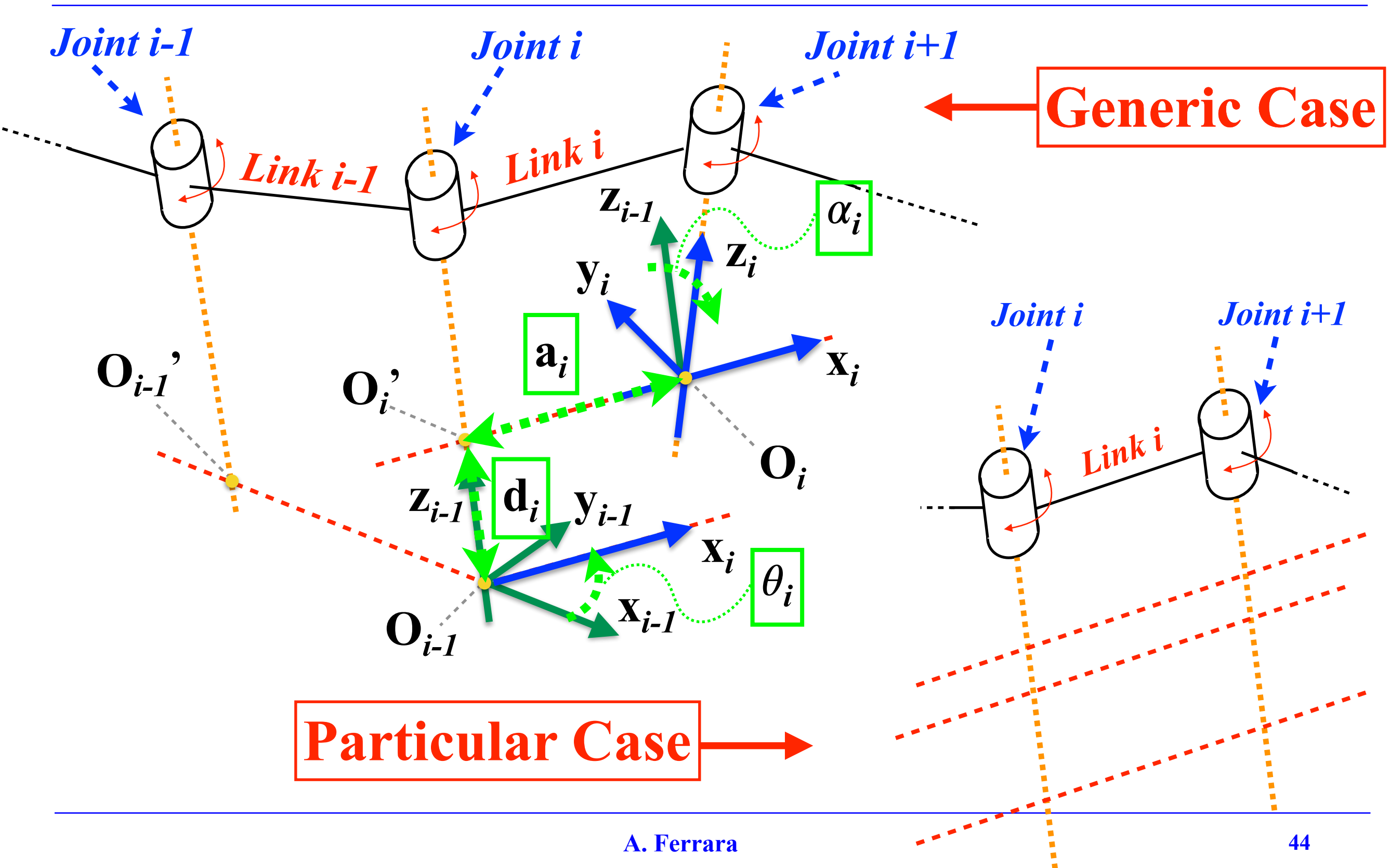
D-H Convention: Particular Cases



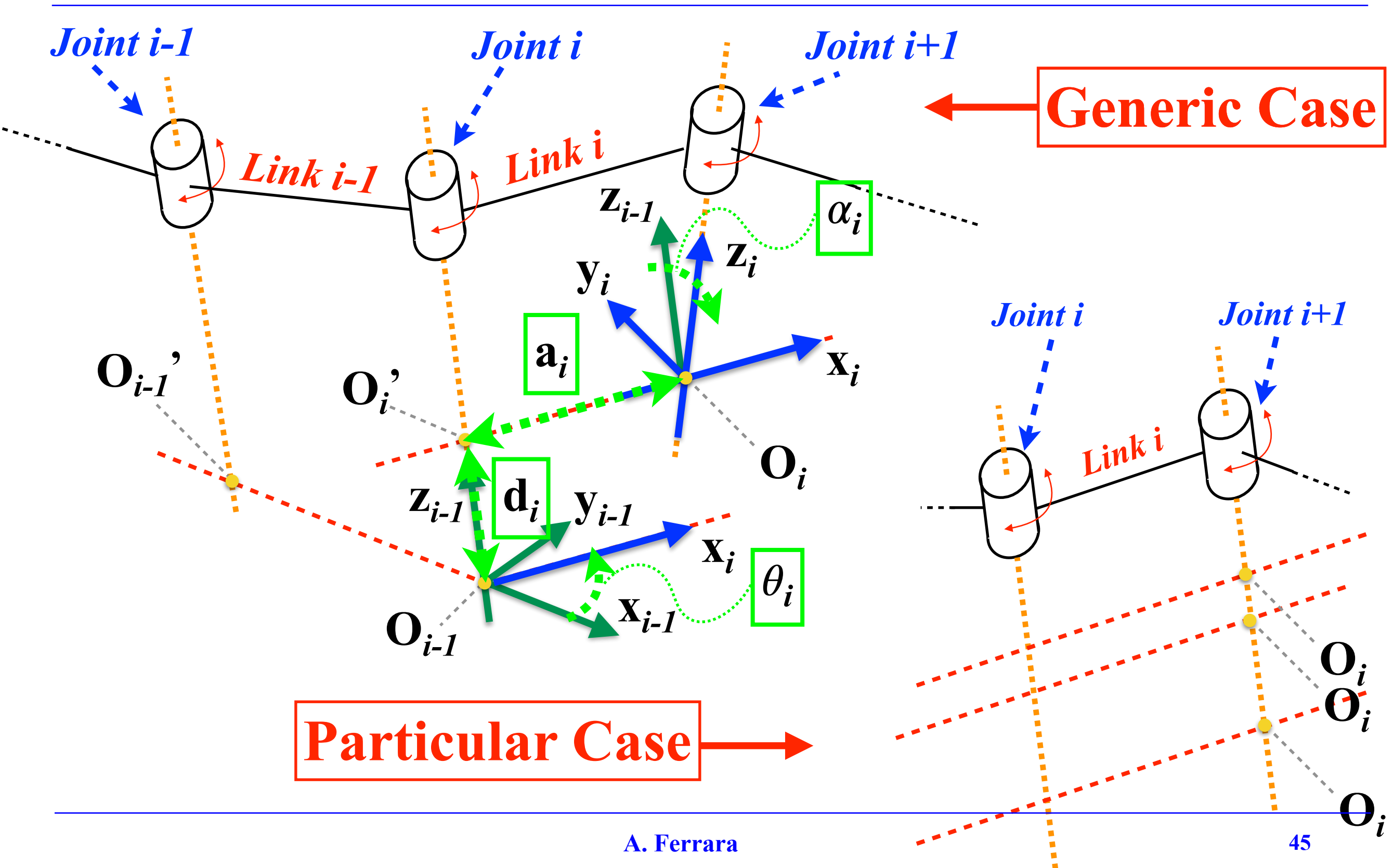
D-H Convention: Particular Cases



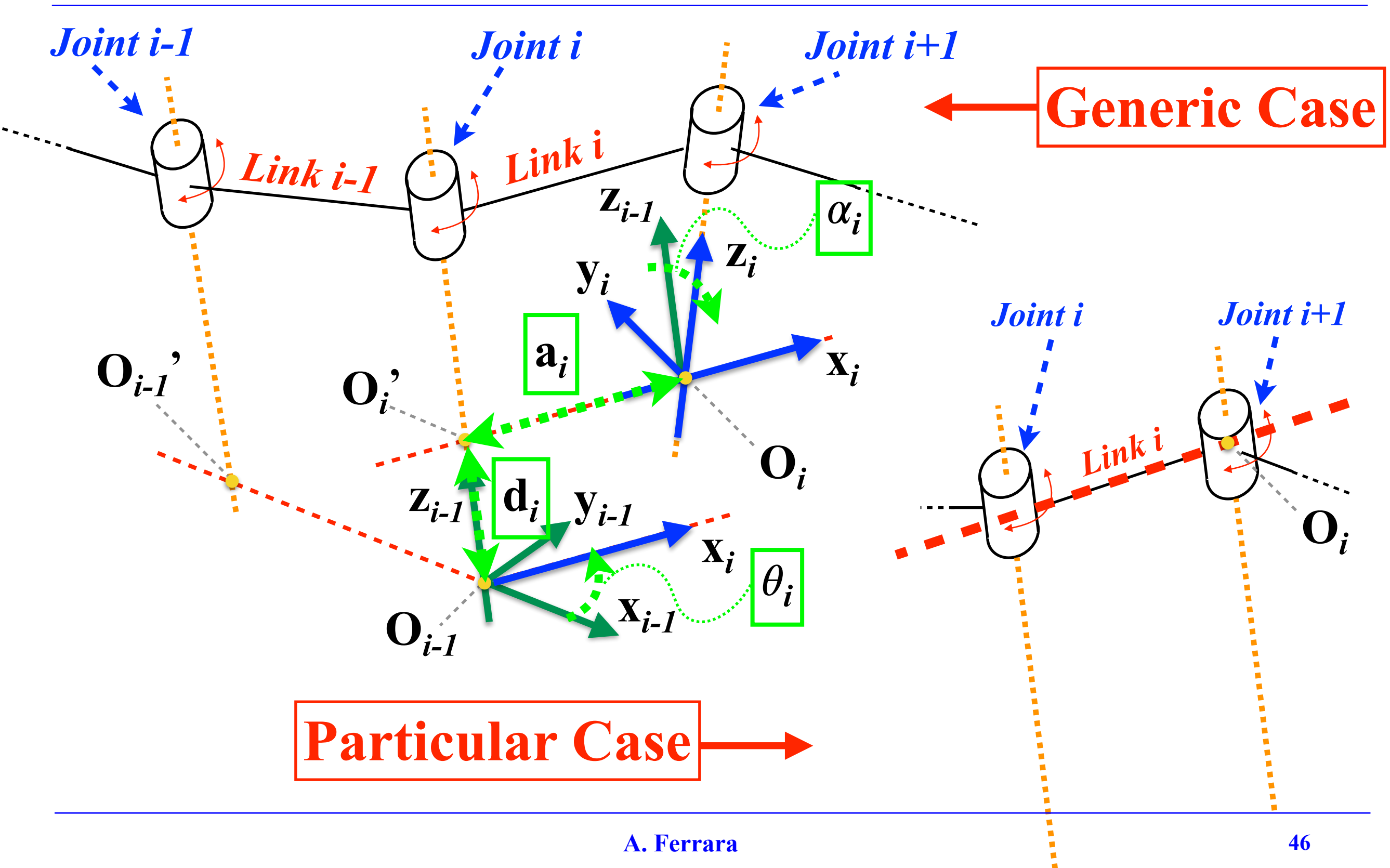
D-H Convention: Particular Cases



D-H Convention: Particular Cases



D-H Convention: Particular Cases





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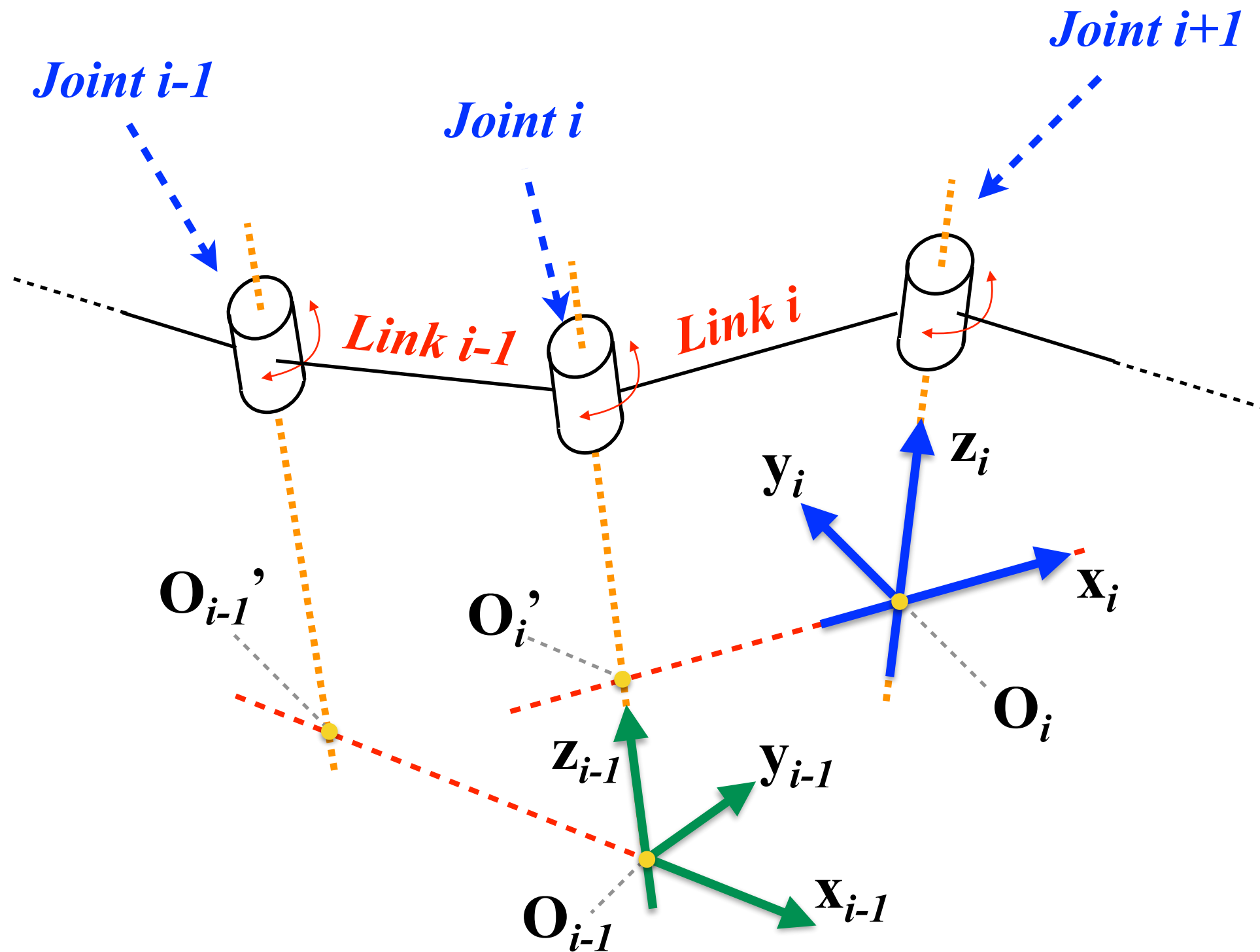
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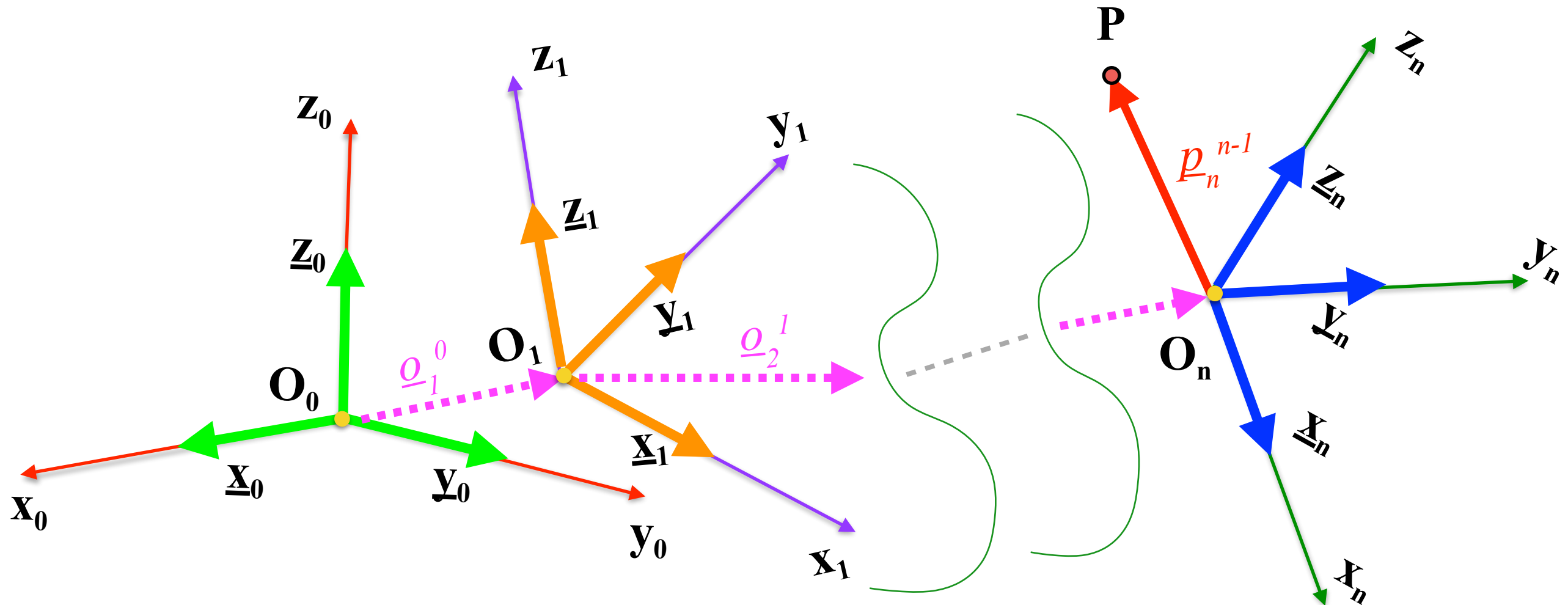
D-H Convention Based Approach

How to Describe the Direct Kinematics: Denavit-Hartenberg Convention Based Approach



In a previous lecture:

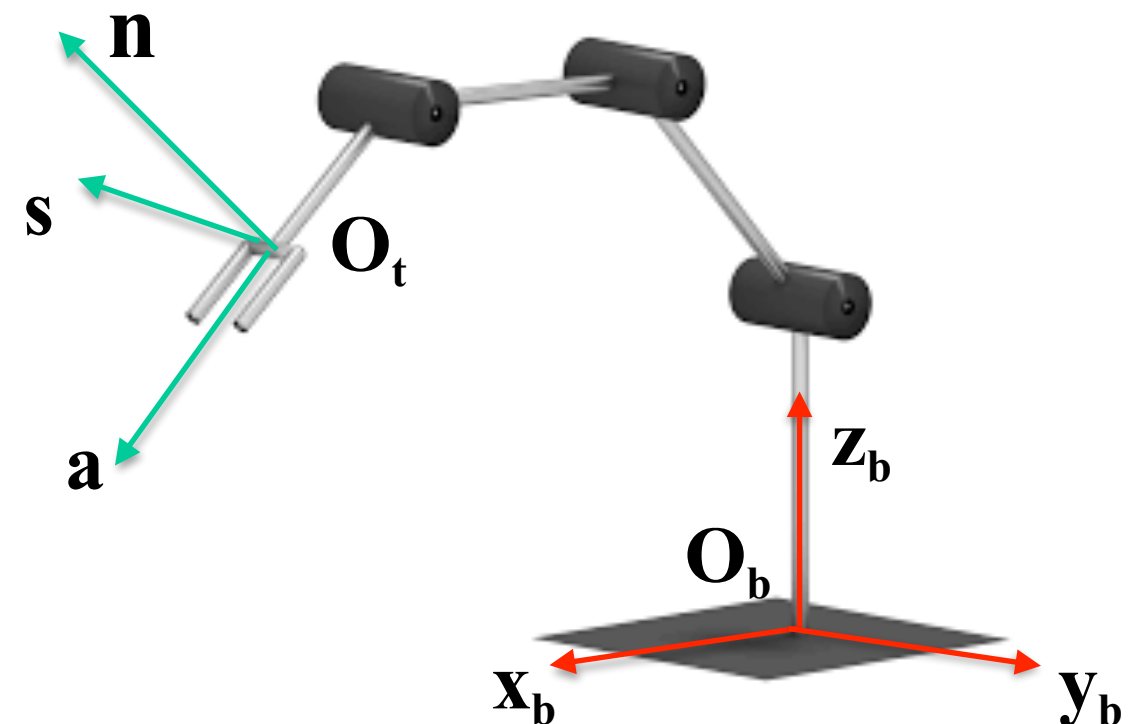
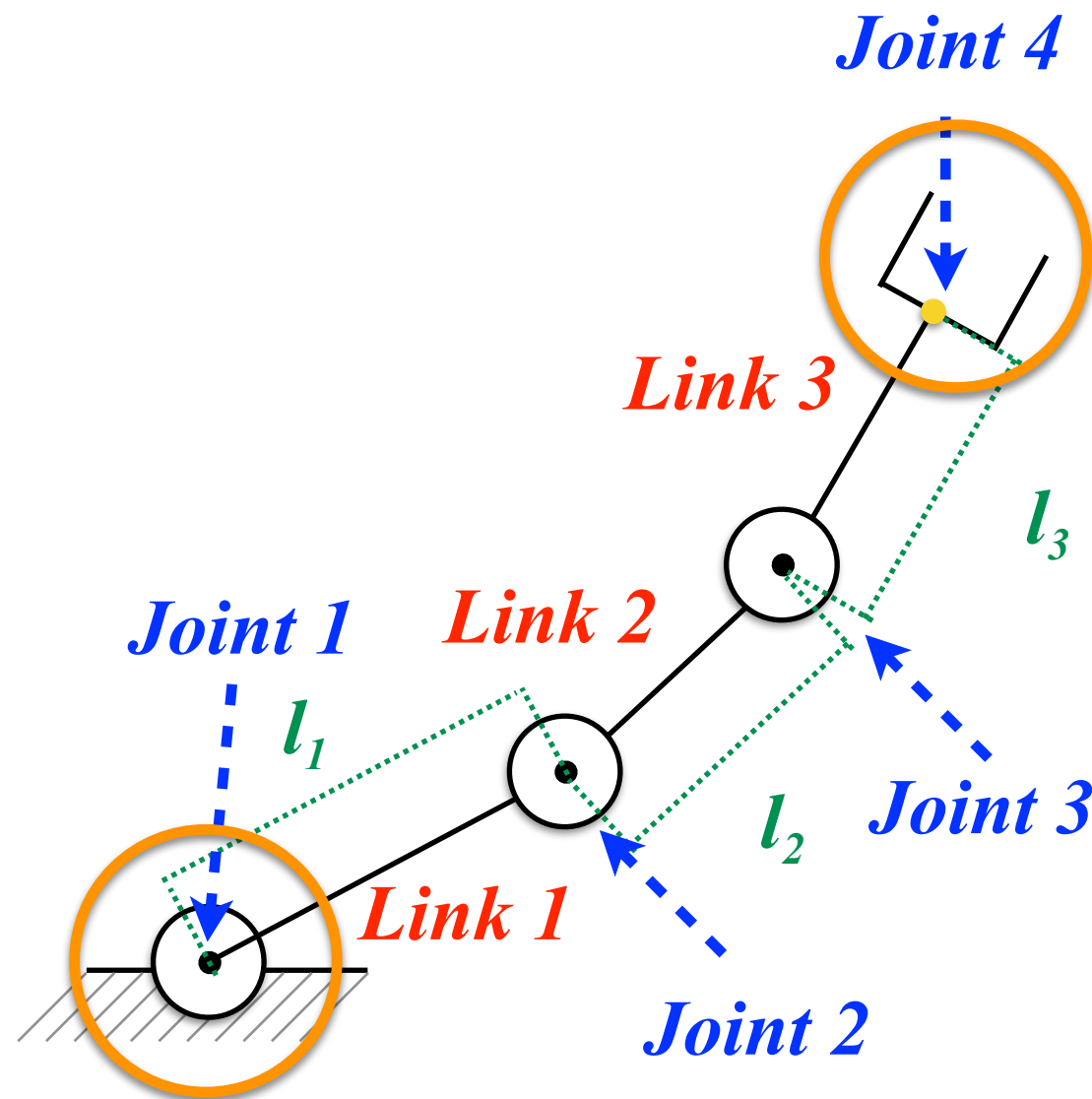
Composition of n Roto-Translations



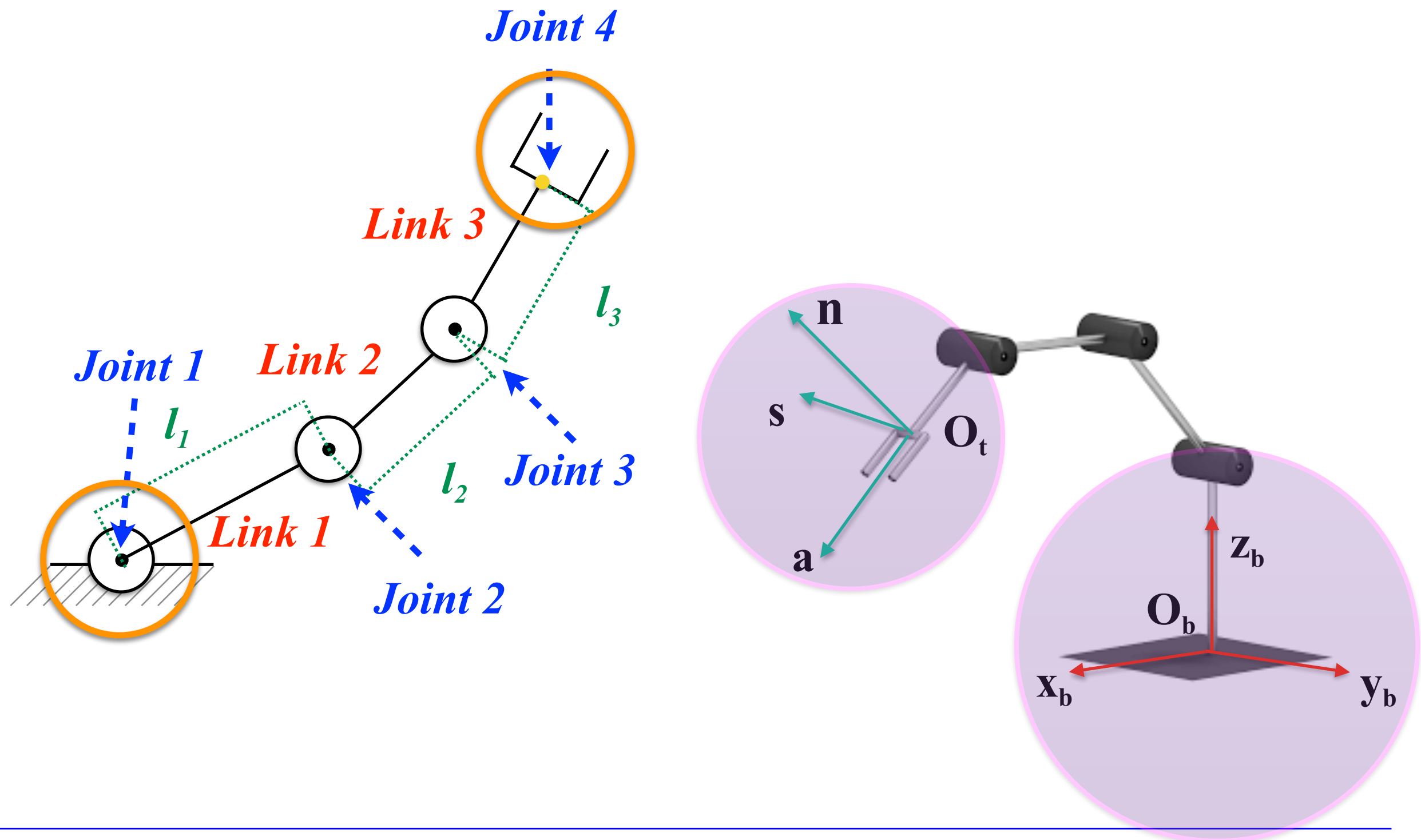
$$\underline{\tilde{p}}^0 = A_1^0 \cdot A_2^1 \cdots A_{n-1}^{n-2} \cdot A_n^{n-1} \cdot \underline{\tilde{p}}^n$$

$$A_n^0$$

How to Describe the Direct Kinematics: Denavit-Hartenberg Convention Based Approach

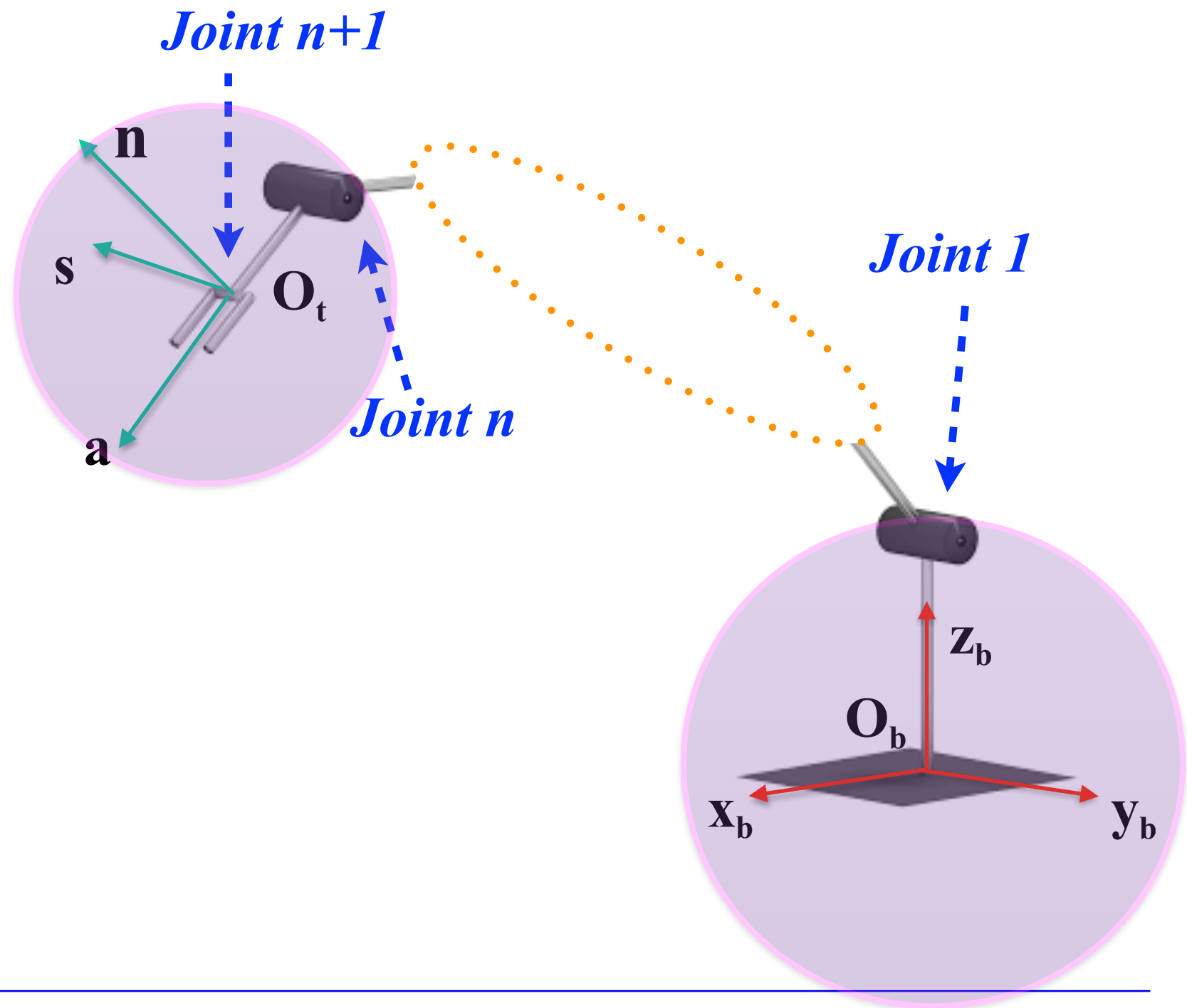


How to Describe the Direct Kinematics: Denavit-Hartenberg Convention Based Approach



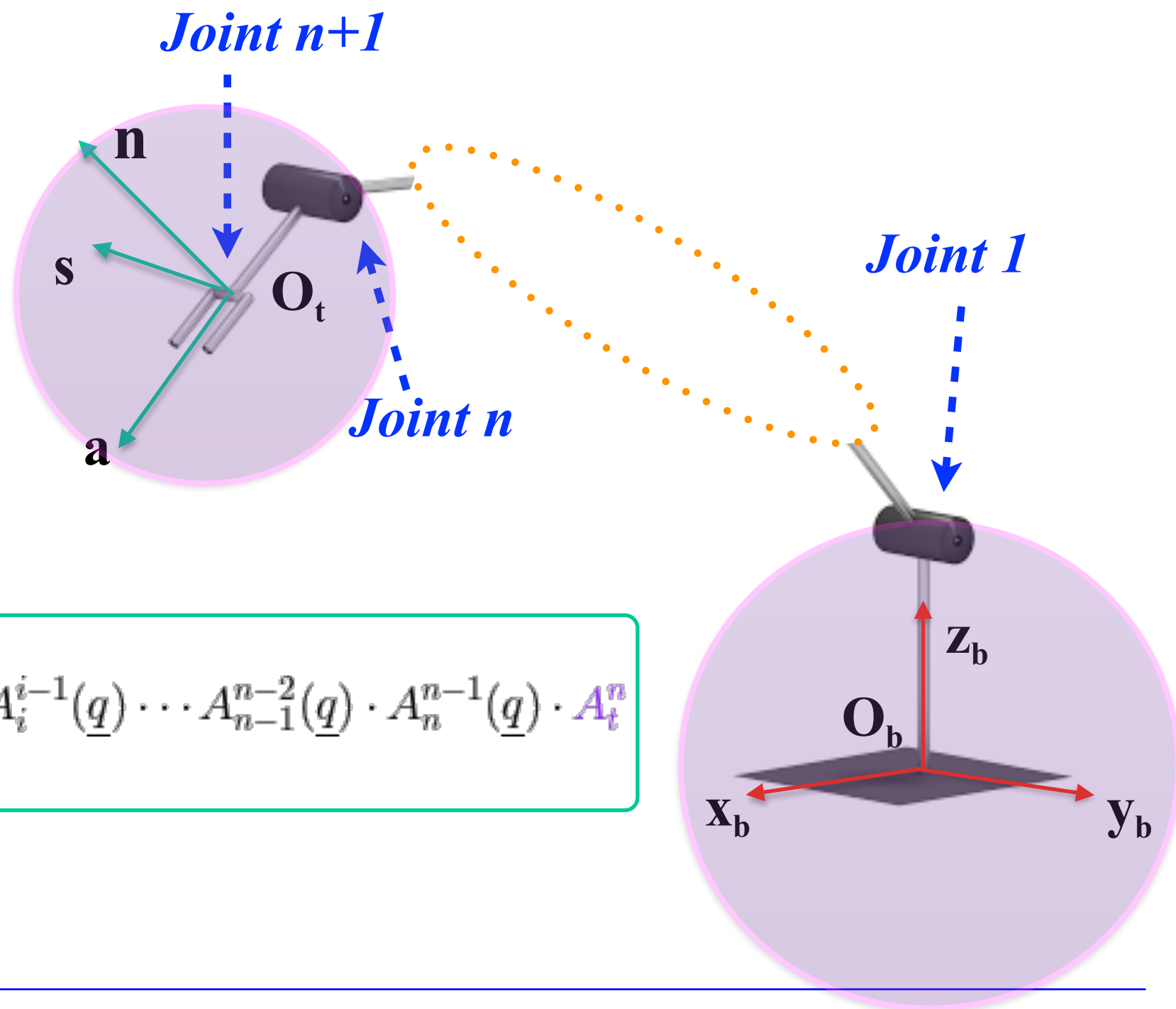
How to Describe the Direct Kinematics: Denavit-Hartenberg Convention Based Approach

More general case



How to Describe the Direct Kinematics: Denavit-Hartenberg Convention Based Approach

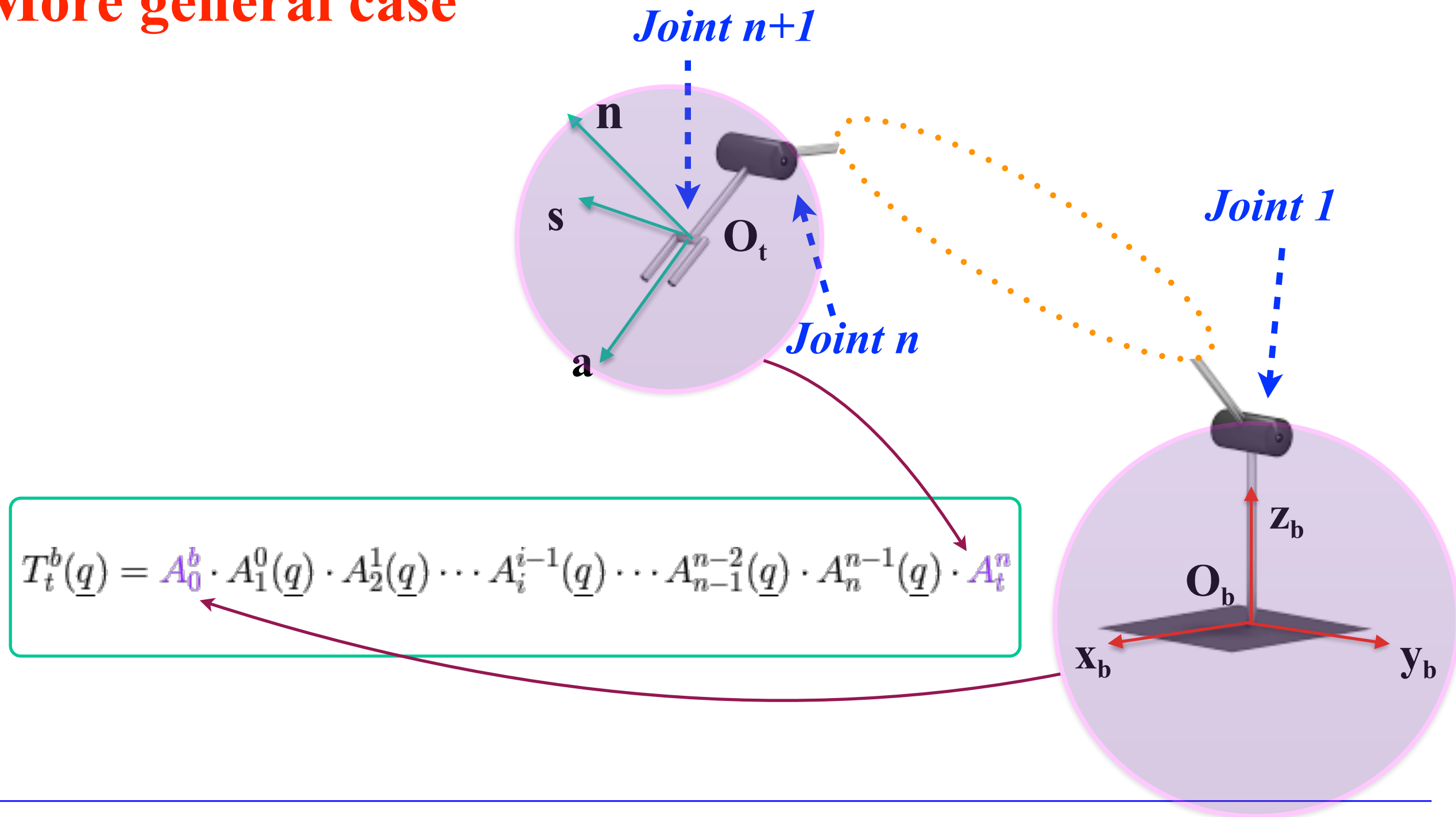
More general case



$$T_t^b(\underline{q}) = A_0^b \cdot A_1^0(\underline{q}) \cdot A_2^1(\underline{q}) \cdots A_i^{i-1}(\underline{q}) \cdots A_{n-1}^{n-2}(\underline{q}) \cdot A_n^{n-1}(\underline{q}) \cdot A_t^n$$

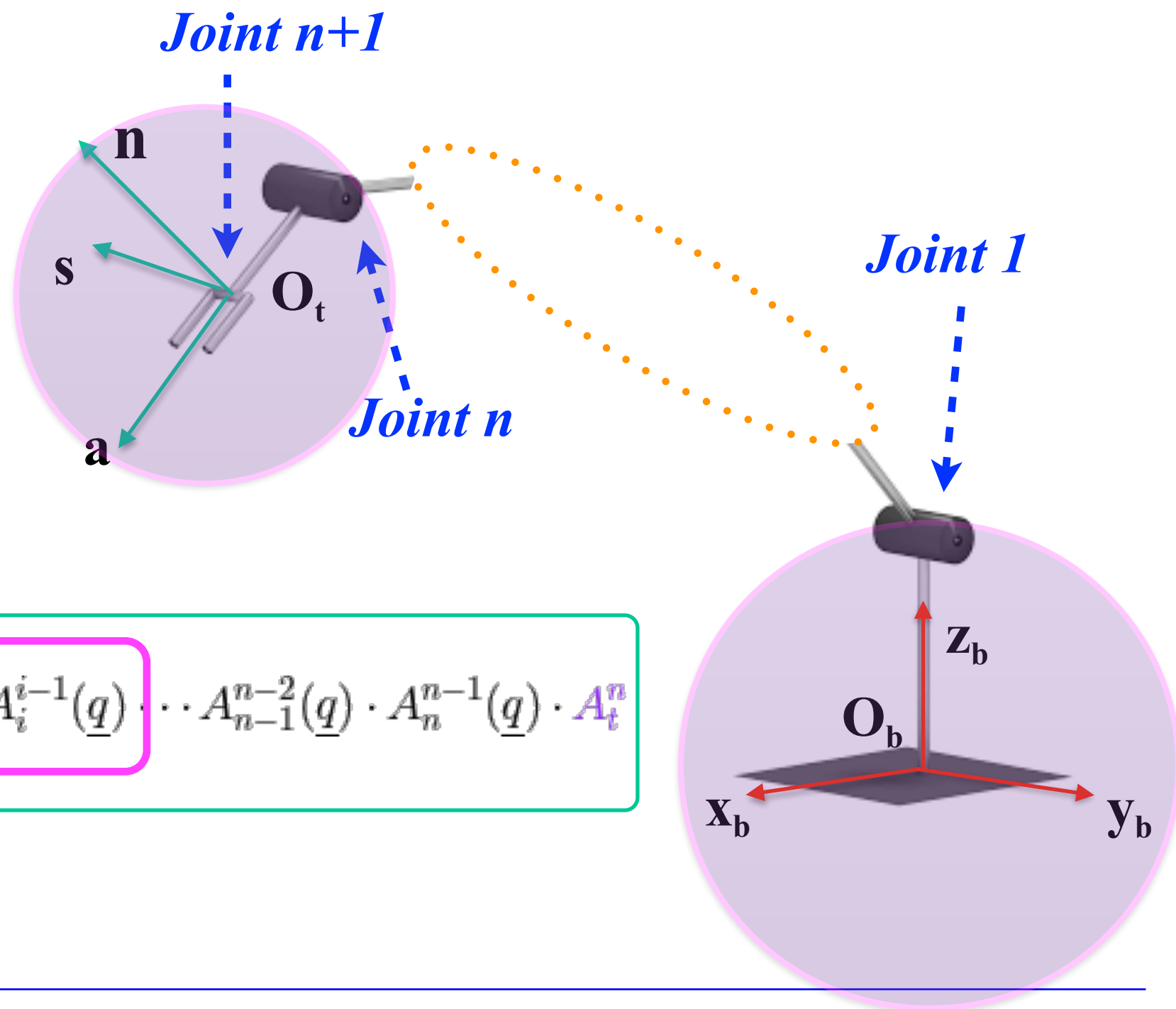
How to Describe the Direct Kinematics: Denavit-Hartenberg Convention Based Approach

More general case



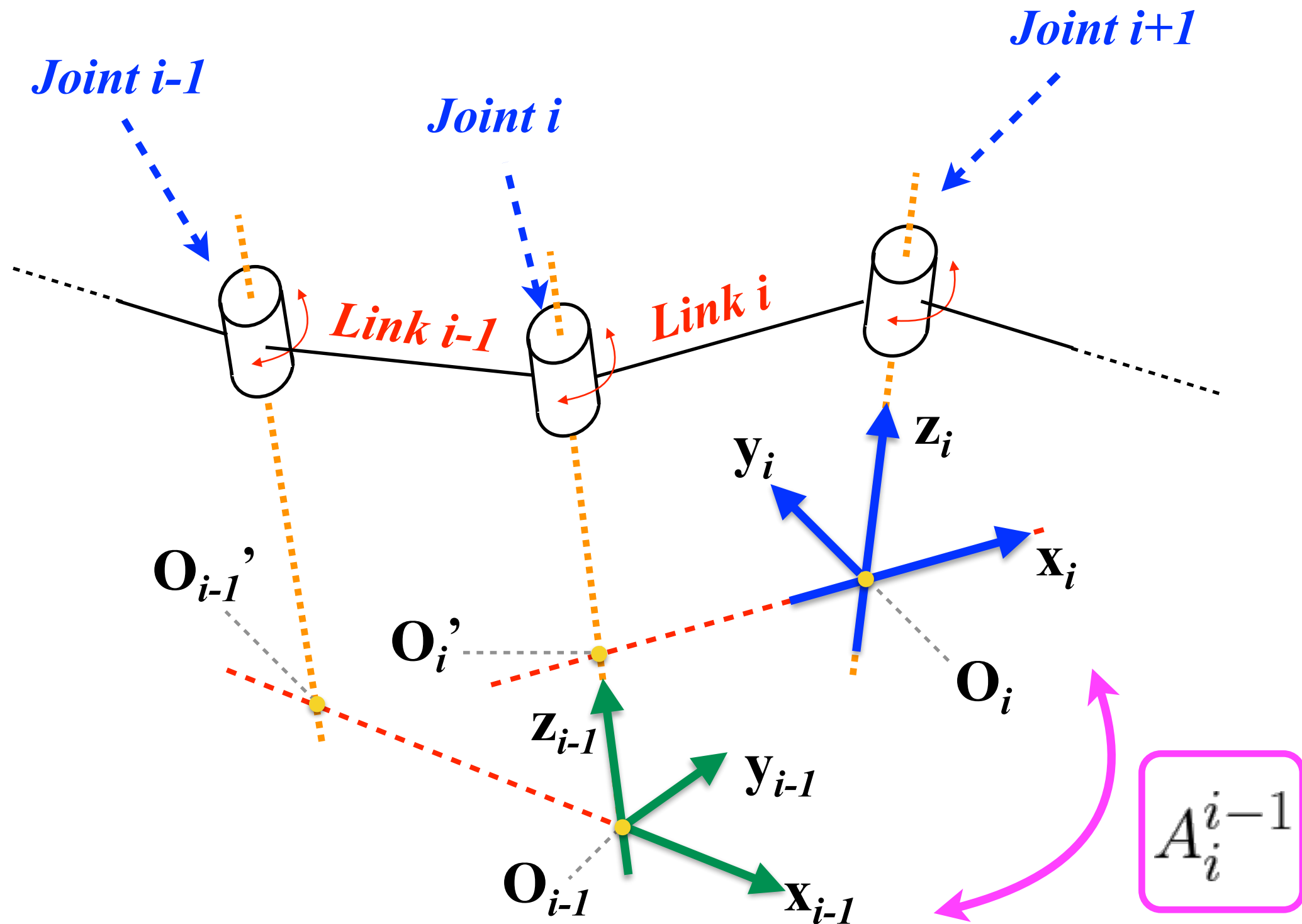
How to Describe the Direct Kinematics: Denavit-Hartenberg Convention Based Approach

More general case

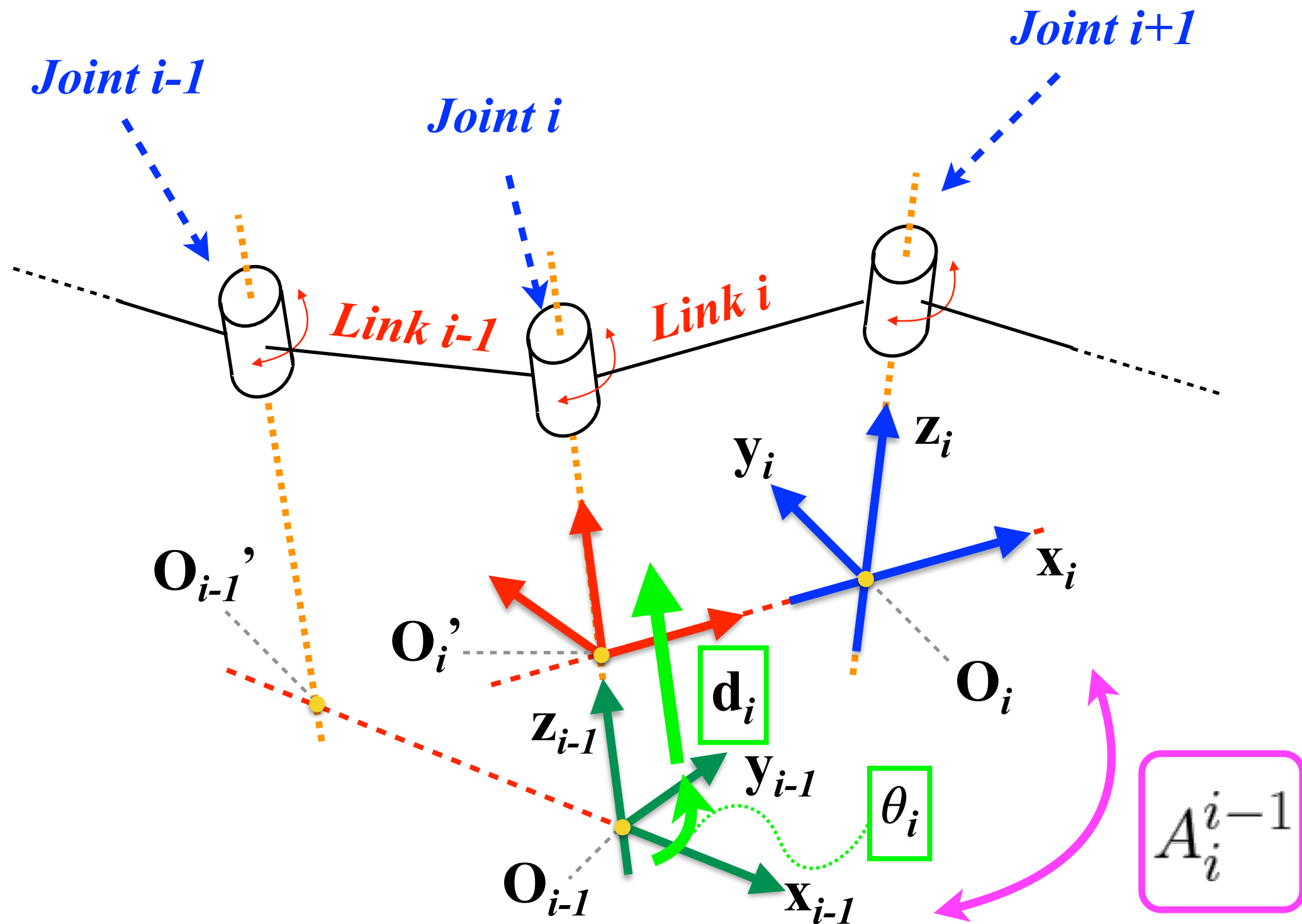


$$T_t^b(\underline{q}) = A_0^b \cdot A_1^0(\underline{q}) \cdot A_2^1(\underline{q}) \cdots A_i^{i-1}(\underline{q}) \cdots A_{n-1}^{n-2}(\underline{q}) \cdot A_n^{n-1}(\underline{q}) \cdot A_t^n$$

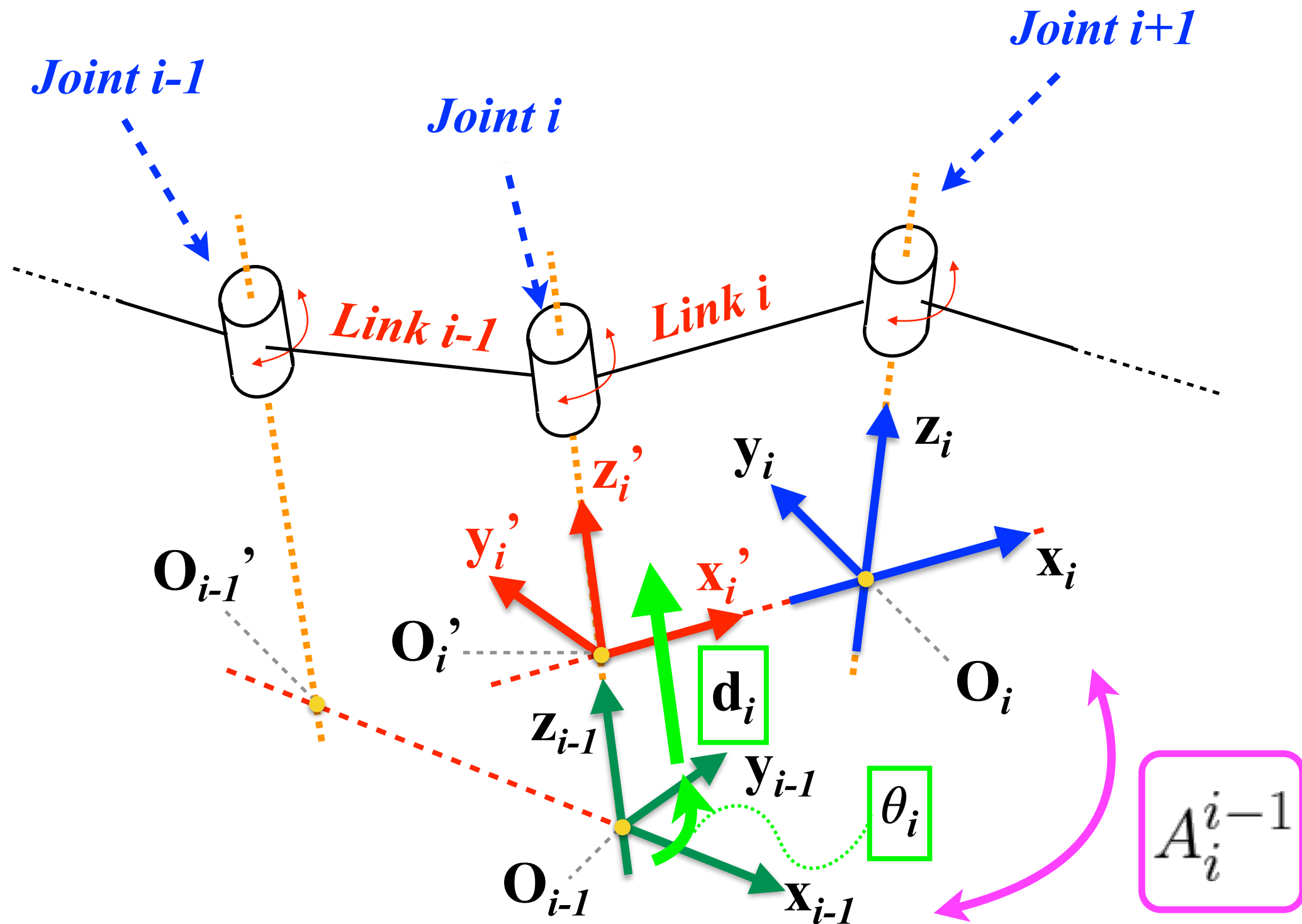
How to compute: A_i^{i-1}



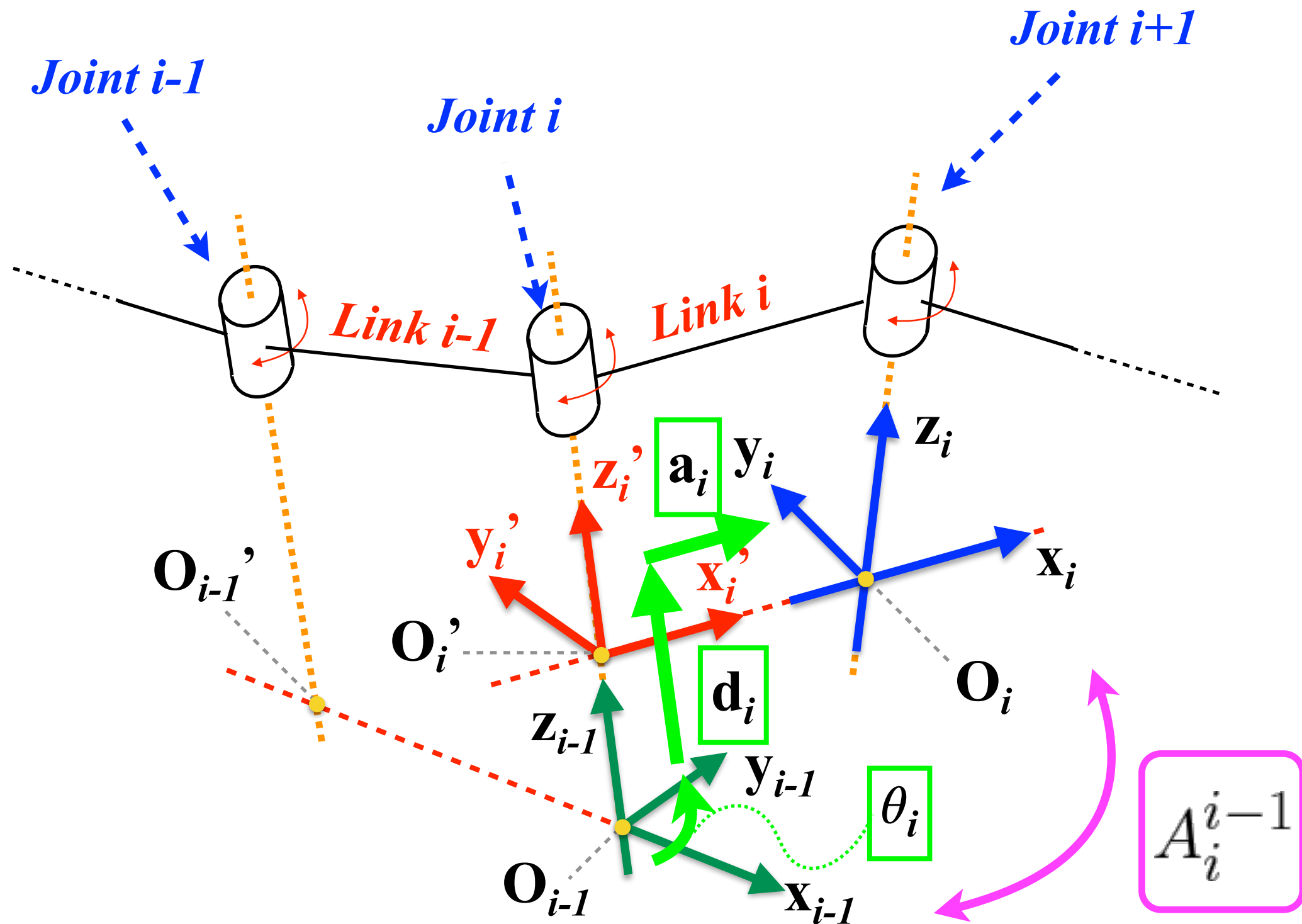
How to compute: A_i^{i-1}



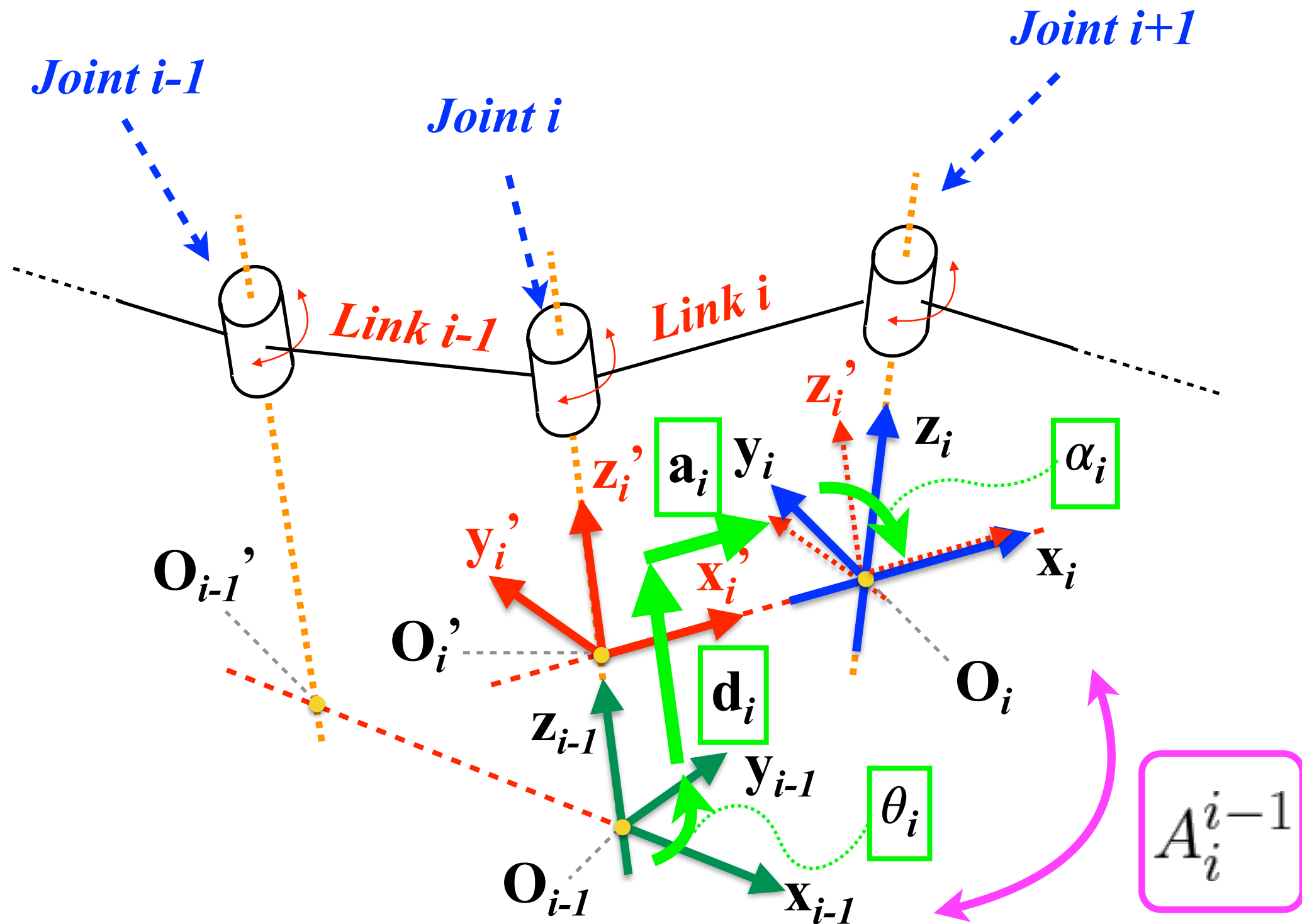
How to compute: A_i^{i-1}



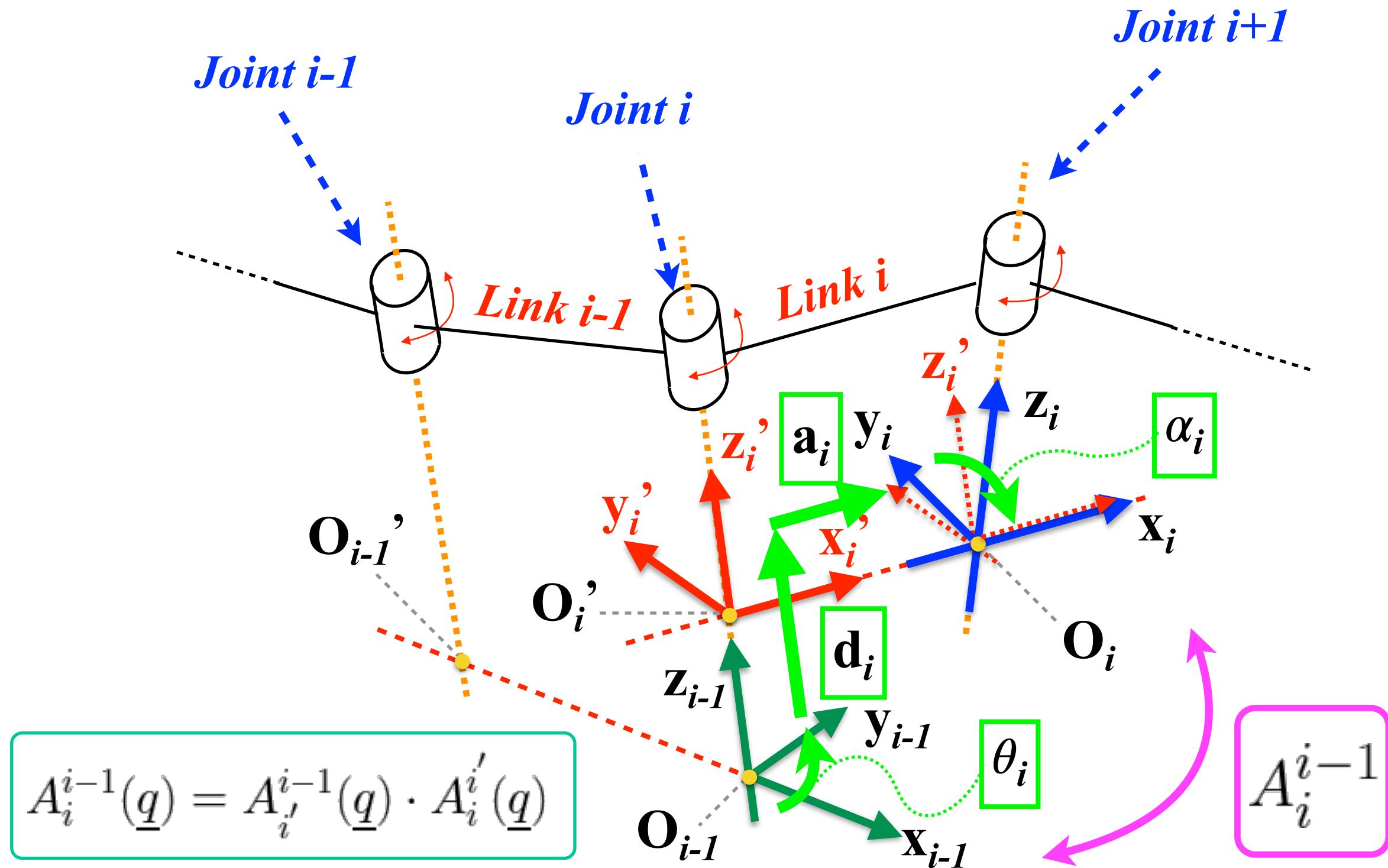
How to compute: A_i^{i-1}



How to compute: A_i^{i-1}



How to compute: A_i^{i-1}



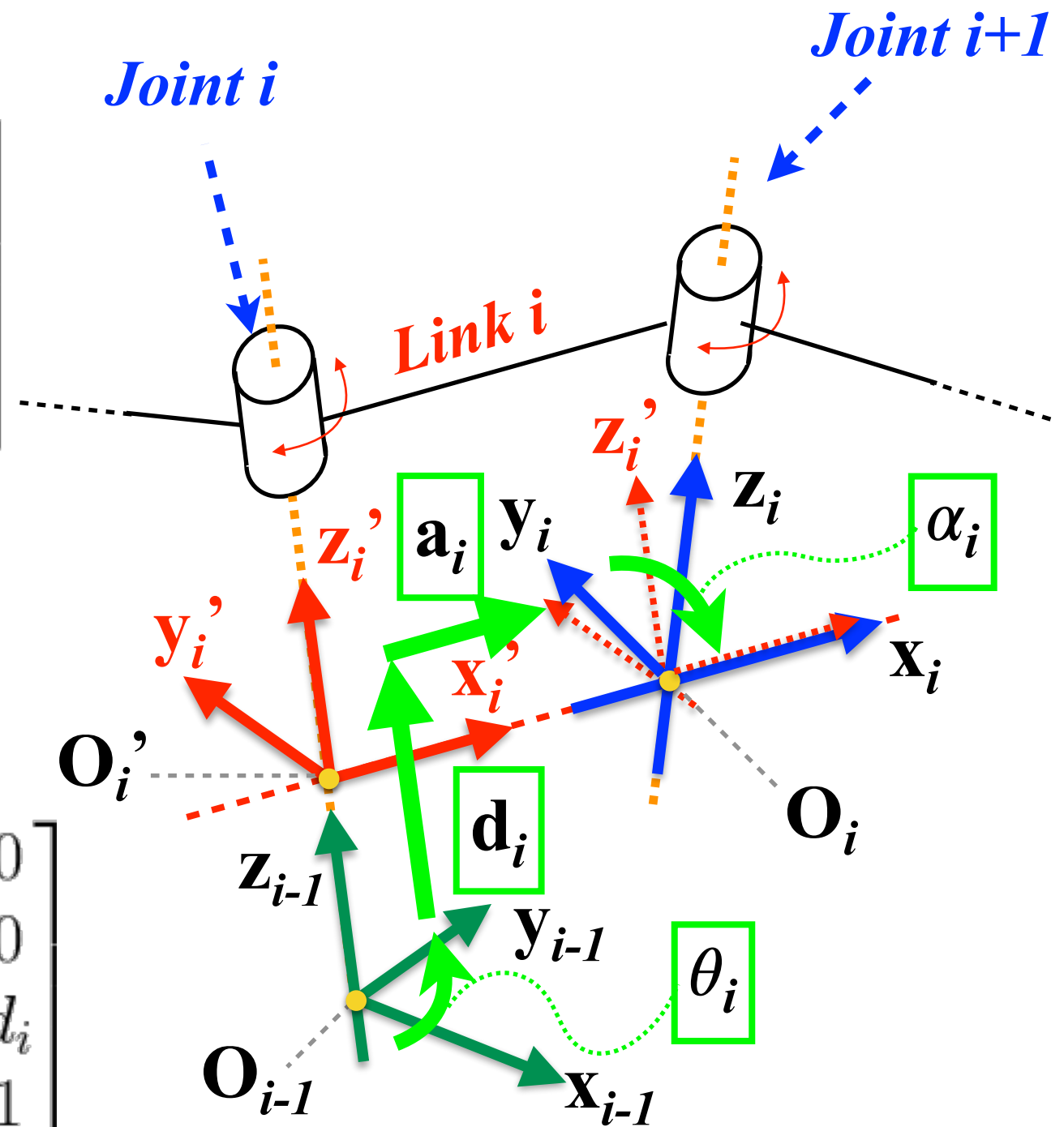
How to compute: A_i^{i-1}

$$A_i^{i-1}(\underline{q}) = A_{i'}^{i-1}(\underline{q}) \cdot A_i^{i'}(\underline{q})$$

$$A_{i'}^{i-1}(\underline{q}) = \begin{bmatrix} R_{i'}^{i-1}(\underline{q}) & \underline{o}_{i'}^{i-1}(\underline{q}) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} R_z(\theta_i) & \underline{o}_{i'}^{i-1}(\underline{q}) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

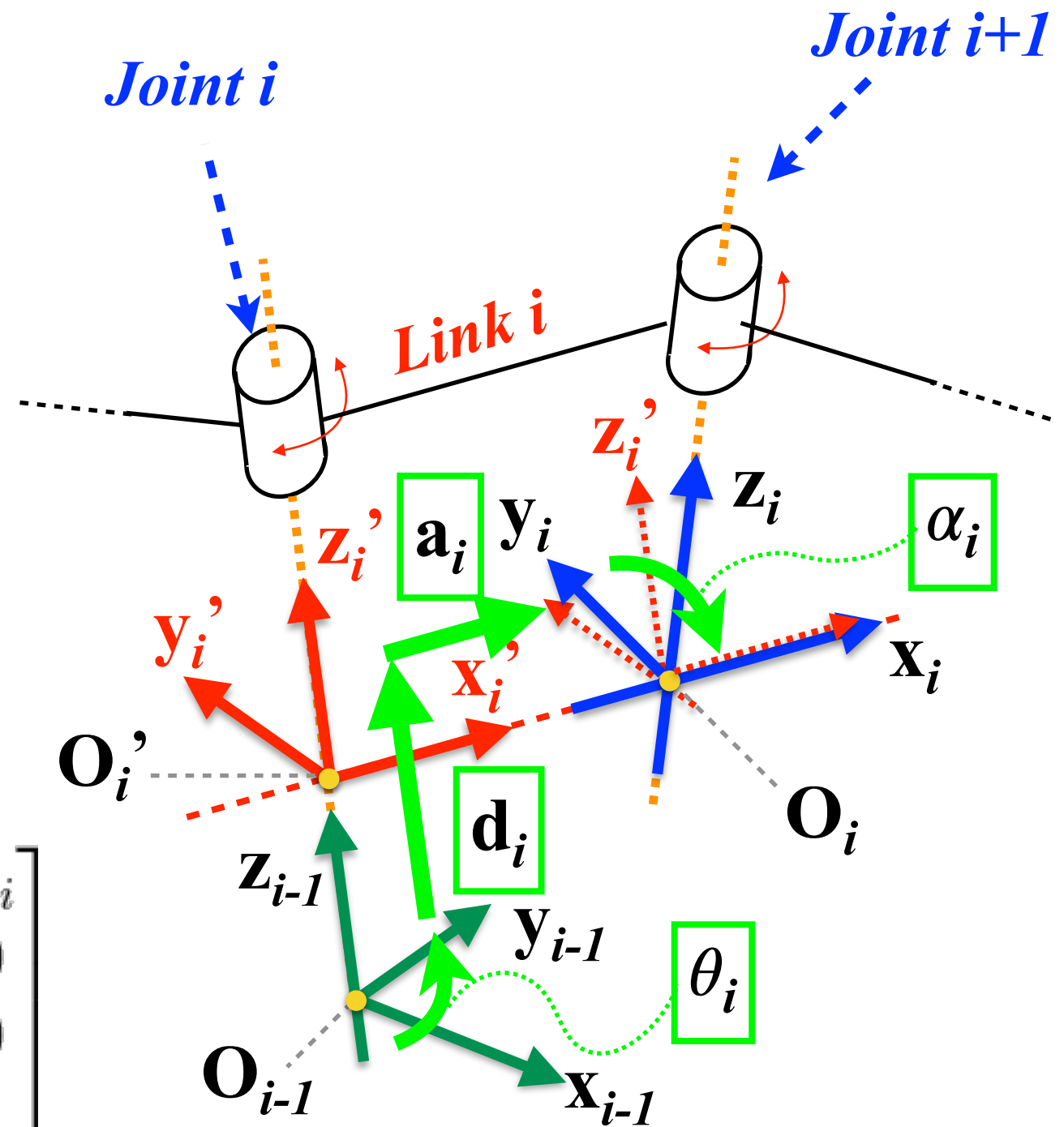
$$= \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) & 0 & 0 \\ \sin(\theta_i) & \cos(\theta_i) & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



How to compute: A_i^{i-1}

$$A_i^{i-1}(\underline{q}) = A_i^{i-1}(\underline{q}) \cdot A_i^{i'}(\underline{q})$$

$$A_i^{i'}(\underline{q}) = \begin{bmatrix} R_i^{i'}(\underline{q}) & \underline{o}_i^{i'}(\underline{q}) \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} R_x(\alpha_i) & \underline{o}_i^{i'}(\underline{q}) \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & \cos(\alpha_i) & -\sin(\alpha_i) & 0 \\ 0 & \sin(\alpha_i) & \cos(\alpha_i) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



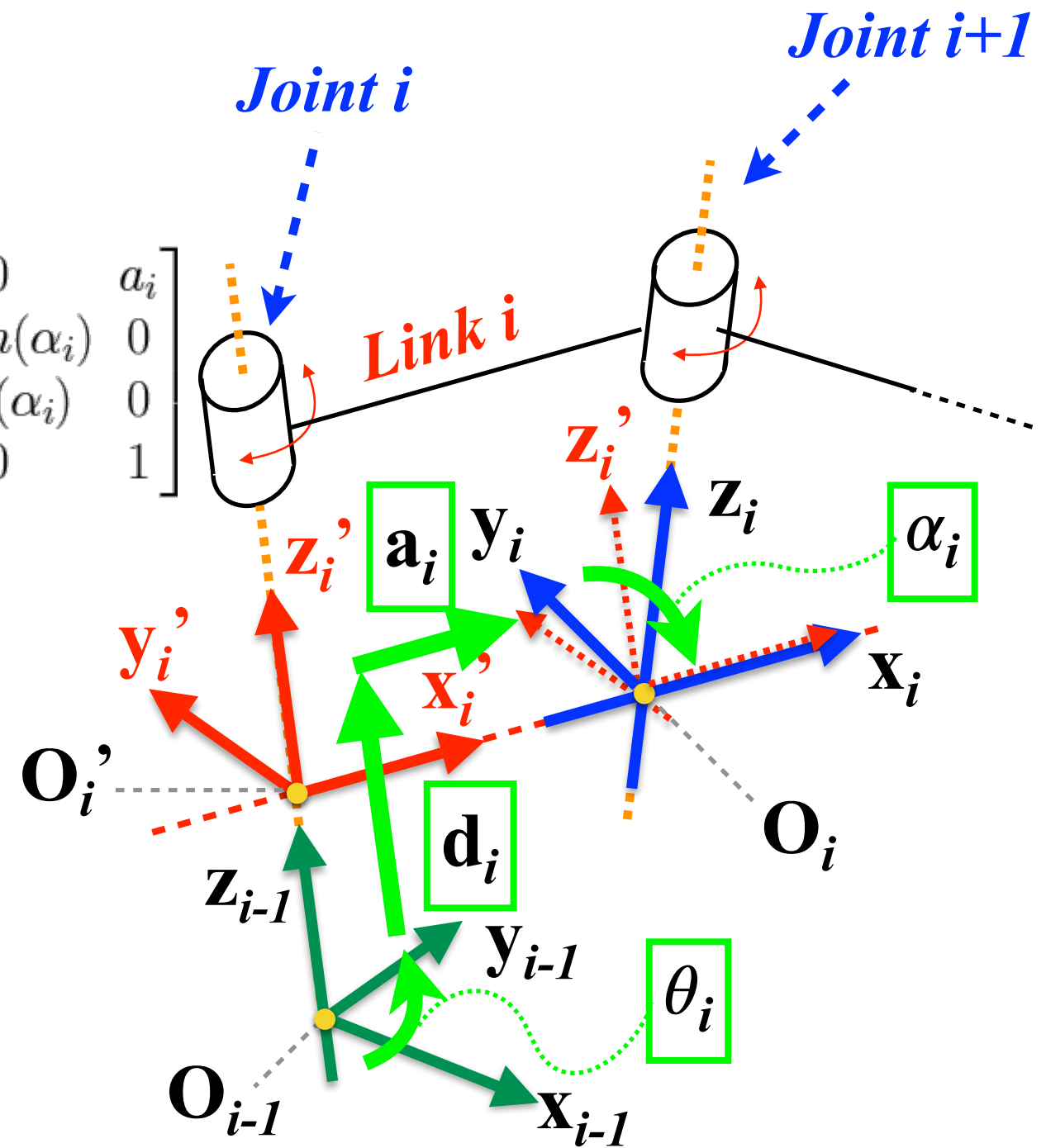
How to compute: A_i^{i-1}

$$A_i^{i-1}(\underline{q}) = A_{i'}^{i-1}(\underline{q}) \cdot A_i^{i'}(\underline{q})$$

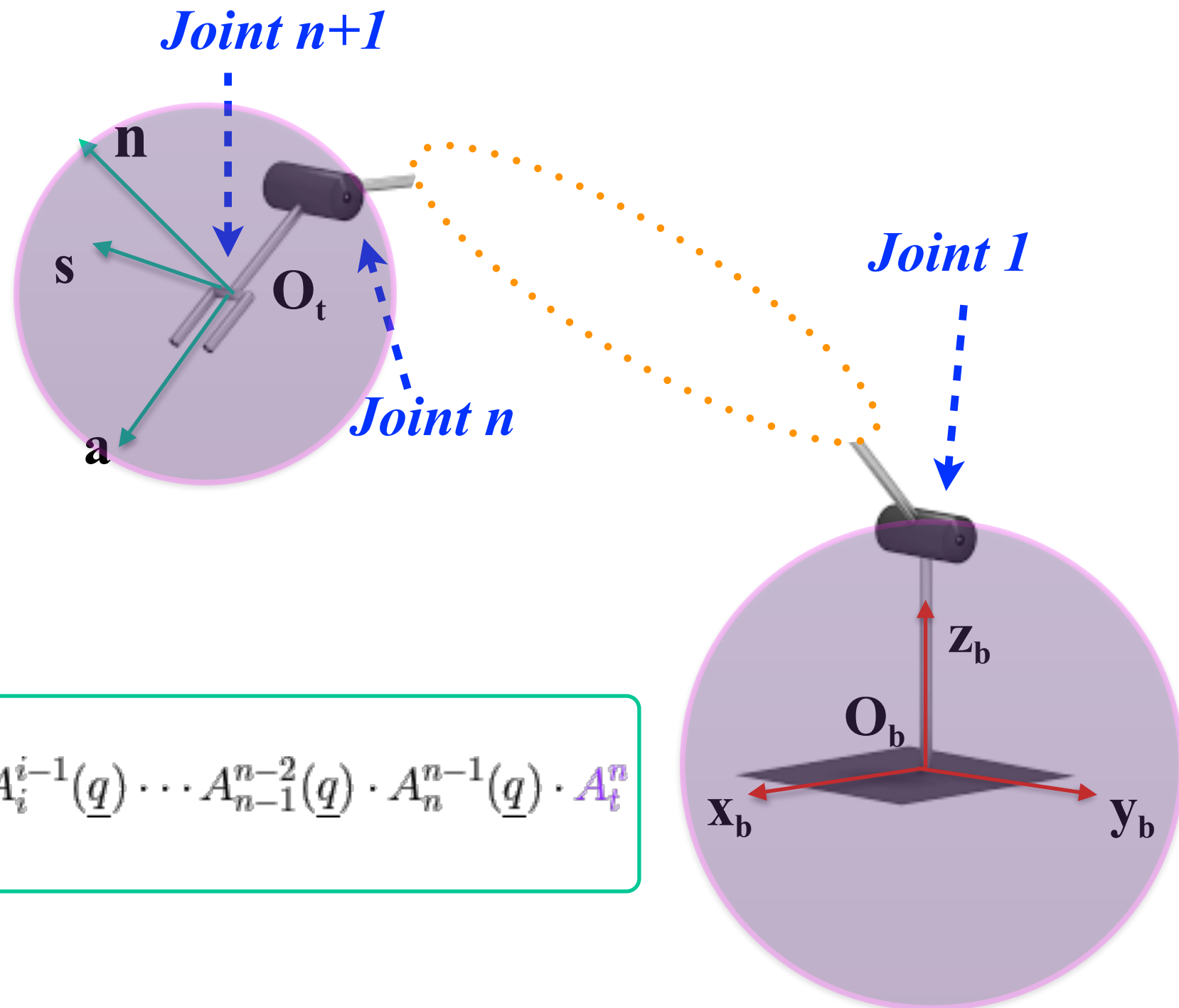
$$A_i^{i-1}(\underline{q}) = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) & 0 & 0 \\ \sin(\theta_i) & \cos(\theta_i) & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & \cos(\alpha_i) & -\sin(\alpha_i) & 0 \\ 0 & \sin(\alpha_i) & \cos(\alpha_i) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_i^{i-1}(\underline{q}) = \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} & 0 & 0 \\ s_{\theta_i} & c_{\theta_i} & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & c_{\alpha_i} & -s_{\alpha_i} & 0 \\ 0 & s_{\alpha_i} & c_{\alpha_i} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_i^{i-1}(\underline{q}) = \begin{bmatrix} c_{\theta_i} & -s_{\theta_i}c_{\alpha_i} & s_{\theta_i}s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i}c_{\alpha_i} & -c_{\theta_i}s_{\alpha_i} & a_i s_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

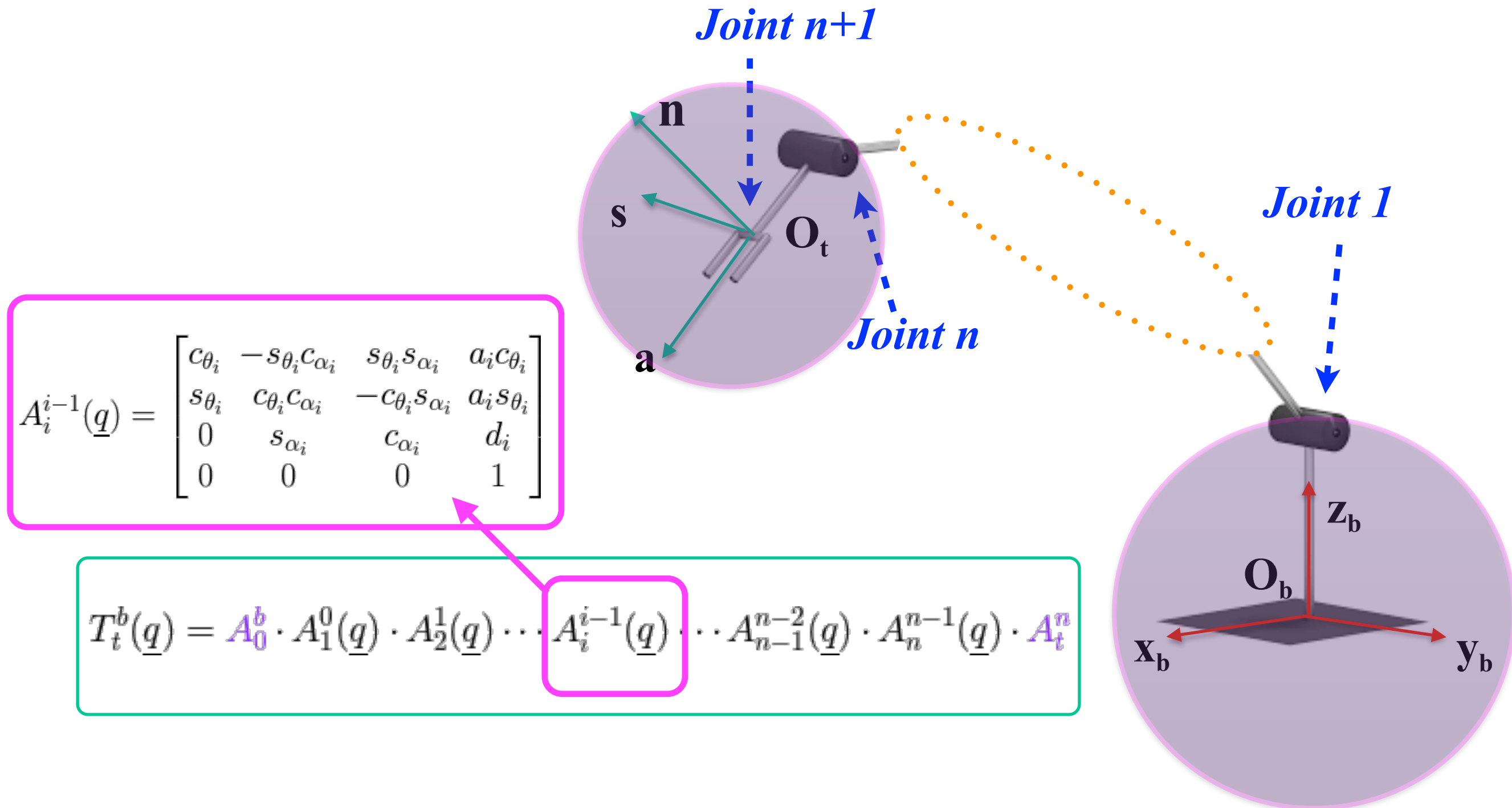


How to Describe the Direct Kinematics: Denavit-Hartenberg Convention Based Approach

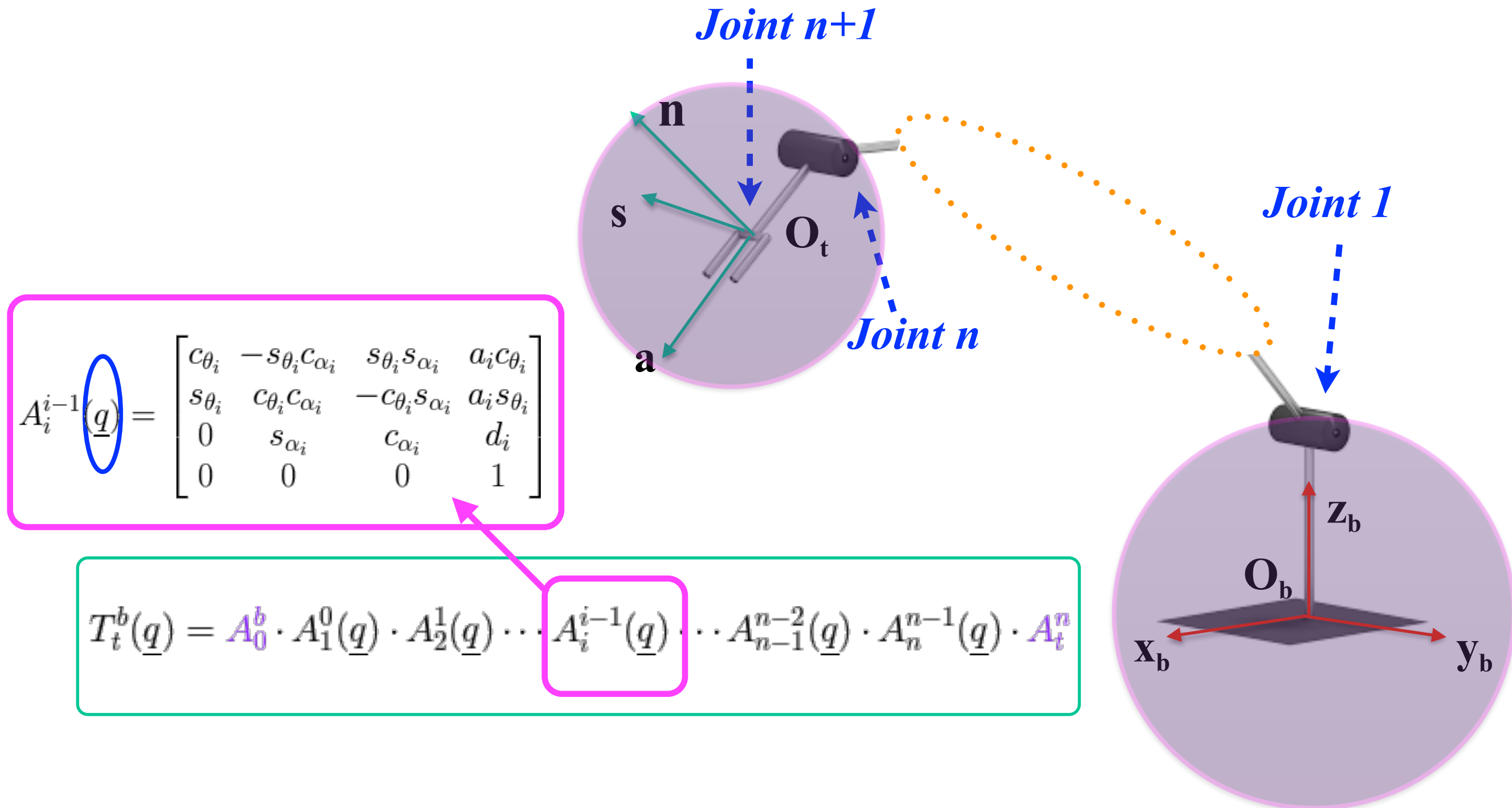


$$T_t^b(\underline{q}) = A_0^b \cdot A_1^0(\underline{q}) \cdot A_2^1(\underline{q}) \cdots A_i^{i-1}(\underline{q}) \cdots A_{n-1}^{n-2}(\underline{q}) \cdot A_n^{n-1}(\underline{q}) \cdot A_t^n$$

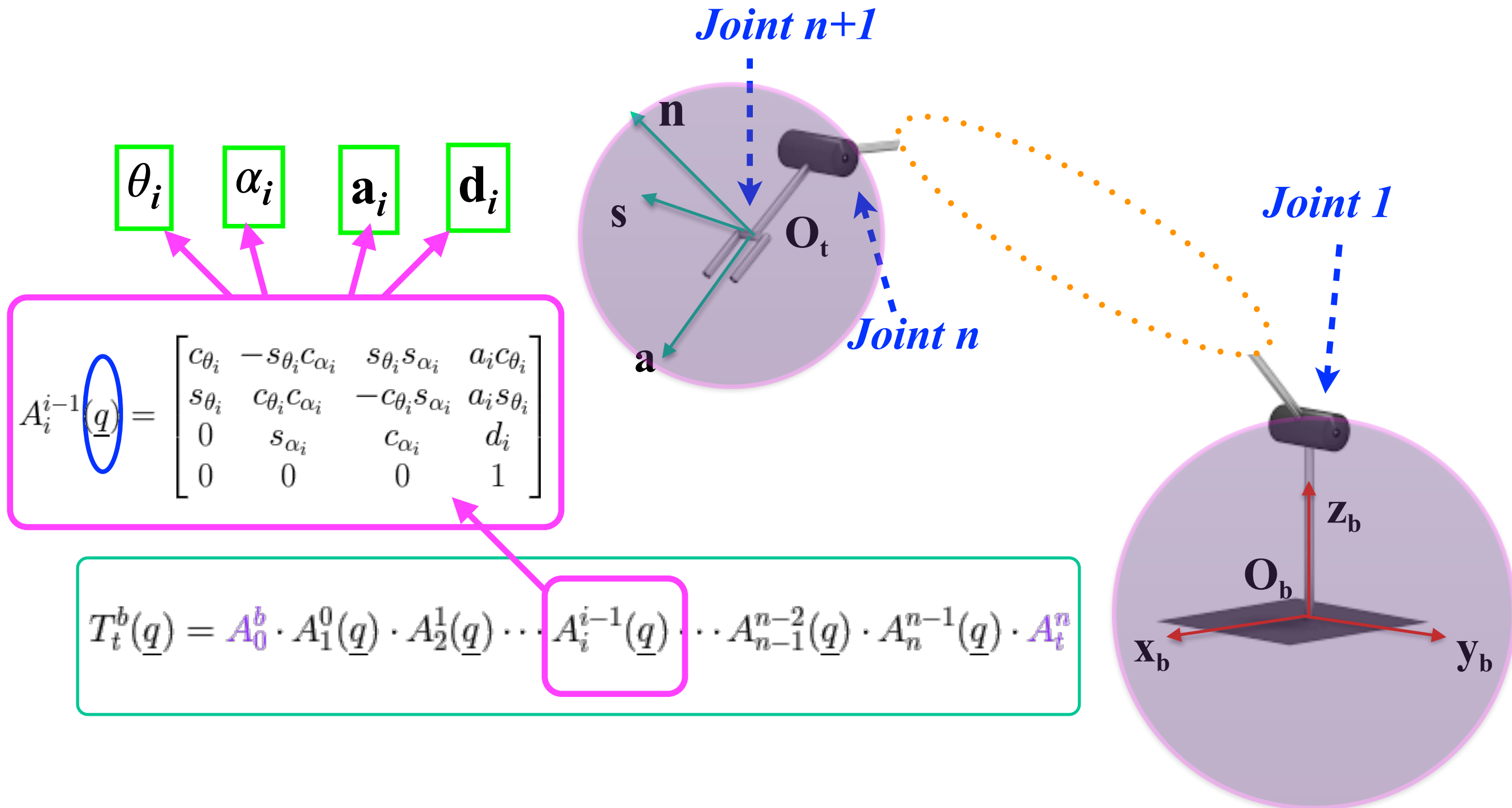
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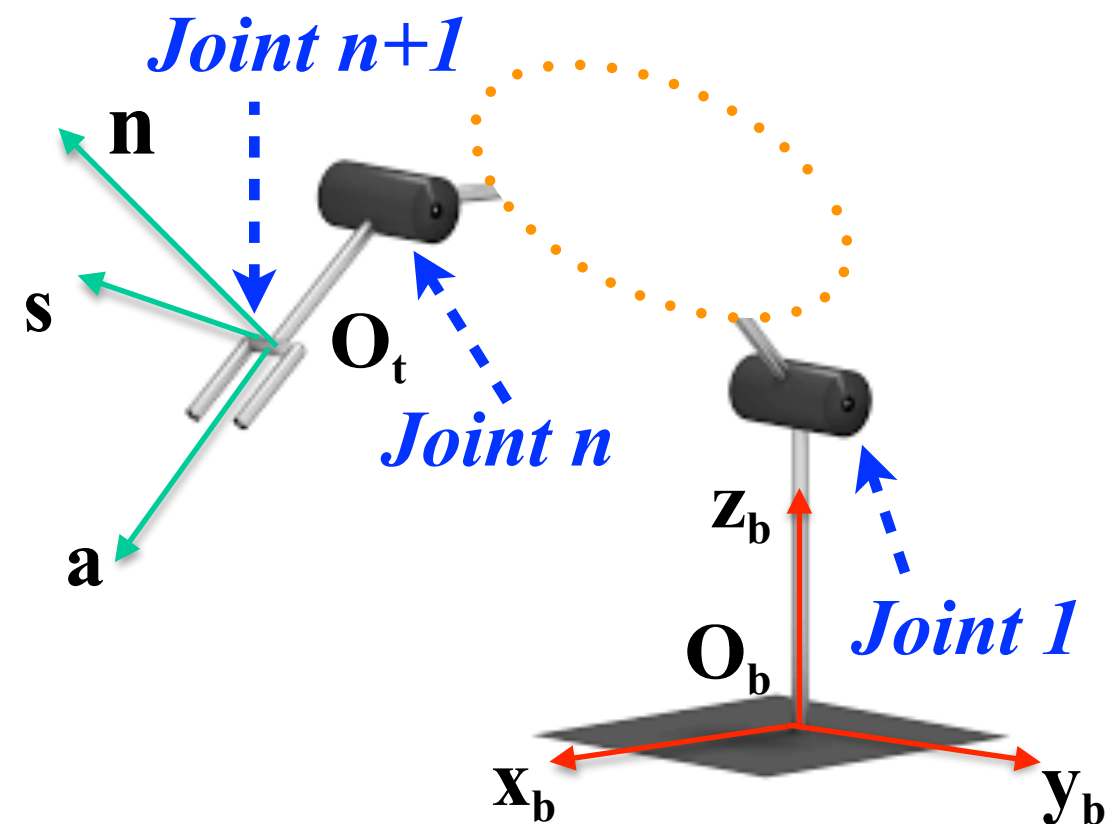
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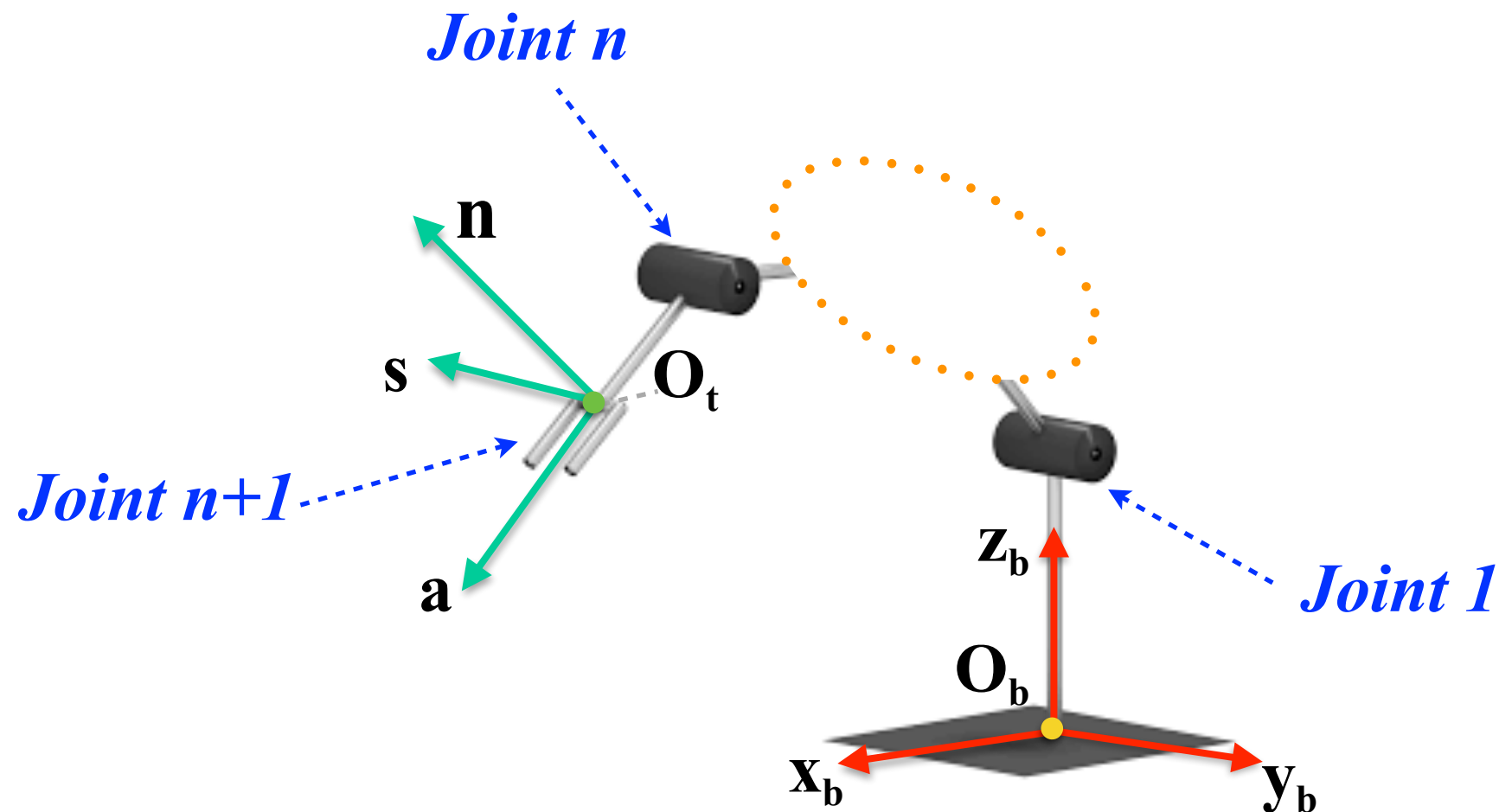


How to Describe the Direct Kinematics: Denavit-Hartenberg Convention Based Approach



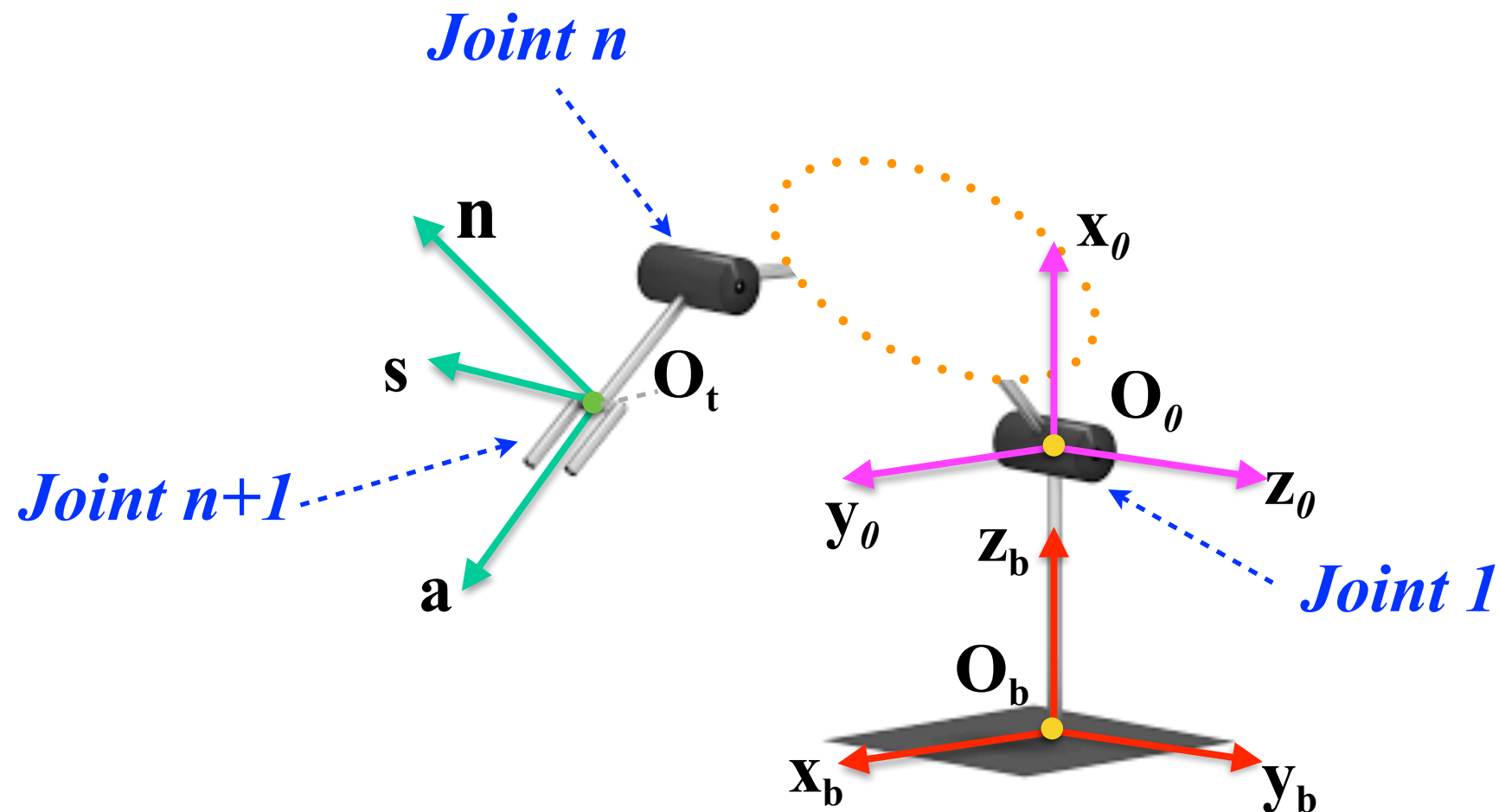
$$T_t^b(\underline{q}) = A_0^b \cdot \begin{bmatrix} c_{\theta_1} & -s_{\theta_1} c_{\alpha_1} & s_{\theta_1} s_{\alpha_1} & a_1 c_{\theta_1} \\ s_{\theta_1} & c_{\theta_1} c_{\alpha_1} & -c_{\theta_1} s_{\alpha_1} & a_1 s_{\theta_1} \\ 0 & s_{\alpha_1} & c_{\alpha_1} & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} c_{\alpha_i} & s_{\theta_i} s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i} c_{\alpha_i} & -c_{\theta_i} s_{\alpha_i} & a_i s_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} c_{\theta_n} & -s_{\theta_n} c_{\alpha_n} & s_{\theta_n} s_{\alpha_n} & a_n c_{\theta_n} \\ s_{\theta_n} & c_{\theta_n} c_{\alpha_n} & -c_{\theta_n} s_{\alpha_n} & a_n s_{\theta_n} \\ 0 & s_{\alpha_n} & c_{\alpha_n} & d_n \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot A_t^n$$

How to compute: A_0^b and A_t^n



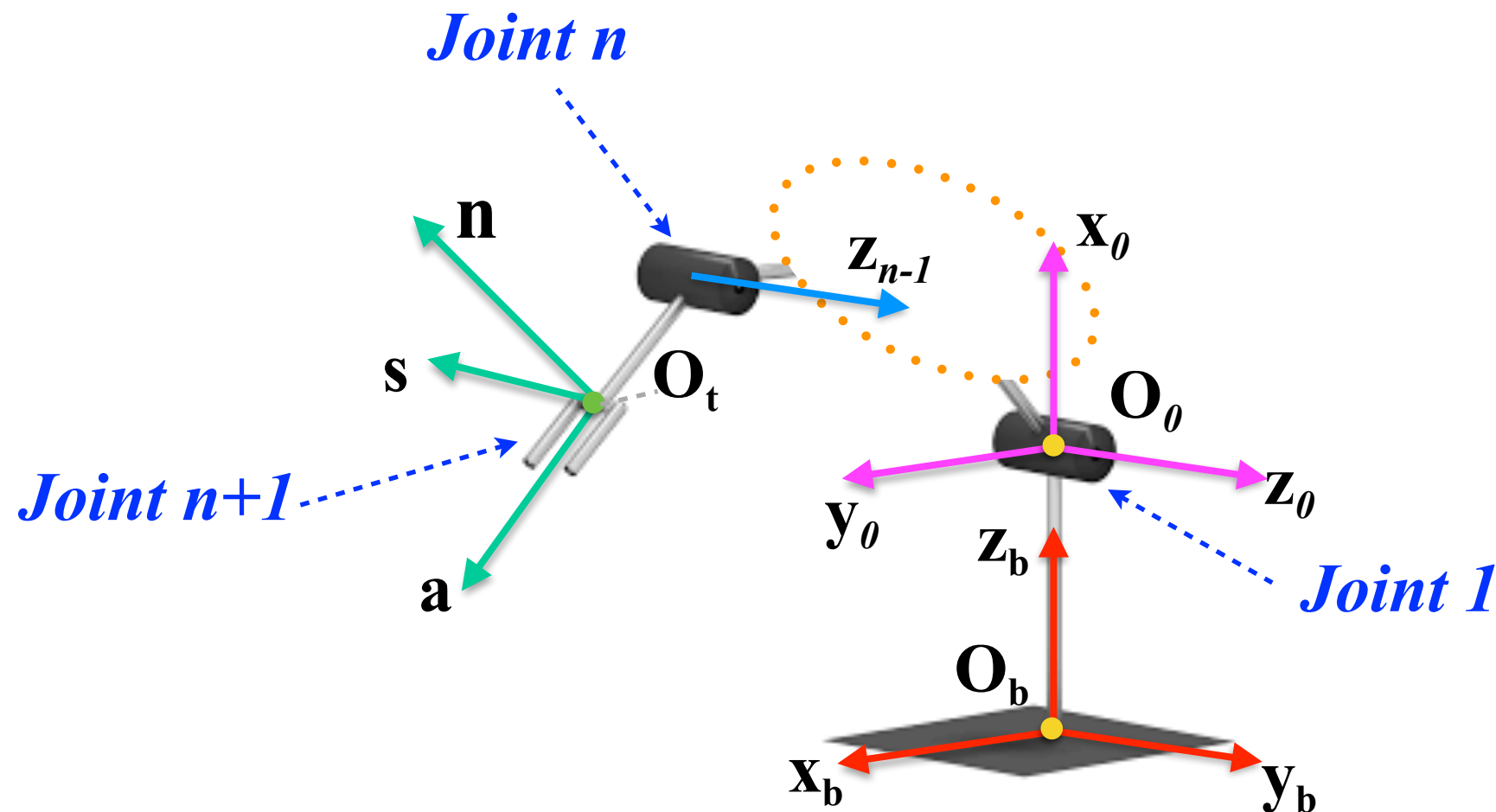
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How to compute: A_0^b and A_t^n



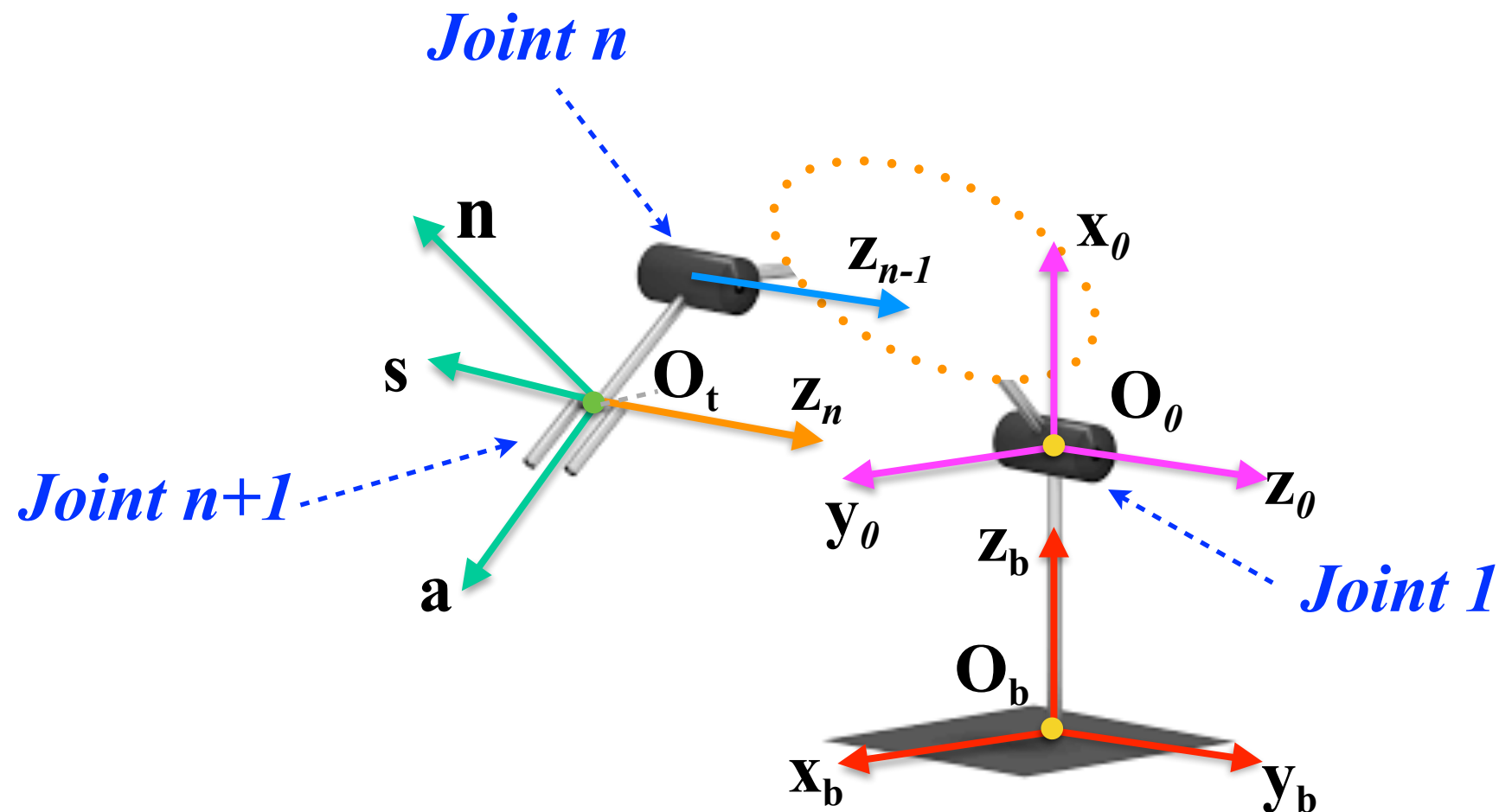
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How to compute: A_0^b and A_t^n



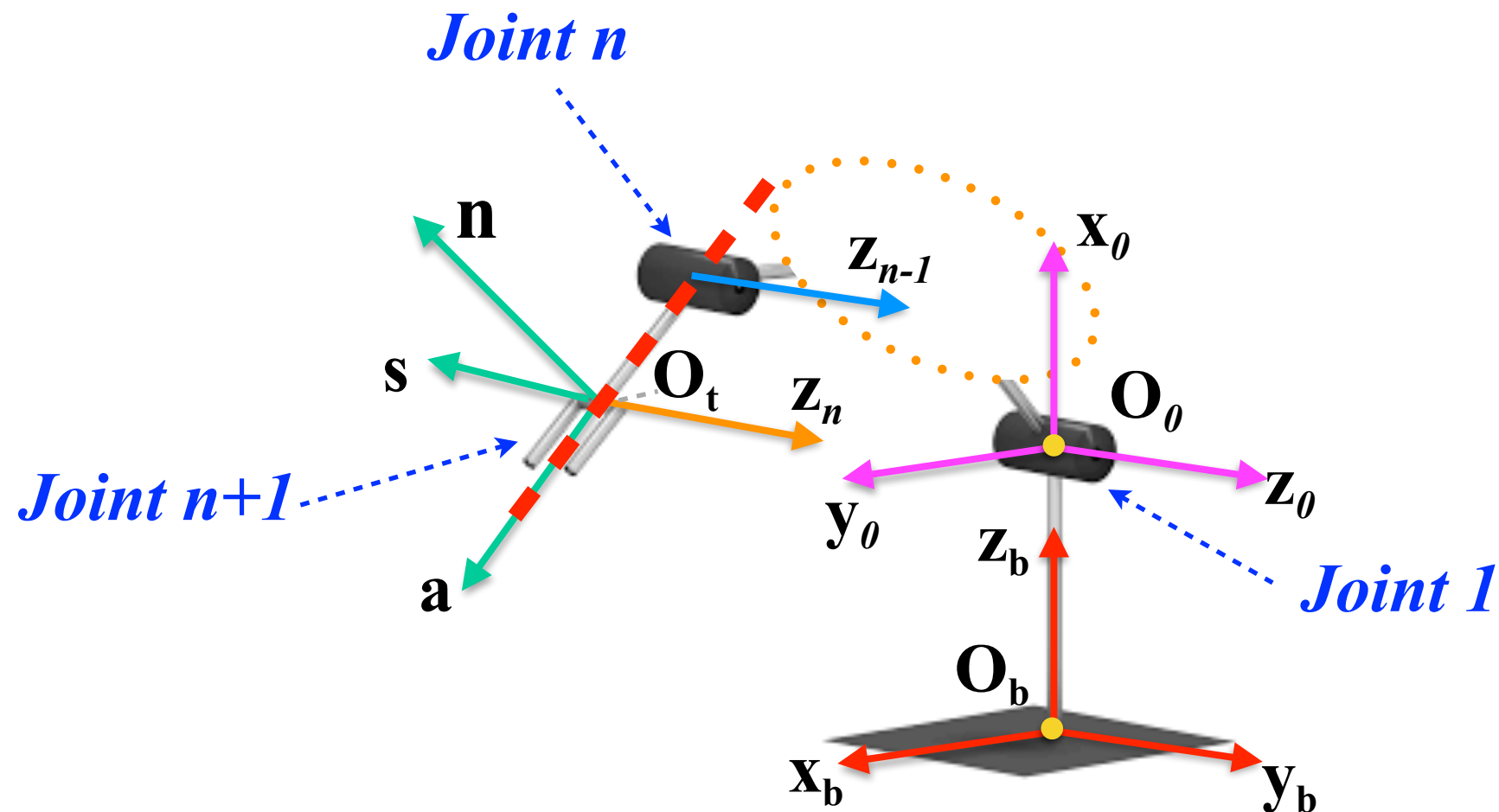
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How to compute: A_0^b and A_t^n



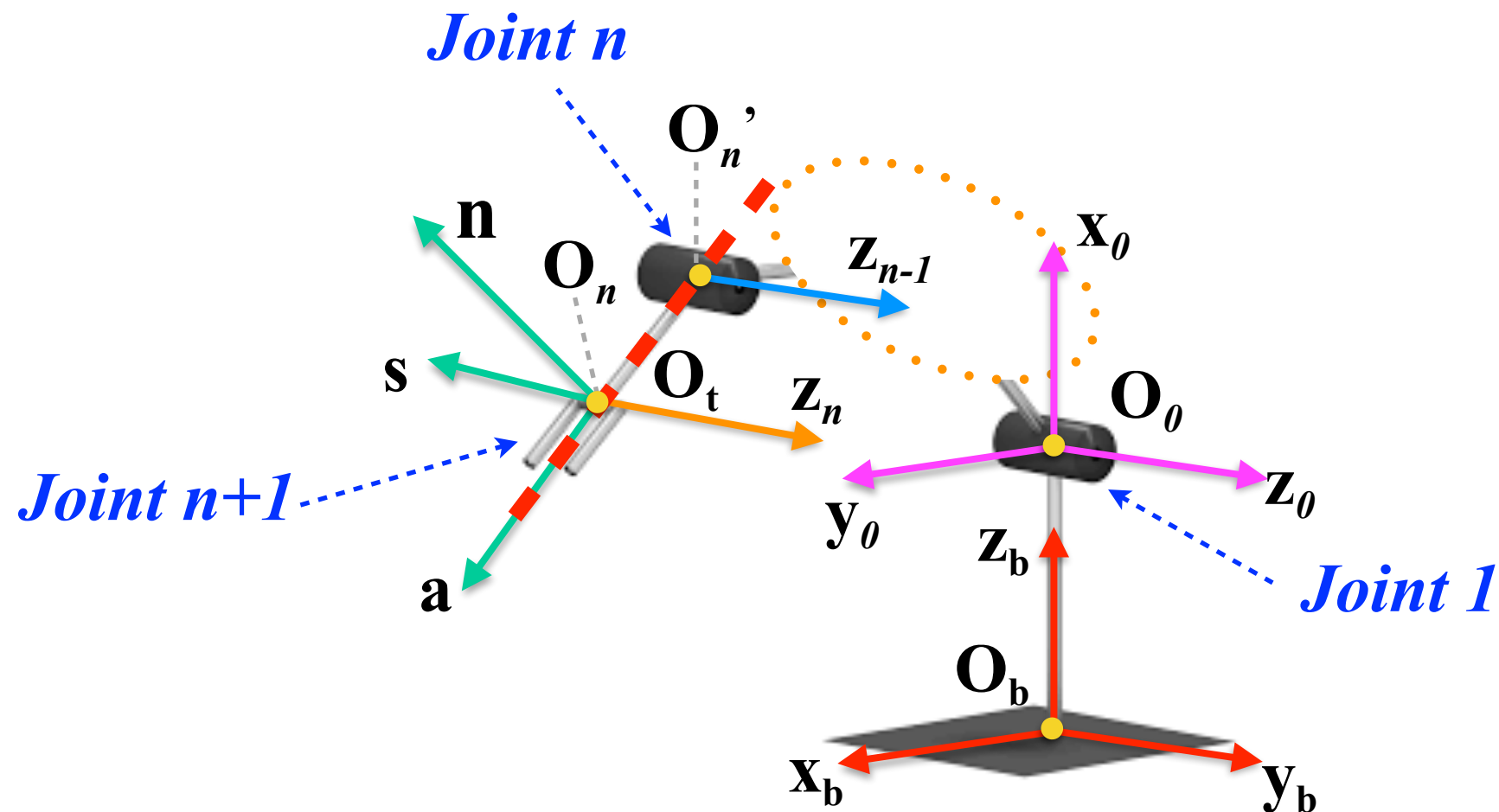
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How to compute: A_0^b and A_t^n



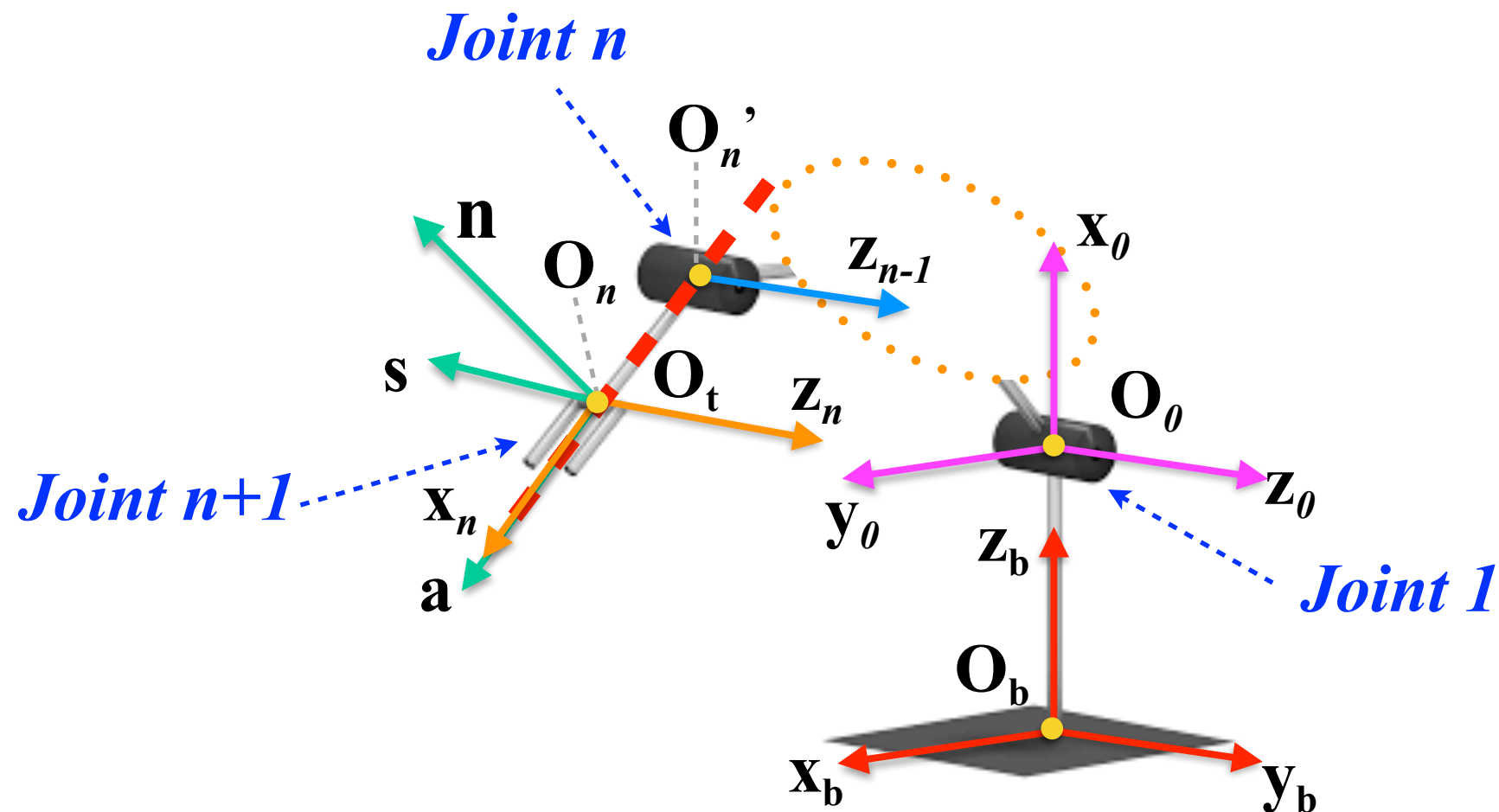
$$T_t^b(\underline{q}) = \underbrace{A_0^b}_{\text{blue oval}} \cdot \begin{bmatrix} c_{\theta_1} & -s_{\theta_1} c_{\alpha_1} & s_{\theta_1} s_{\alpha_1} & a_1 c_{\theta_1} \\ s_{\theta_1} & c_{\theta_1} c_{\alpha_1} & -c_{\theta_1} s_{\alpha_1} & a_1 s_{\theta_1} \\ 0 & s_{\alpha_1} & c_{\alpha_1} & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} c_{\alpha_i} & s_{\theta_i} s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i} c_{\alpha_i} & -c_{\theta_i} s_{\alpha_i} & a_i s_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} c_{\theta_n} & -s_{\theta_n} c_{\alpha_n} & s_{\theta_n} s_{\alpha_n} & a_n c_{\theta_n} \\ s_{\theta_n} & c_{\theta_n} c_{\alpha_n} & -c_{\theta_n} s_{\alpha_n} & a_n s_{\theta_n} \\ 0 & s_{\alpha_n} & c_{\alpha_n} & d_n \\ 0 & 0 & 0 & 1 \end{bmatrix} \underbrace{A_t^n}_{\text{blue oval}}$$

How to compute: A_0^b and A_t^n



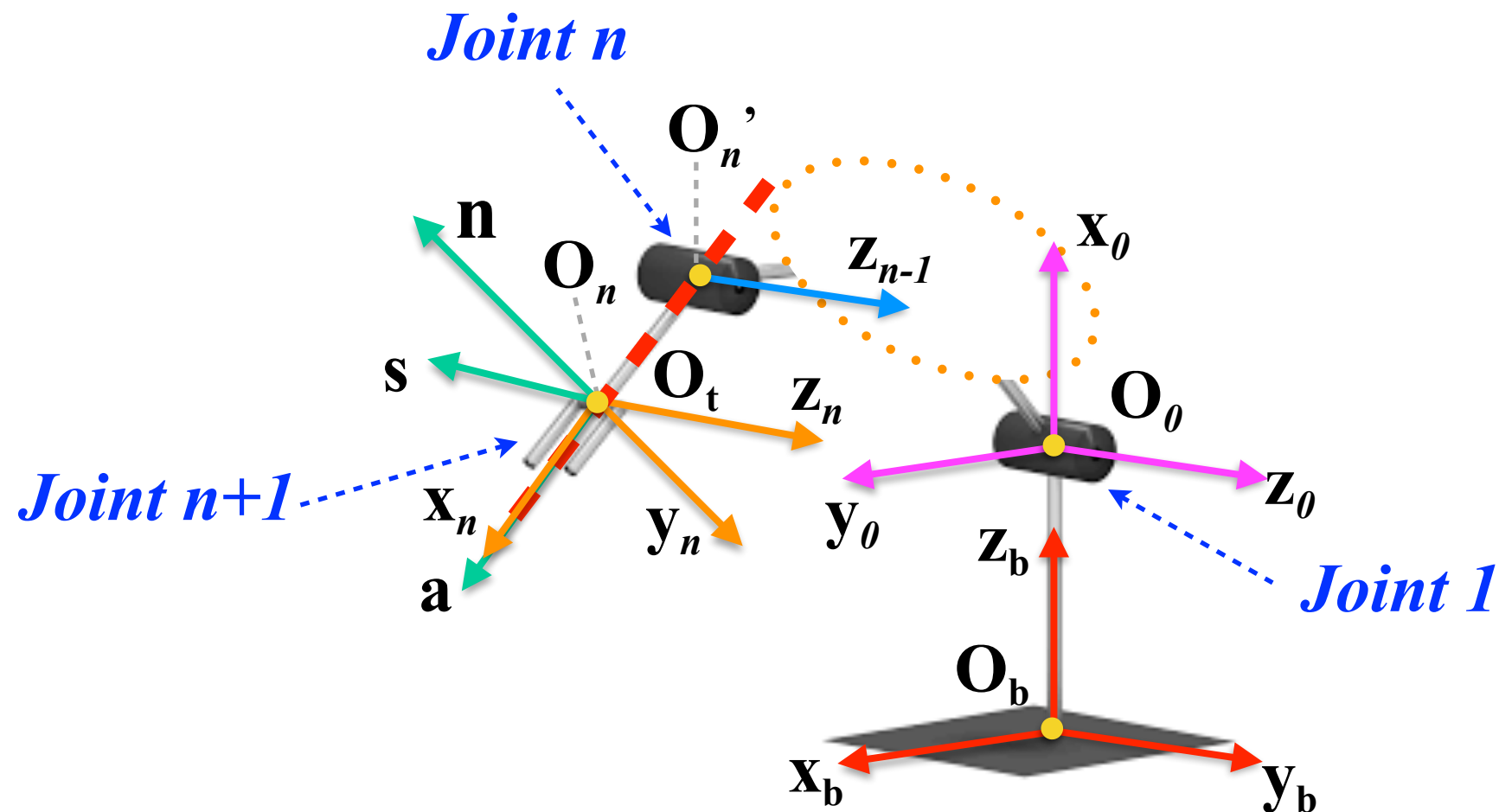
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How to compute: A_0^b and A_t^n



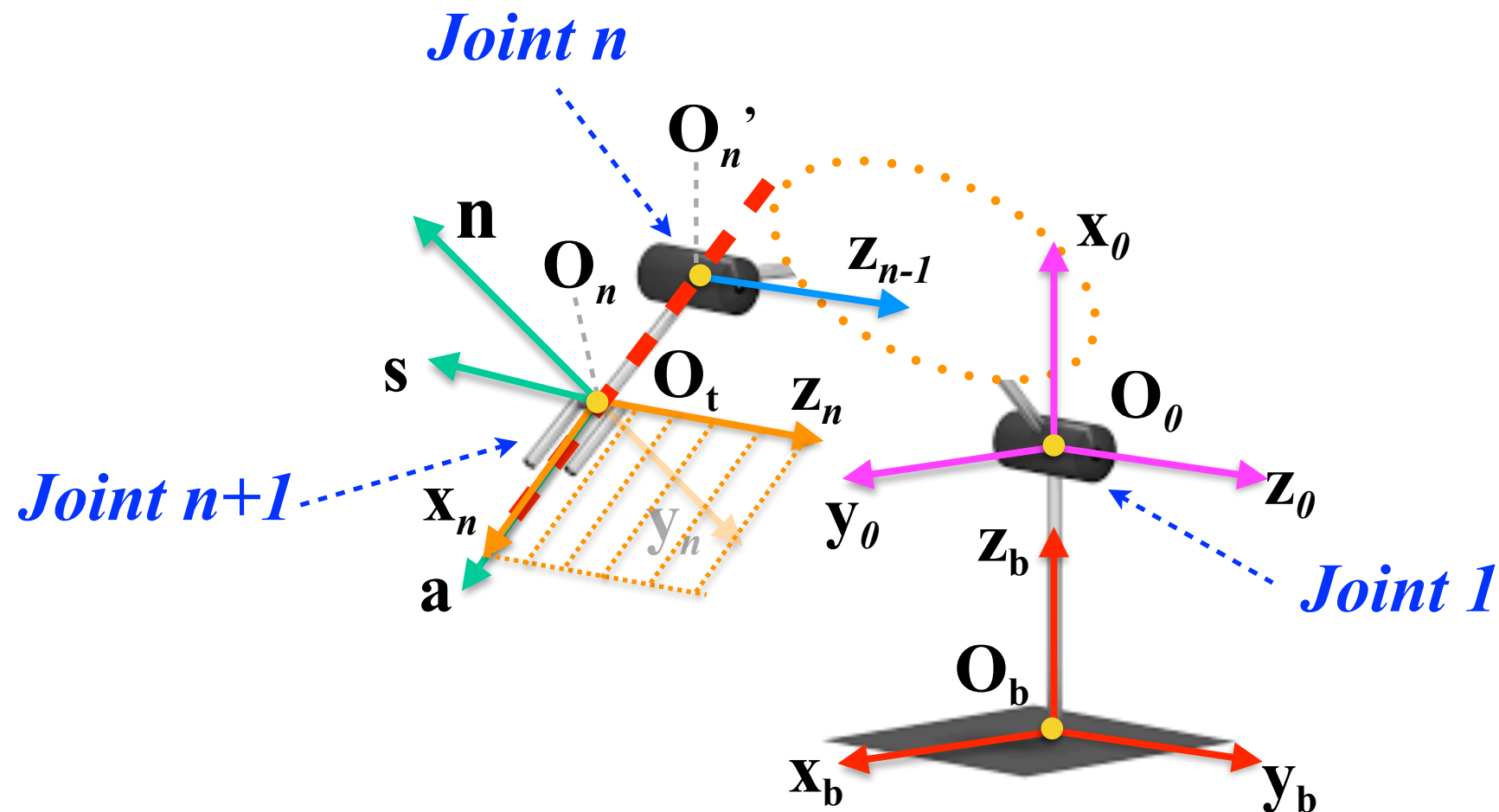
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How to compute: A_0^b and A_t^n



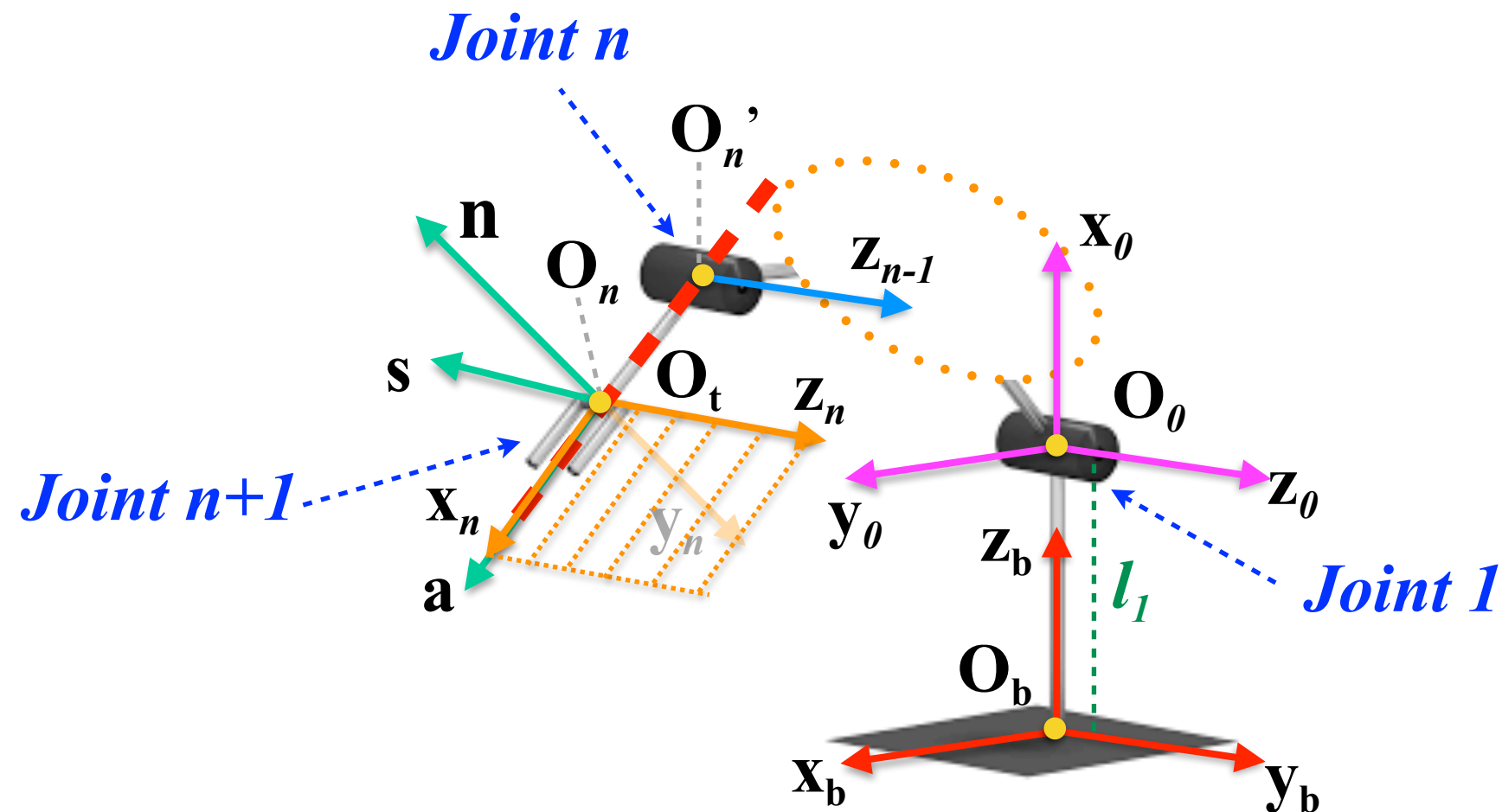
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How to compute: A_0^b and A_t^n



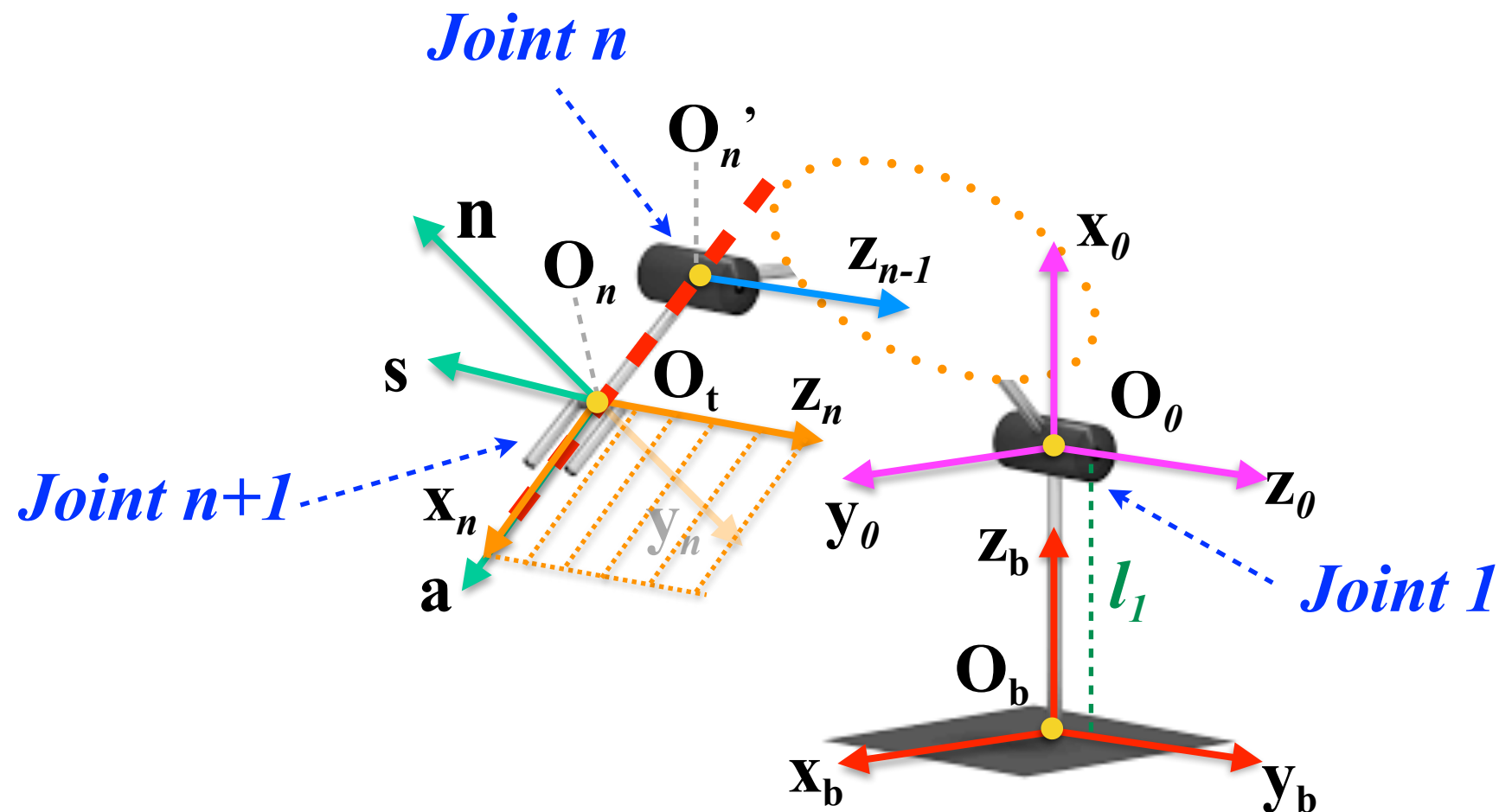
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How to compute: A_0^b and A_t^n



$$A_0^b = \begin{bmatrix} & & & \\ & R_0^b & & \underline{O}_0^b \\ & & & \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

How to compute: A_0^b and A_t^n



$$A_0^b = \begin{bmatrix} R_0^b & \underline{O}_0^b \\ \hline 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_t^n = \begin{bmatrix} R_t^n & \underline{O}_t^n \\ \hline 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



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Dipartimento di
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Course of Robot Control

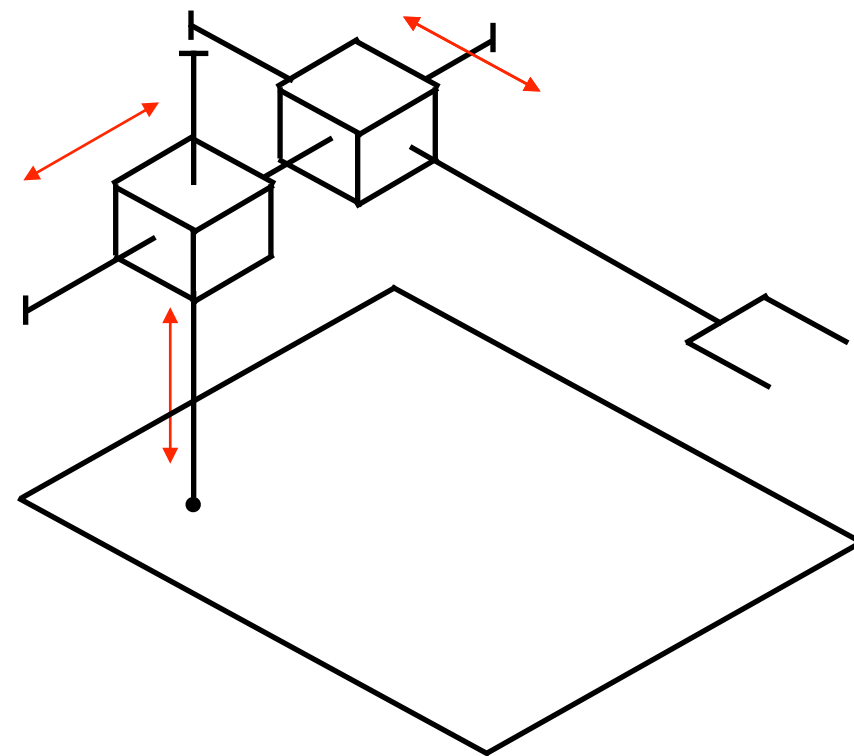
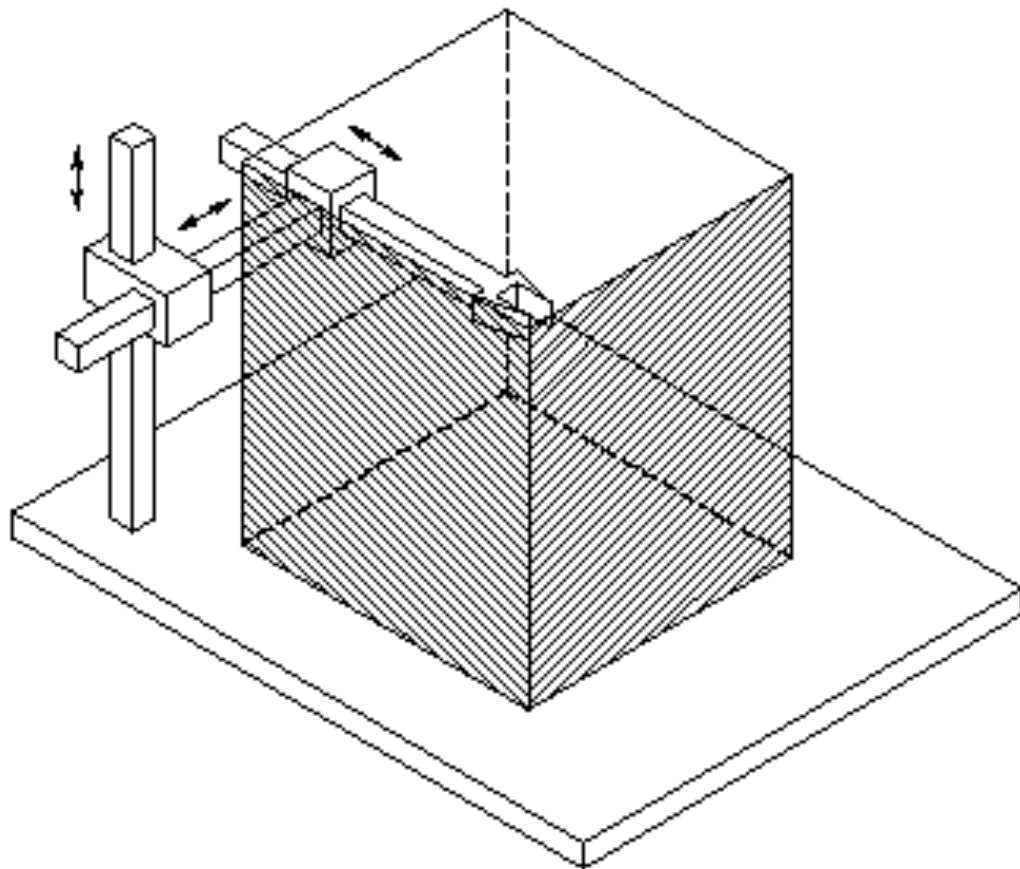
Prof. Antonella Ferrara



D-H Convention Based Approach: Cartesian Robots

How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Cartesian Robots

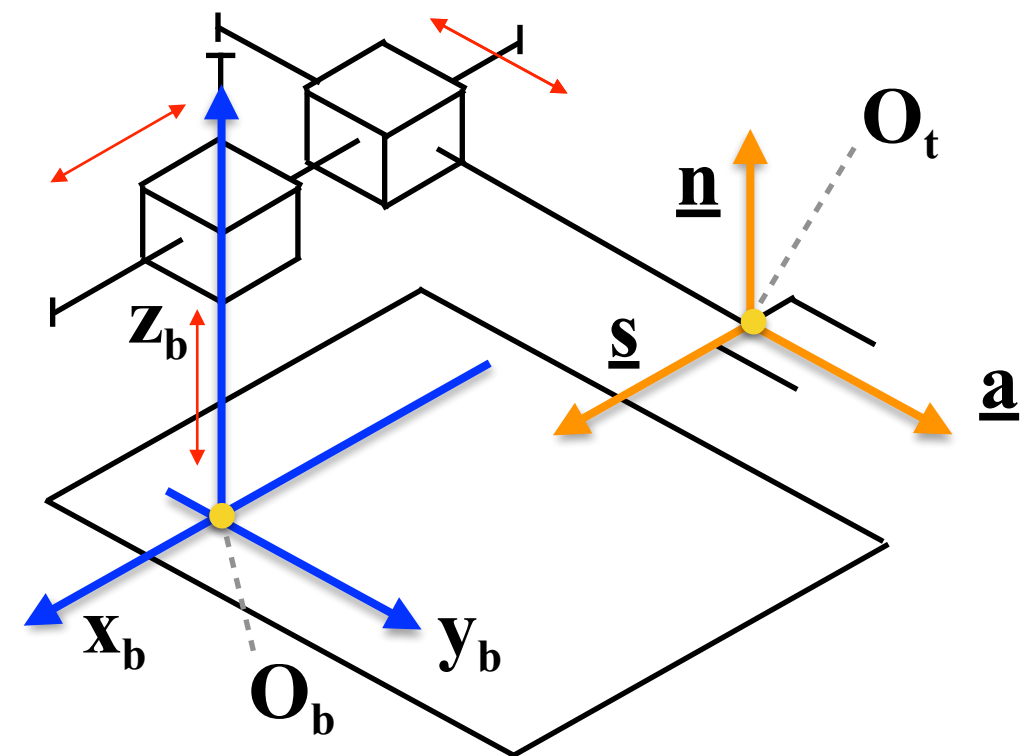
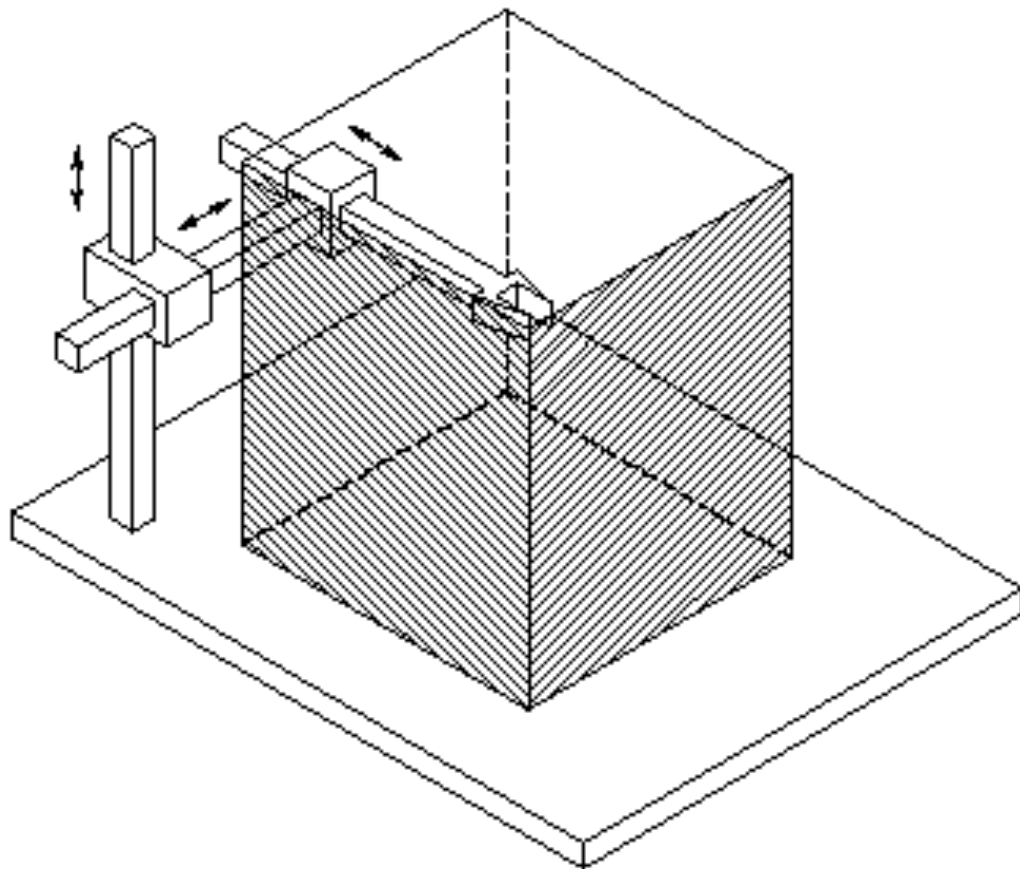


Source: Robotica - Modellistica, Pianificazione E Controllo
Siciliano, Sciavicco, Villani, Oriolo
Mcgraw-Hill (one of the course textbooks)

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Cartesian Robots

Assign the Base Reference Frame and the Tool Reference Frame

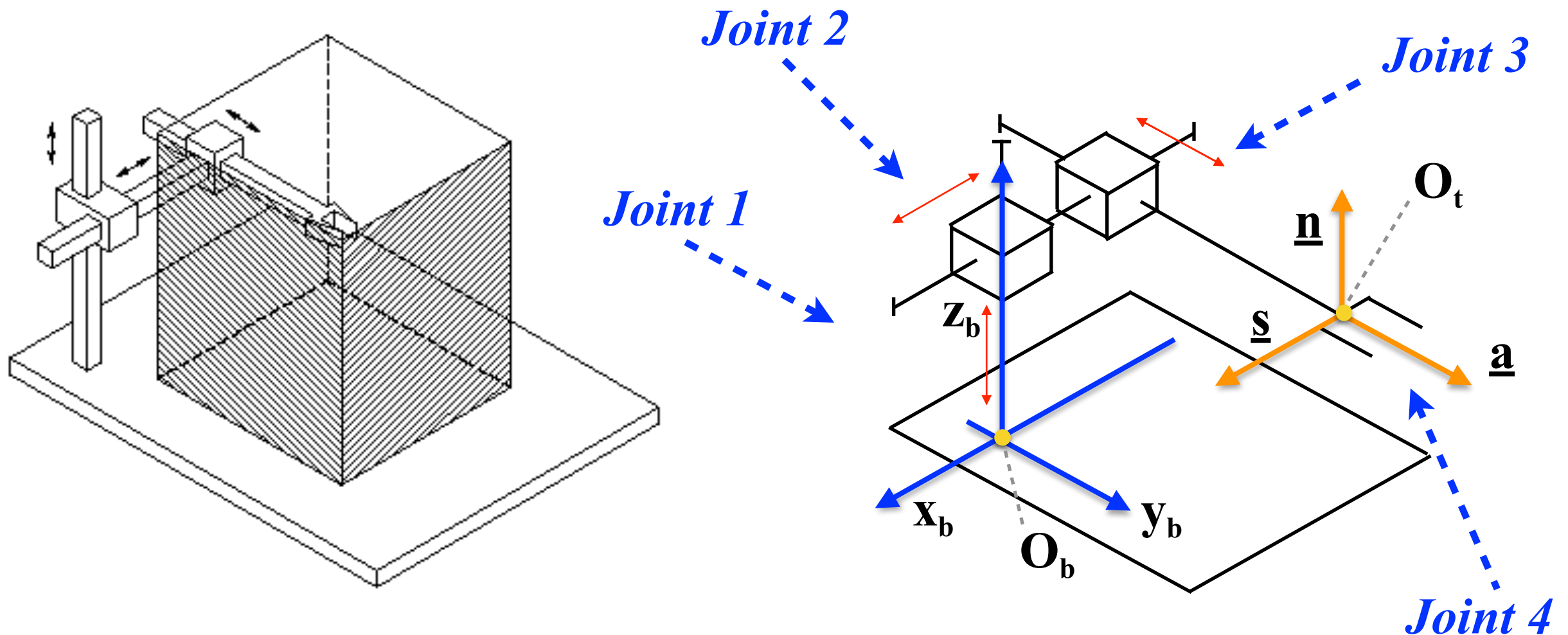


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Cartesian Robots

Number the joints

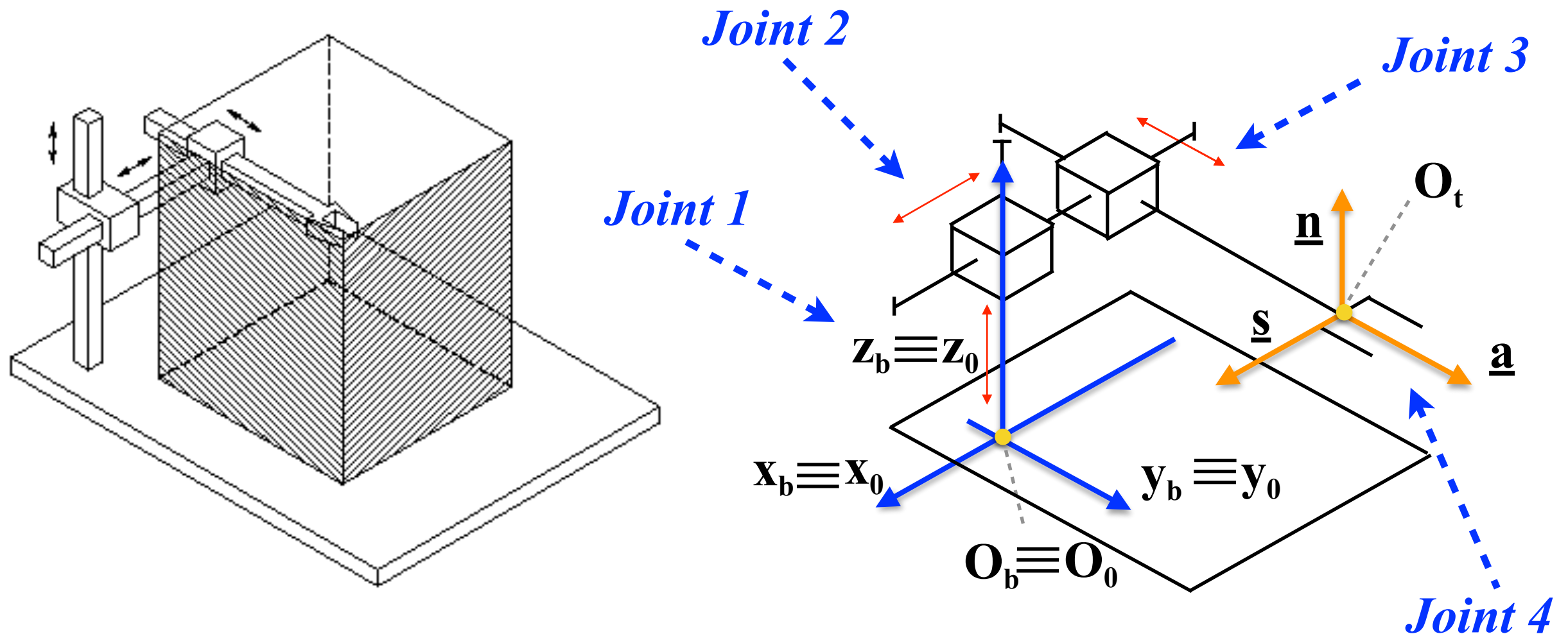


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Cartesian Robots

Assign the D-H Reference Frames from 0 to n

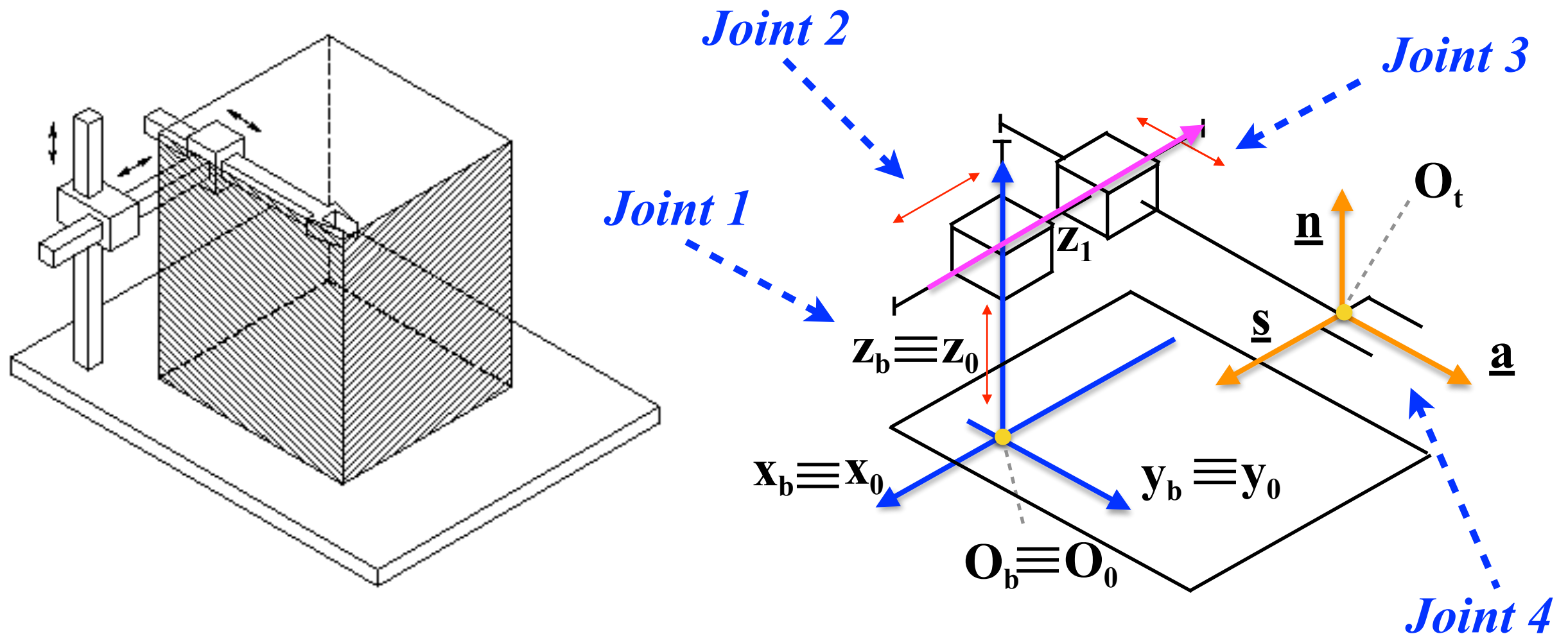


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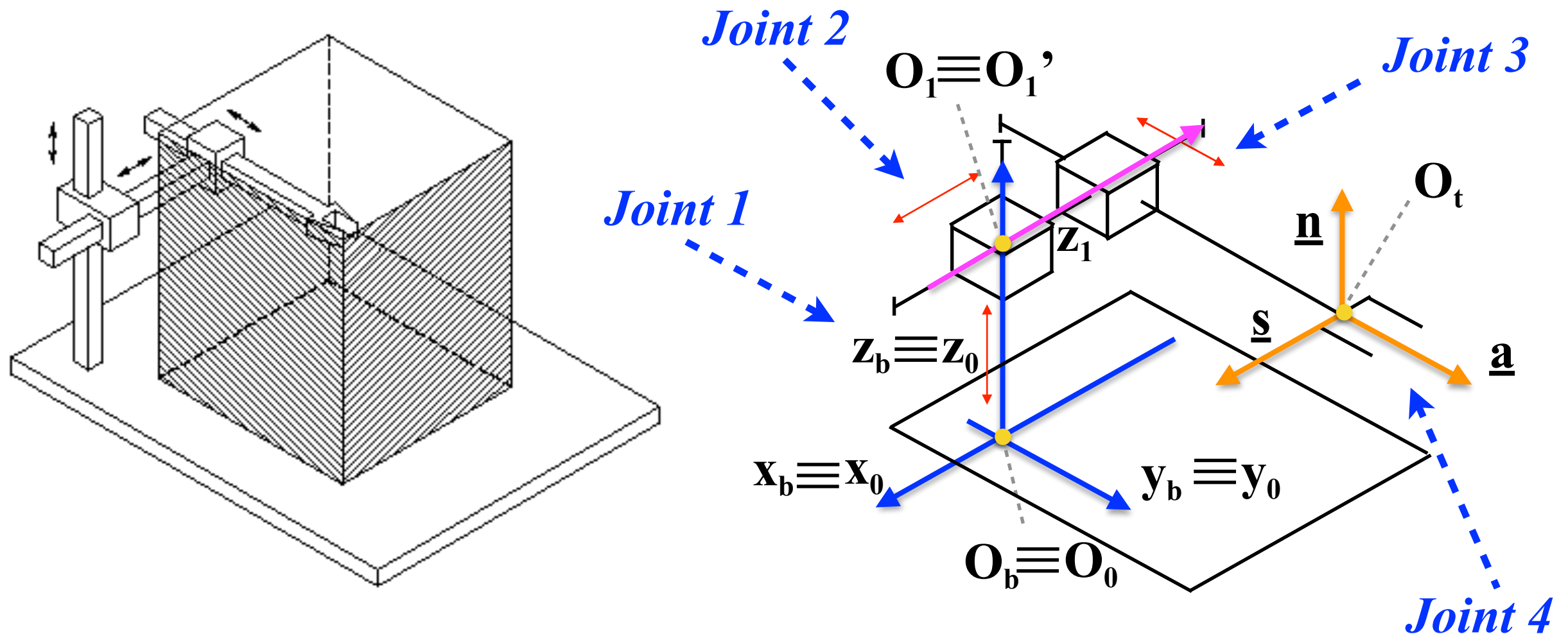


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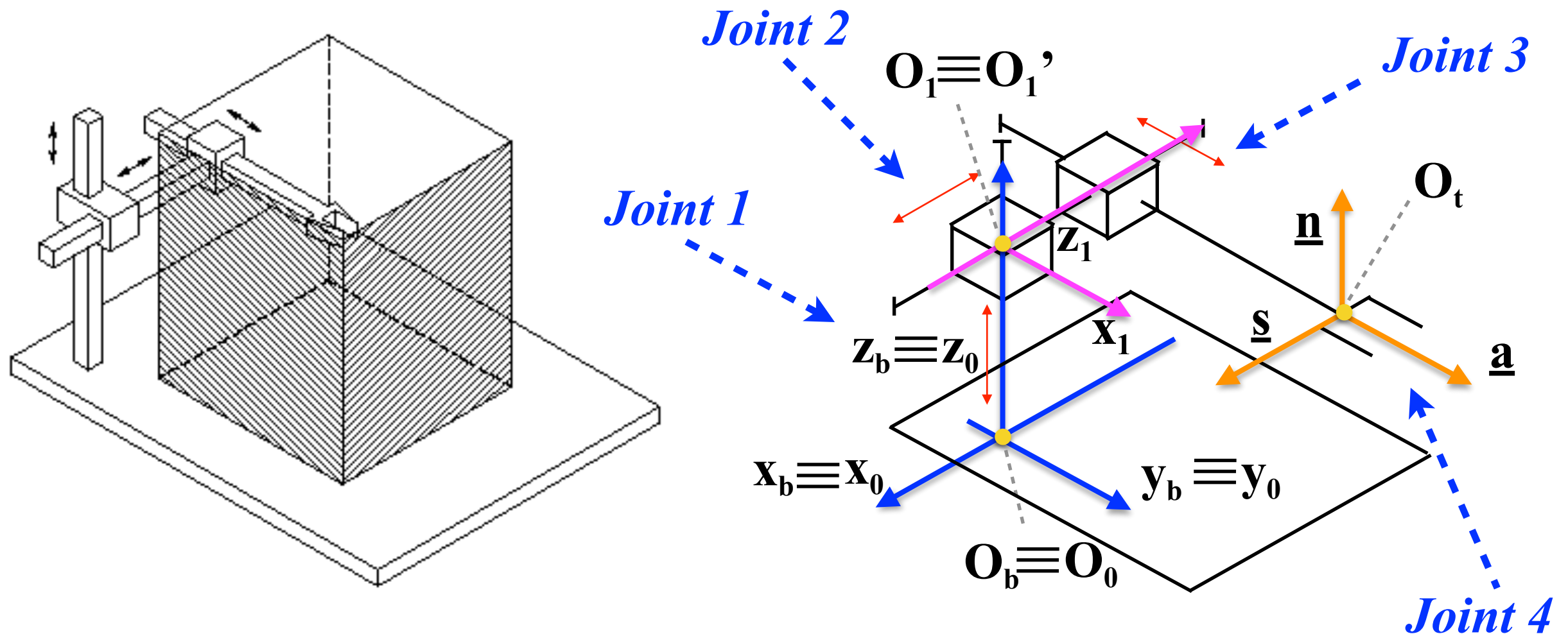


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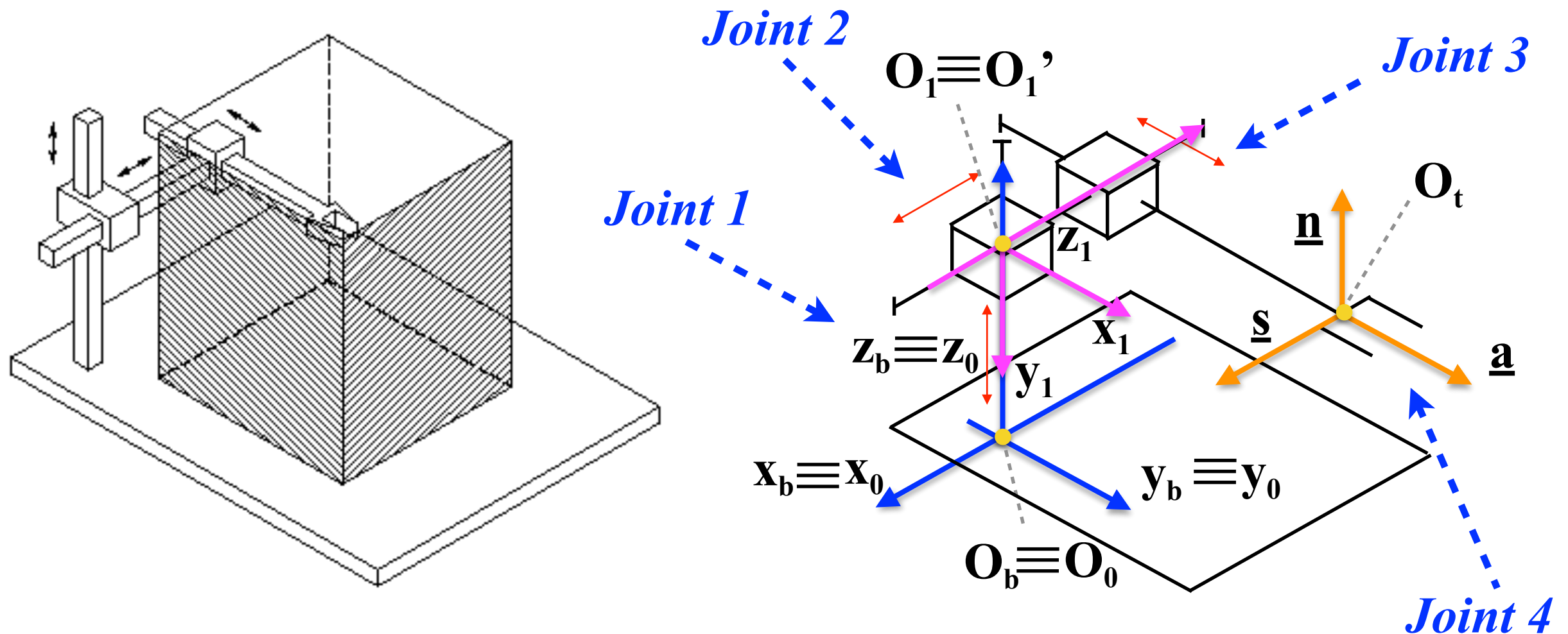


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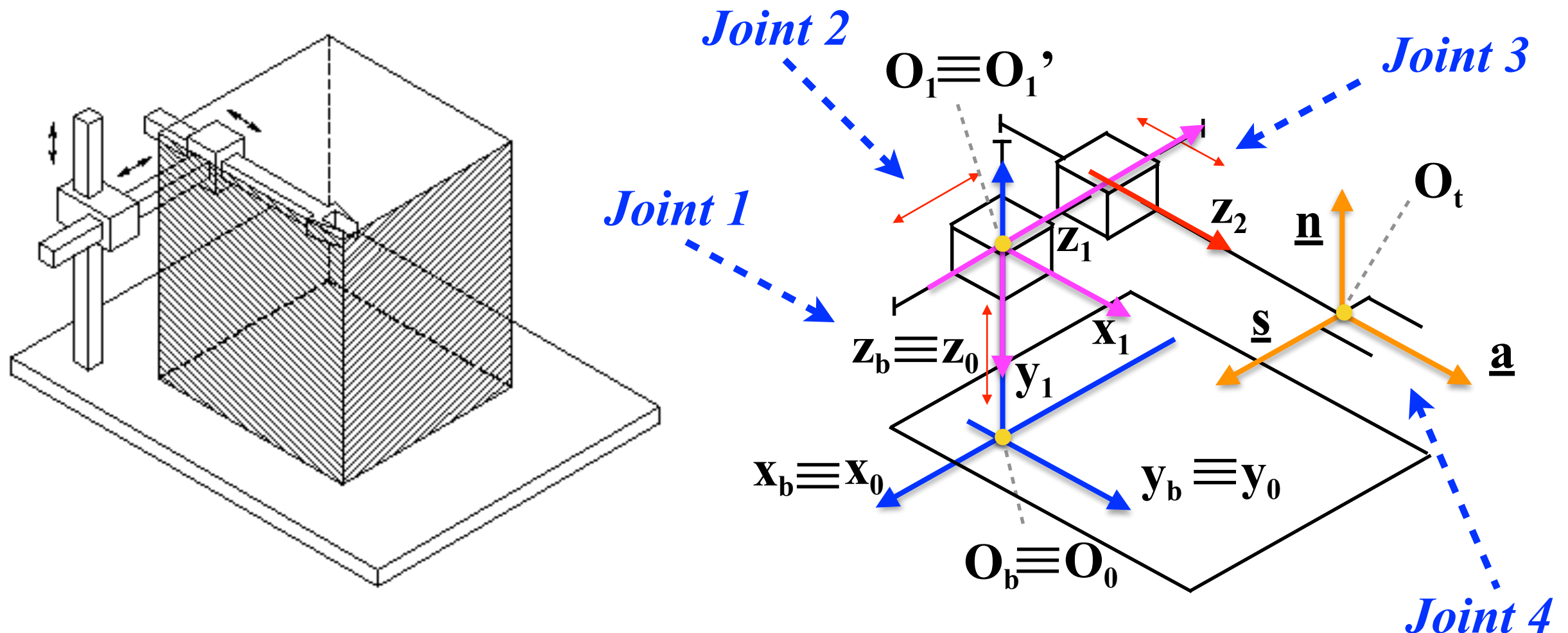


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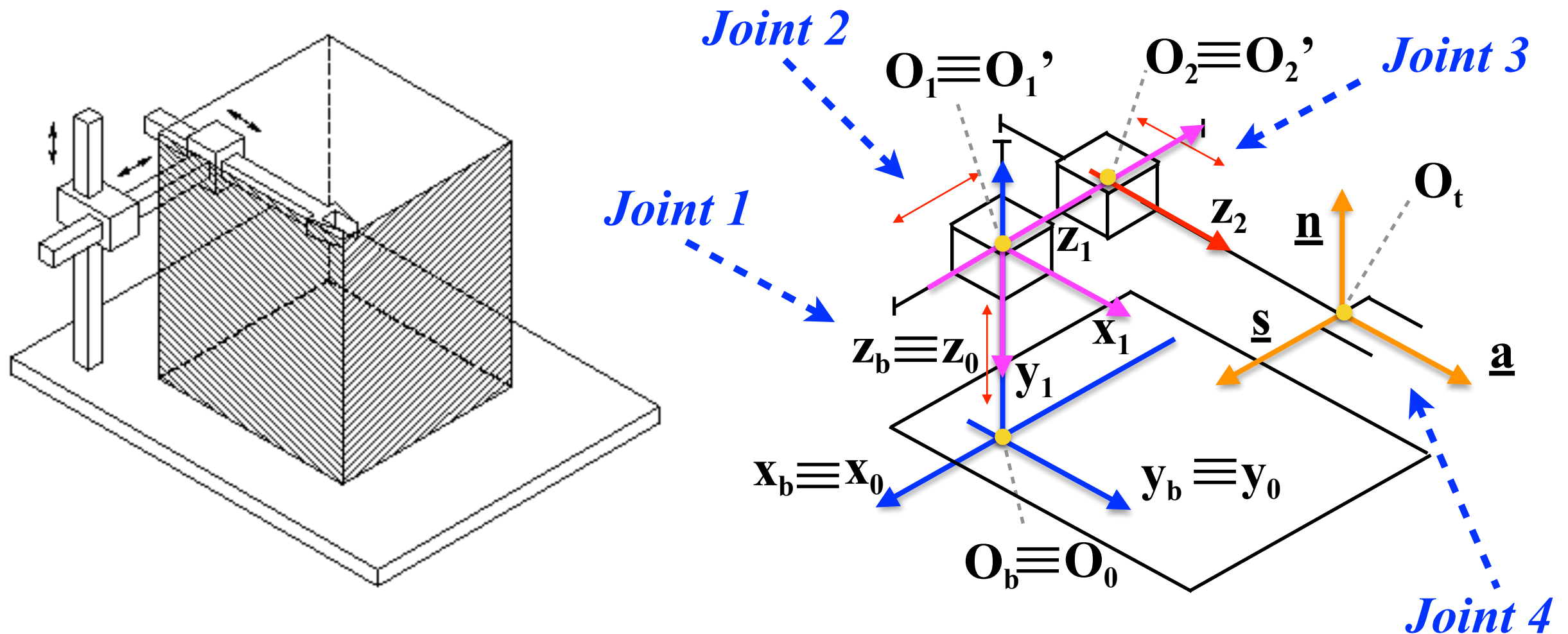


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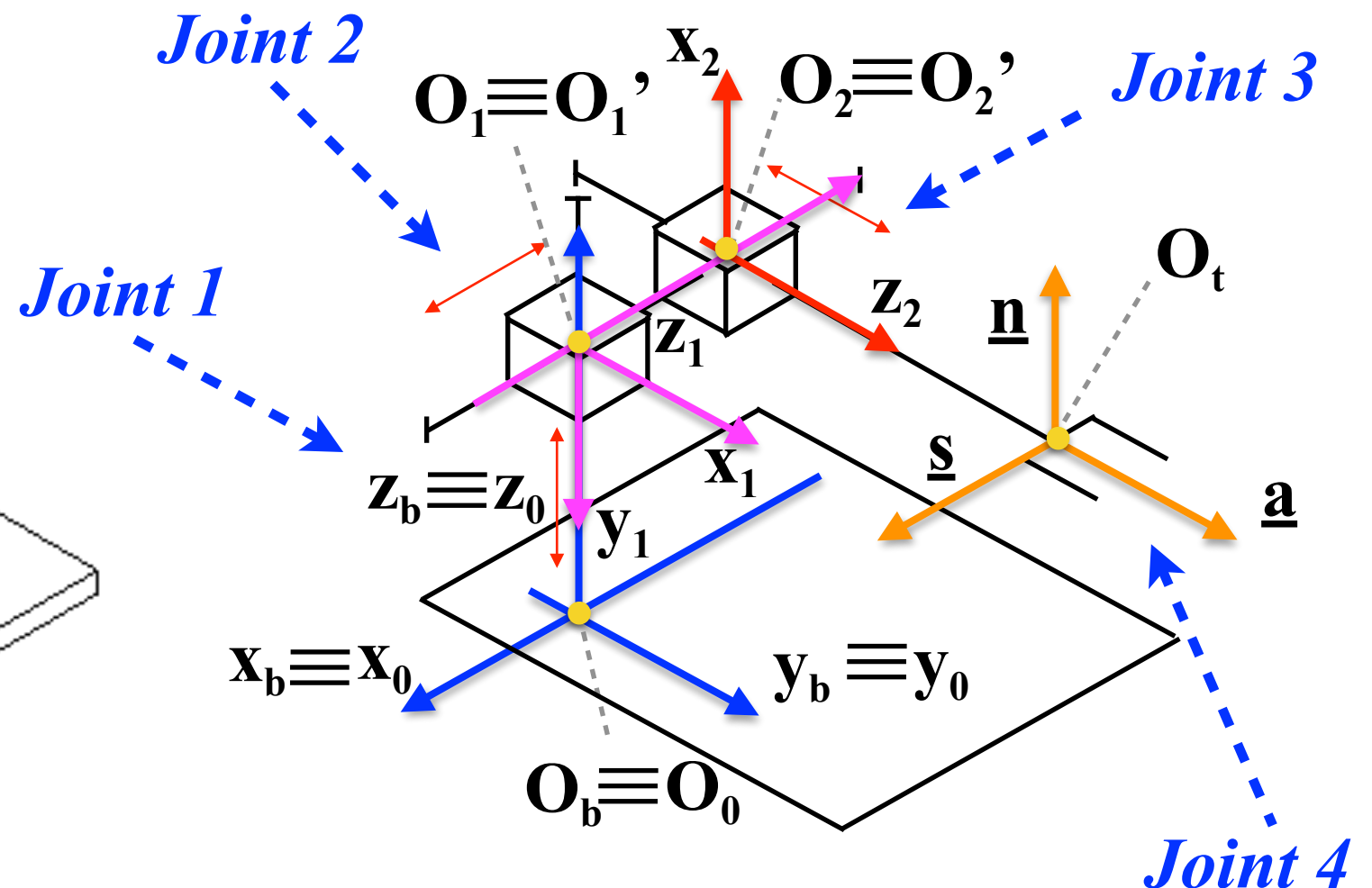
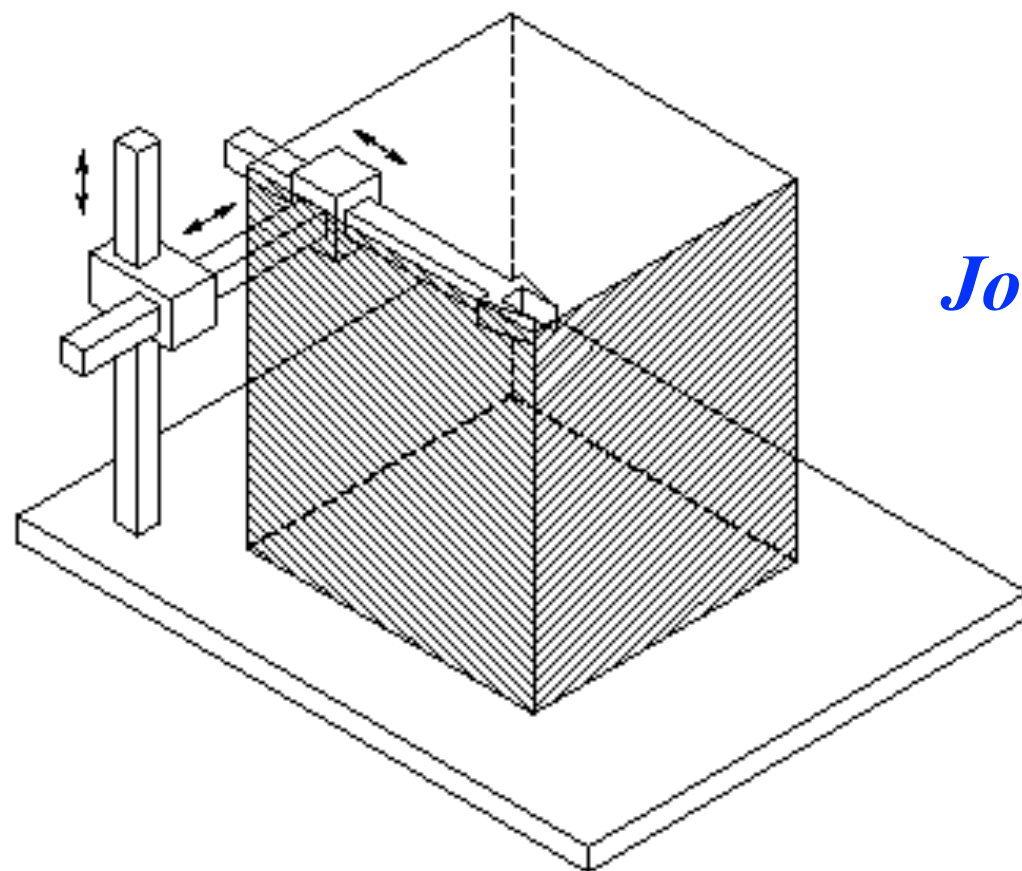


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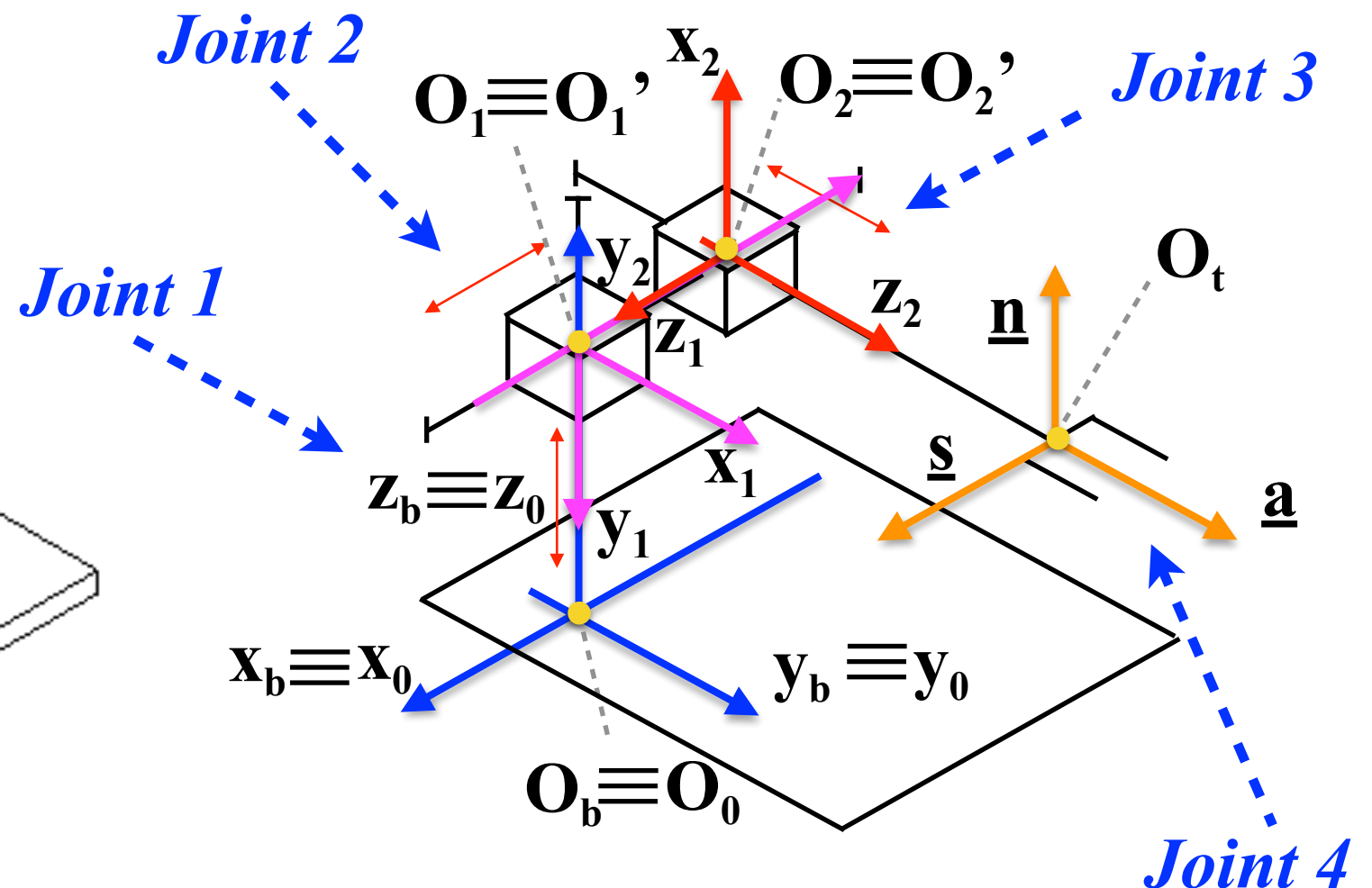
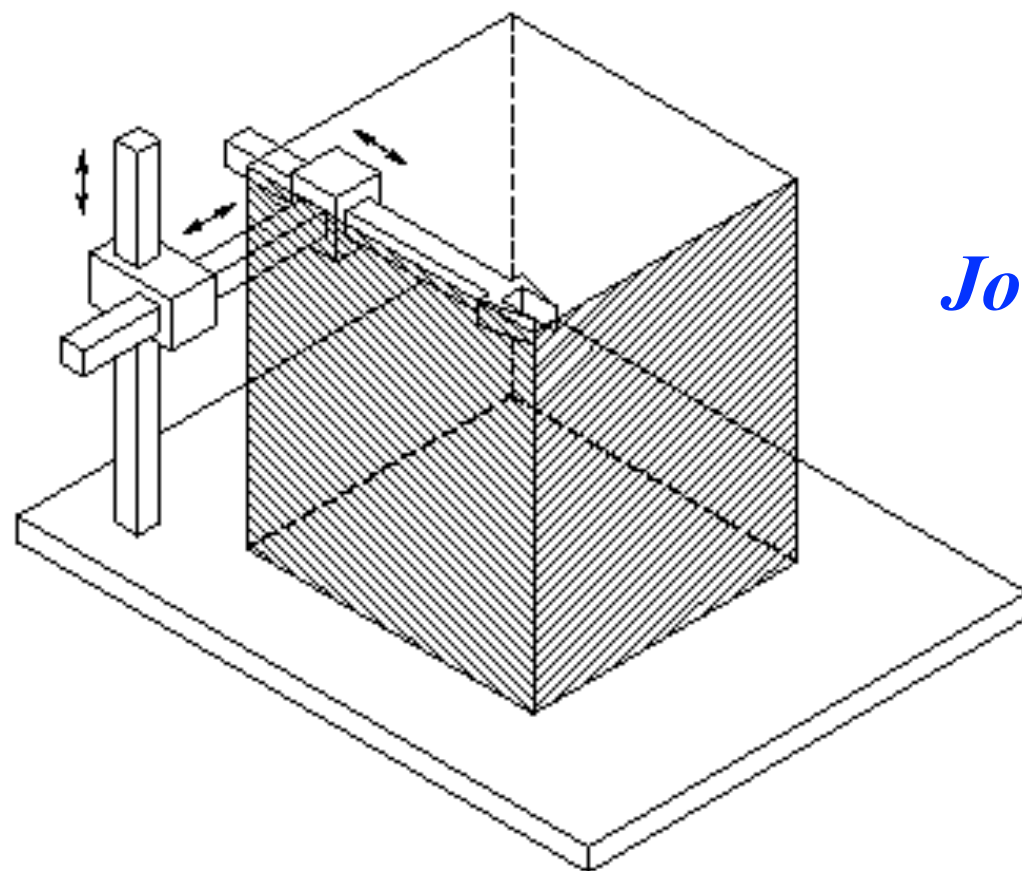


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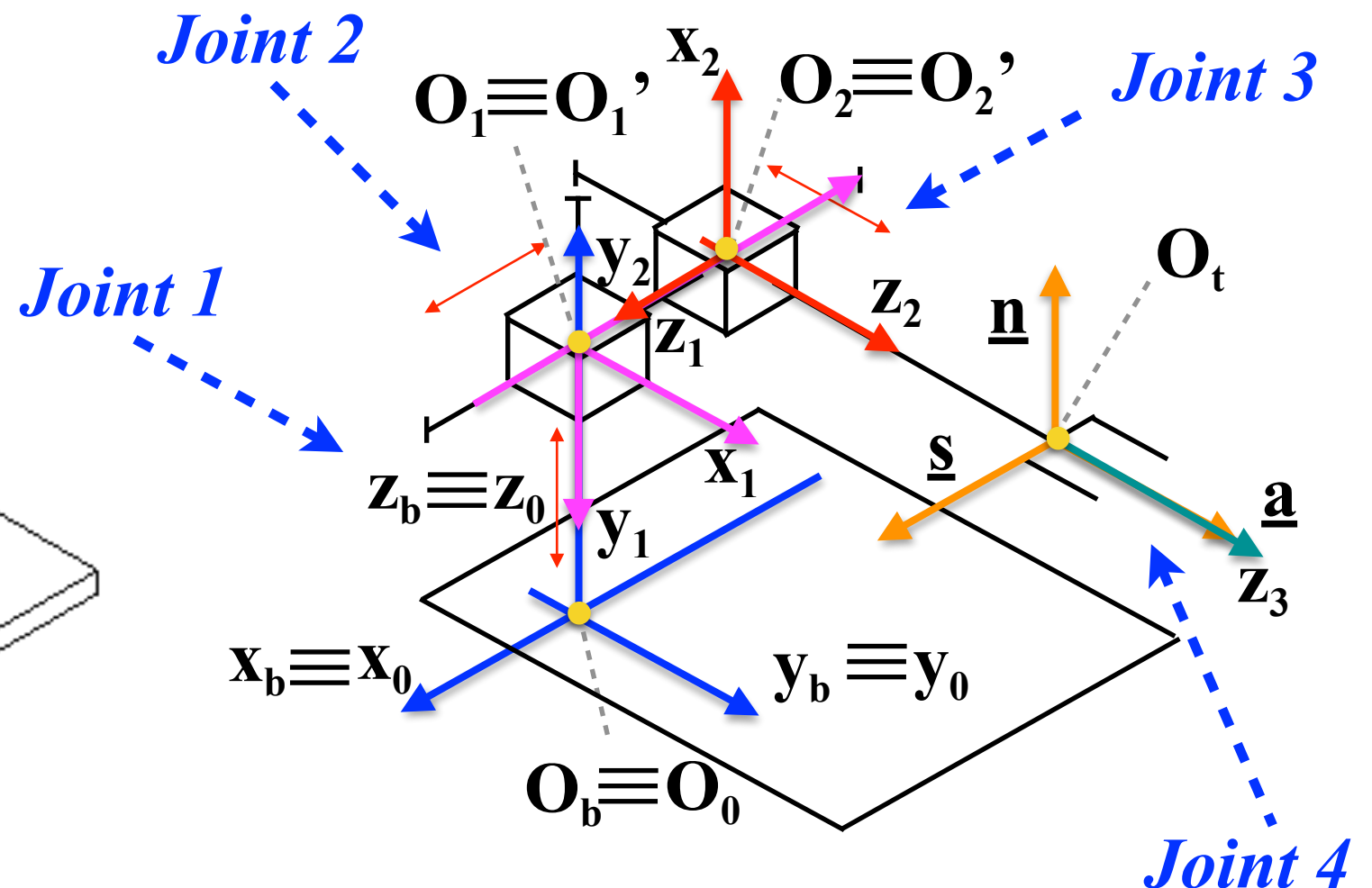
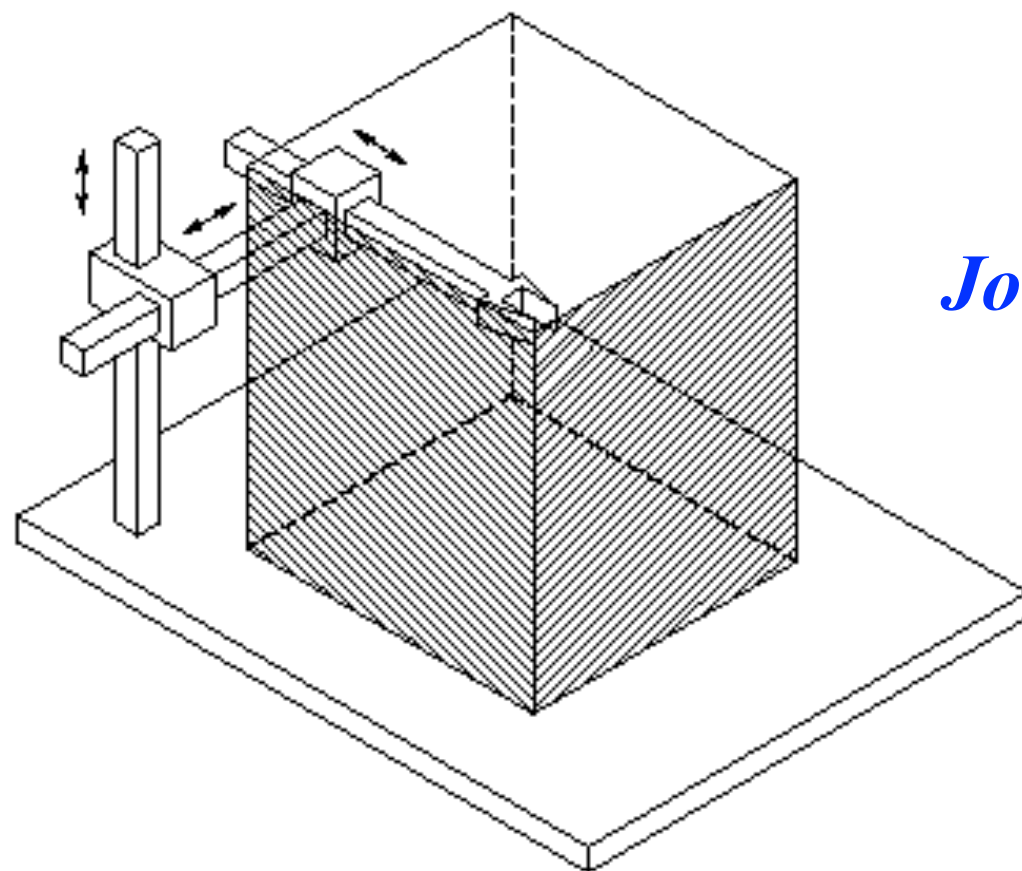


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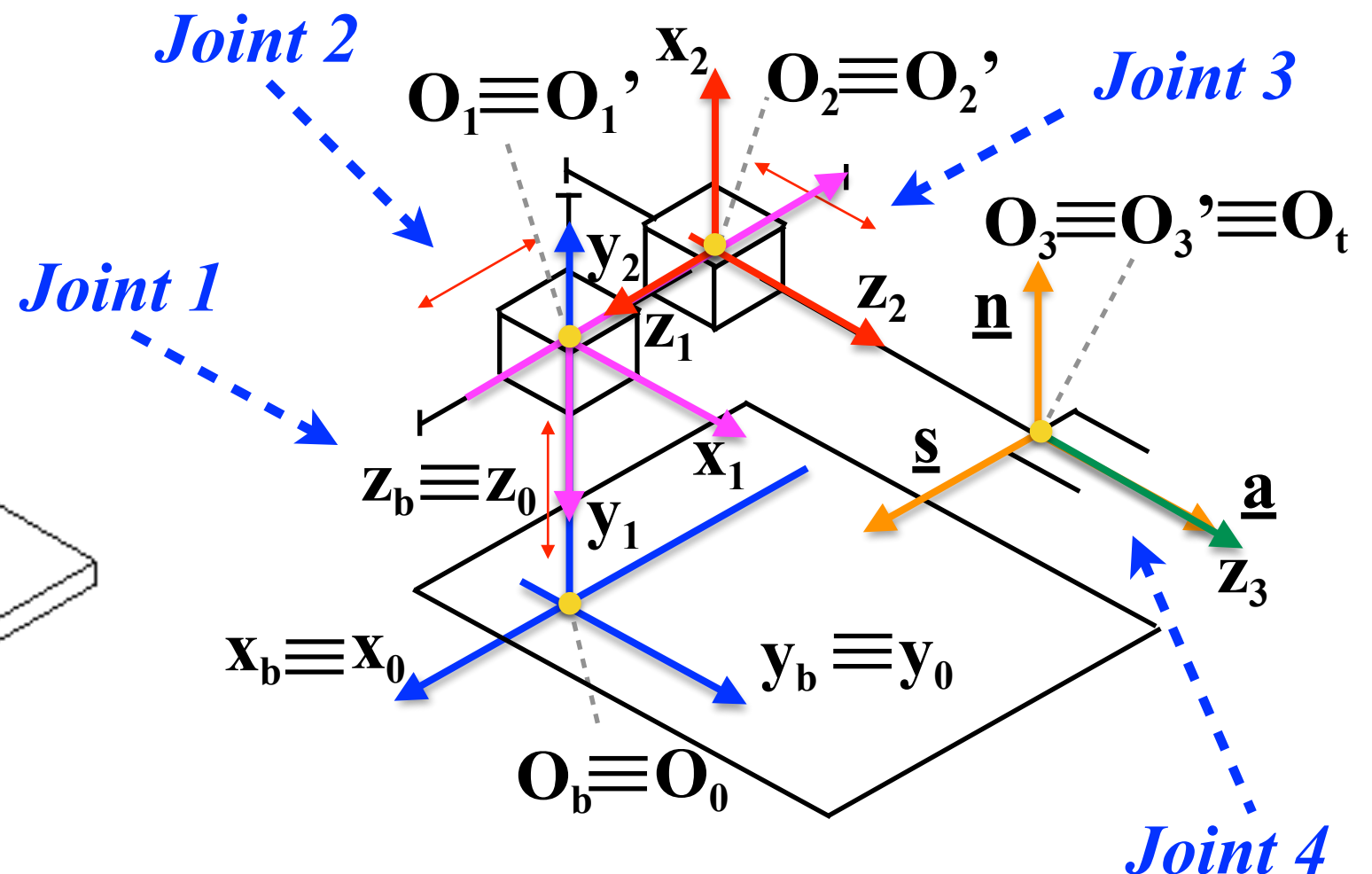
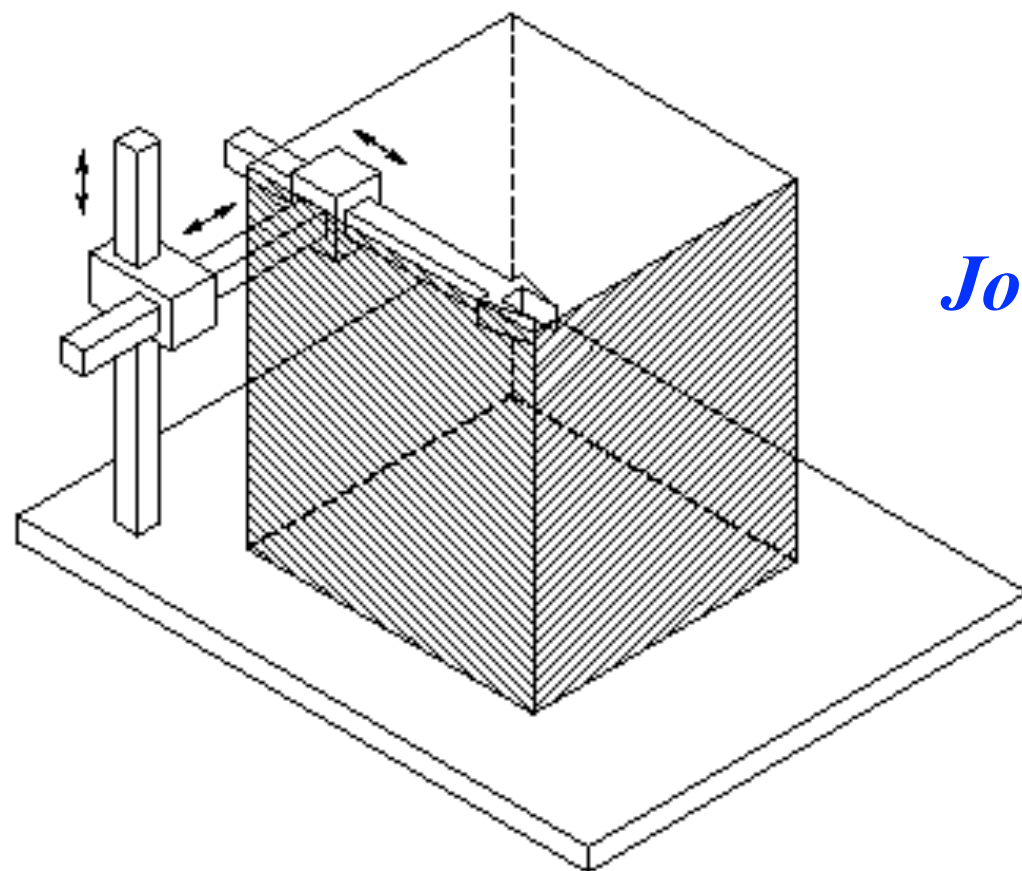


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Cartesian Robots

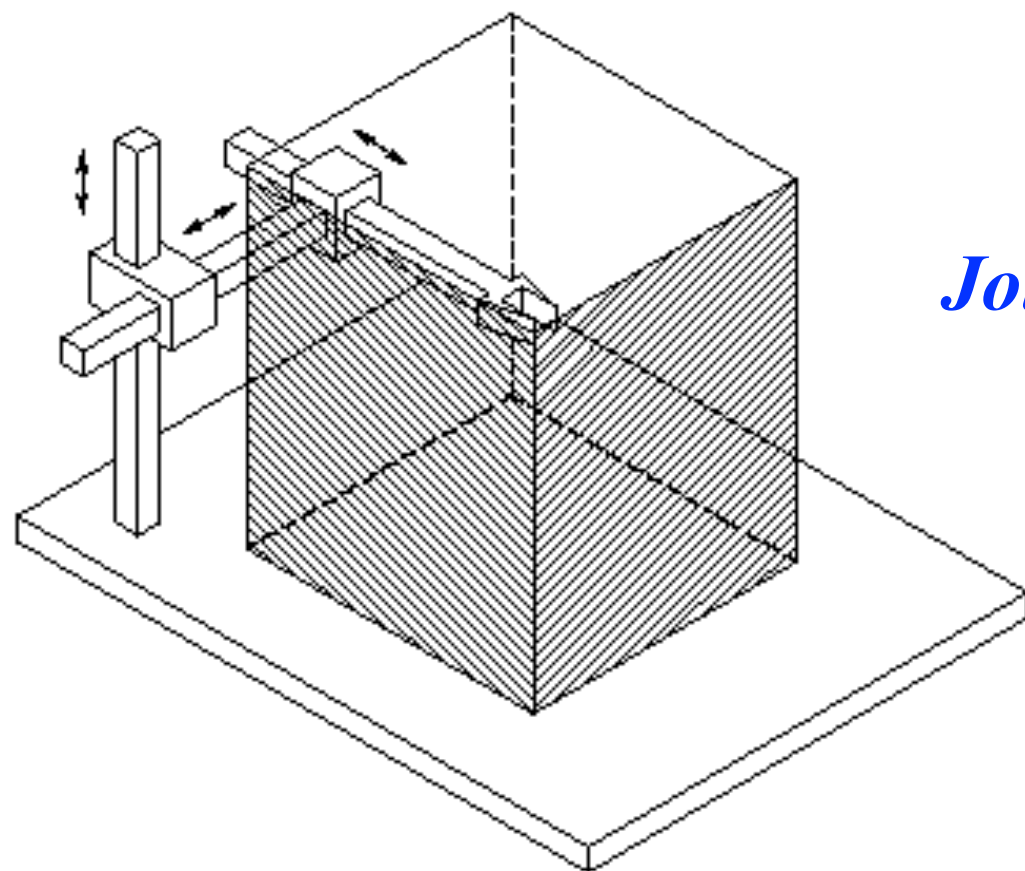
Assign the D-H Reference Frames from 0 to n



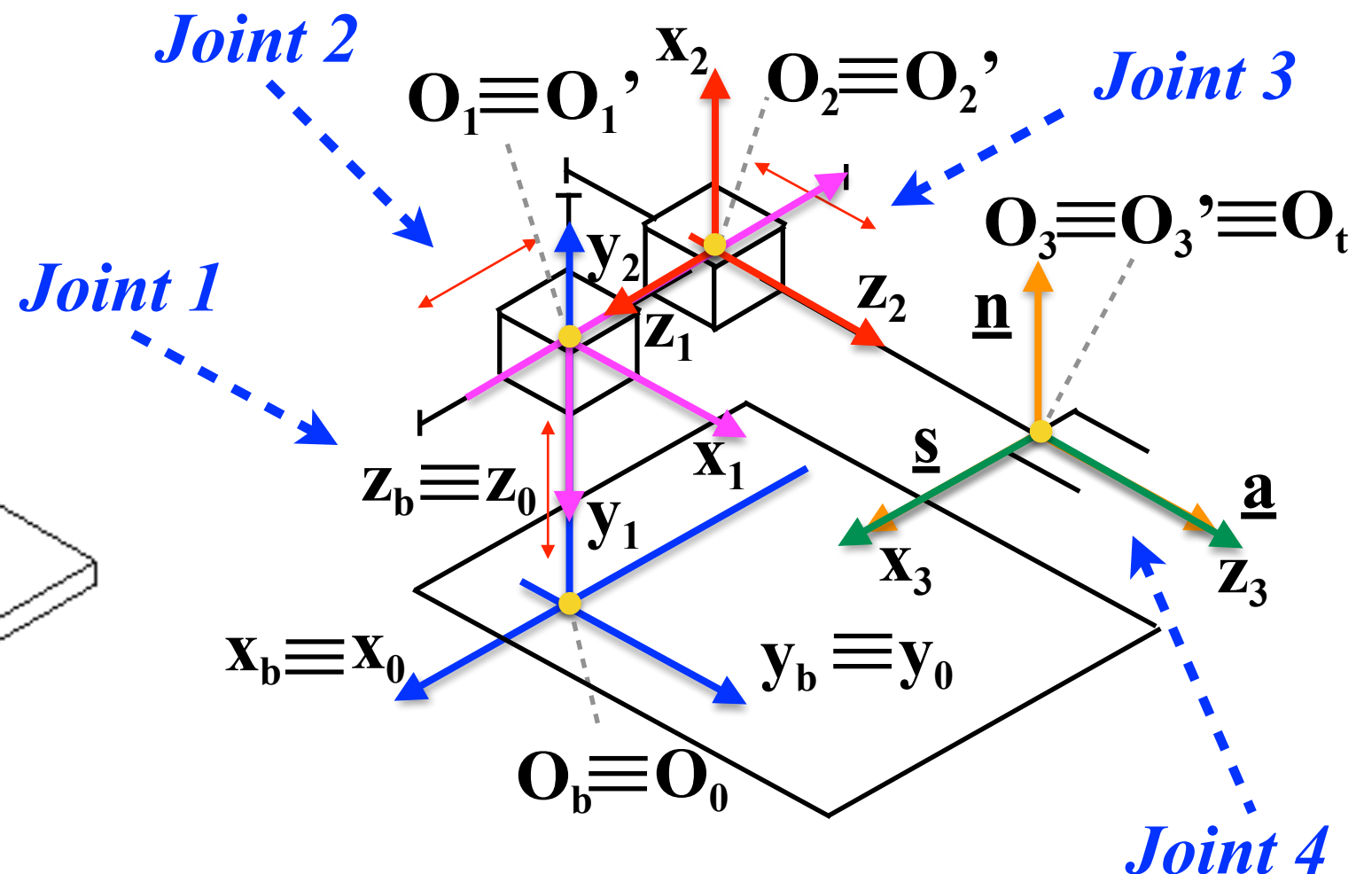
Source: Robotica - Modellistica, Pianificazione E Controllo
Siciliano, Sciavicco, Villani, Oriolo
Mcgraw-Hill (one of the course textbooks)

How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Cartesian Robots



Assign the D-H Reference Frames from 0 to n

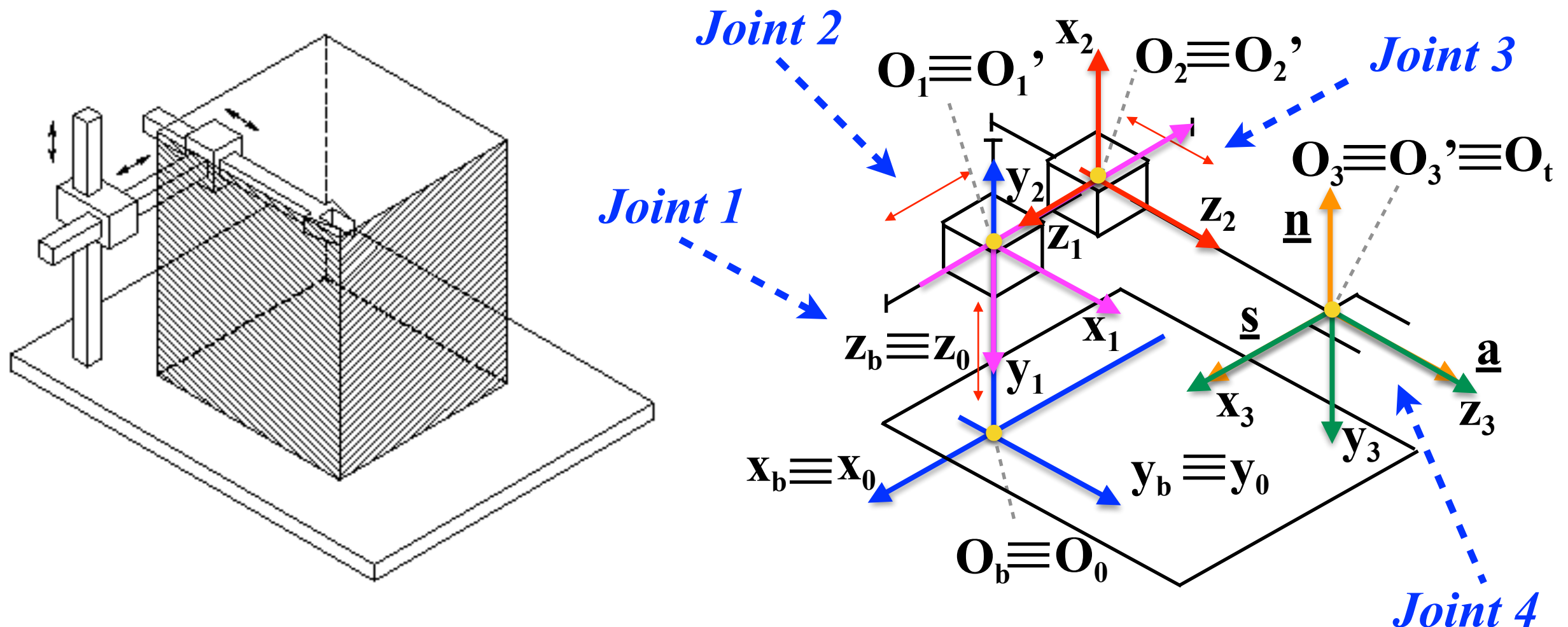


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How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Cartesian Robots

Assign the D-H Reference Frames from 0 to n



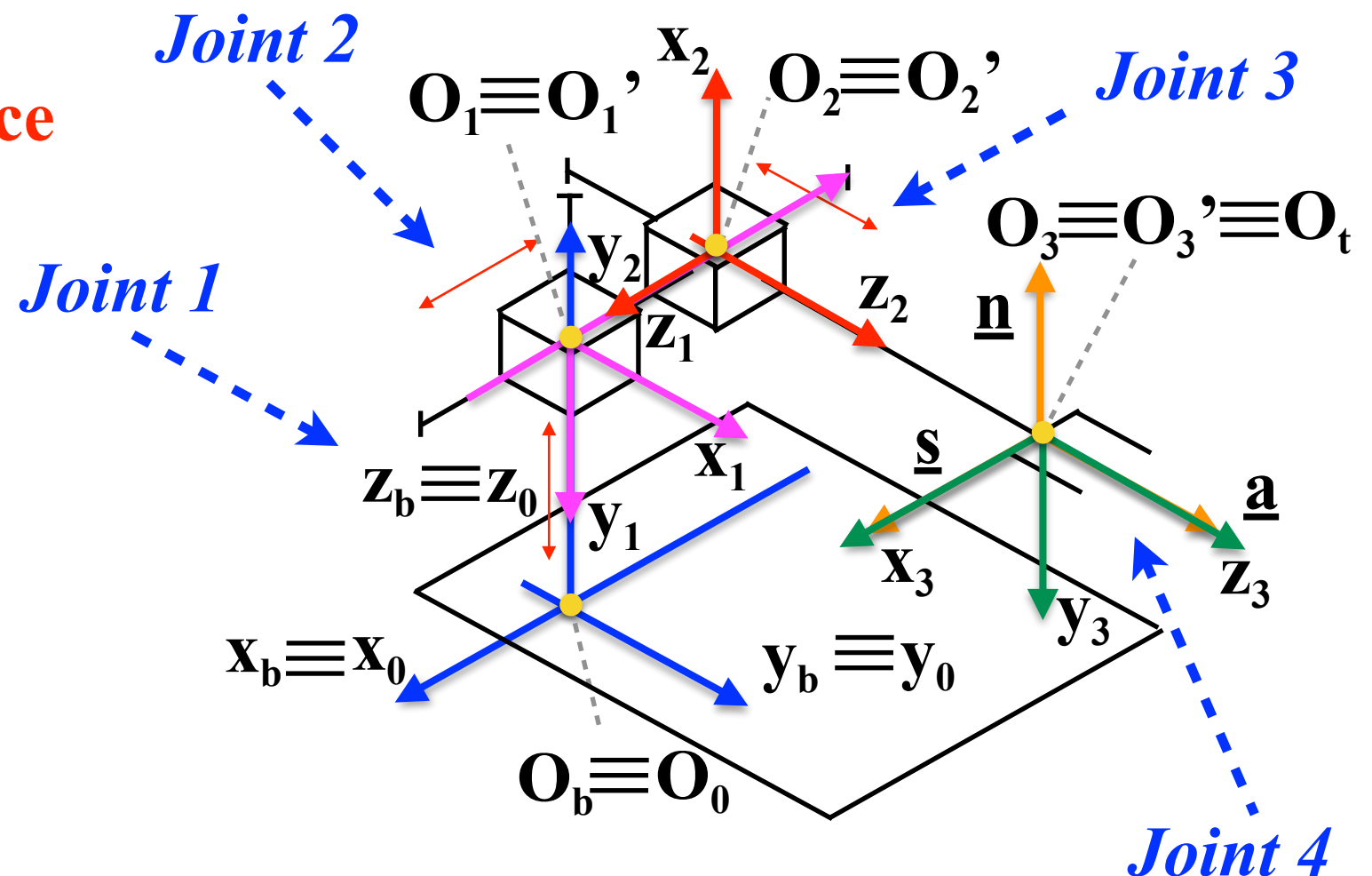
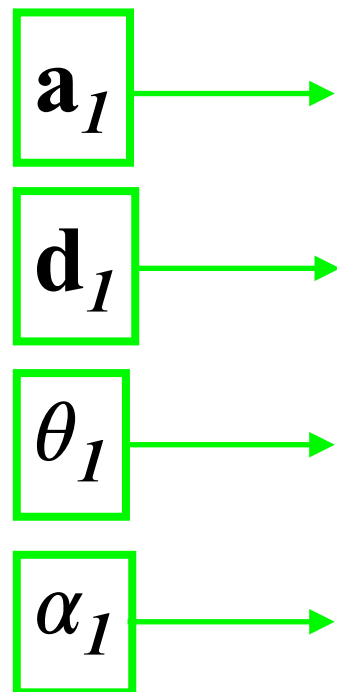
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How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Cartesian Robots

Determine the D-H parameters

D-H parameters for reference frame 1



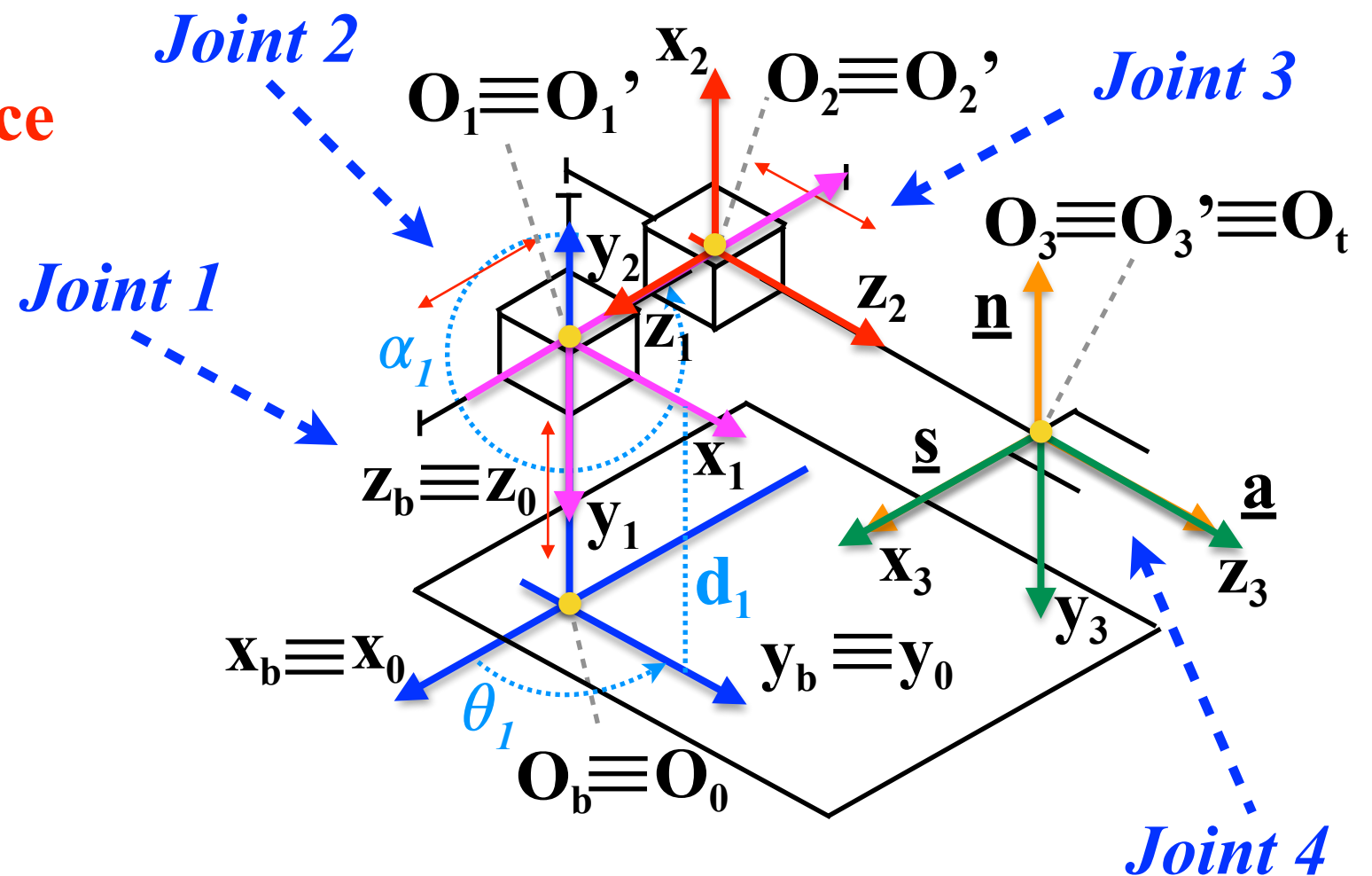
How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Cartesian Robots

Determine the D-H parameters

D-H parameters for reference frame 1

a_1	$\rightarrow$	0 [m]
d_1	$\rightarrow$	see picture
θ_1	$\rightarrow$	90°
α_1	$\rightarrow$	270°

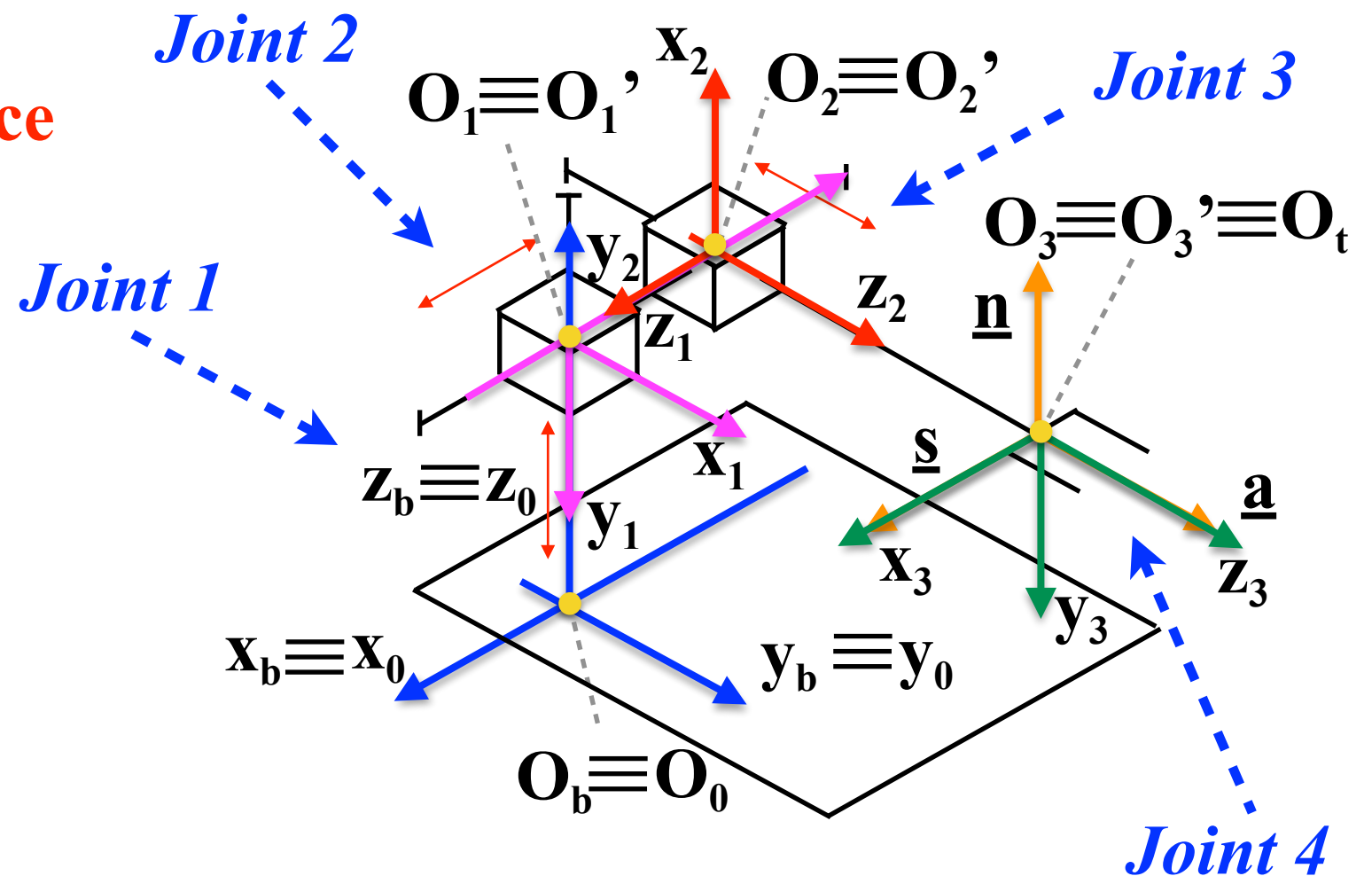
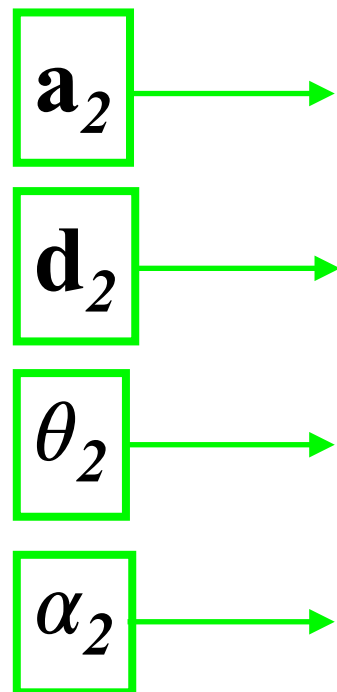


How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Cartesian Robots

Determine the D-H parameters

D-H parameters for reference frame 2



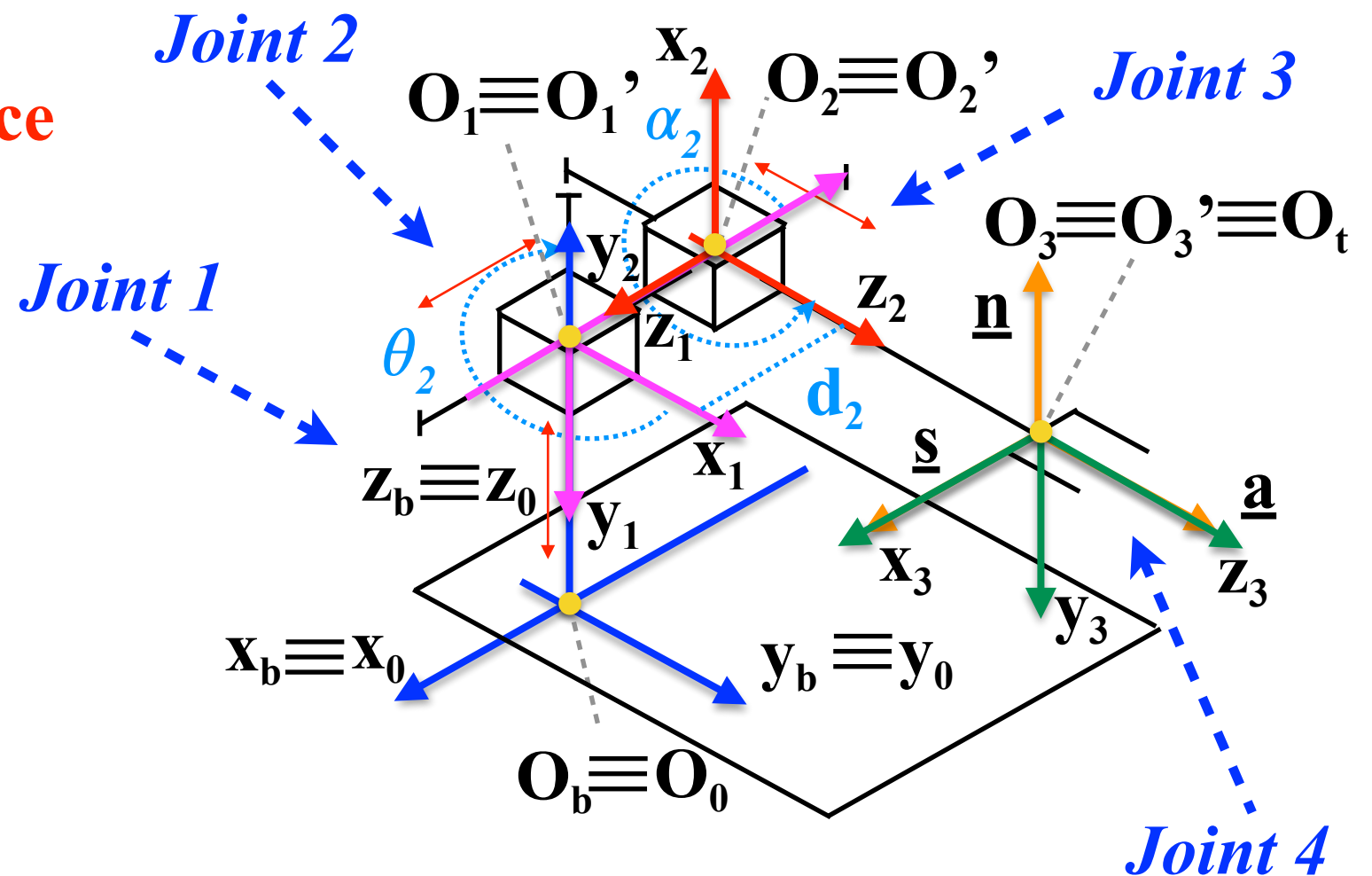
How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Cartesian Robots

Determine the D-H parameters

D-H parameters for reference frame 2

a_2	$\rightarrow$	0 [m]
d_2	$\rightarrow$	see picture
θ_2	$\rightarrow$	270°
α_2	$\rightarrow$	270°

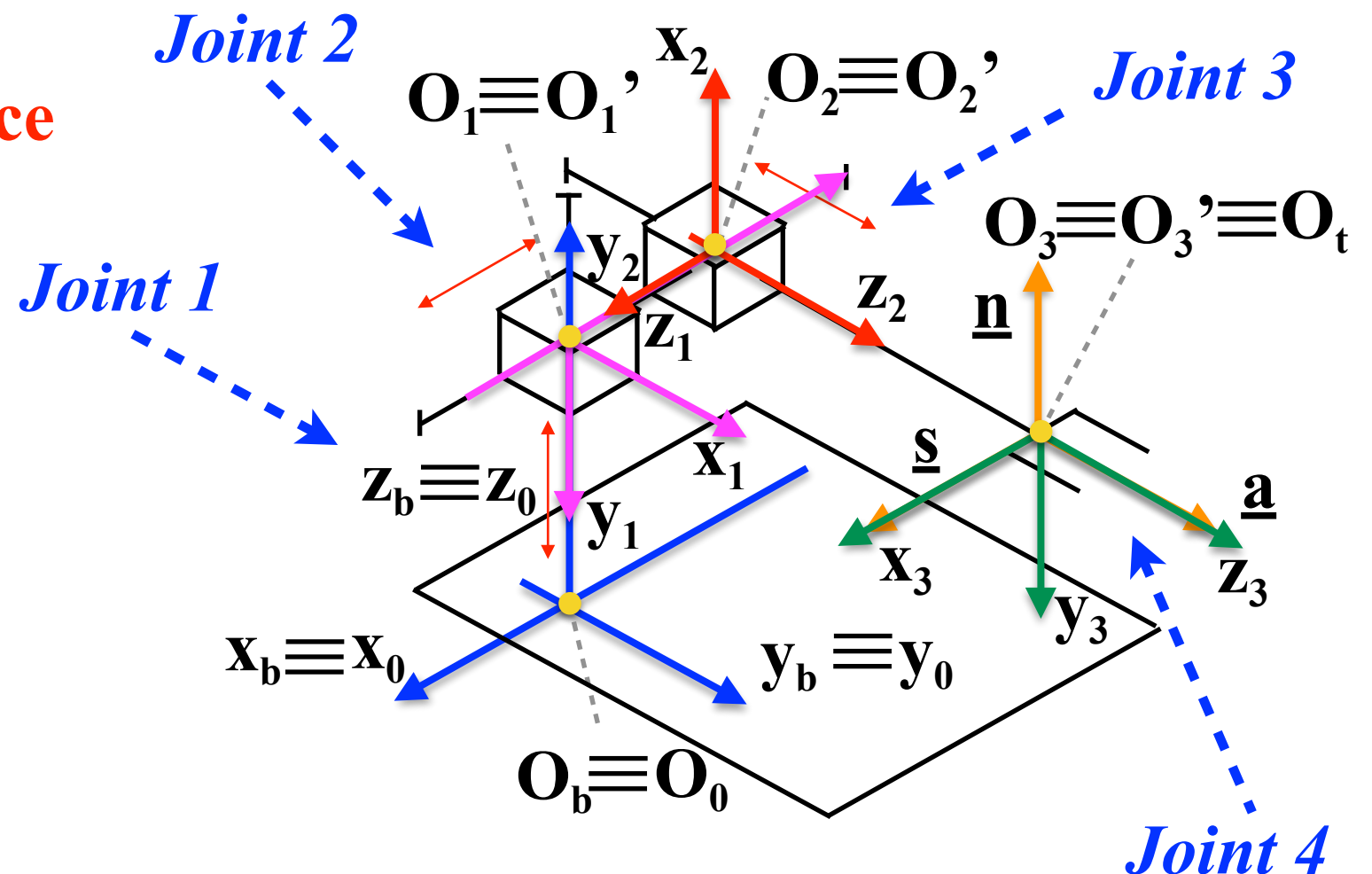
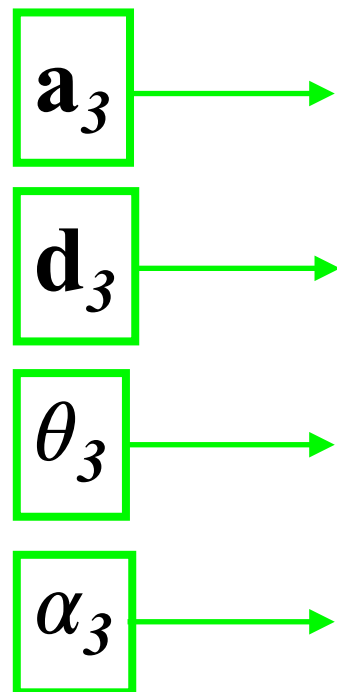


How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Cartesian Robots

Determine the D-H parameters

D-H parameters for reference frame 3



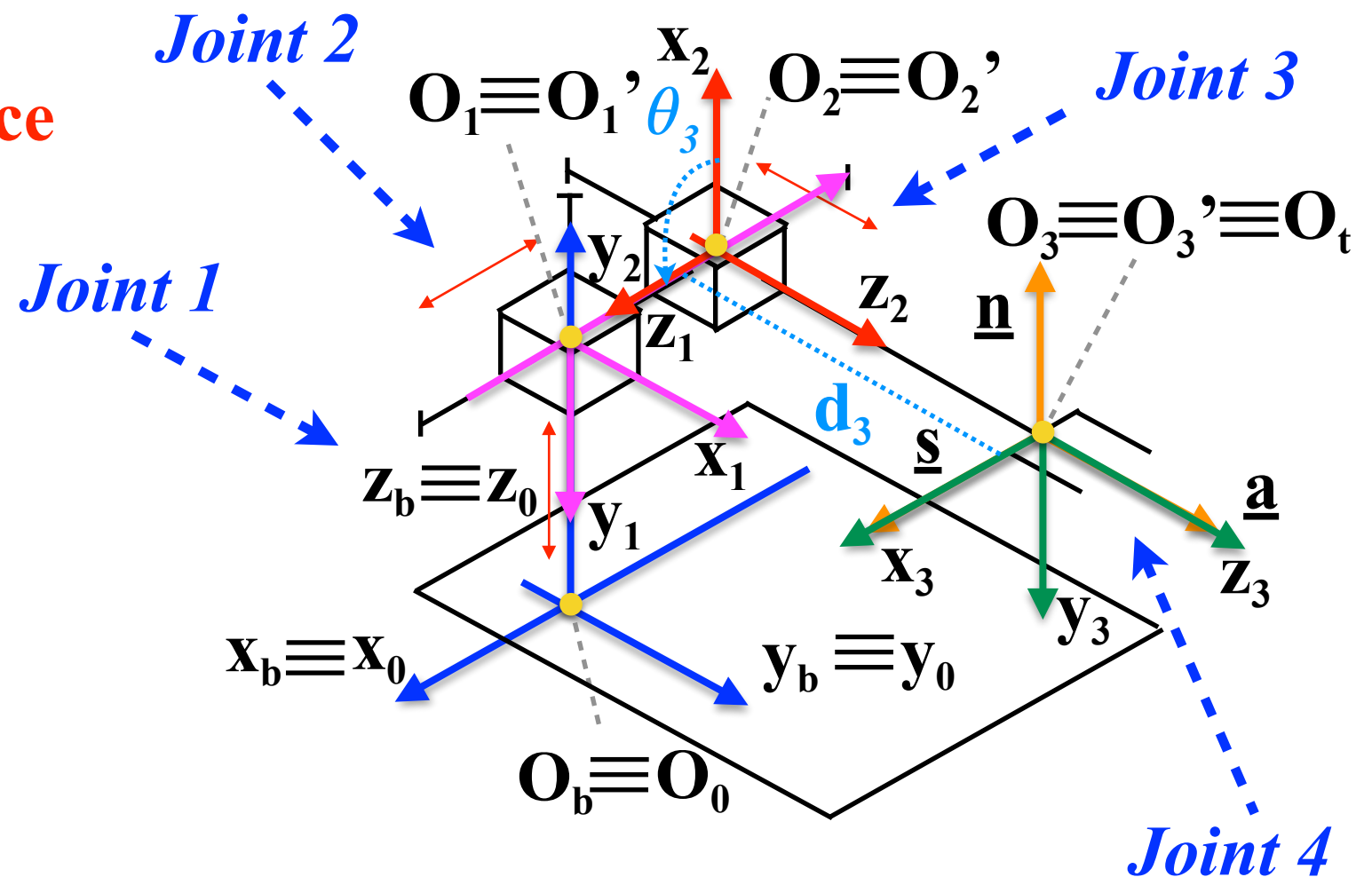
How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Cartesian Robots

Determine the D-H parameters

D-H parameters for reference frame 3

a_3	$\rightarrow$	0 [m]
d_3	$\rightarrow$	see picture
θ_3	$\rightarrow$	90°
α_3	$\rightarrow$	0°

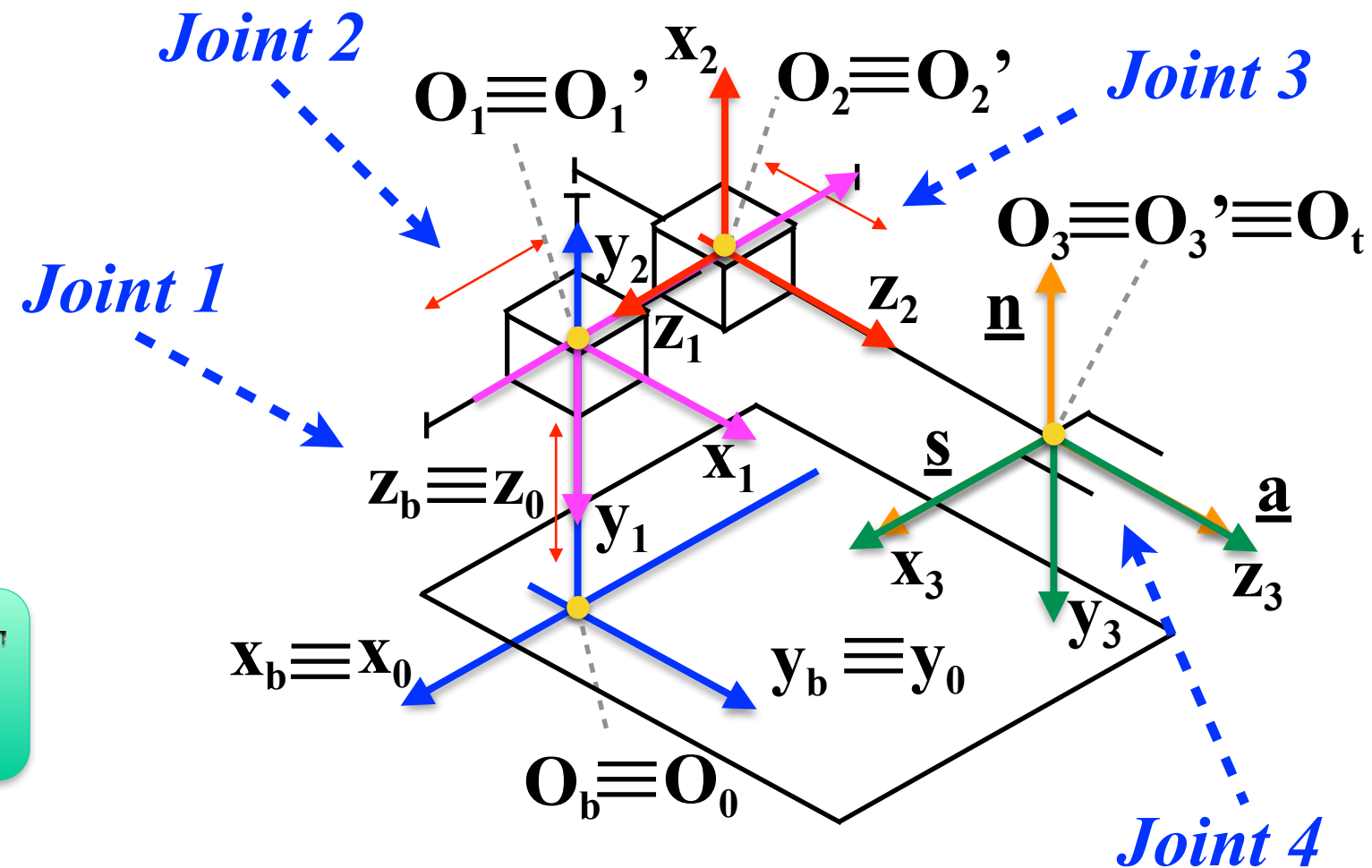


How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Cartesian Robots

Define the vector
of the joint variables

$$\underline{q} = [q_1, q_2, q_3]^T = [x_1, x_2, x_3]^T$$



How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Cartesian Robots

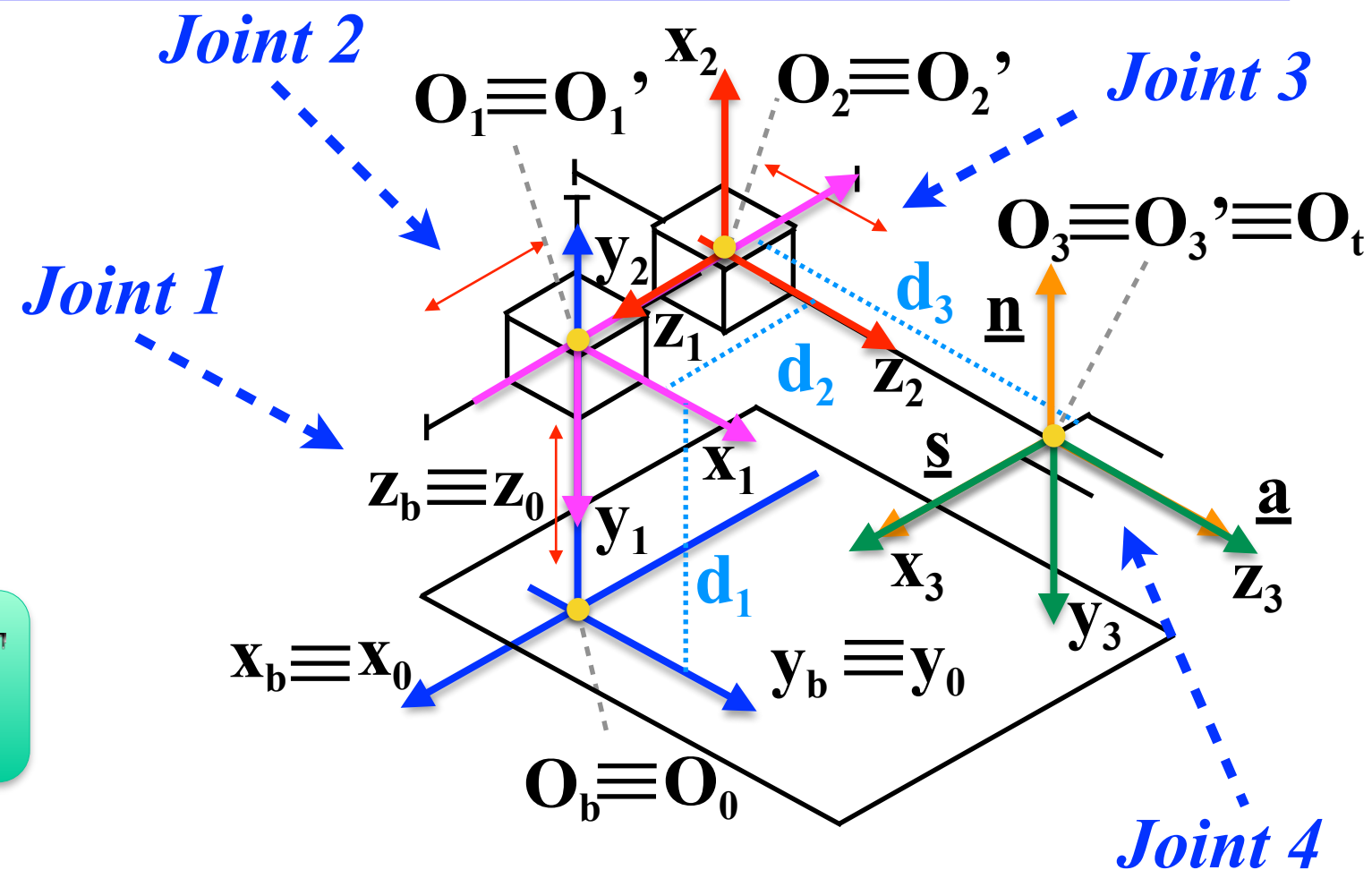
Define the vector
of the joint variables

$$\underline{q} = [q_1, q_2, q_3]^T = [x_1, x_2, x_3]^T$$

d_1

d_2

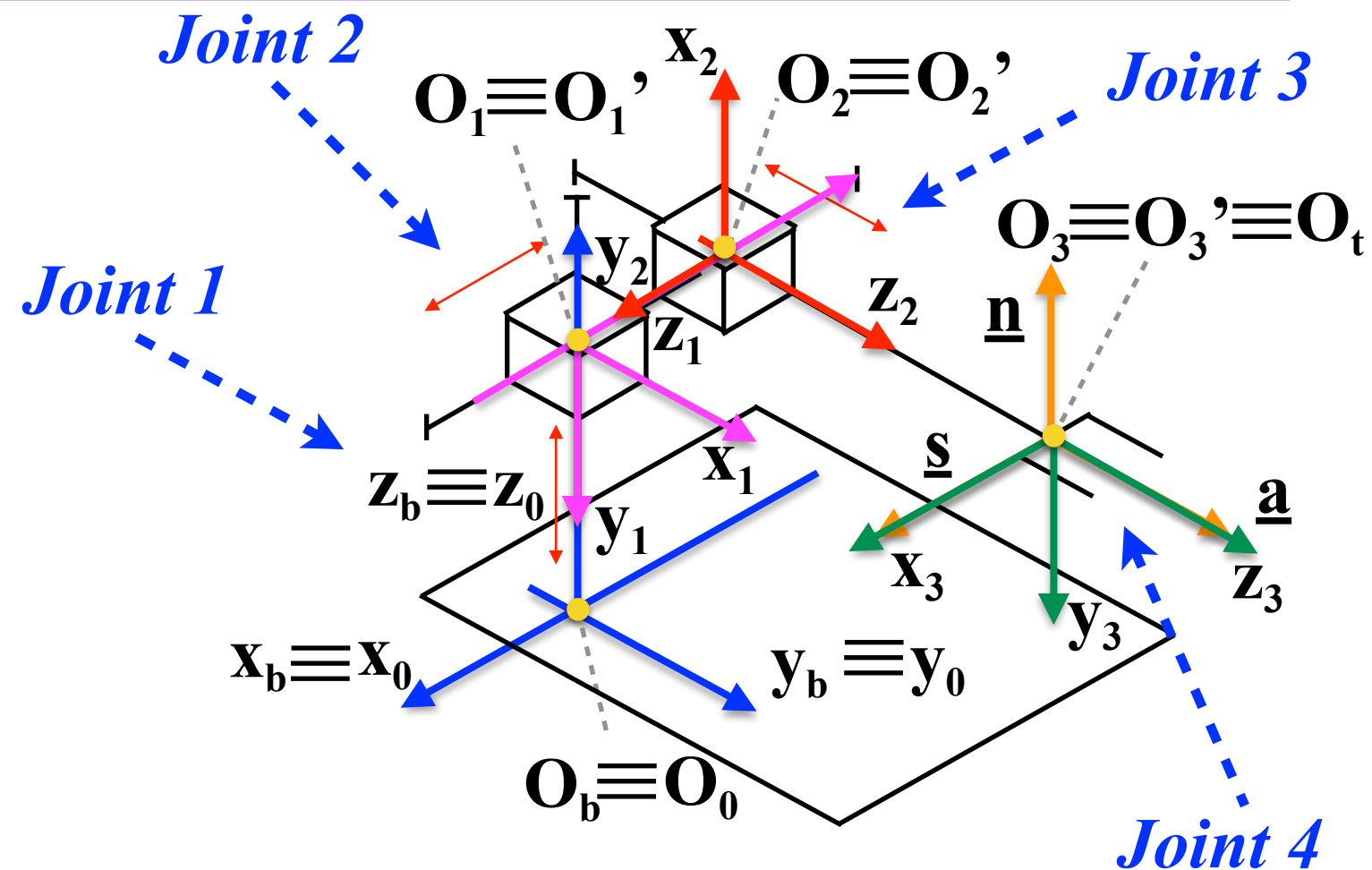
d_3



How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Cartesian Robots

Determine the homogeneous transformation matrix

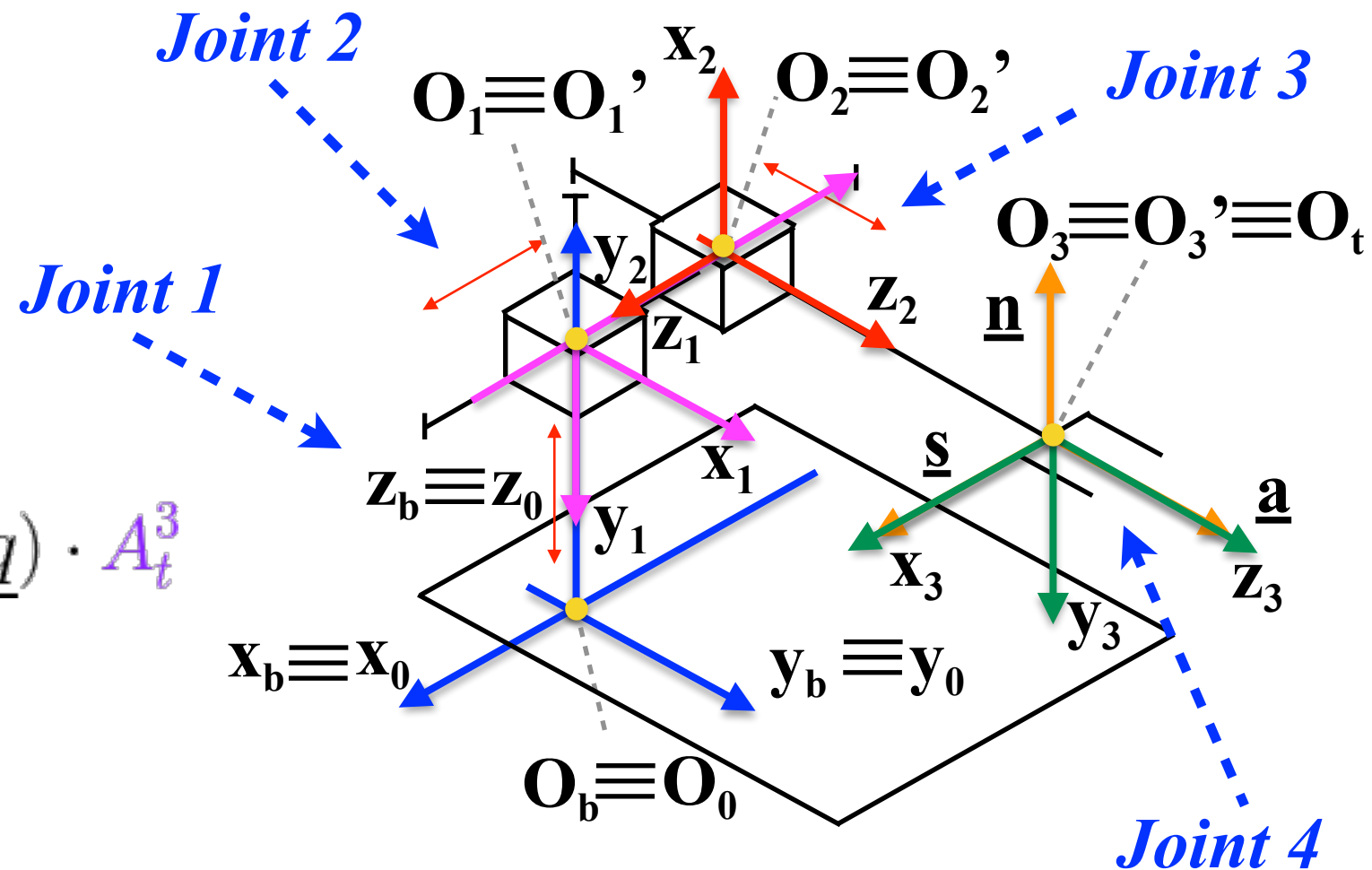


How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Cartesian Robots

Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = A_0^b \cdot A_1^0(\underline{q}) \cdot A_2^1(\underline{q}) \cdot A_3^2(\underline{q}) \cdot A_t^3$$

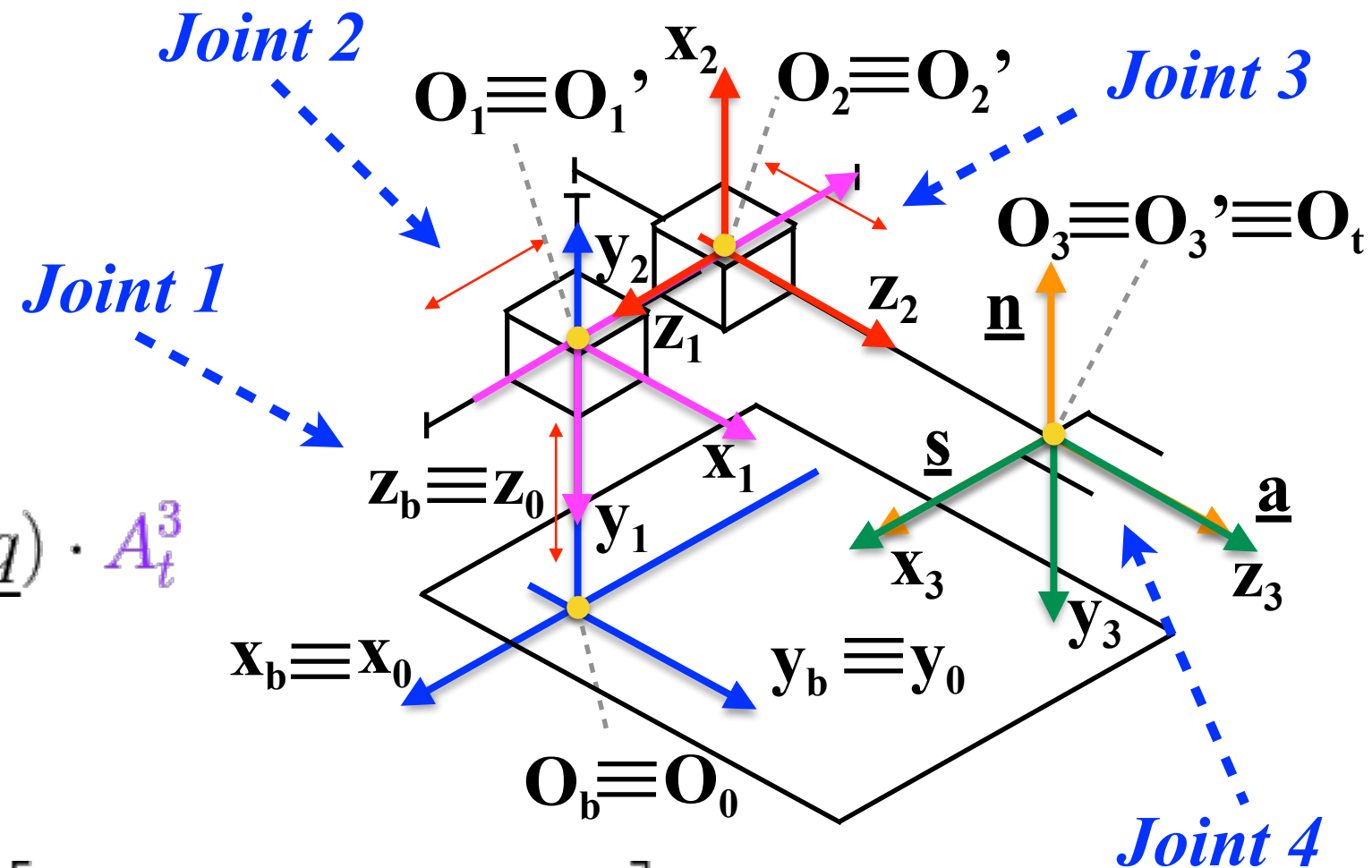


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Cartesian Robots

Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = A_0^b \cdot A_1^0(\underline{q}) \cdot A_2^1(\underline{q}) \cdot A_3^2(\underline{q}) \cdot A_t^3$$



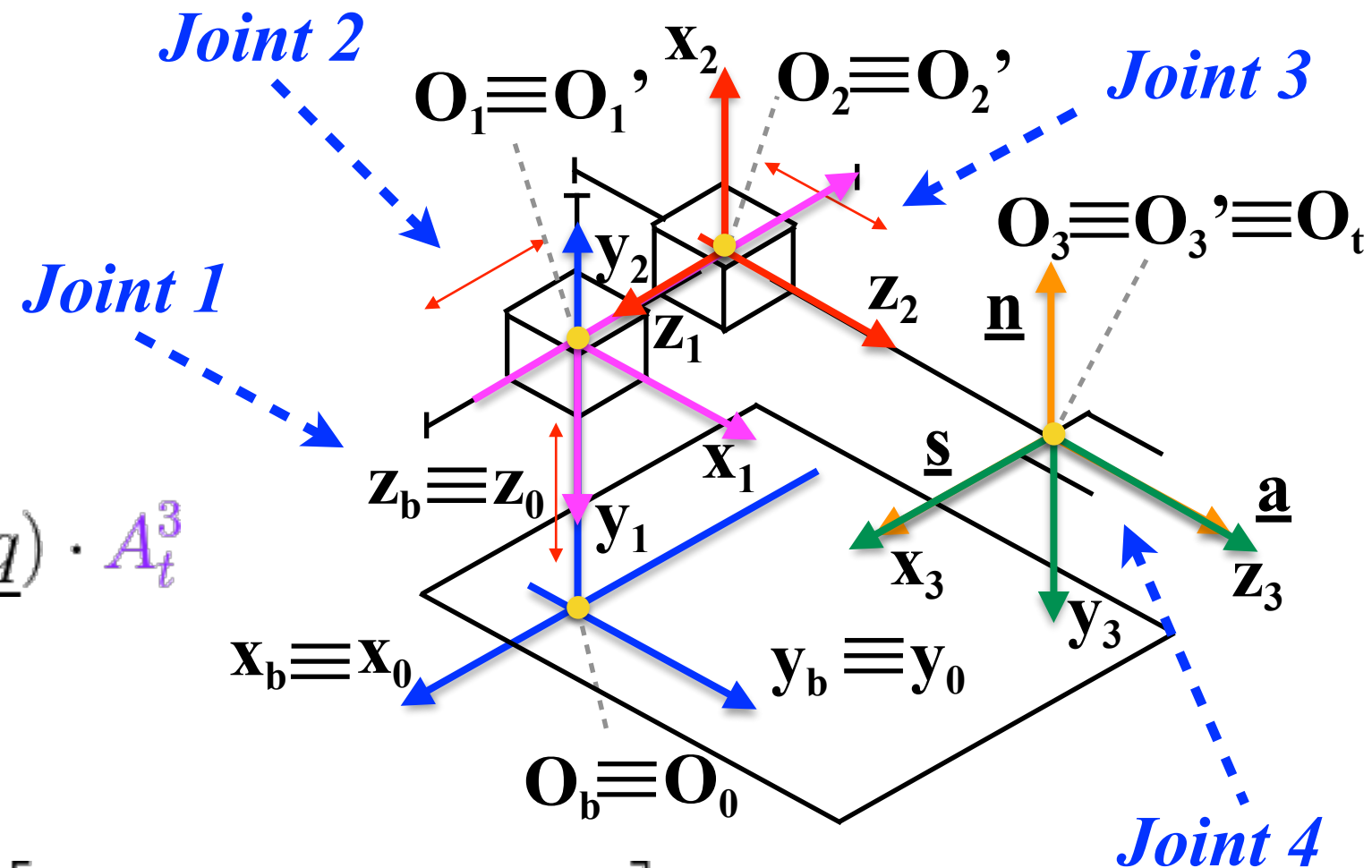
$$T_t^b(\underline{q}) = A_0^b \cdot \begin{bmatrix} c_{\theta_1} & -s_{\theta_1} c_{\alpha_1} & s_{\theta_1} s_{\alpha_1} & a_1 c_{\theta_1} \\ s_{\theta_1} & c_{\theta_1} c_{\alpha_1} & -c_{\theta_1} s_{\alpha_1} & a_1 s_{\theta_1} \\ 0 & s_{\alpha_1} & c_{\alpha_1} & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_2} & -s_{\theta_2} c_{\alpha_2} & s_{\theta_2} s_{\alpha_2} & a_2 c_{\theta_2} \\ s_{\theta_2} & c_{\theta_2} c_{\alpha_2} & -c_{\theta_2} s_{\alpha_2} & a_2 s_{\theta_2} \\ 0 & s_{\alpha_2} & c_{\alpha_2} & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_3} & -s_{\theta_3} c_{\alpha_3} & s_{\theta_3} s_{\alpha_3} & a_3 c_{\theta_3} \\ s_{\theta_3} & c_{\theta_3} c_{\alpha_3} & -c_{\theta_3} s_{\alpha_3} & a_3 s_{\theta_3} \\ 0 & s_{\alpha_3} & c_{\alpha_3} & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot A_t^3$$

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Cartesian Robots

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$$T_t^b(\underline{q}) = A_0^b \cdot A_1^0(\underline{q}) \cdot A_2^1(\underline{q}) \cdot A_3^2(\underline{q}) \cdot A_t^3$$



$$T_t^b(\underline{q}) = A_0^b \cdot \begin{bmatrix} c_{\theta_1} & -s_{\theta_1} c_{\alpha_1} & s_{\theta_1} s_{\alpha_1} & a_1 c_{\theta_1} \\ s_{\theta_1} & c_{\theta_1} c_{\alpha_1} & -c_{\theta_1} s_{\alpha_1} & a_1 s_{\theta_1} \\ 0 & s_{\alpha_1} & c_{\alpha_1} & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_2} & -s_{\theta_2} c_{\alpha_2} & s_{\theta_2} s_{\alpha_2} & a_2 c_{\theta_2} \\ s_{\theta_2} & c_{\theta_2} c_{\alpha_2} & -c_{\theta_2} s_{\alpha_2} & a_2 s_{\theta_2} \\ 0 & s_{\alpha_2} & c_{\alpha_2} & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_3} & -s_{\theta_3} c_{\alpha_3} & s_{\theta_3} s_{\alpha_3} & a_3 c_{\theta_3} \\ s_{\theta_3} & c_{\theta_3} c_{\alpha_3} & -c_{\theta_3} s_{\alpha_3} & a_3 s_{\theta_3} \\ 0 & s_{\alpha_3} & c_{\alpha_3} & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot A_t^3$$

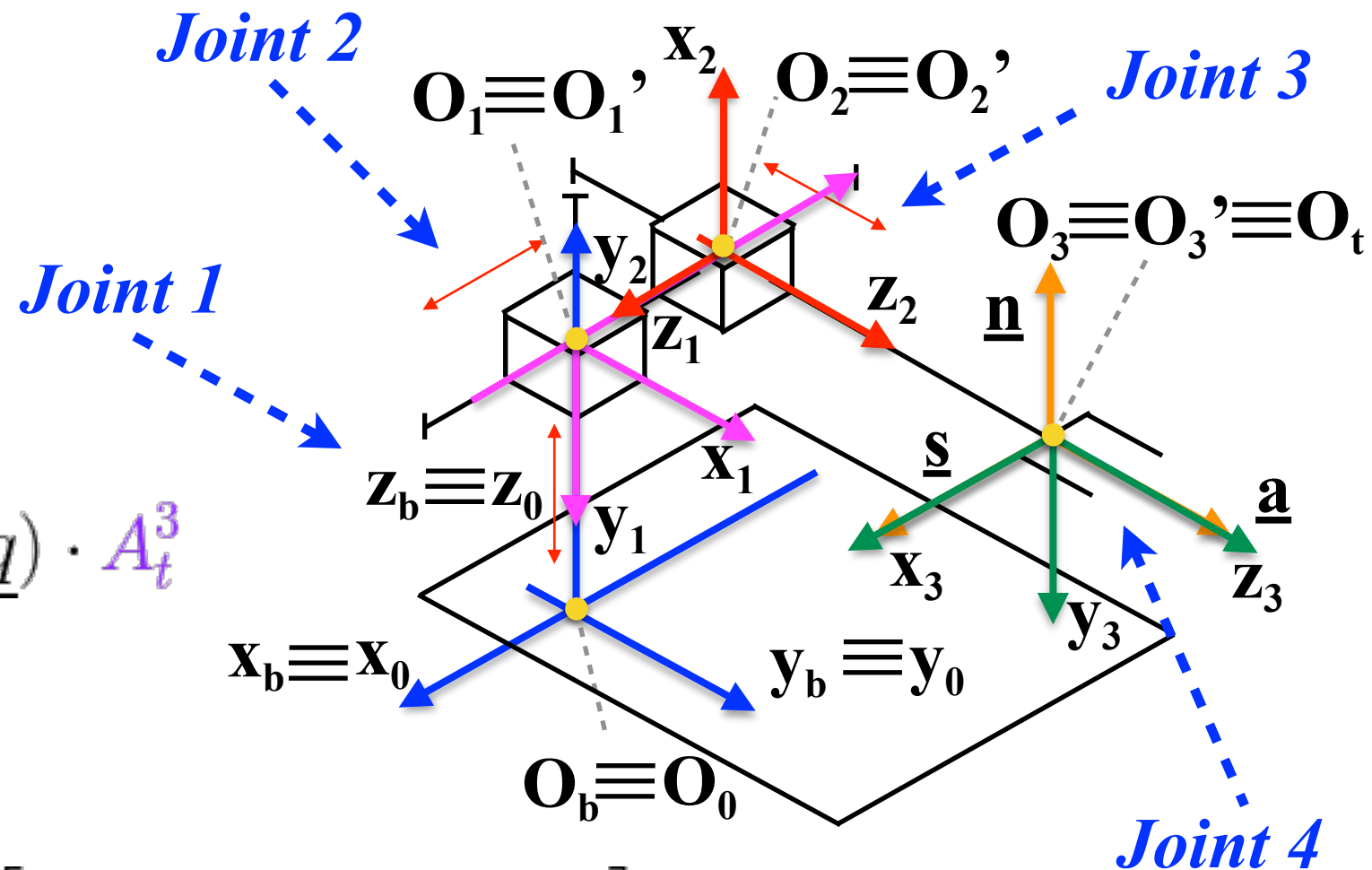
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Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = A_0^b \cdot A_1^0(\underline{q}) \cdot A_2^1(\underline{q}) \cdot A_3^2(\underline{q}) \cdot A_t^3$$

$$T_t^b(\underline{q}) = I_{4 \times 4} \cdot \begin{bmatrix} c_{\theta_1} & -s_{\theta_1} c_{\alpha_1} & s_{\theta_1} s_{\alpha_1} & a_1 c_{\theta_1} \\ s_{\theta_1} & c_{\theta_1} c_{\alpha_1} & -c_{\theta_1} s_{\alpha_1} & a_1 s_{\theta_1} \\ 0 & s_{\alpha_1} & c_{\alpha_1} & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_2} & -s_{\theta_2} c_{\alpha_2} & s_{\theta_2} s_{\alpha_2} & a_2 c_{\theta_2} \\ s_{\theta_2} & c_{\theta_2} c_{\alpha_2} & -c_{\theta_2} s_{\alpha_2} & a_2 s_{\theta_2} \\ 0 & s_{\alpha_2} & c_{\alpha_2} & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_3} & -s_{\theta_3} c_{\alpha_3} & s_{\theta_3} s_{\alpha_3} & a_3 c_{\theta_3} \\ s_{\theta_3} & c_{\theta_3} c_{\alpha_3} & -c_{\theta_3} s_{\alpha_3} & a_3 s_{\theta_3} \\ 0 & s_{\alpha_3} & c_{\alpha_3} & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot A_t^3$$

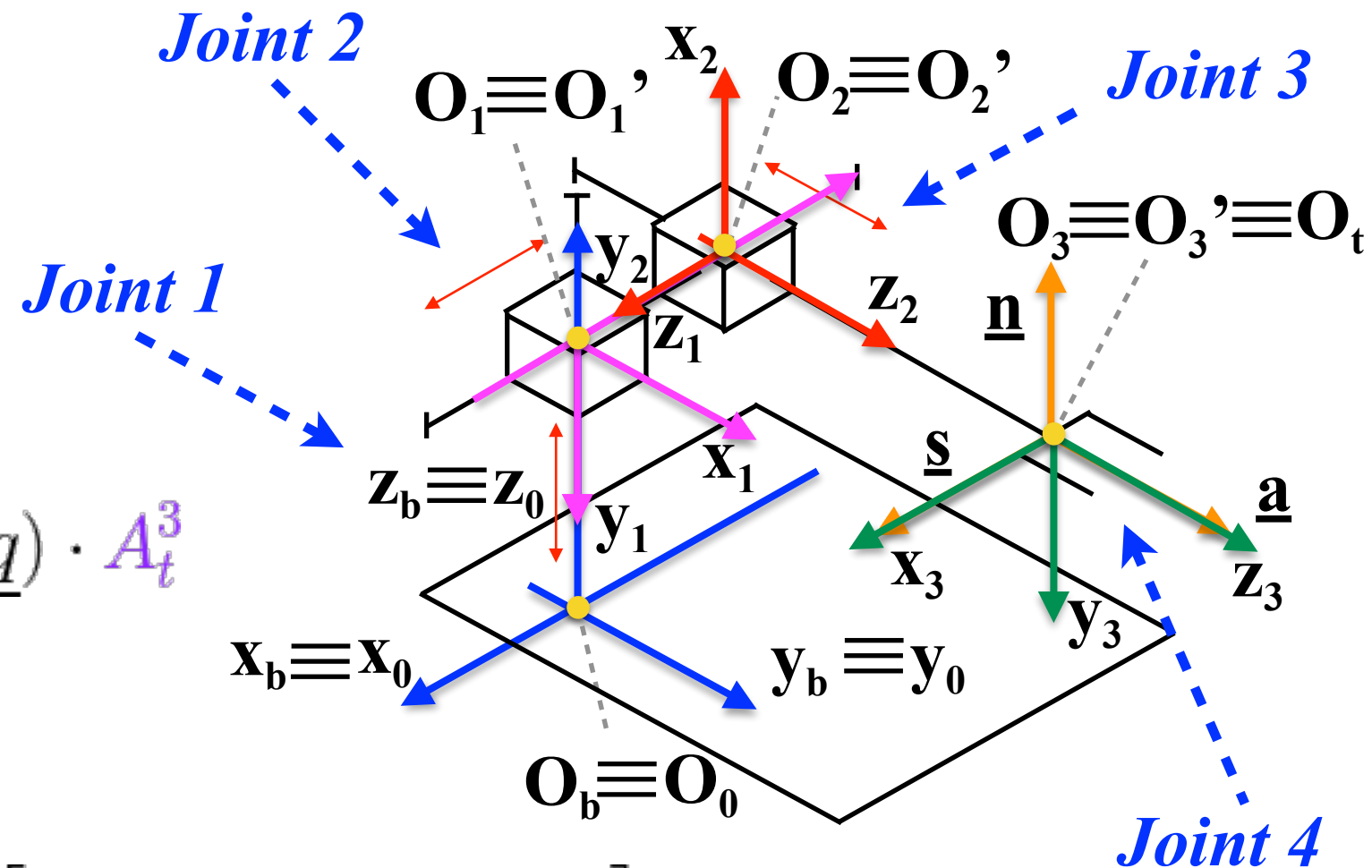


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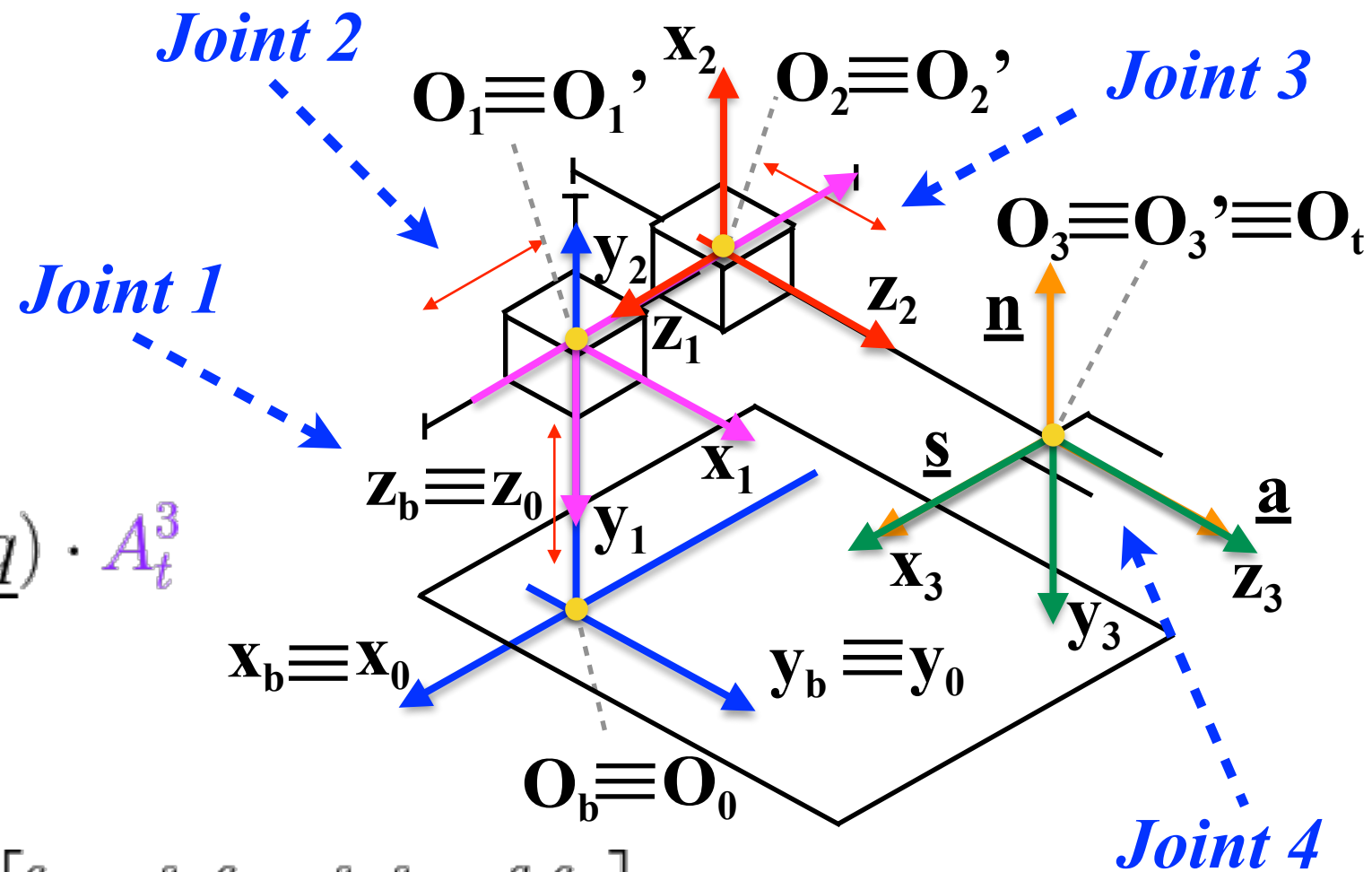
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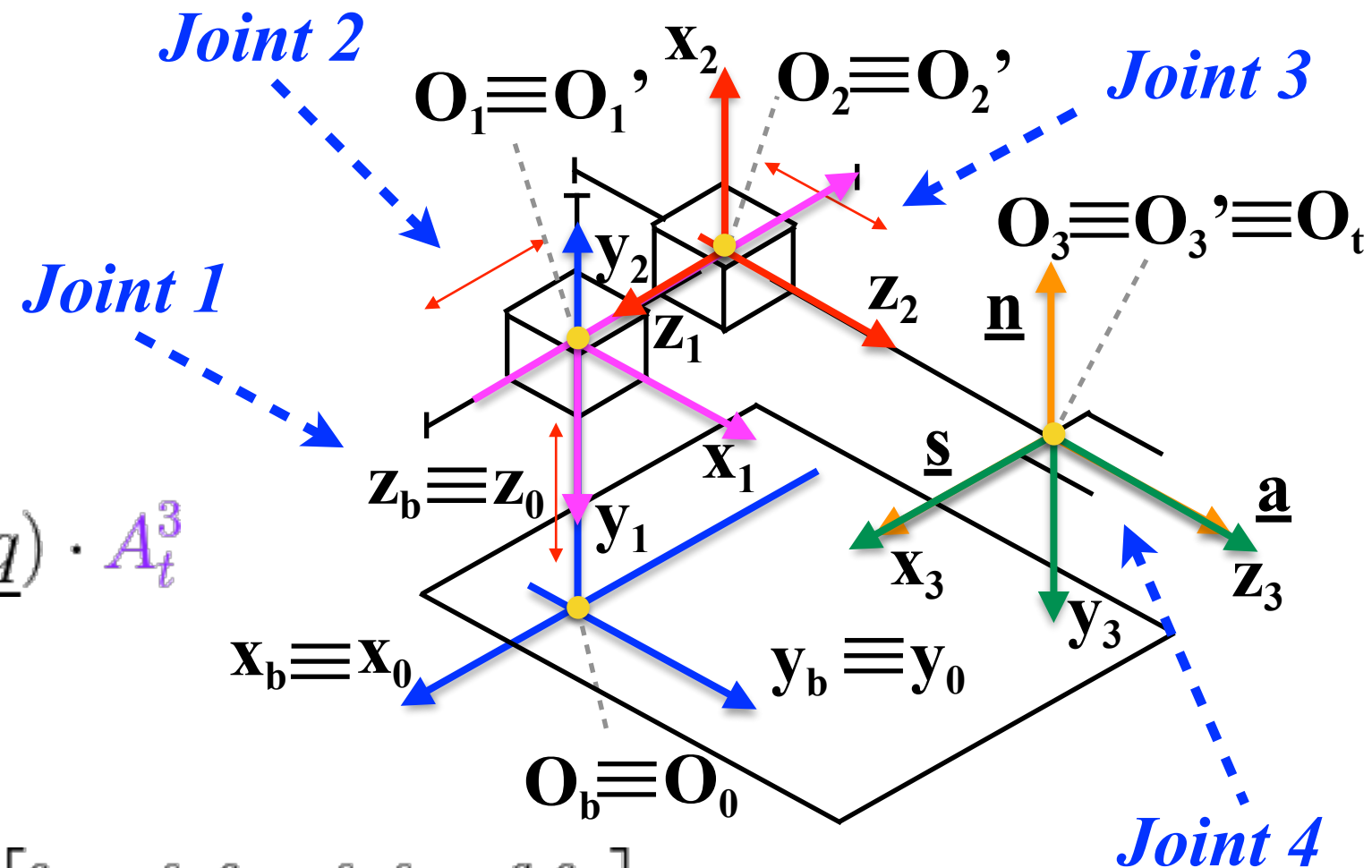


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Cartesian Robots

Determine the homogeneous transformation matrix

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Complete the computations as homework



UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

Course of Robot Control

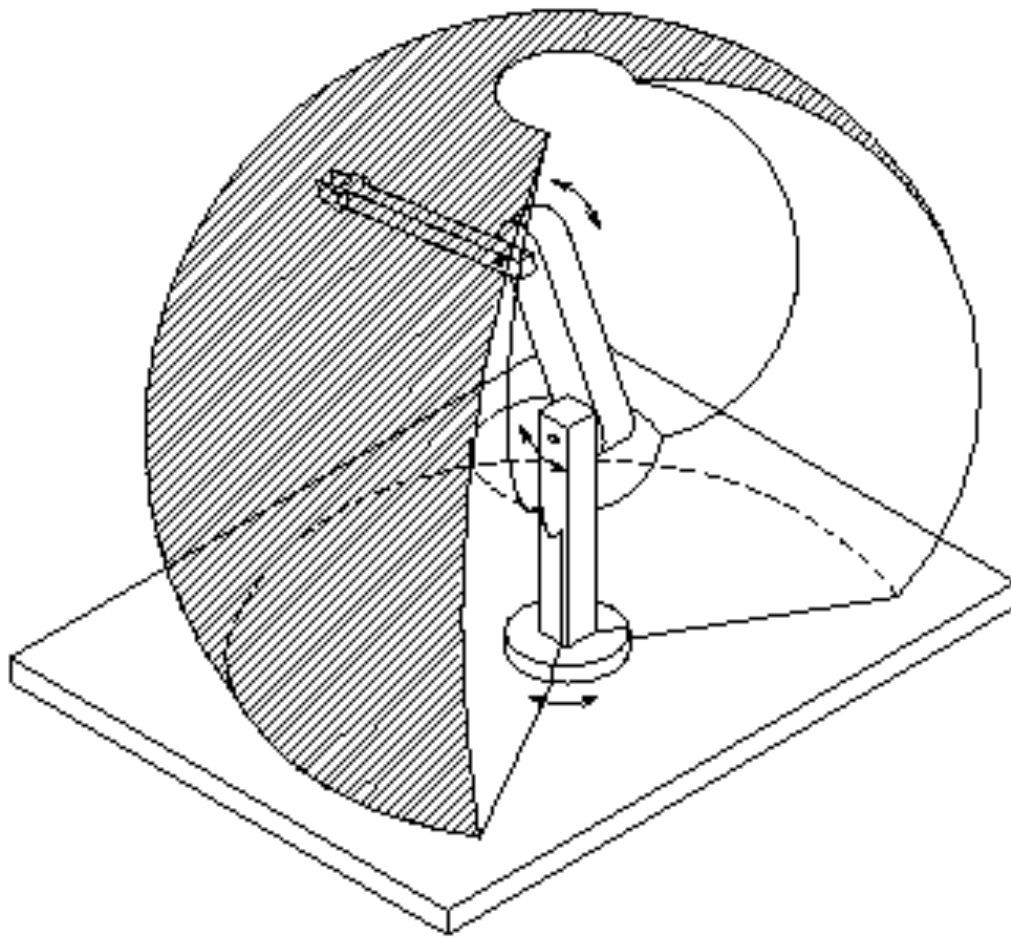
Prof. Antonella Ferrara



D-H Convention Based Approach: Anthropomorphic Robots

How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Anthropomorphic Robots

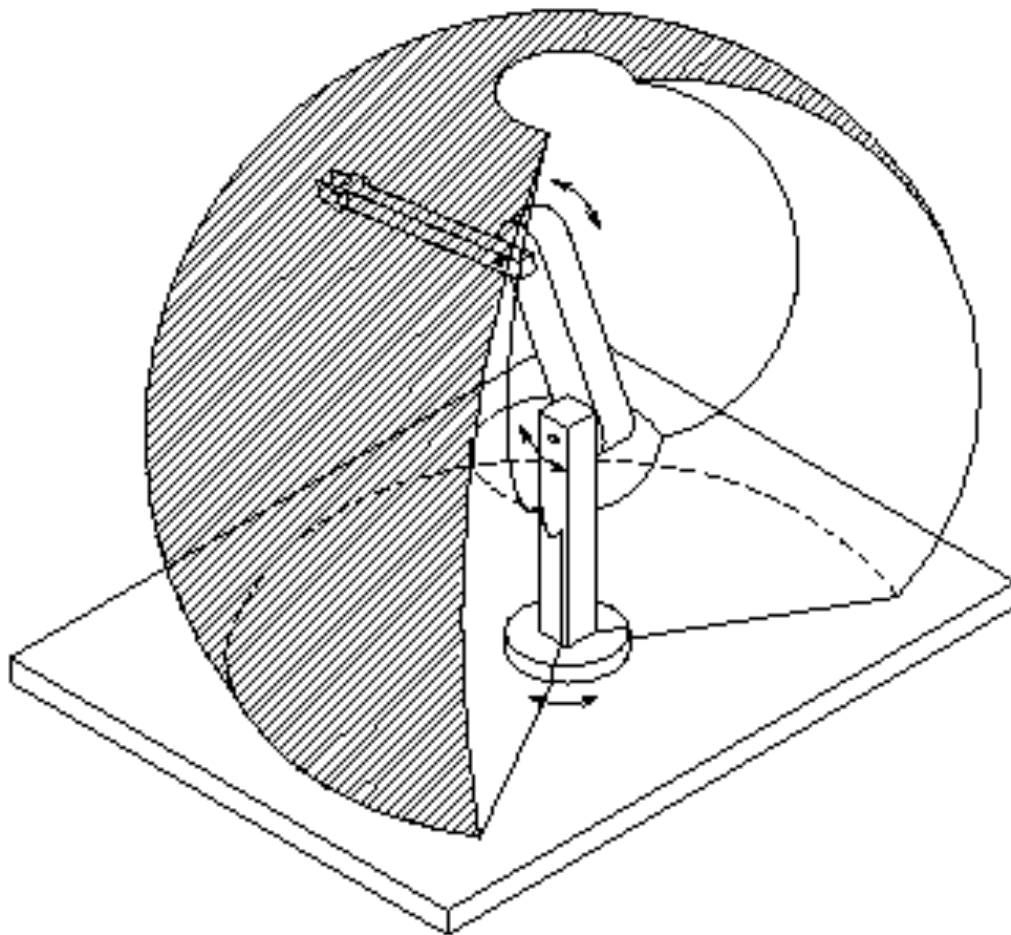


Source: Robotica - Modellistica, Pianificazione E Controllo
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Mcgraw-Hill (one of the course textbooks)

How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Anthropomorphic Robots

Assign the Base Reference Frame and the Tool Reference Frame

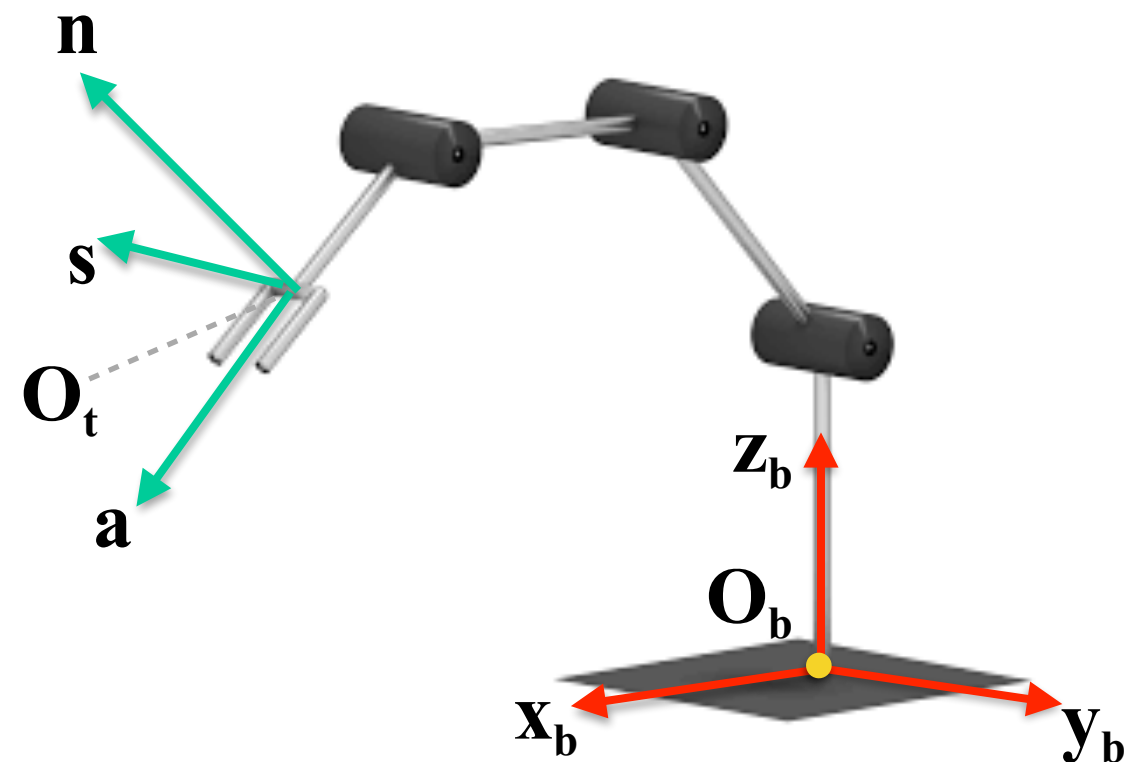
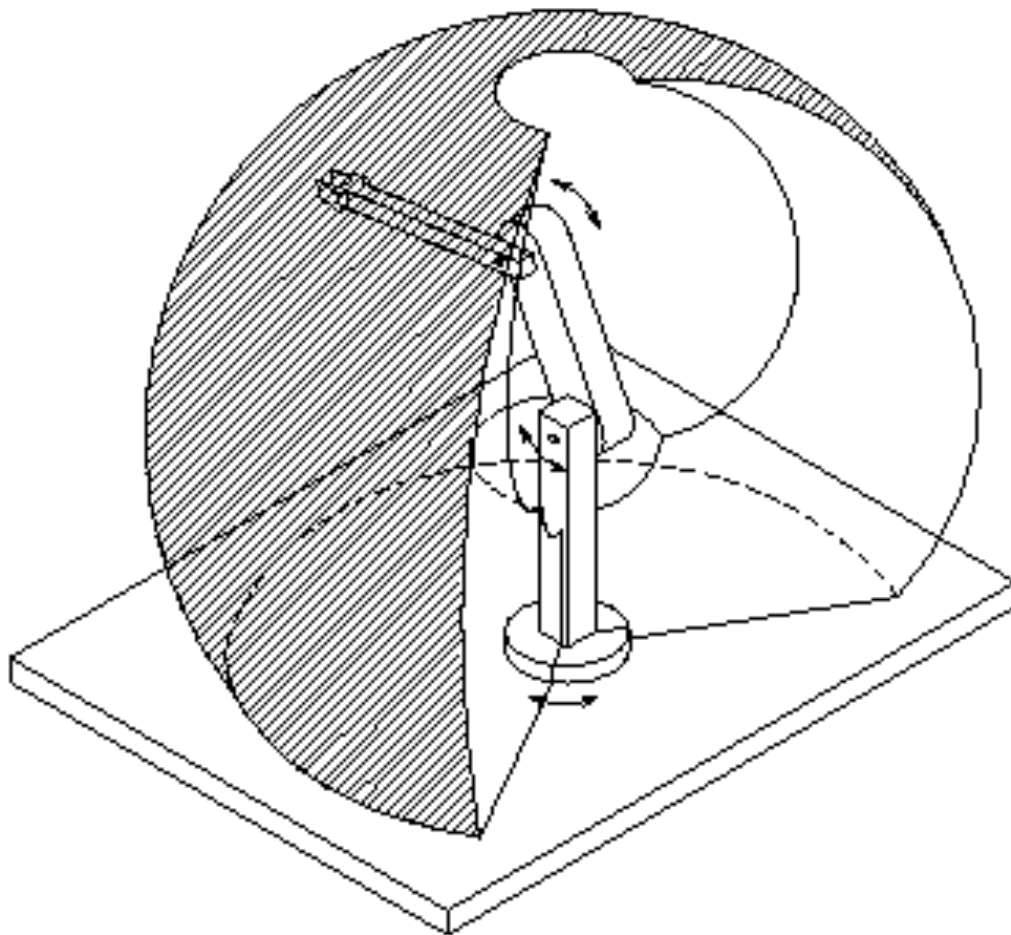


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How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Anthropomorphic Robots

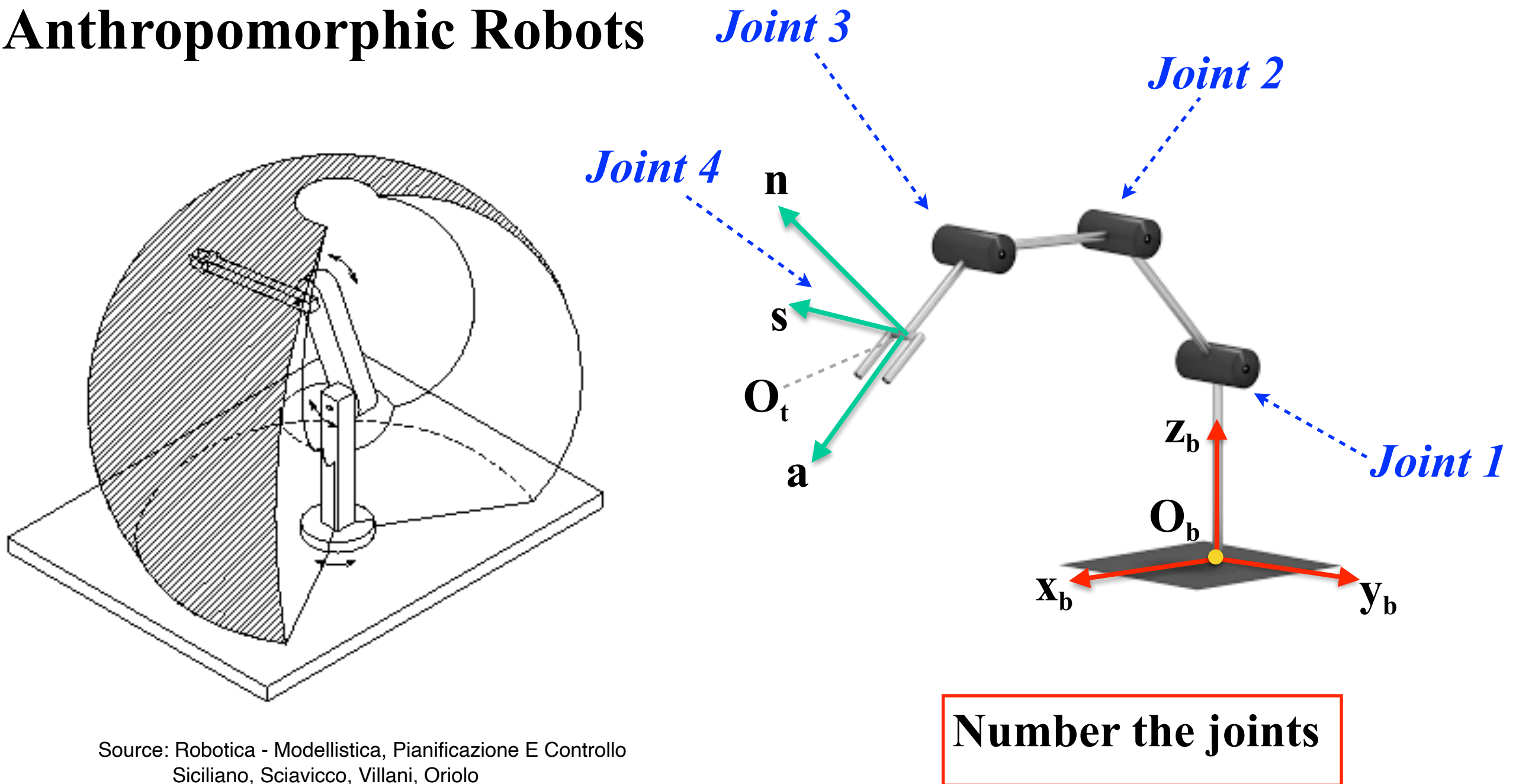
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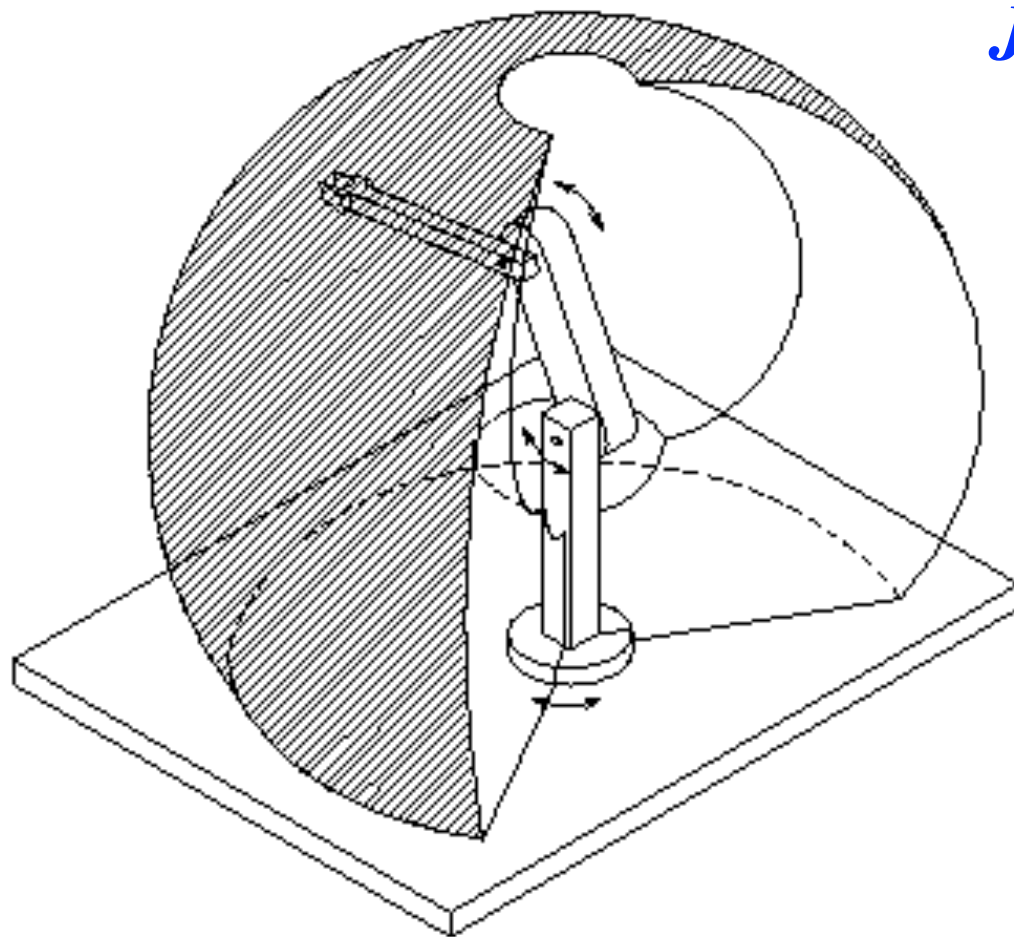
Anthropomorphic Robots



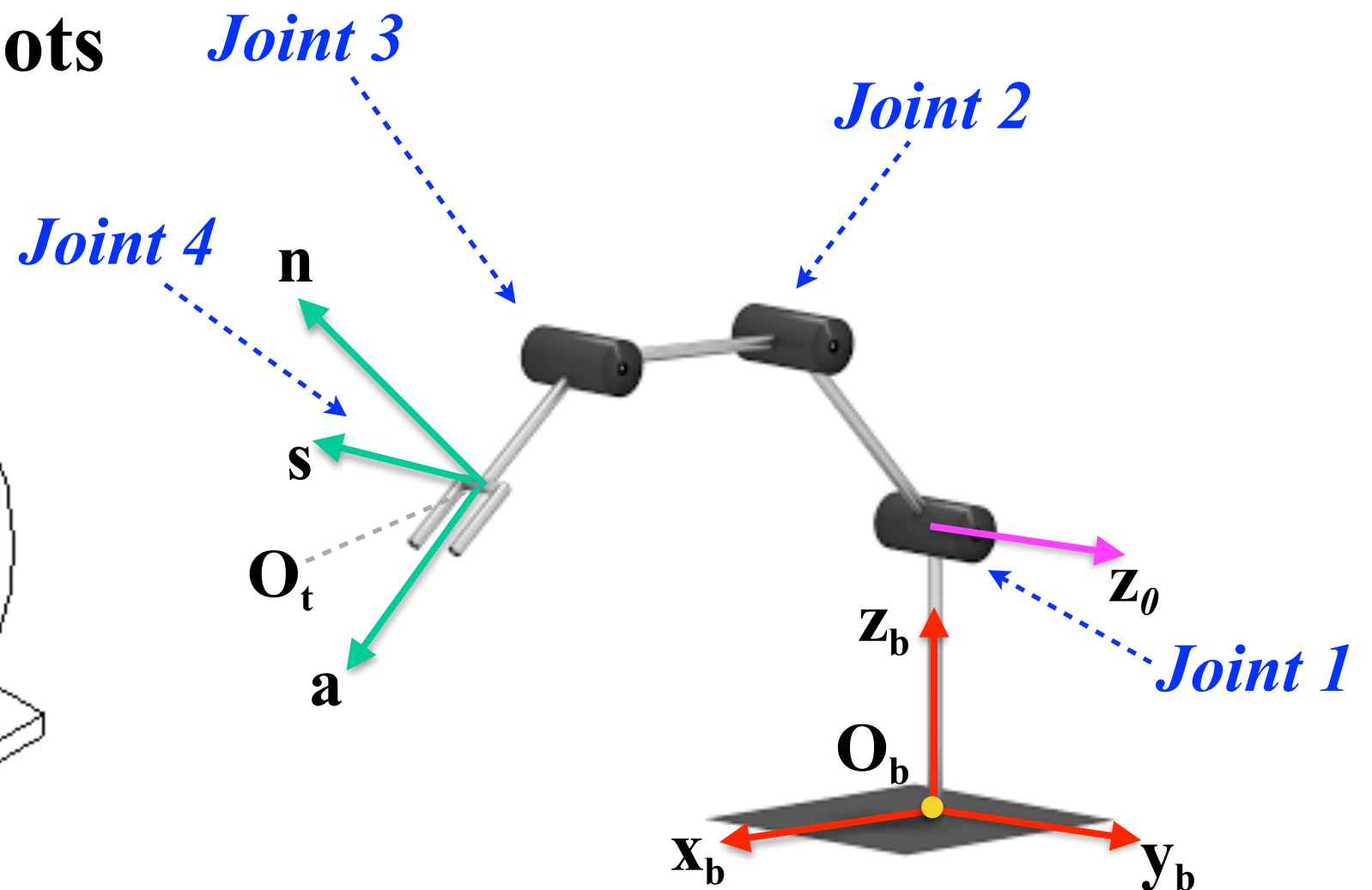
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Anthropomorphic Robots



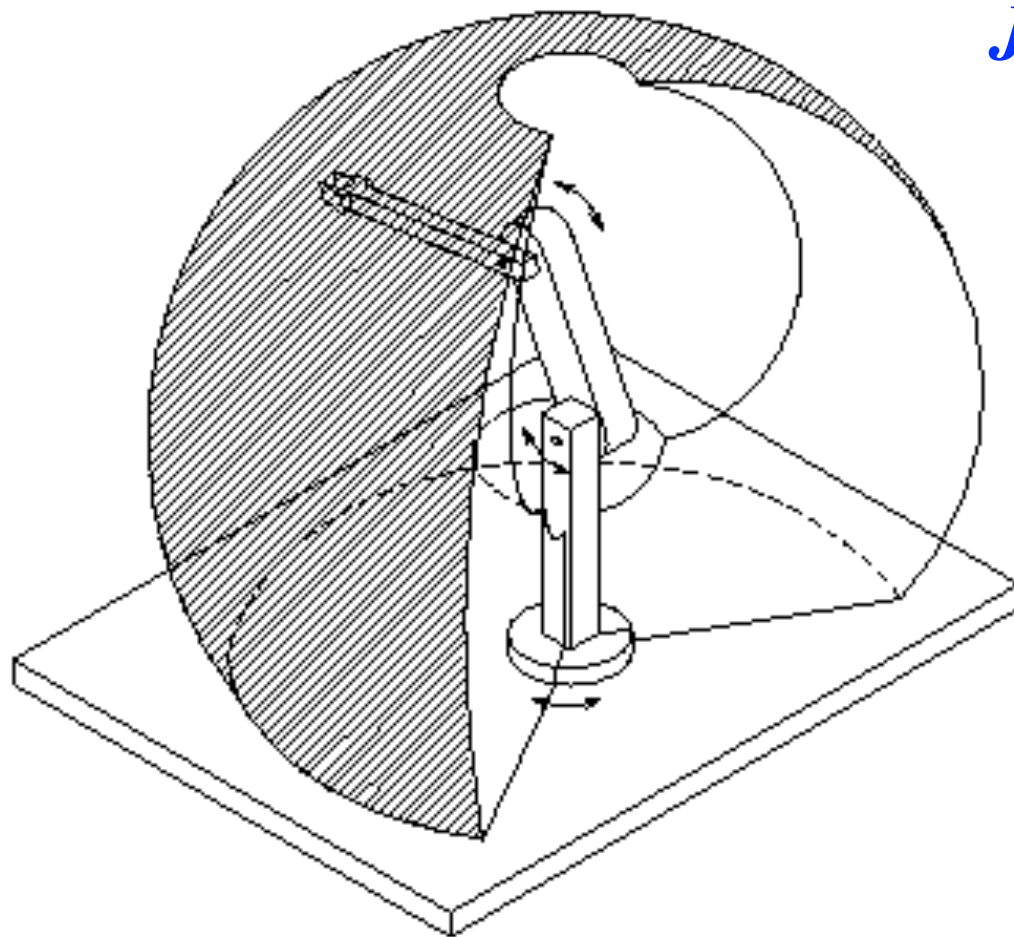
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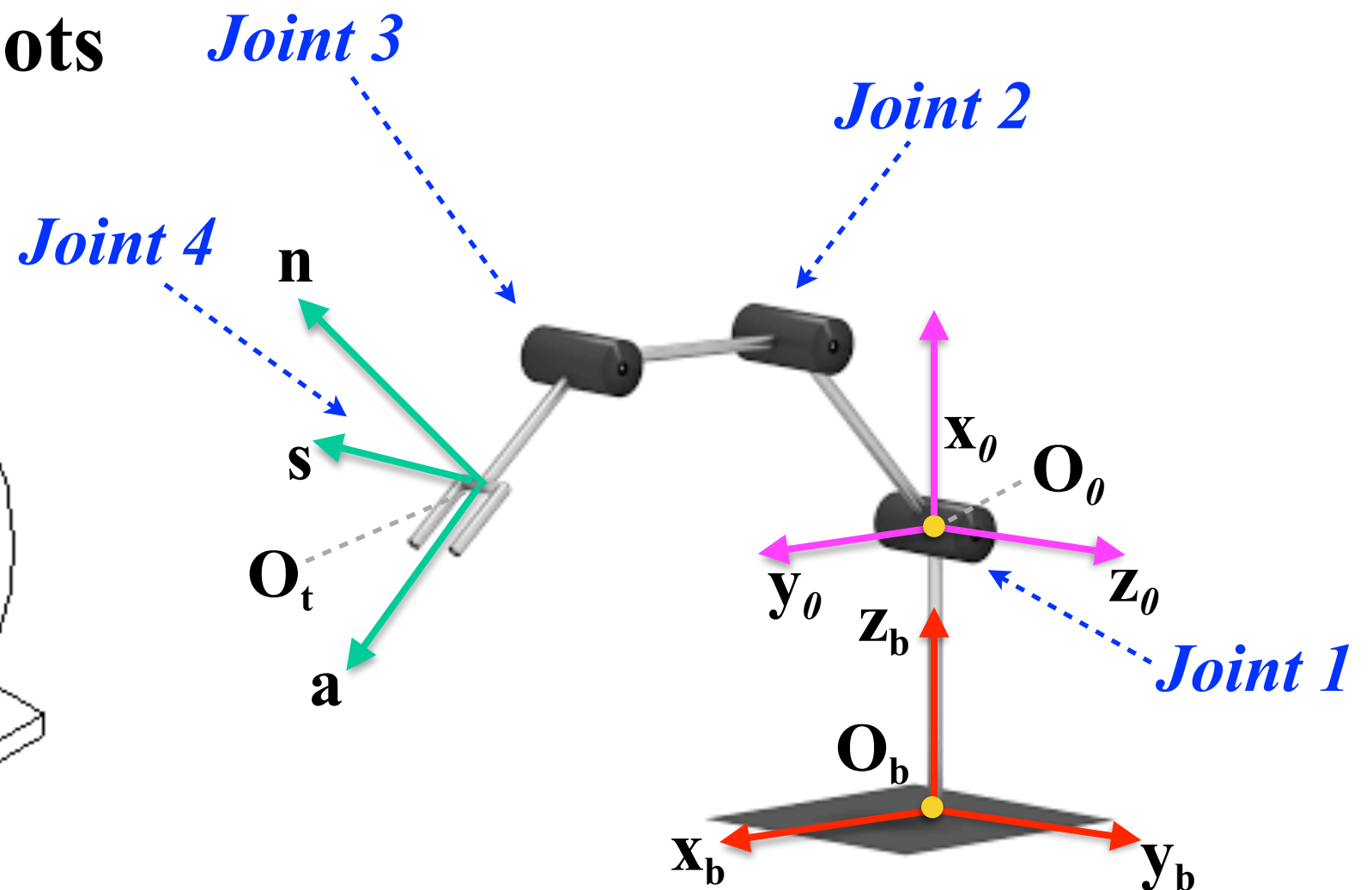
**Assign the D-H Reference
Frames from 0 to n**

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Anthropomorphic Robots



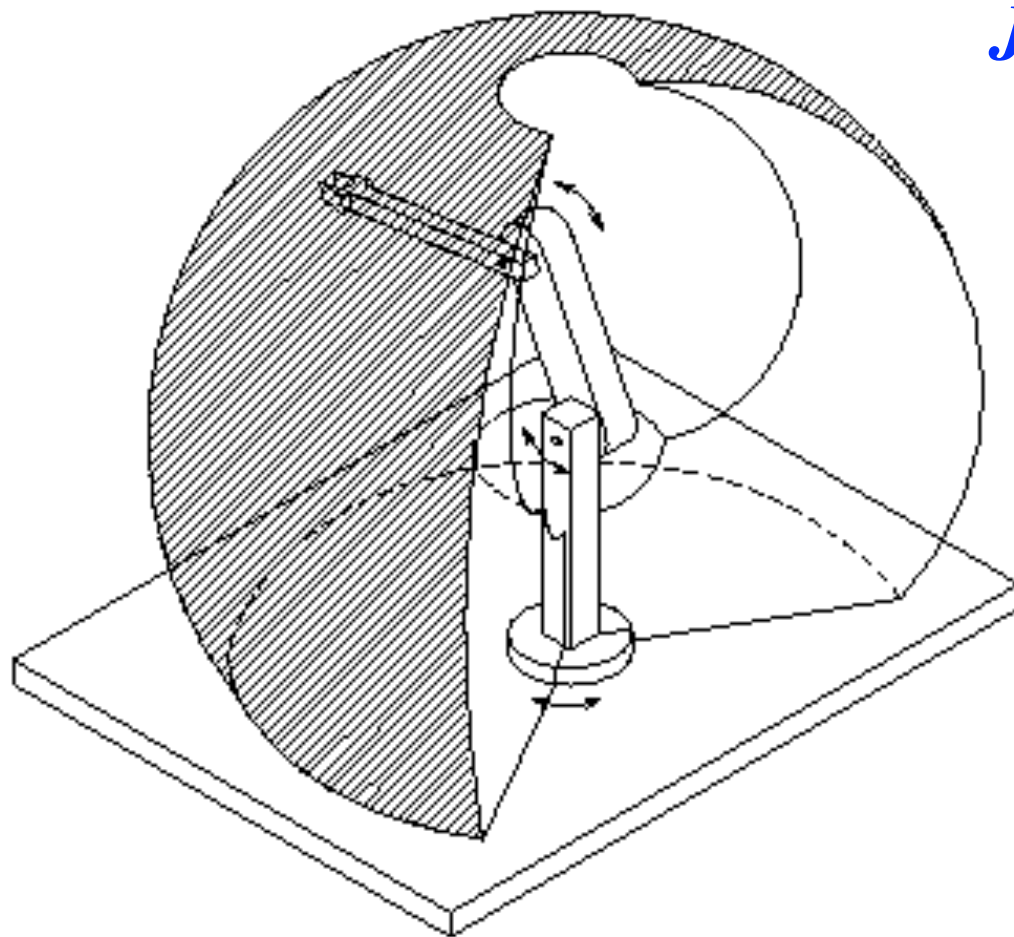
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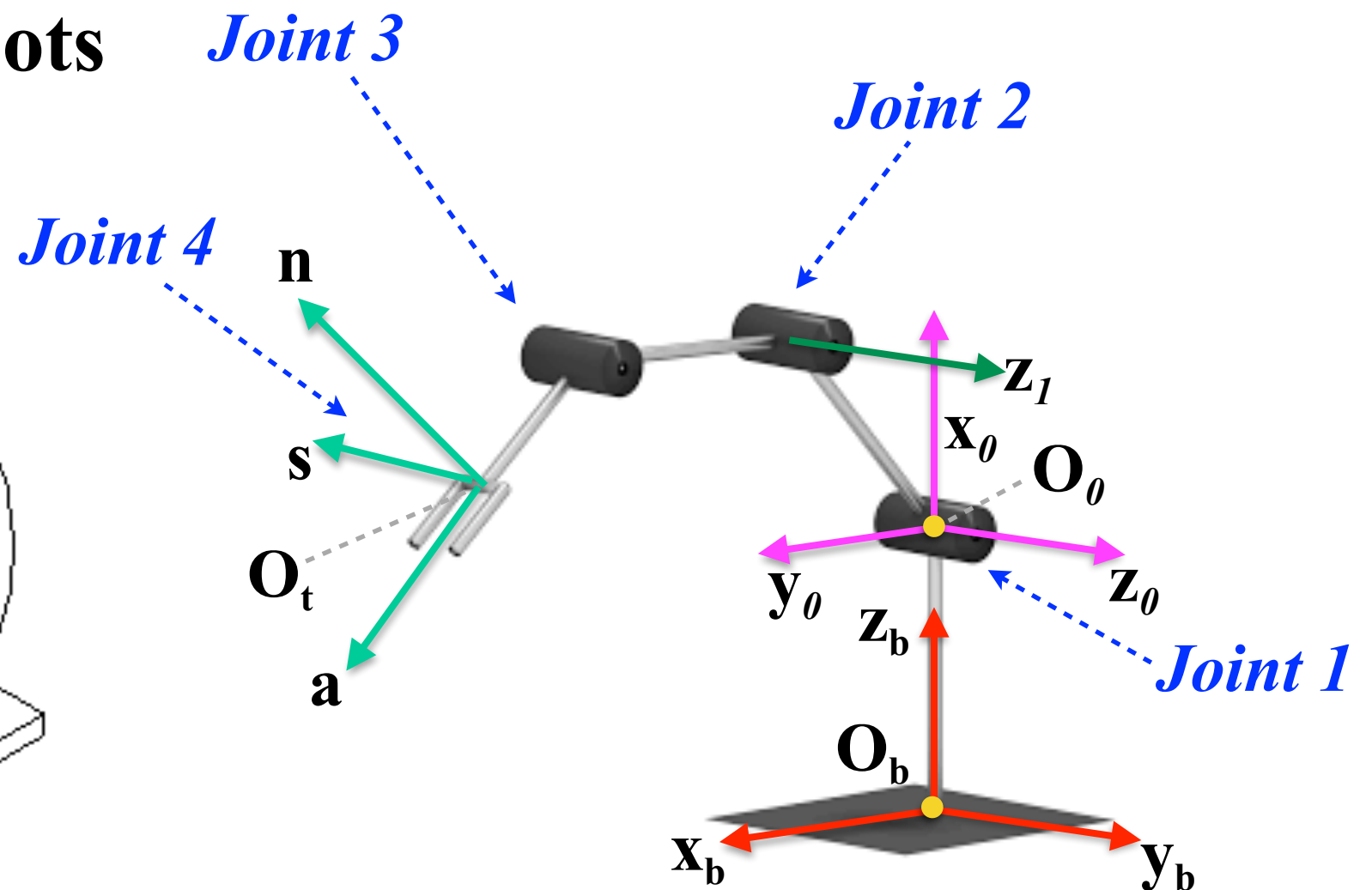
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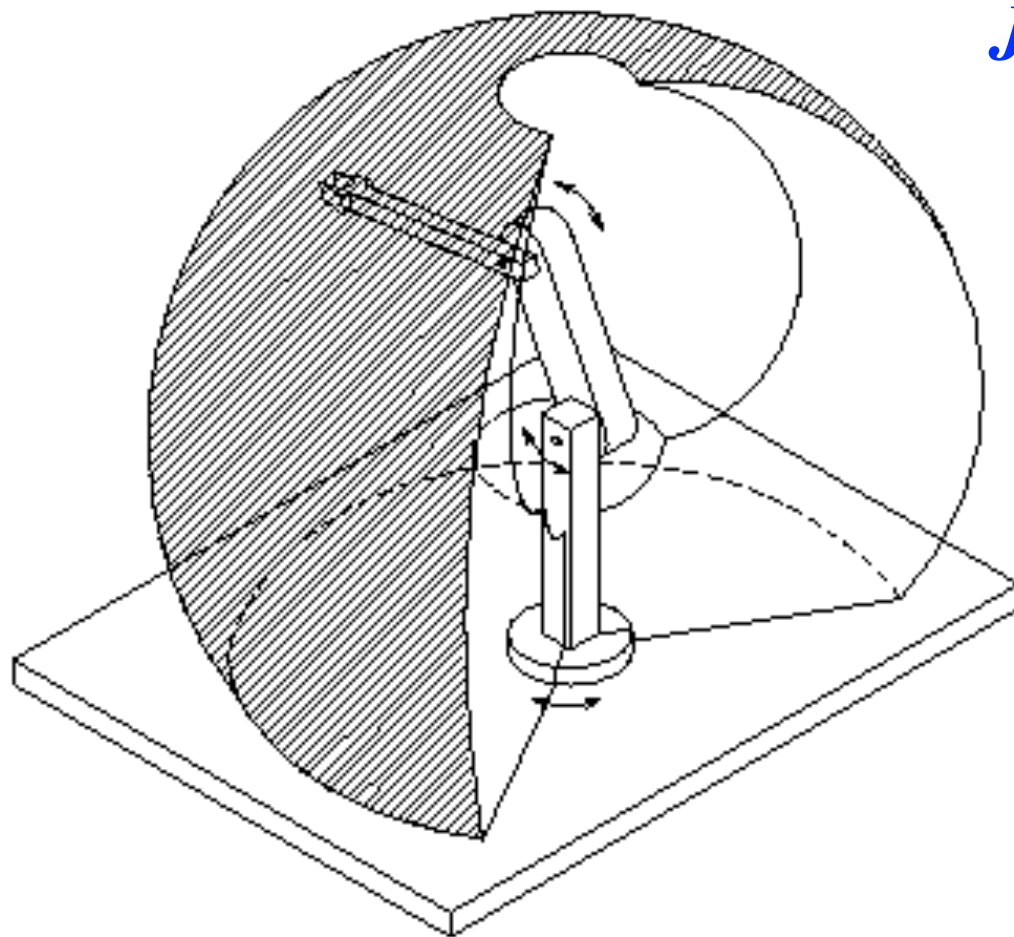
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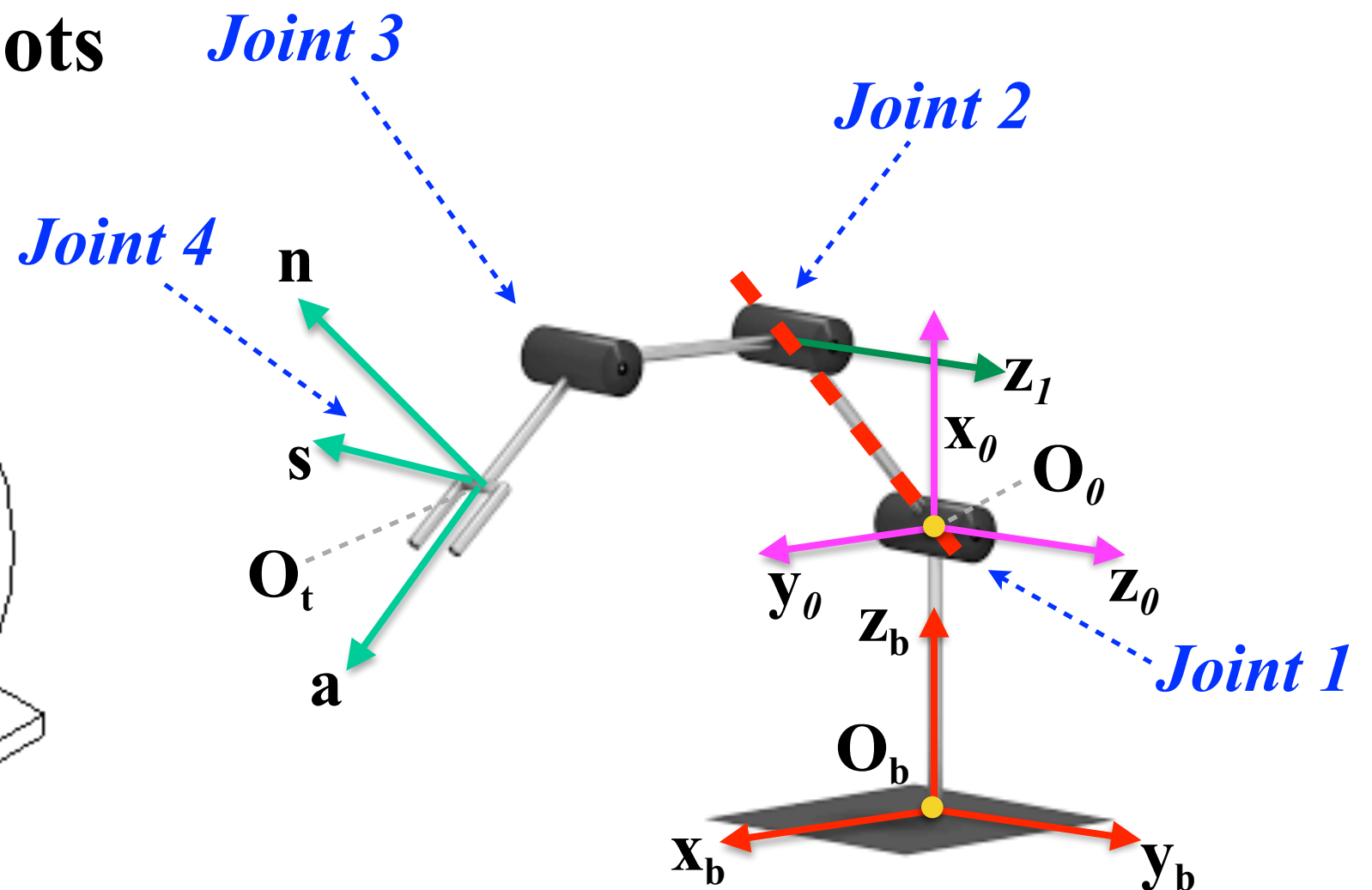
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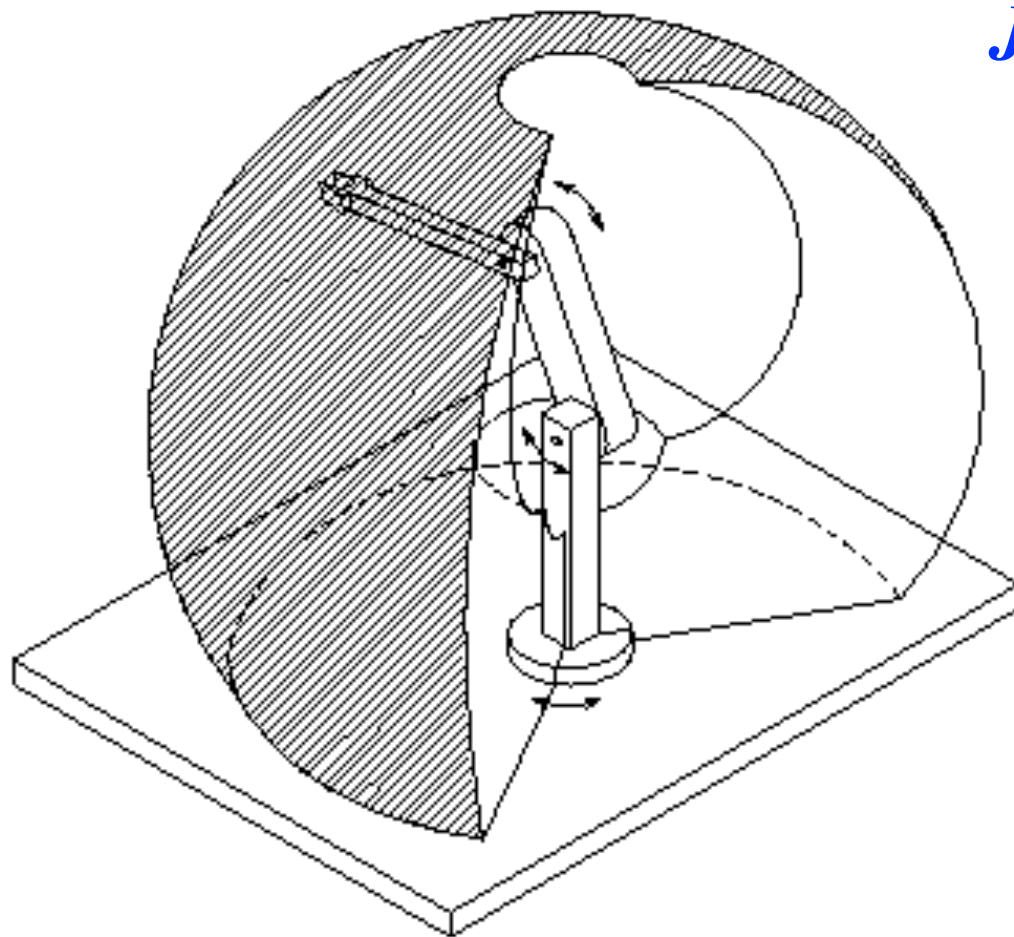
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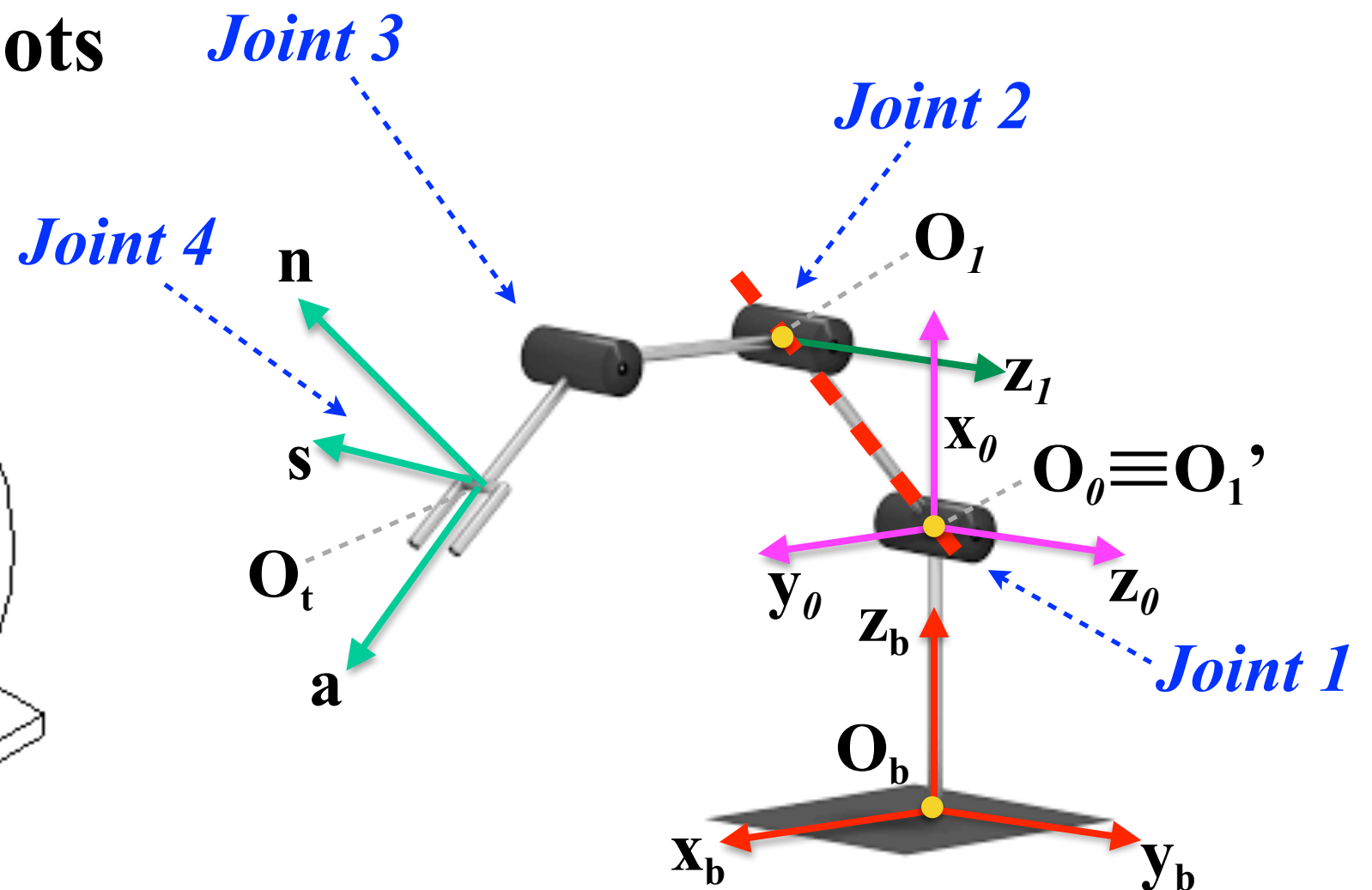
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How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Anthropomorphic Robots



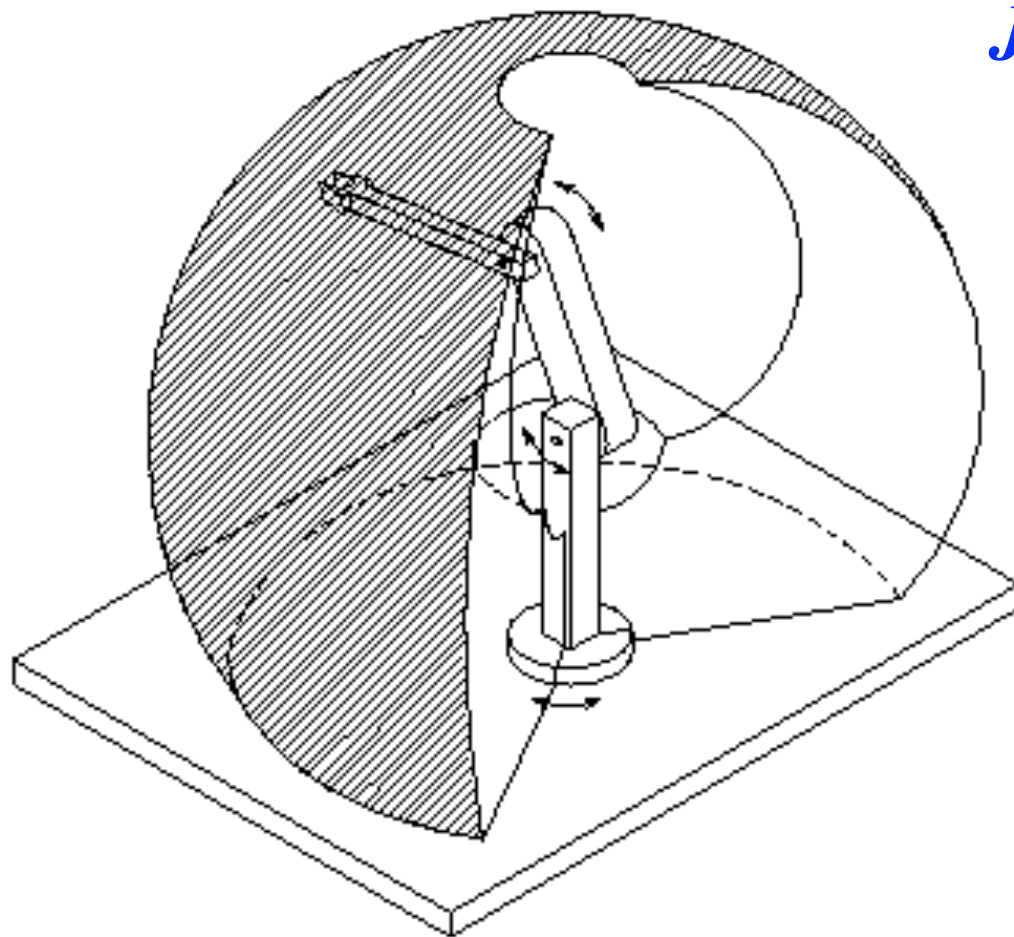
Source: Robotica - Modellistica, Pianificazione E Controllo
Siciliano, Sciavicco, Villani, Oriolo
Mcgraw-Hill (one of the course textbooks)



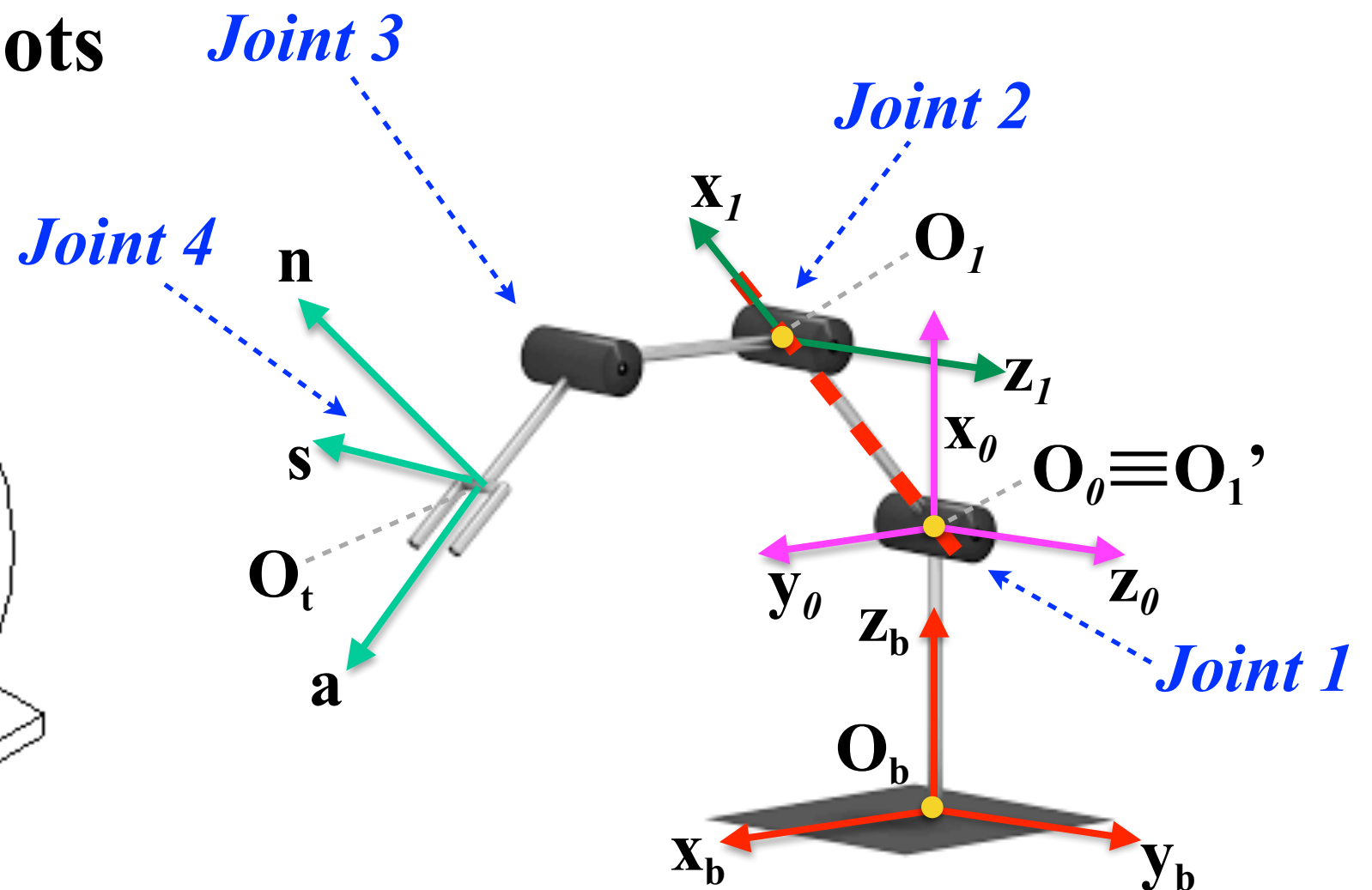
**Assign the D-H Reference
Frames from 0 to n**

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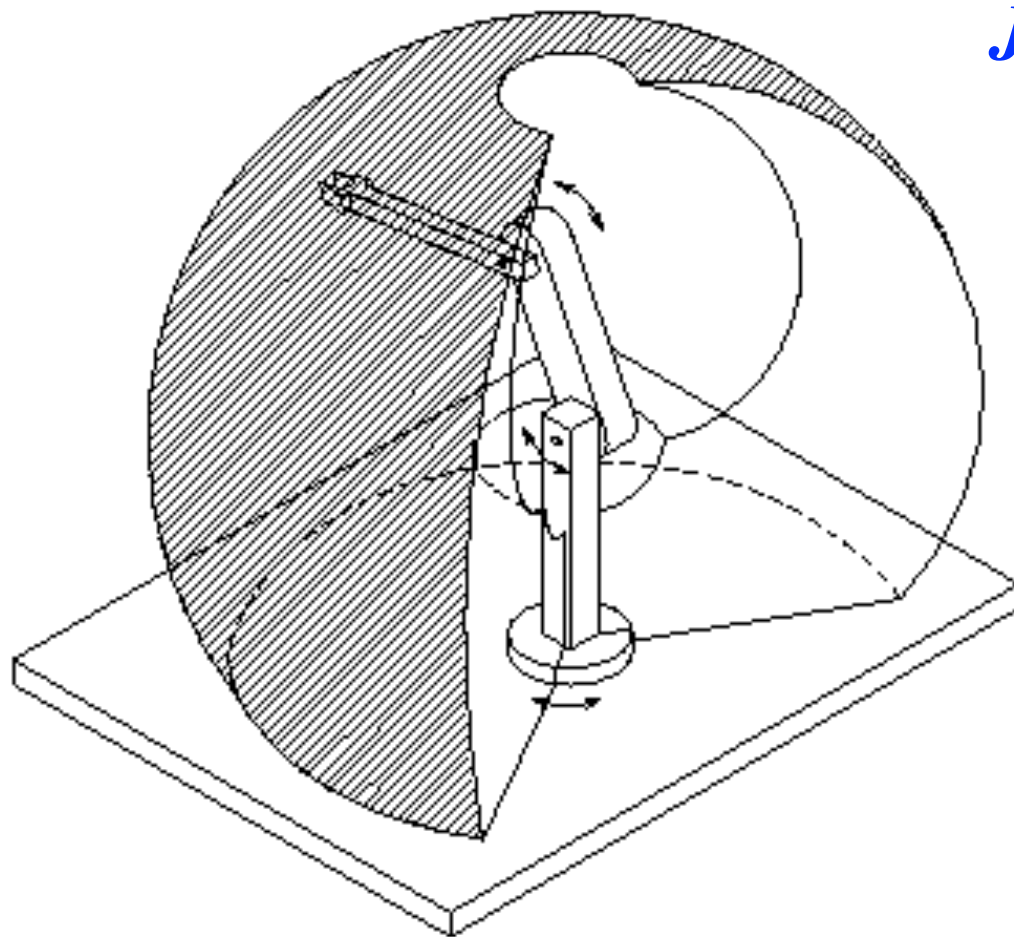
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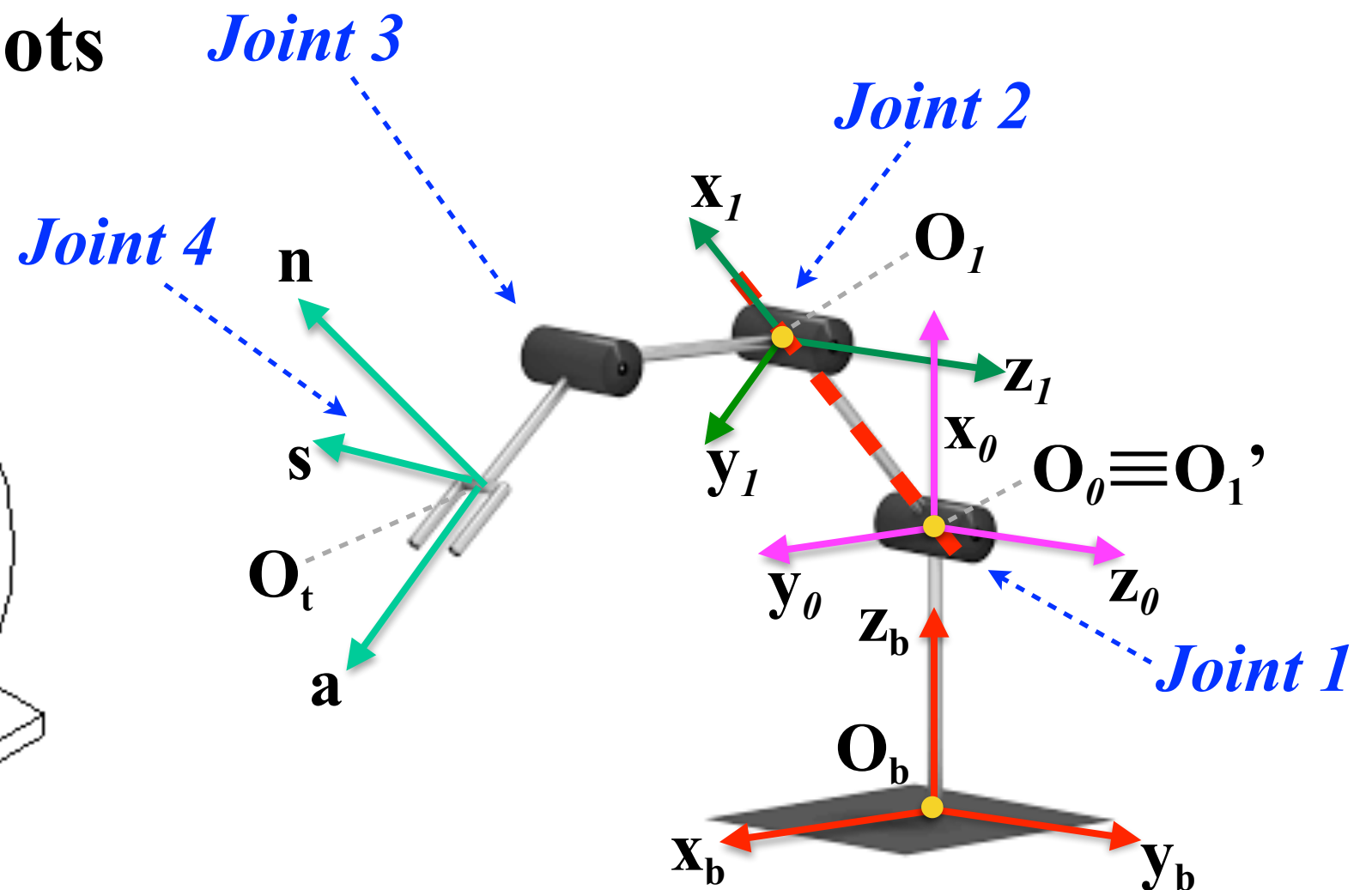
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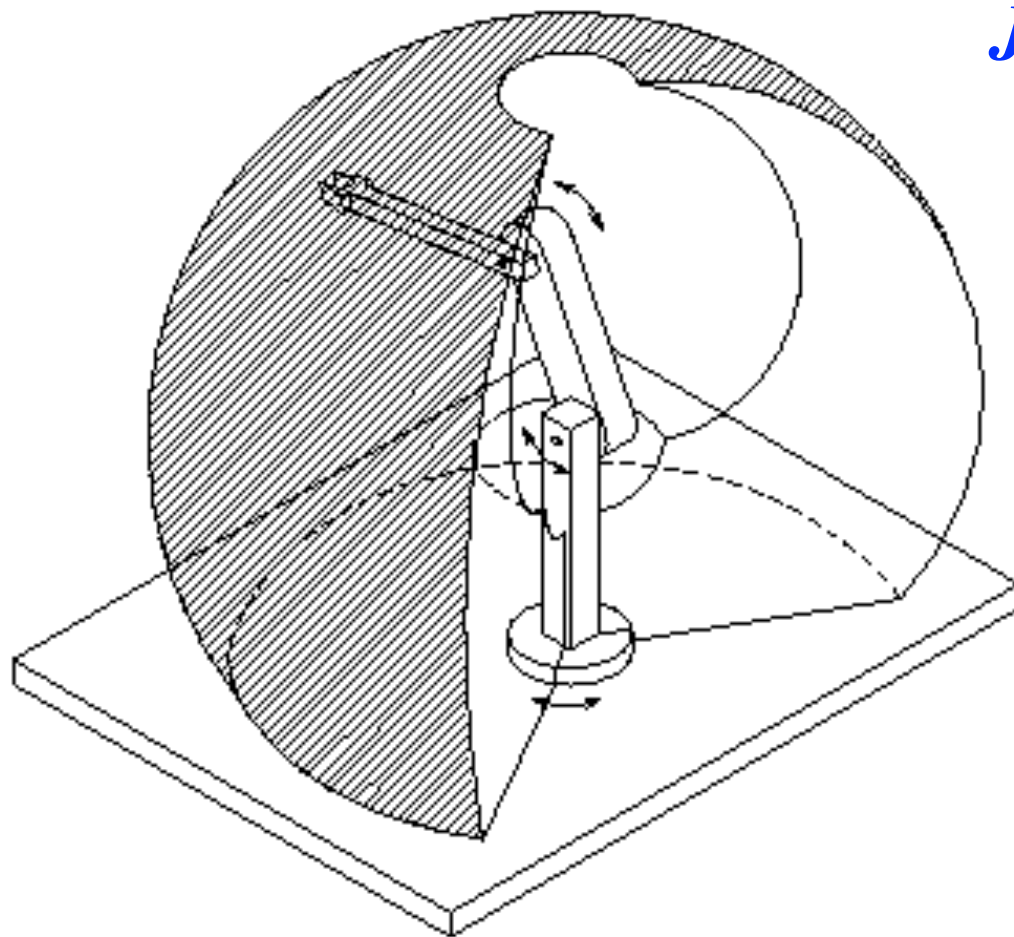
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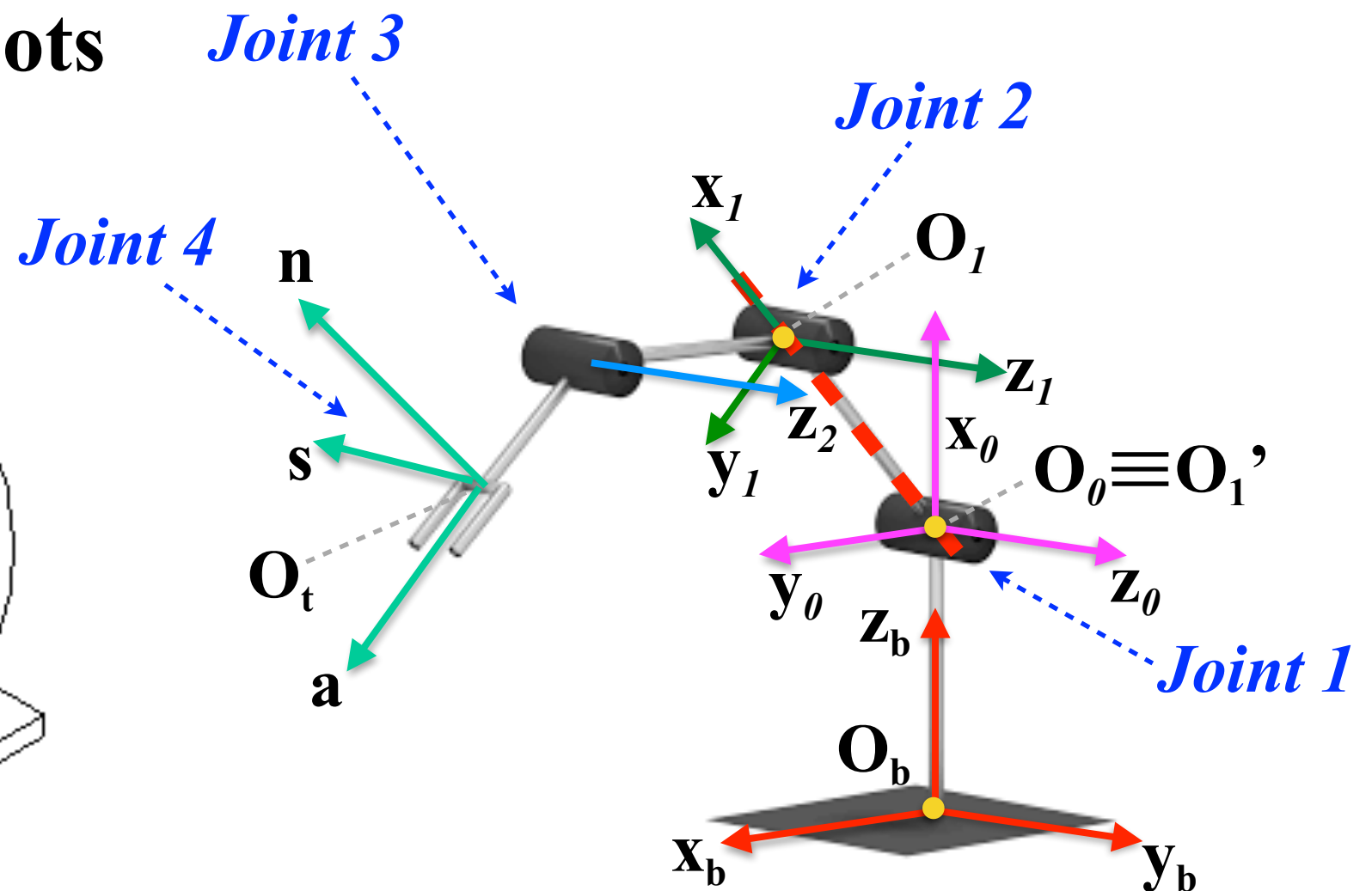
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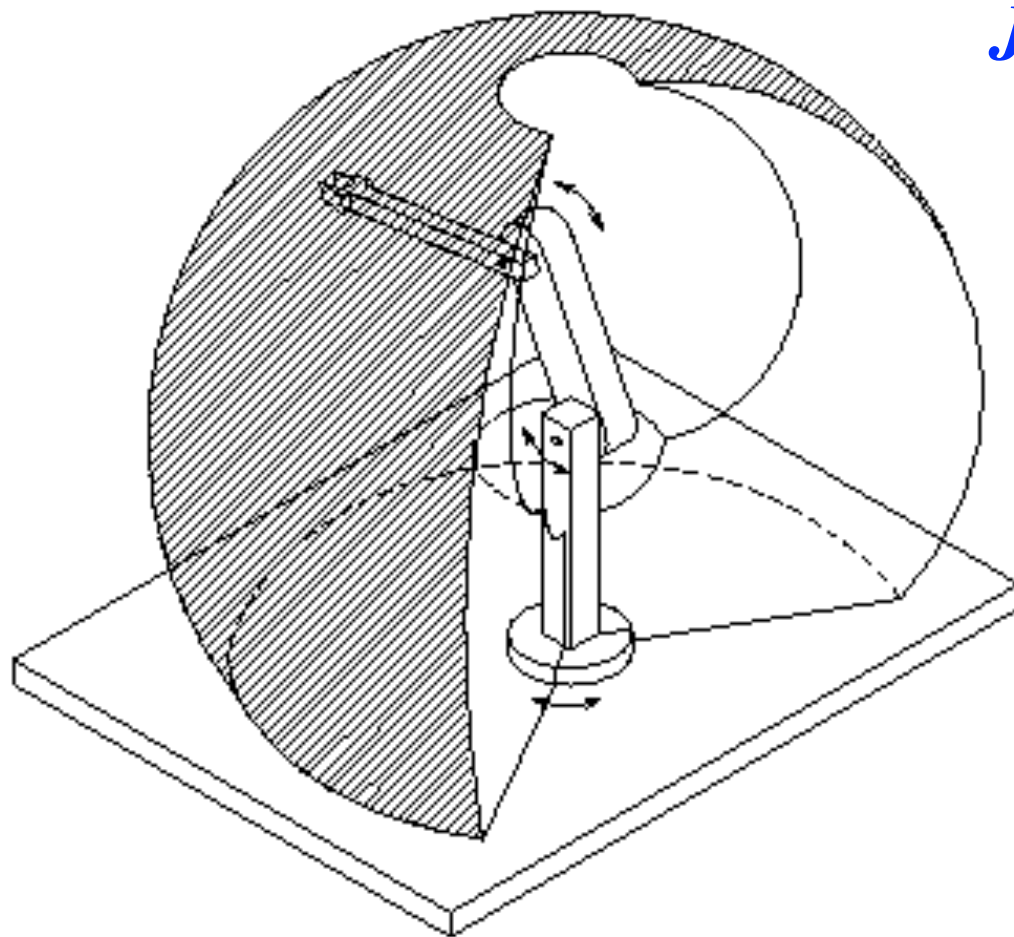
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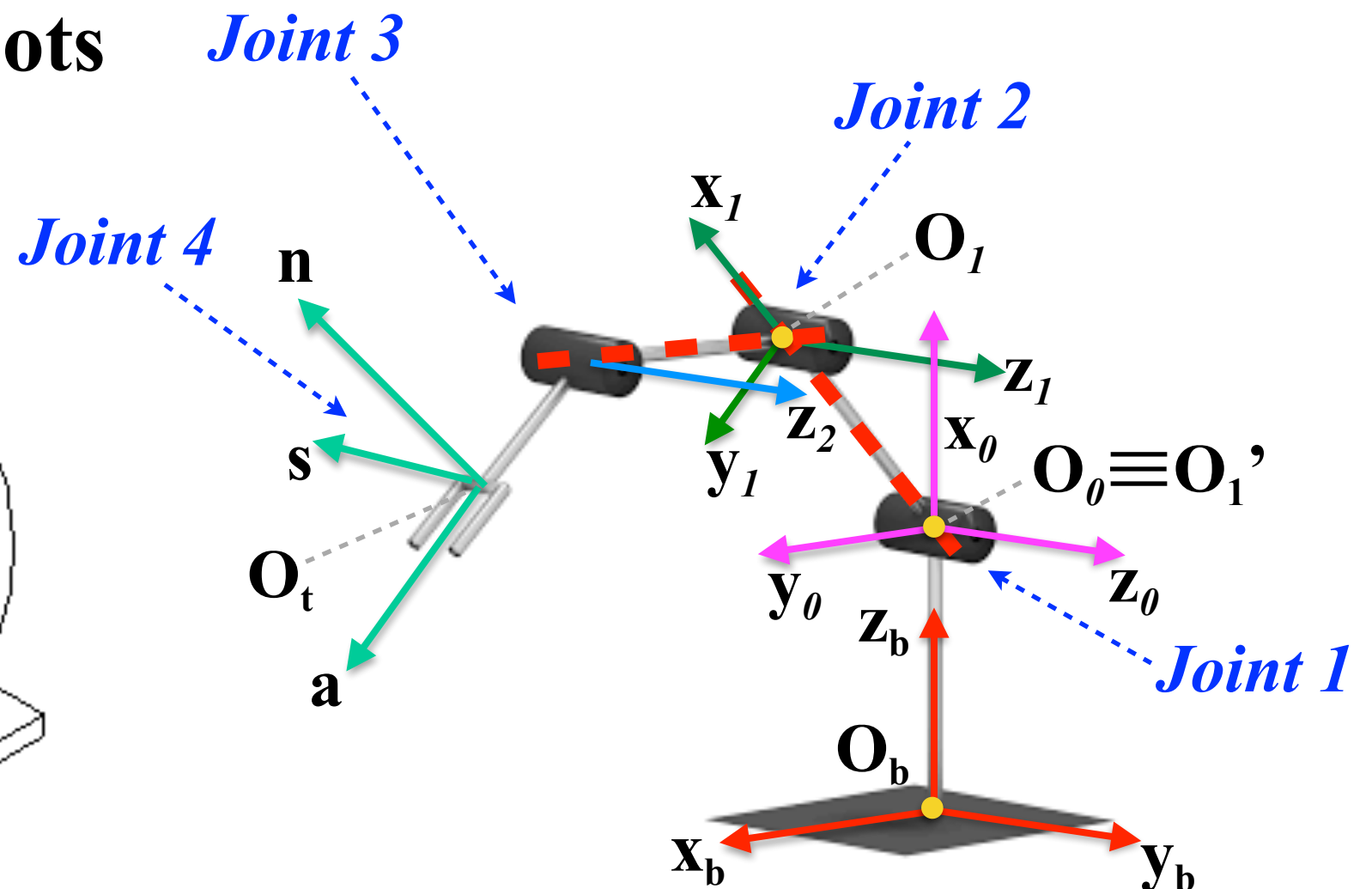
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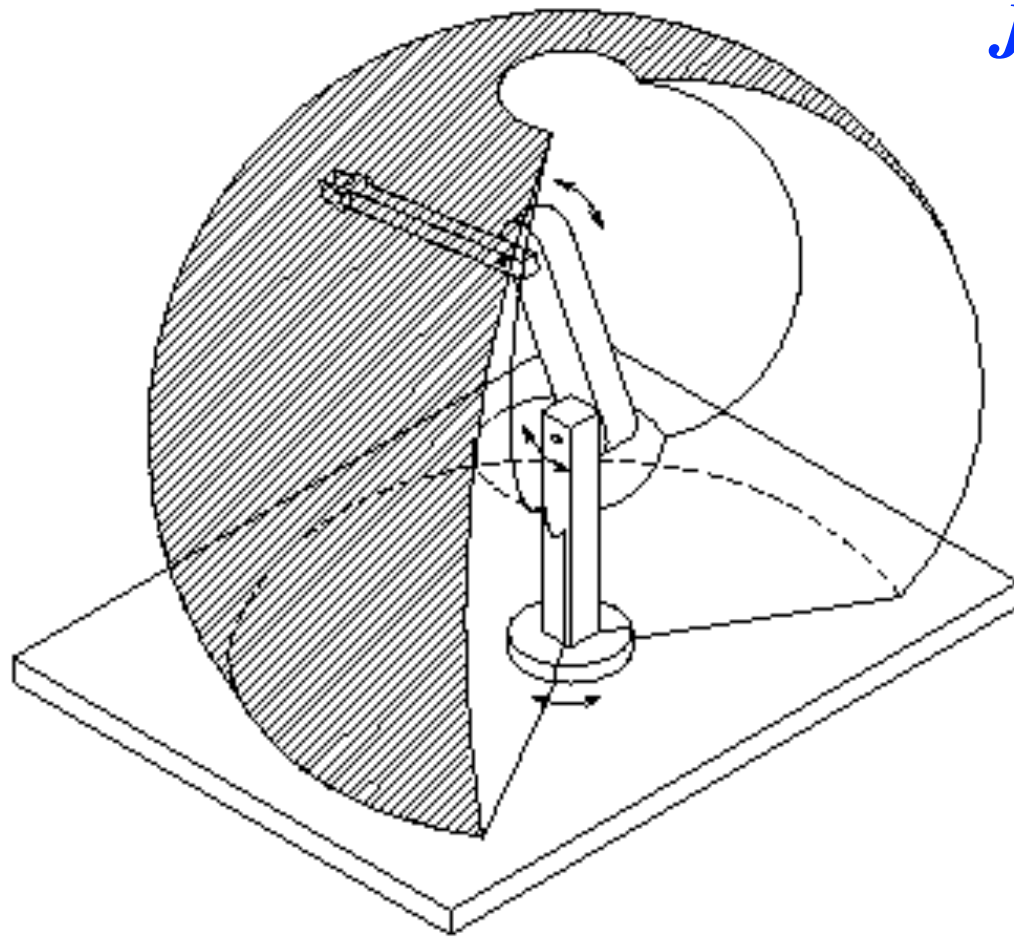
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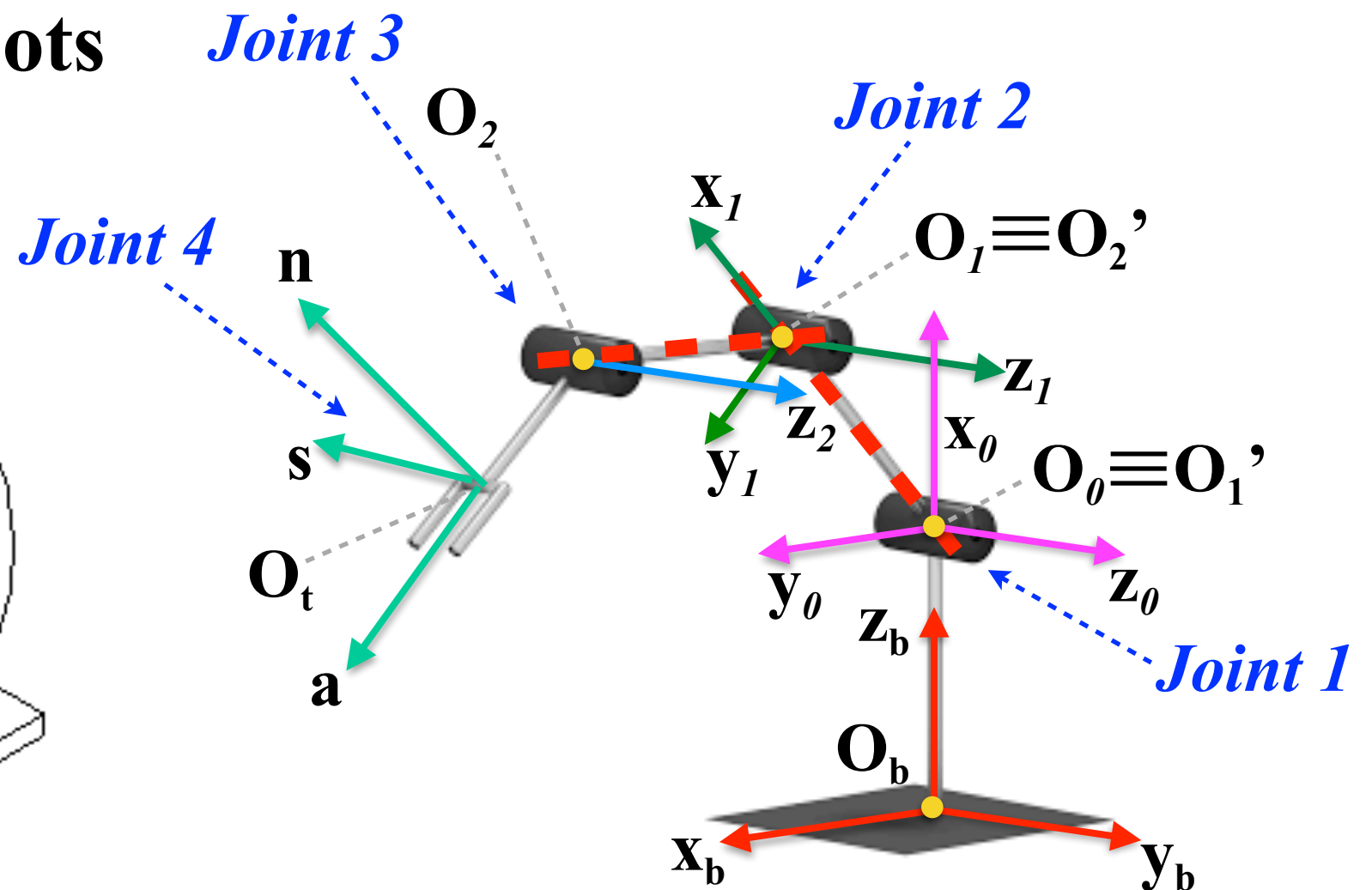
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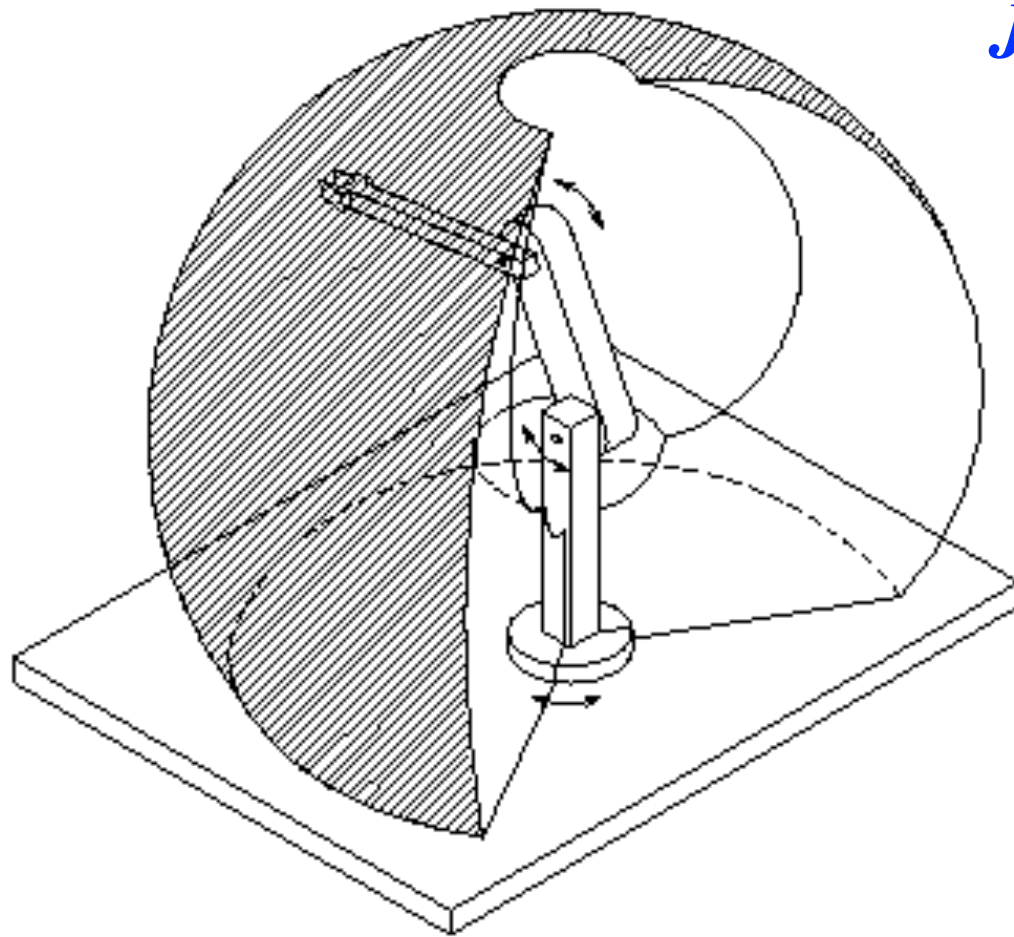
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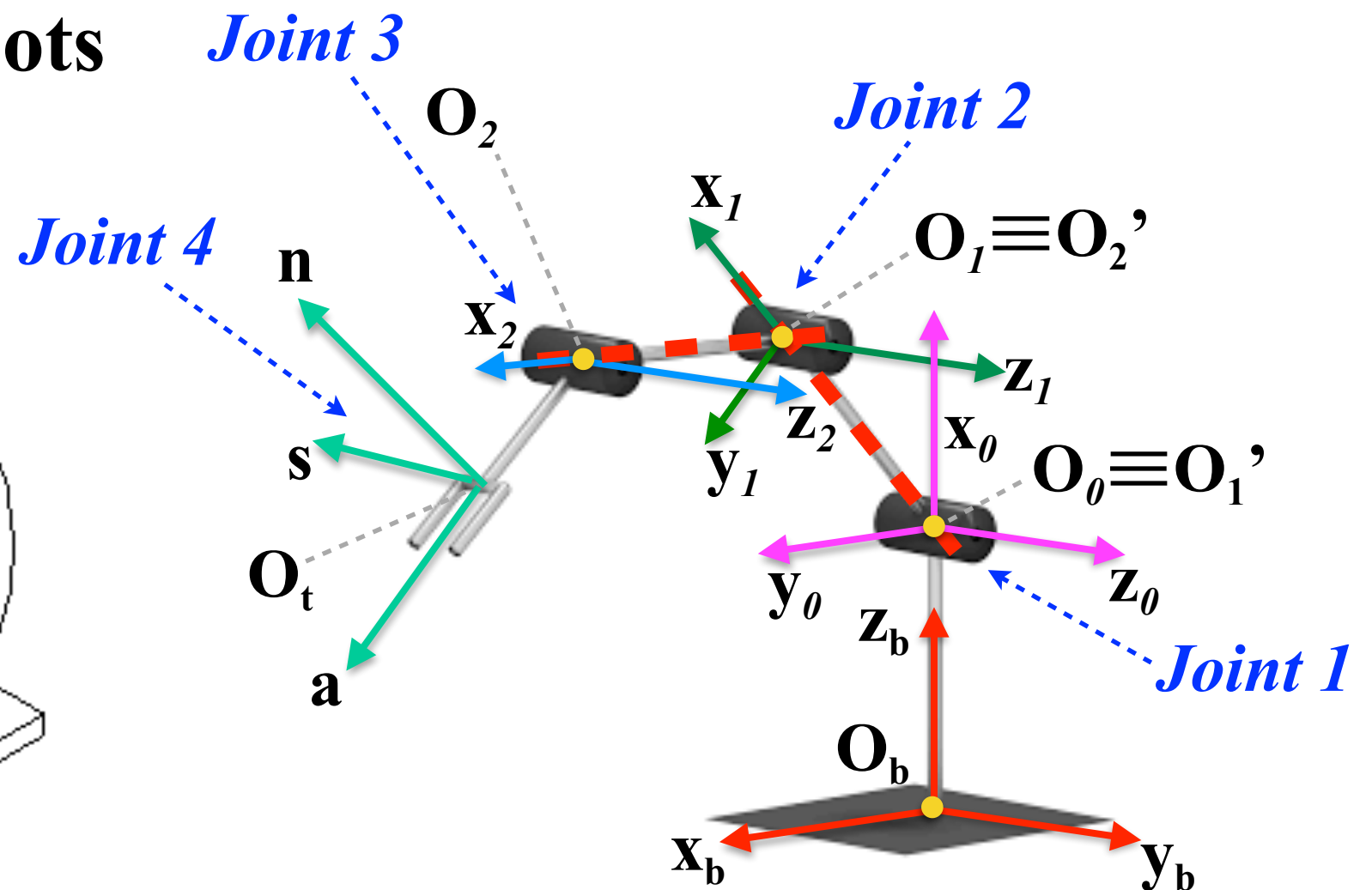
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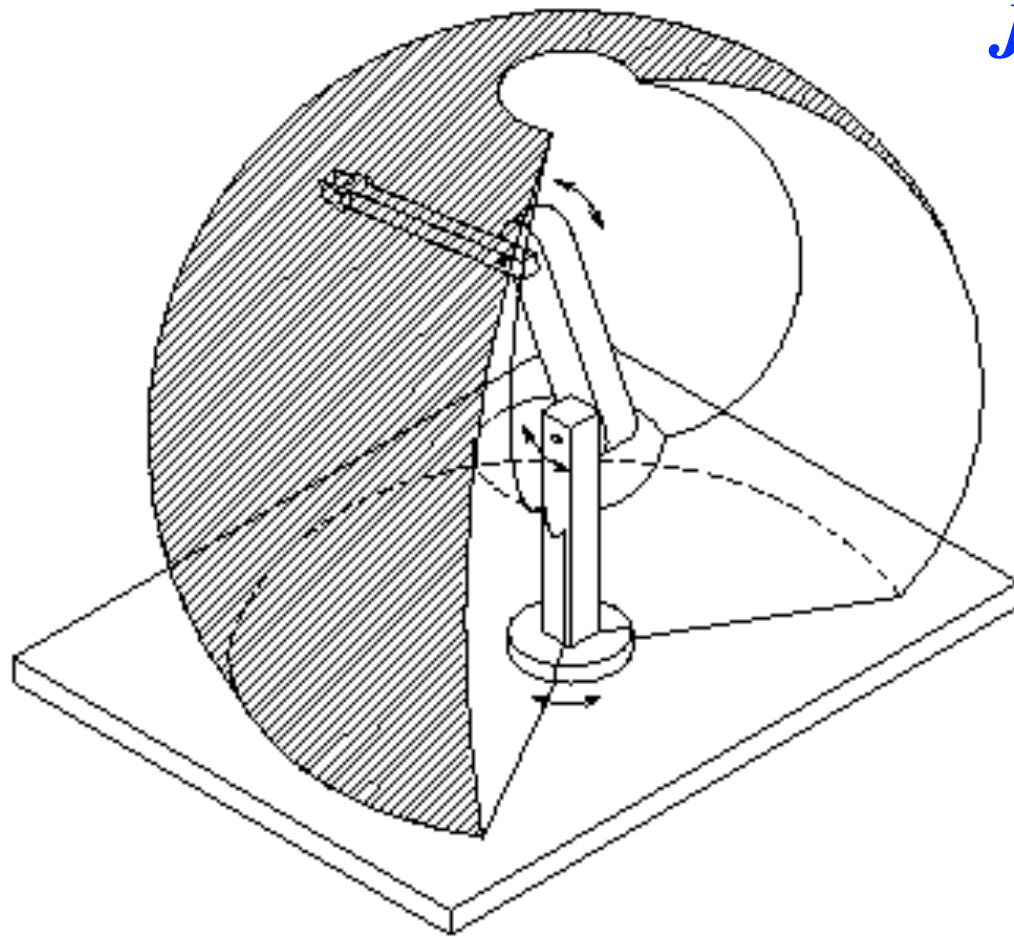
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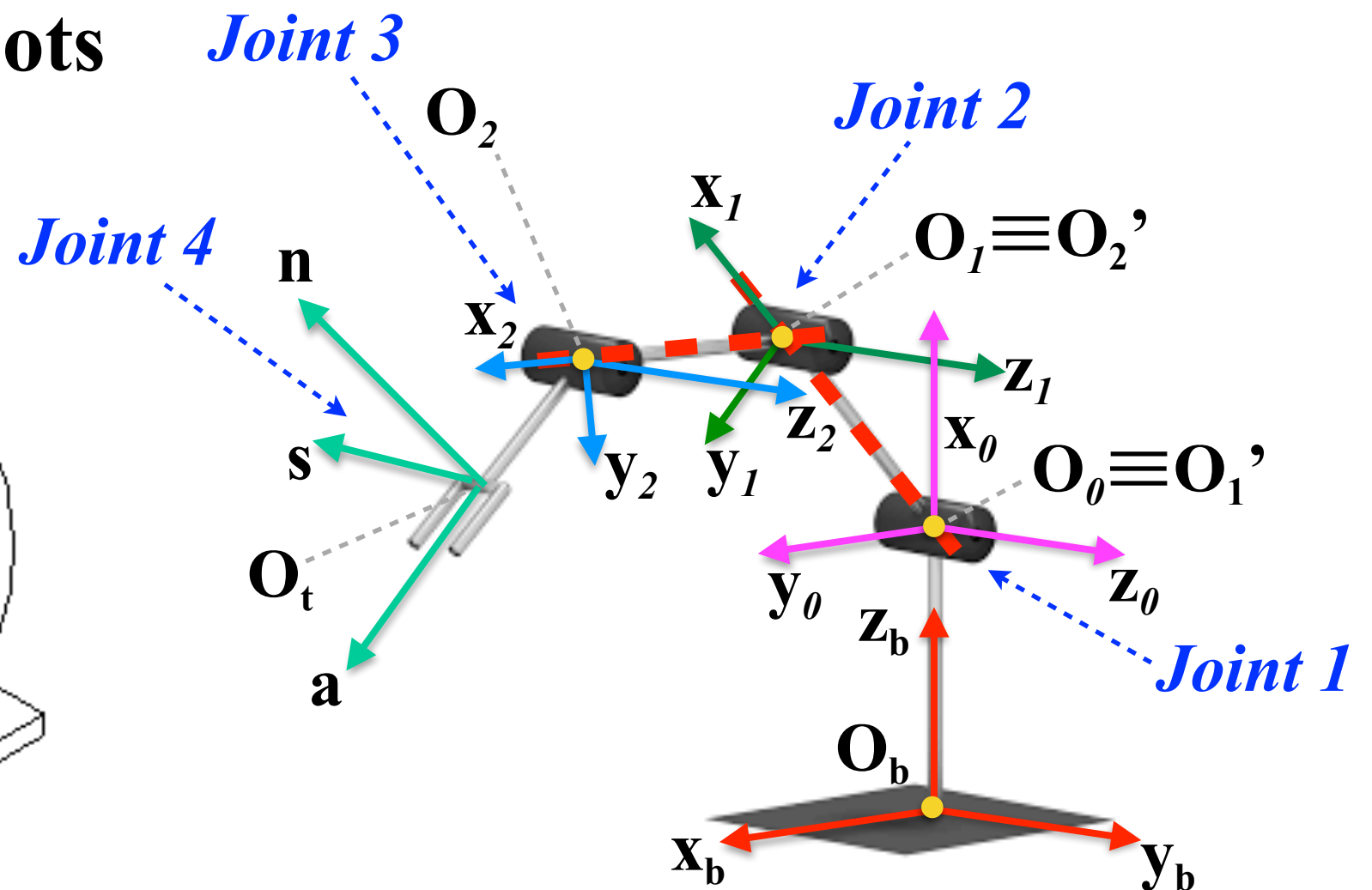
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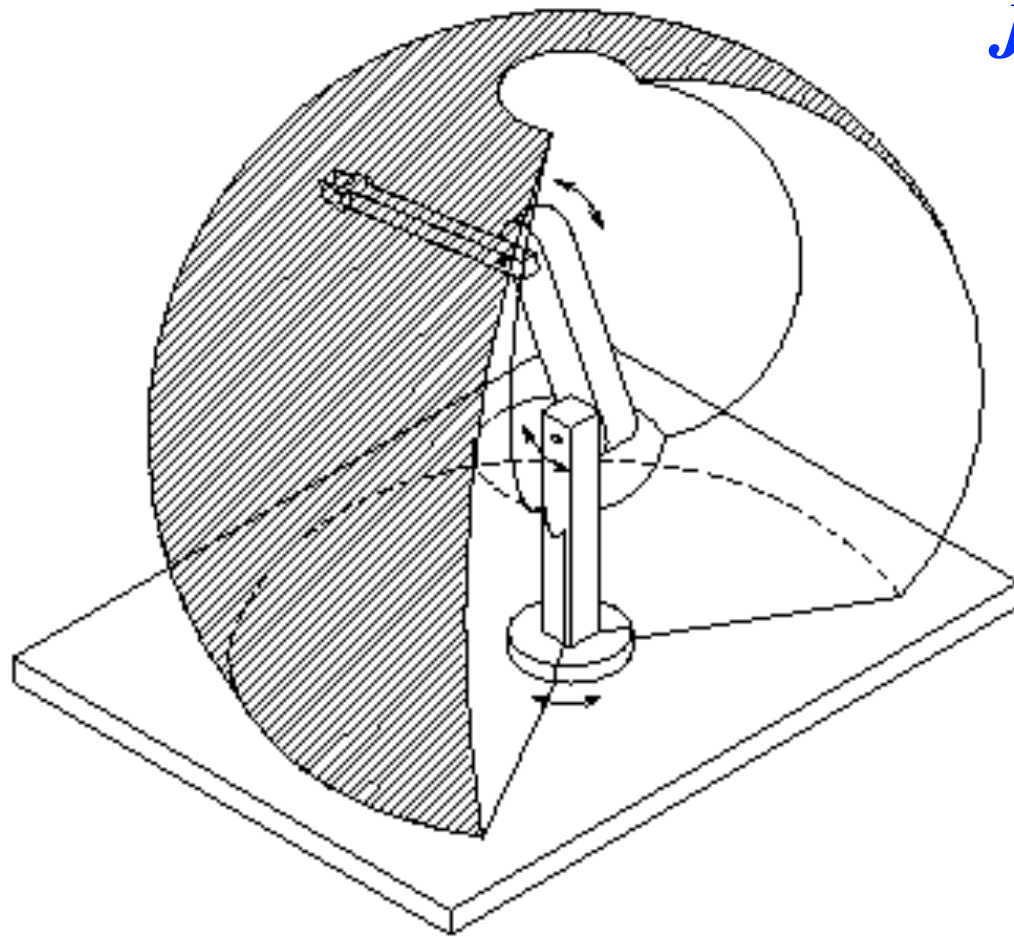
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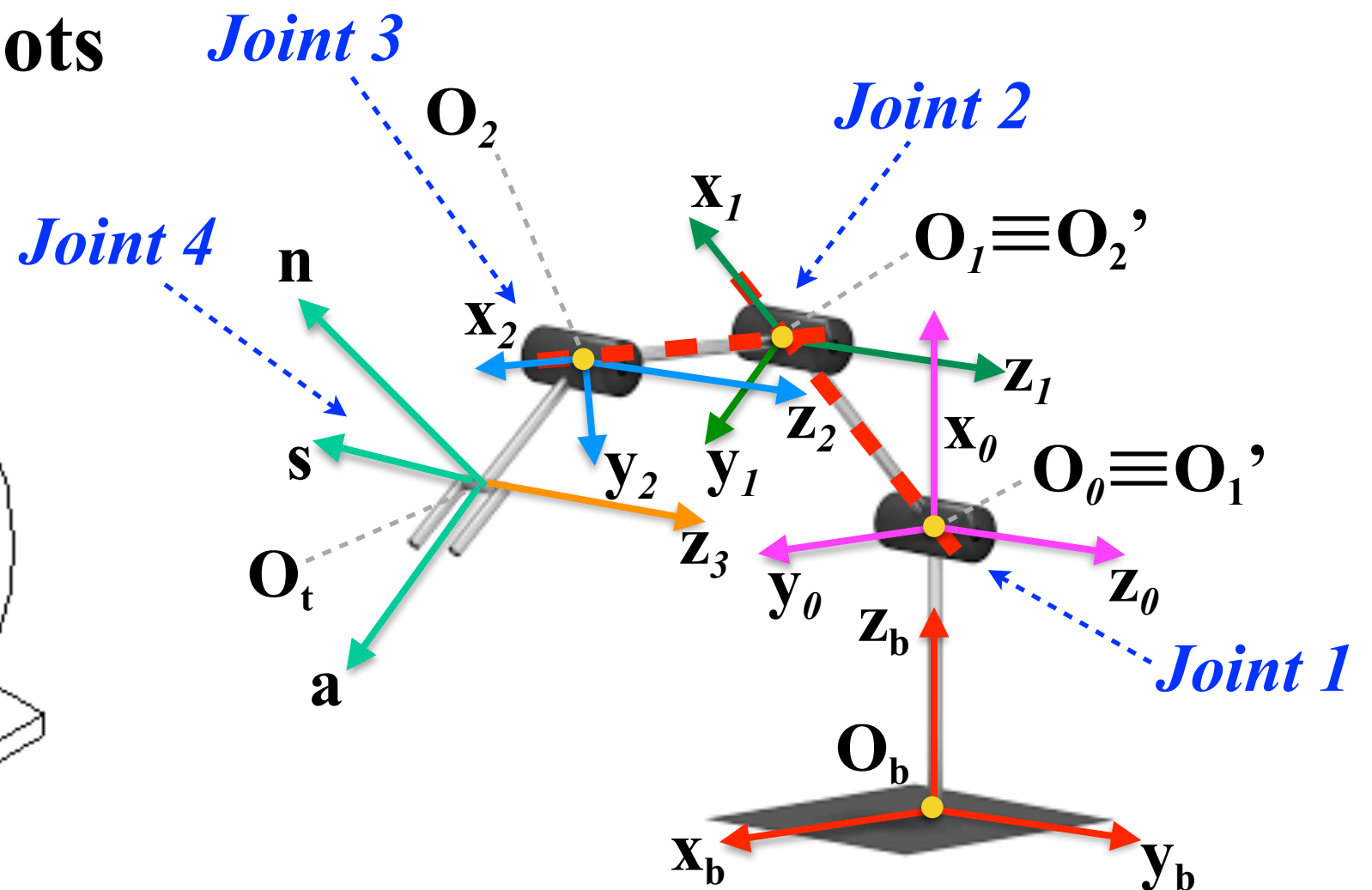
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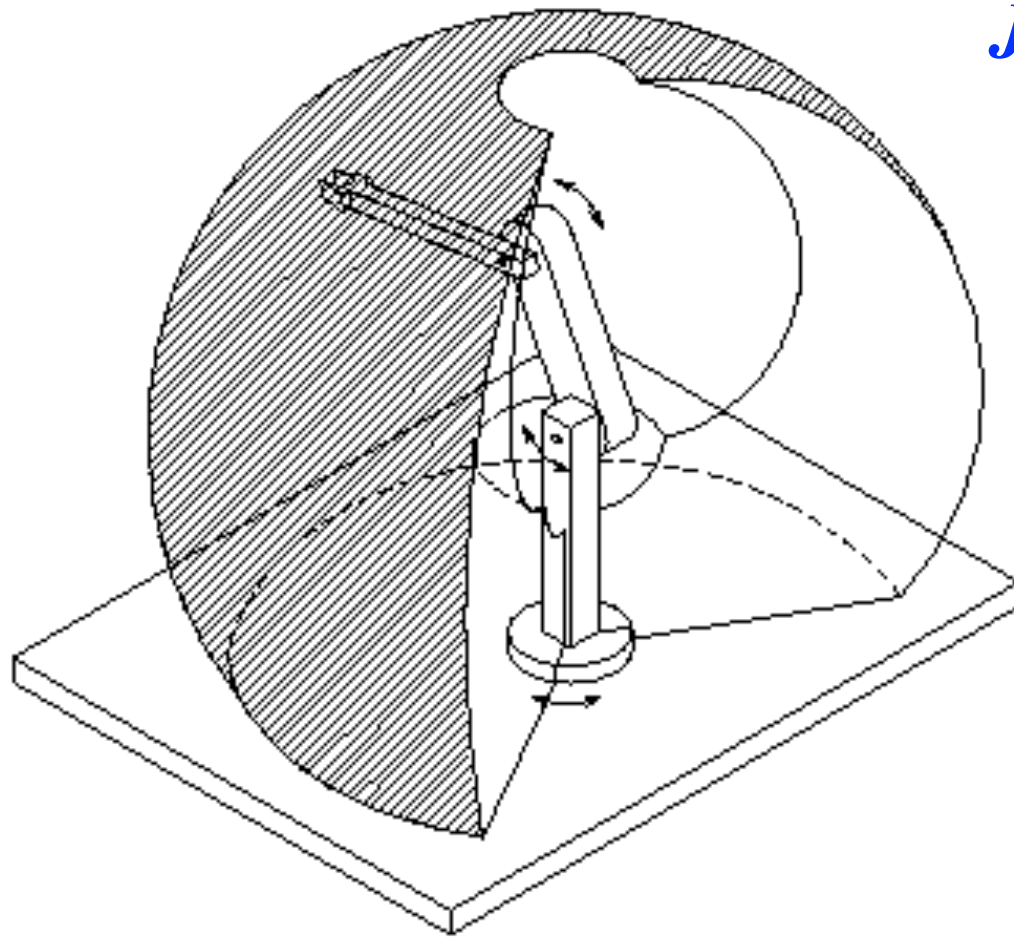
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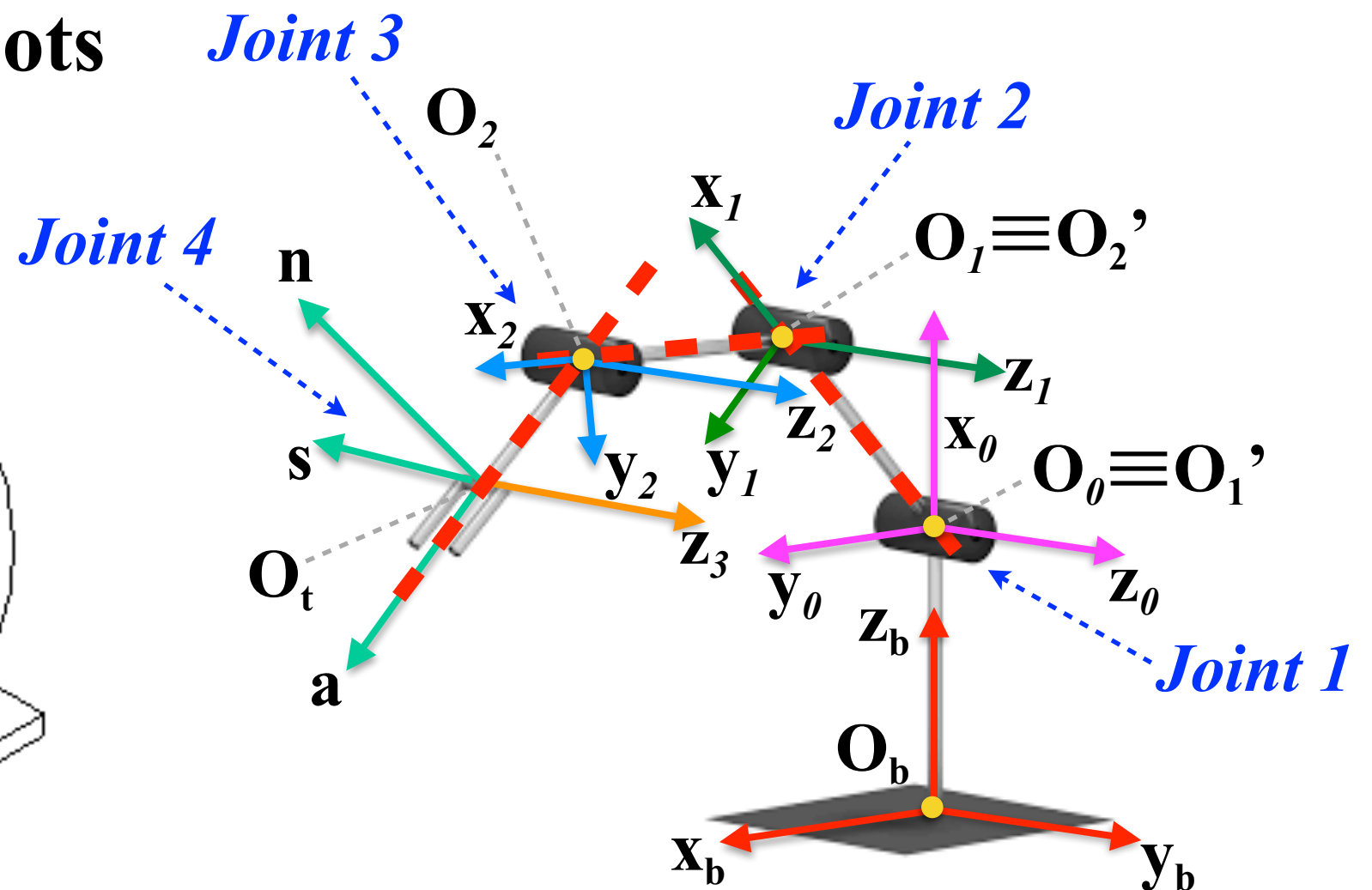
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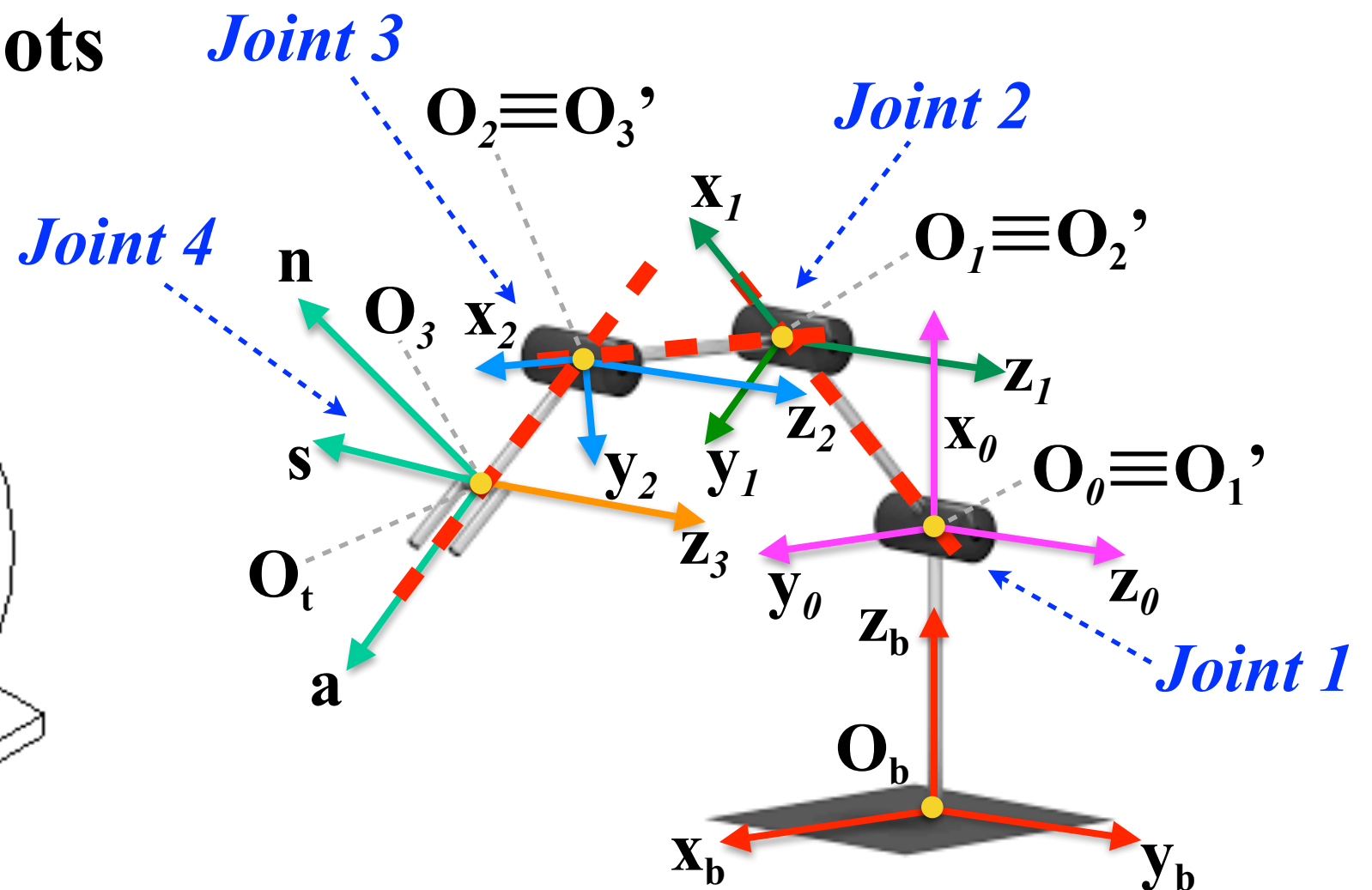


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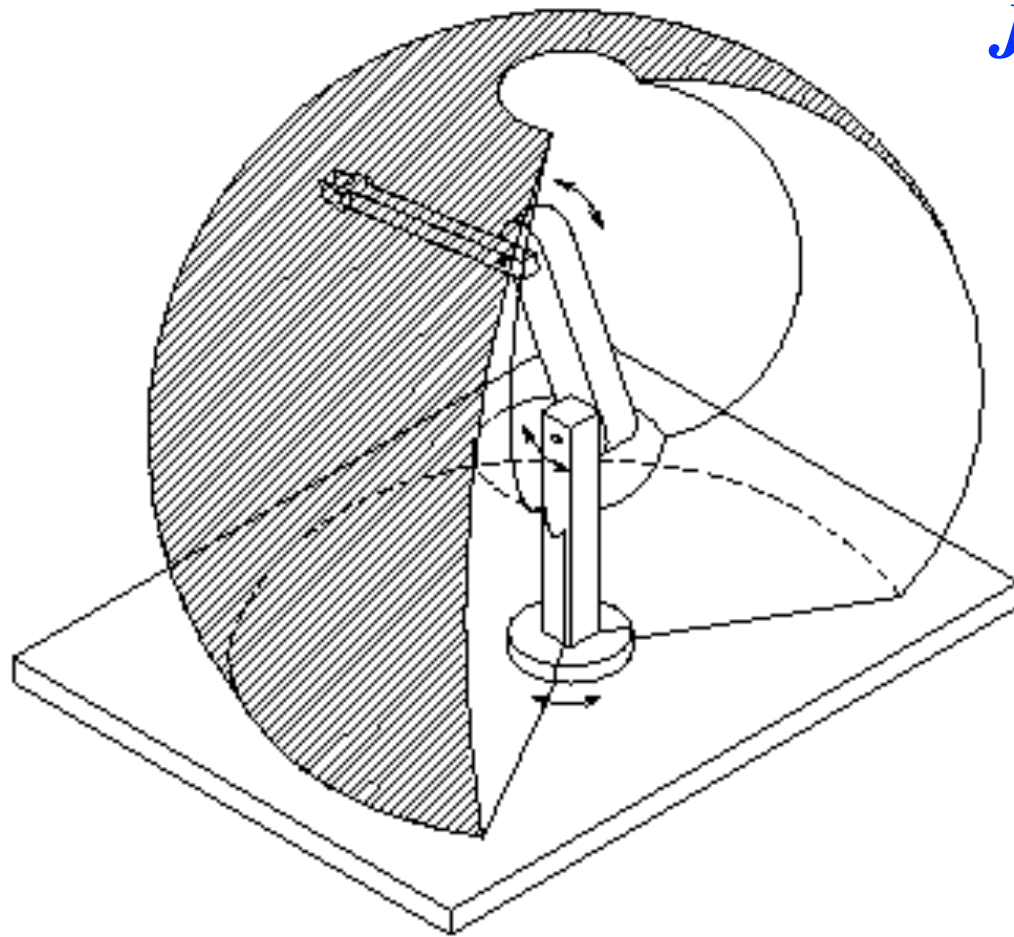
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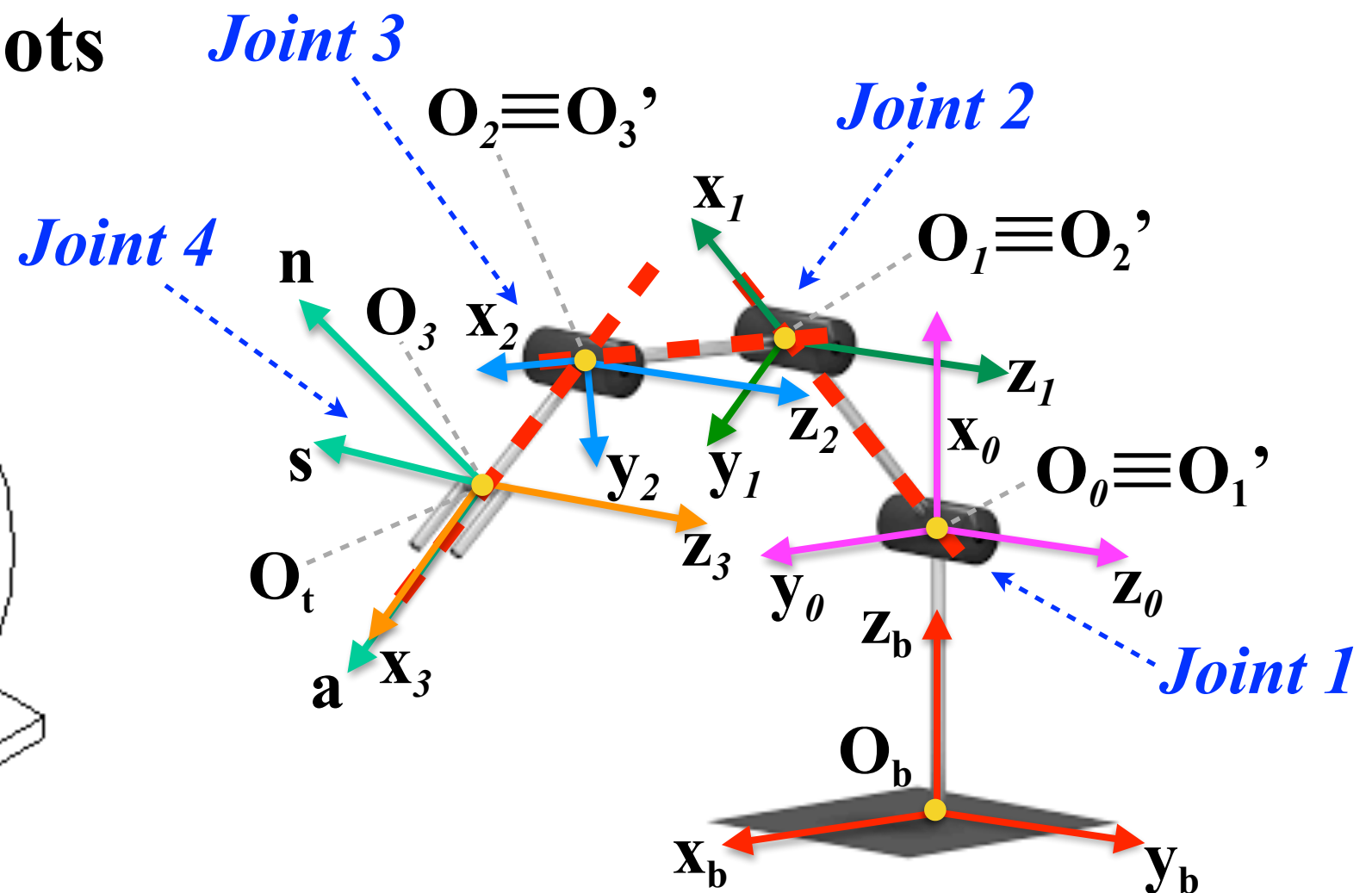
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Anthropomorphic Robots



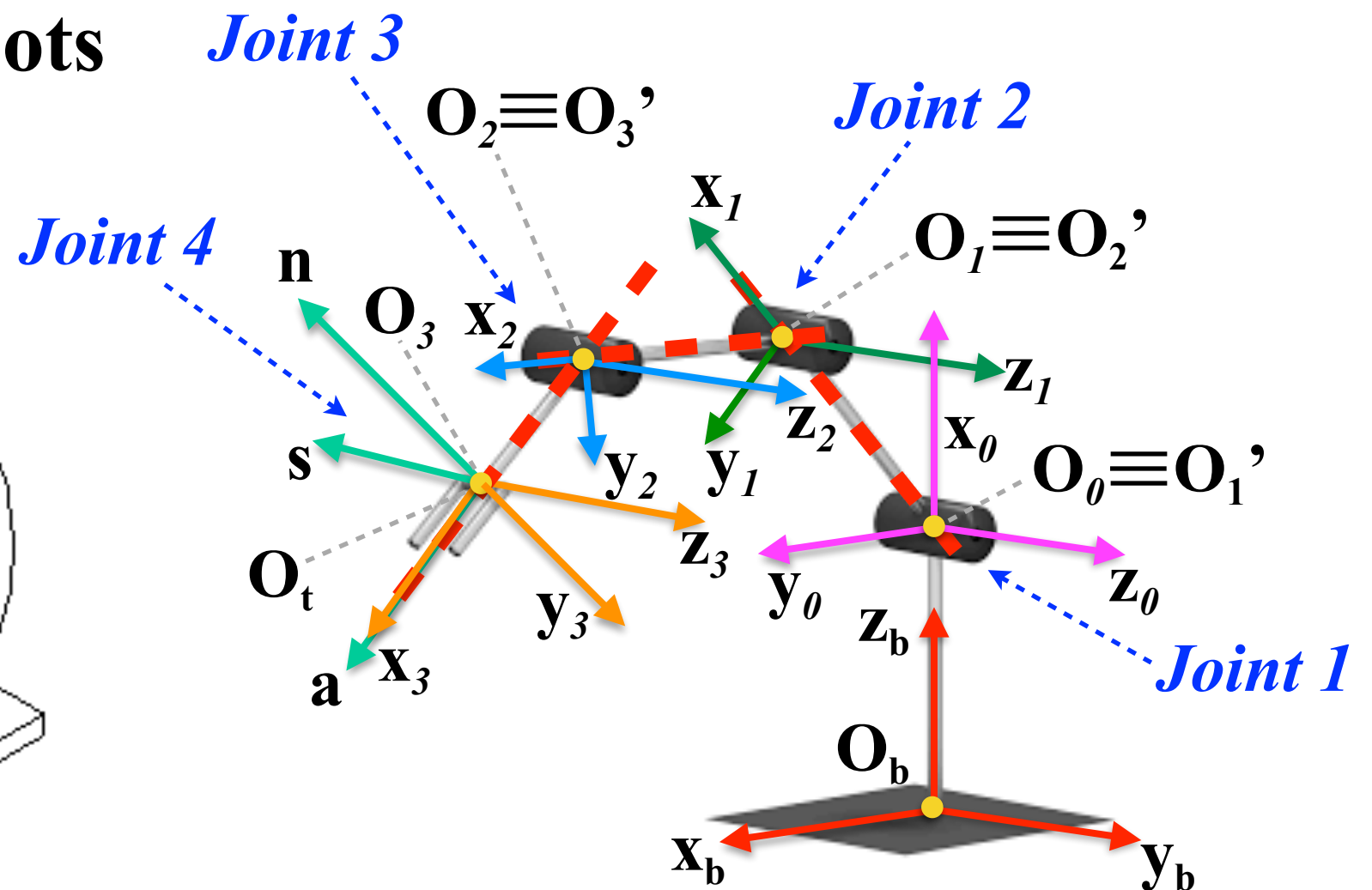
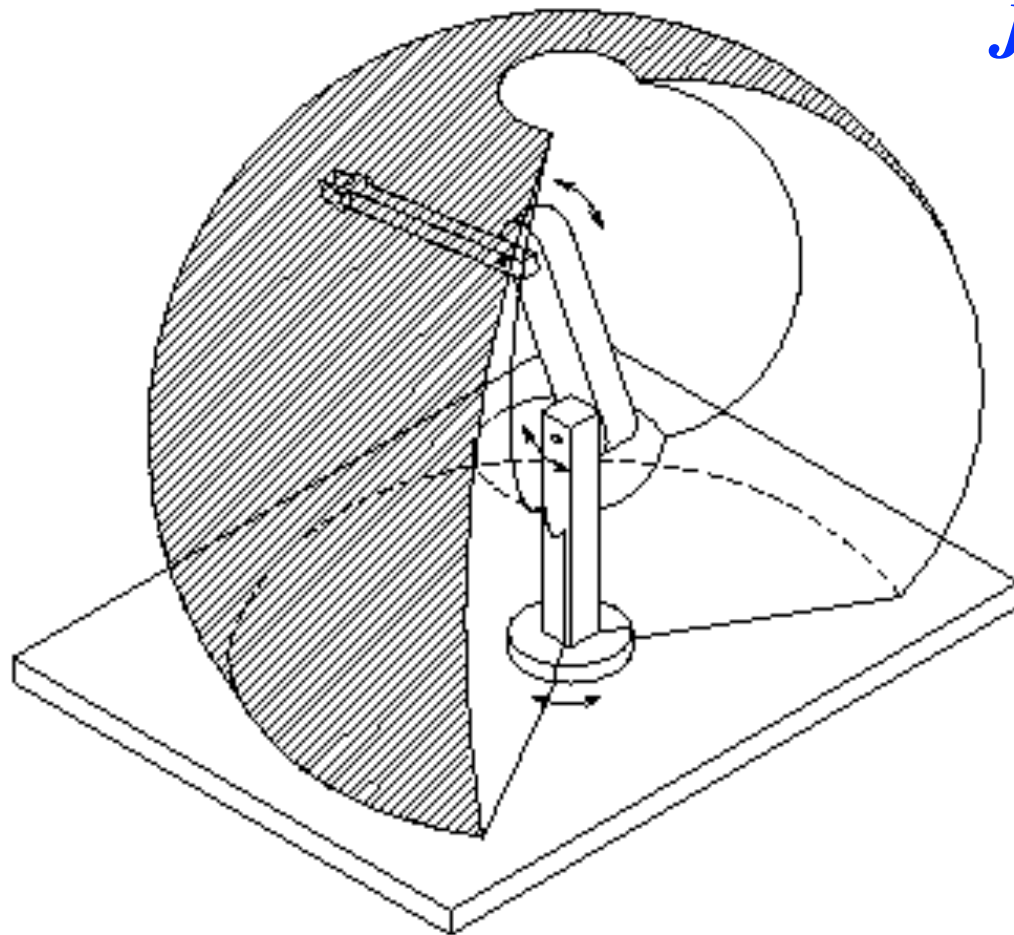
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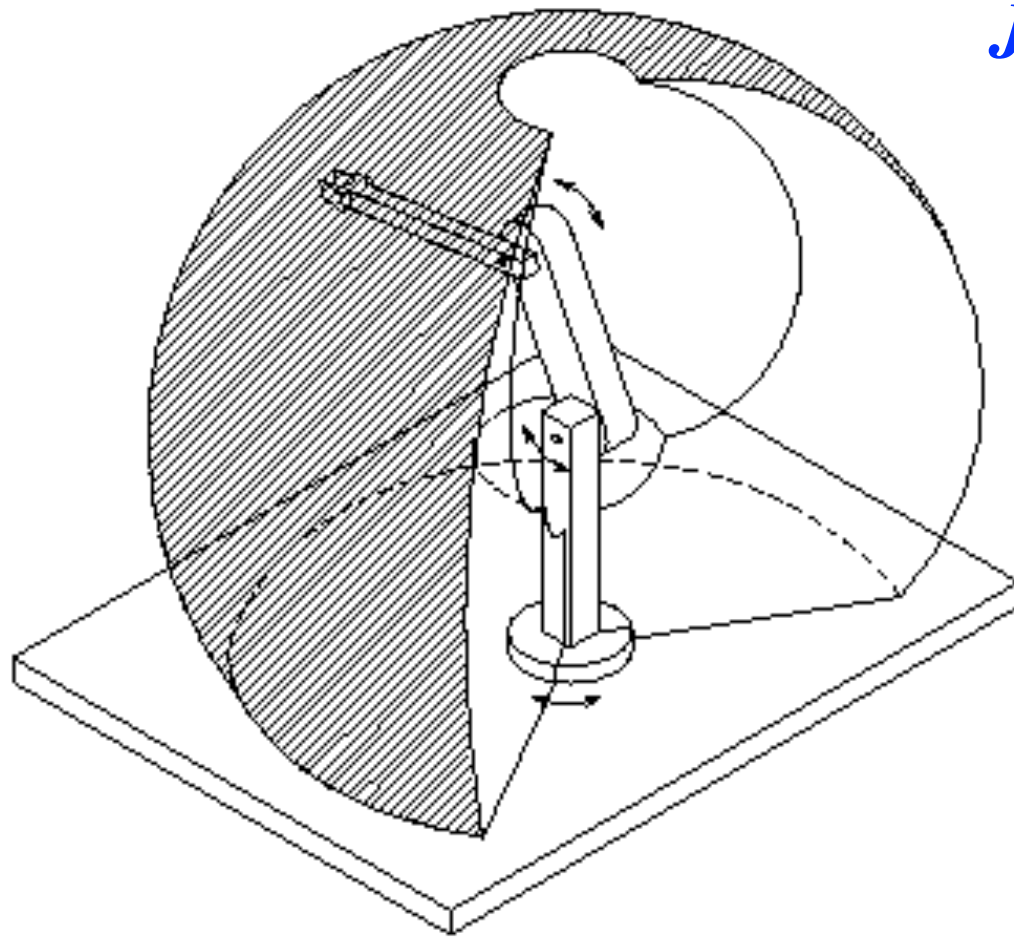


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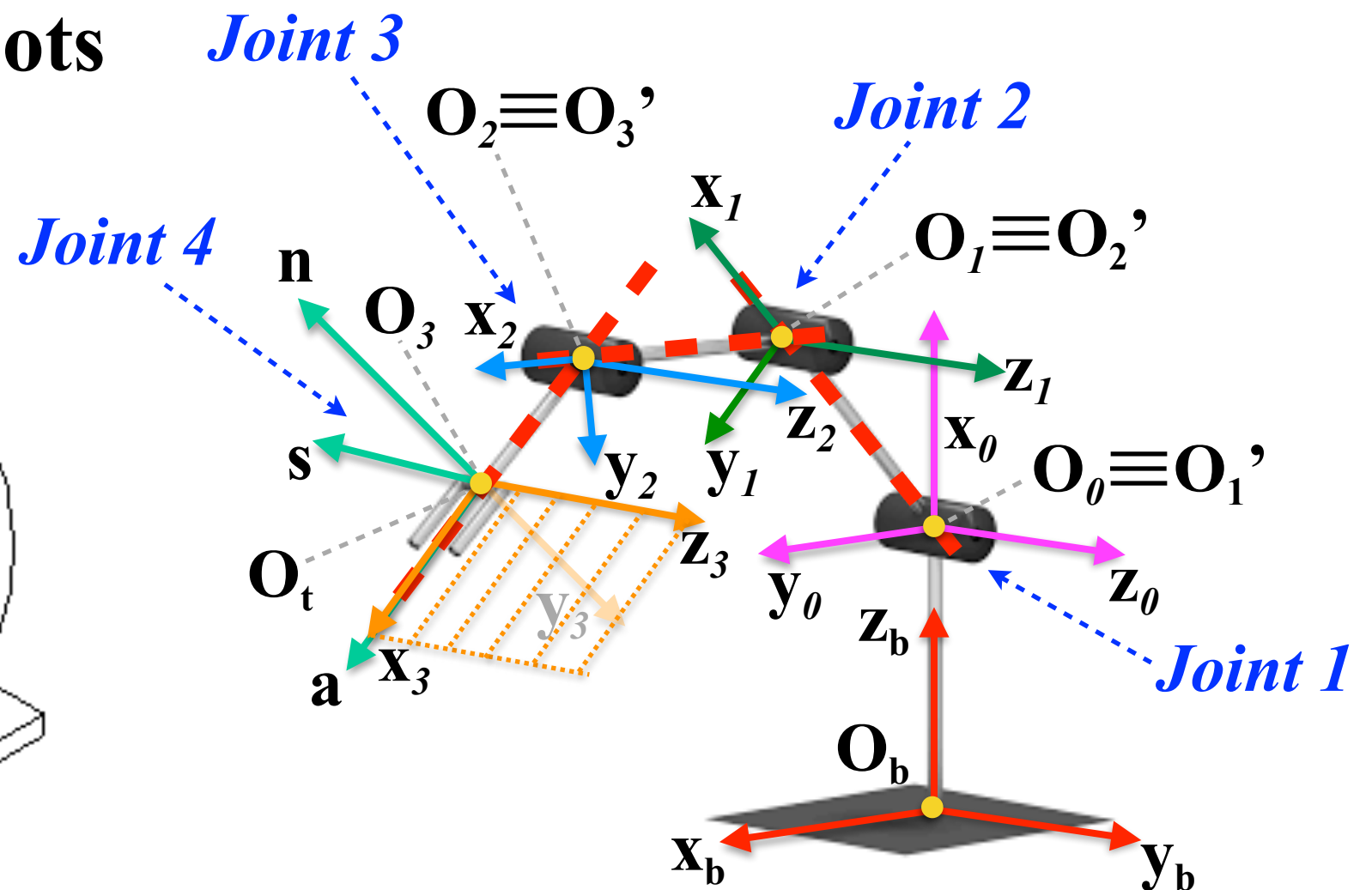
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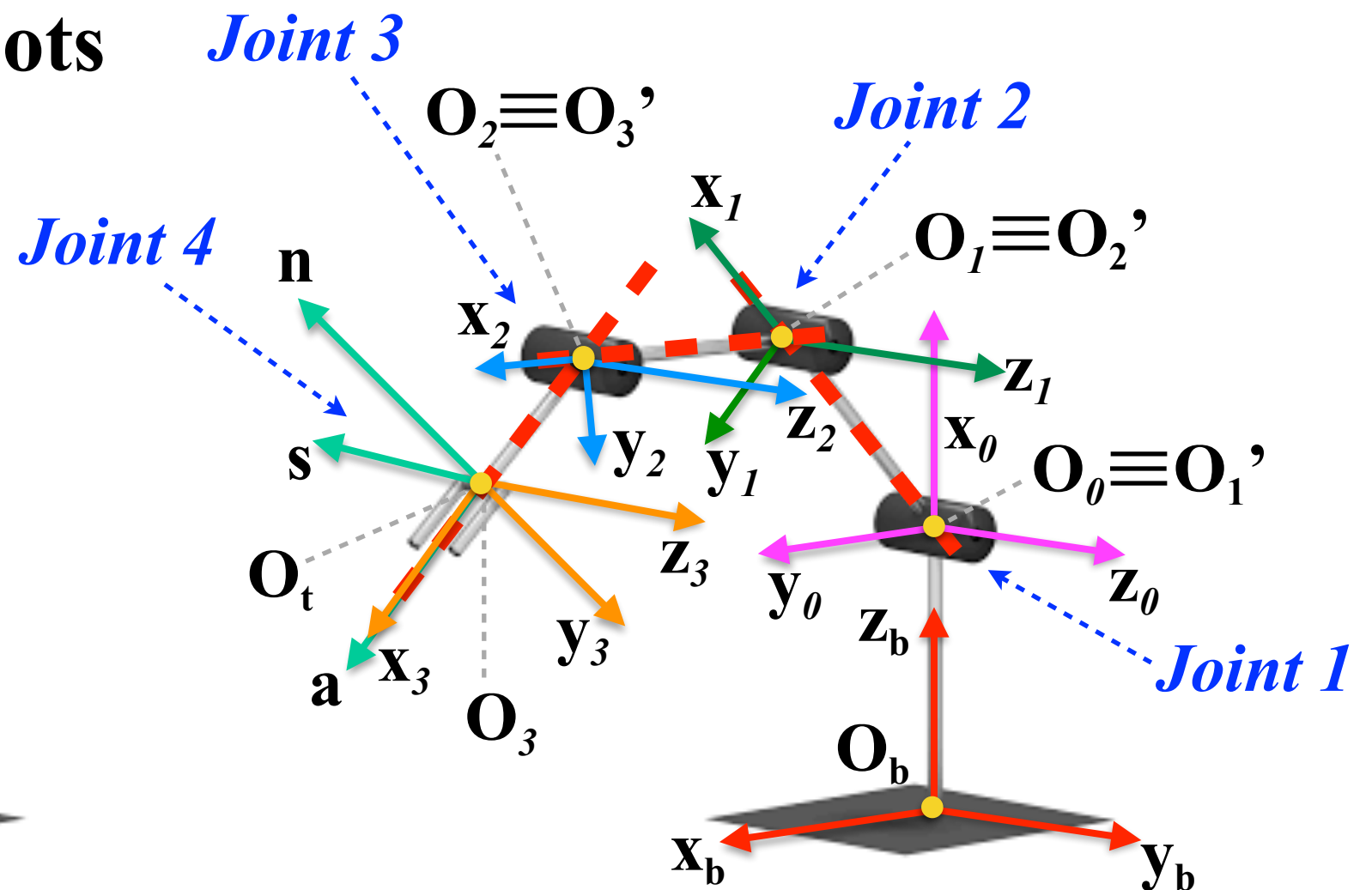
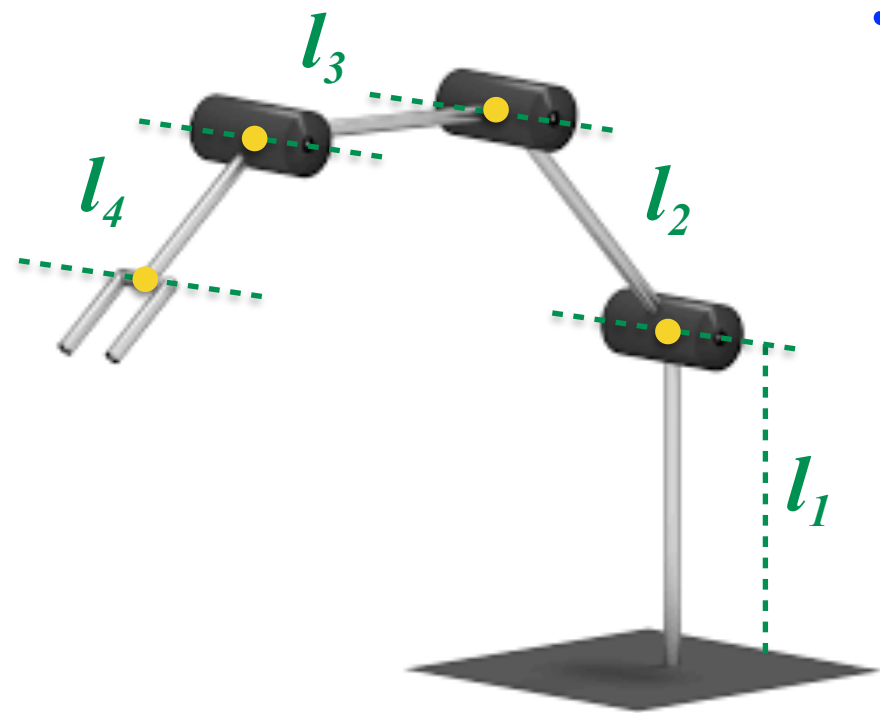
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Assign the D-H Reference
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Anthropomorphic Robots



Determine the D-H parameters

How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Anthropomorphic Robots

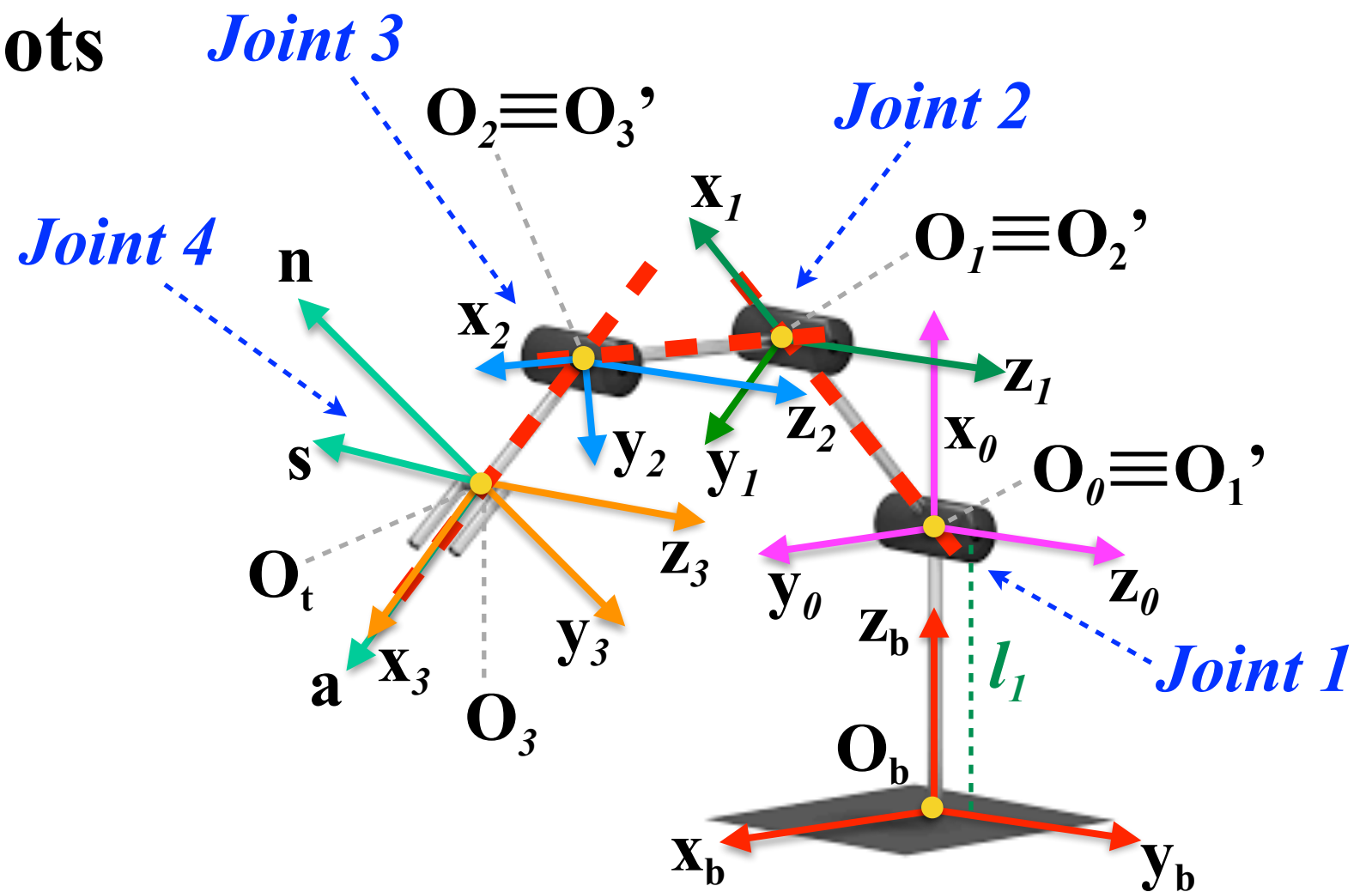
D-H parameters for reference frame 1

a_1

d_1

θ_1

α_1



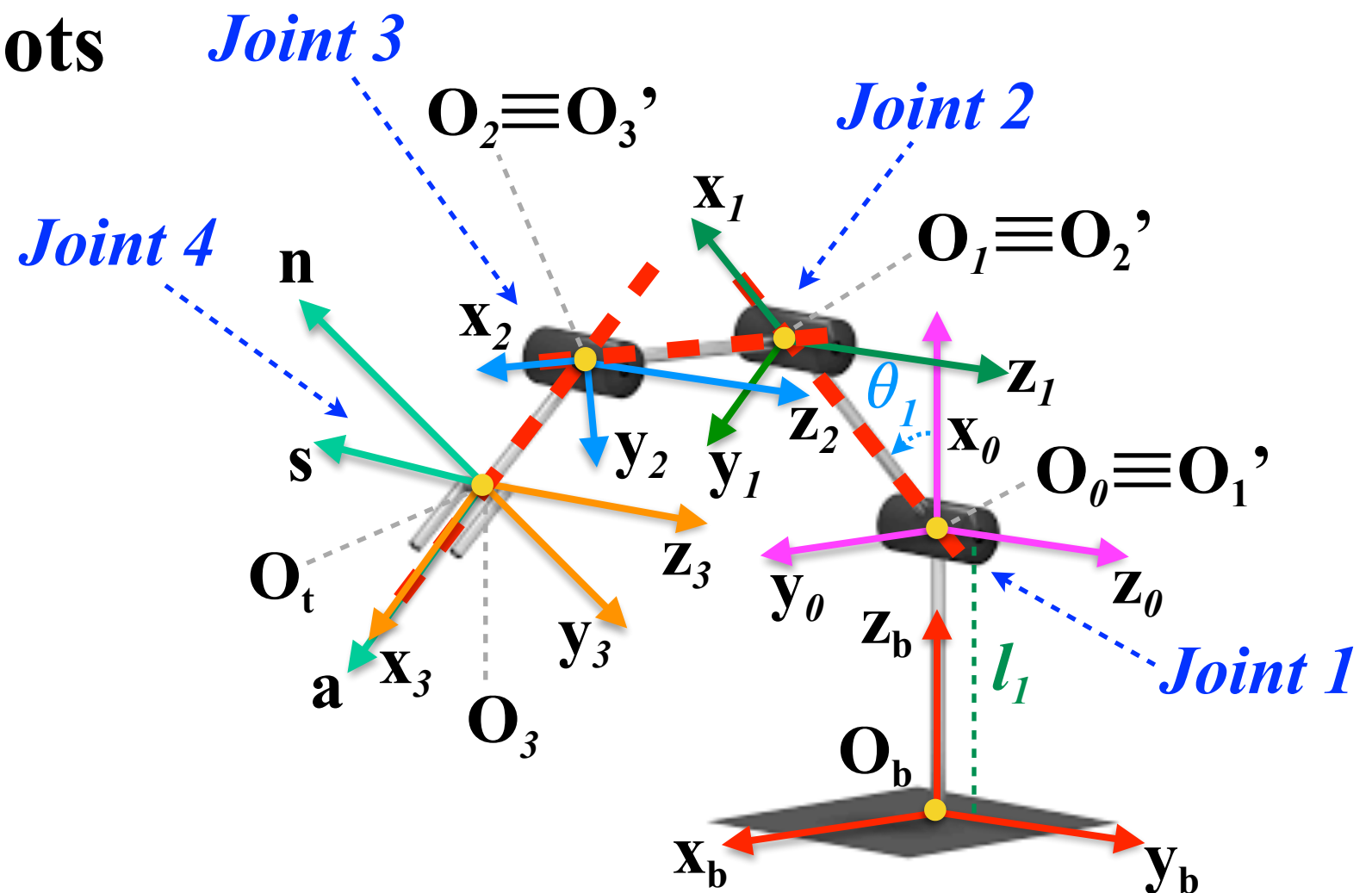
Determine the D-H parameters

How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Anthropomorphic Robots

D-H parameters for reference frame 1

a_1	l_2 [m]
d_1	0 [m]
θ_1	see picture
α_1	0°

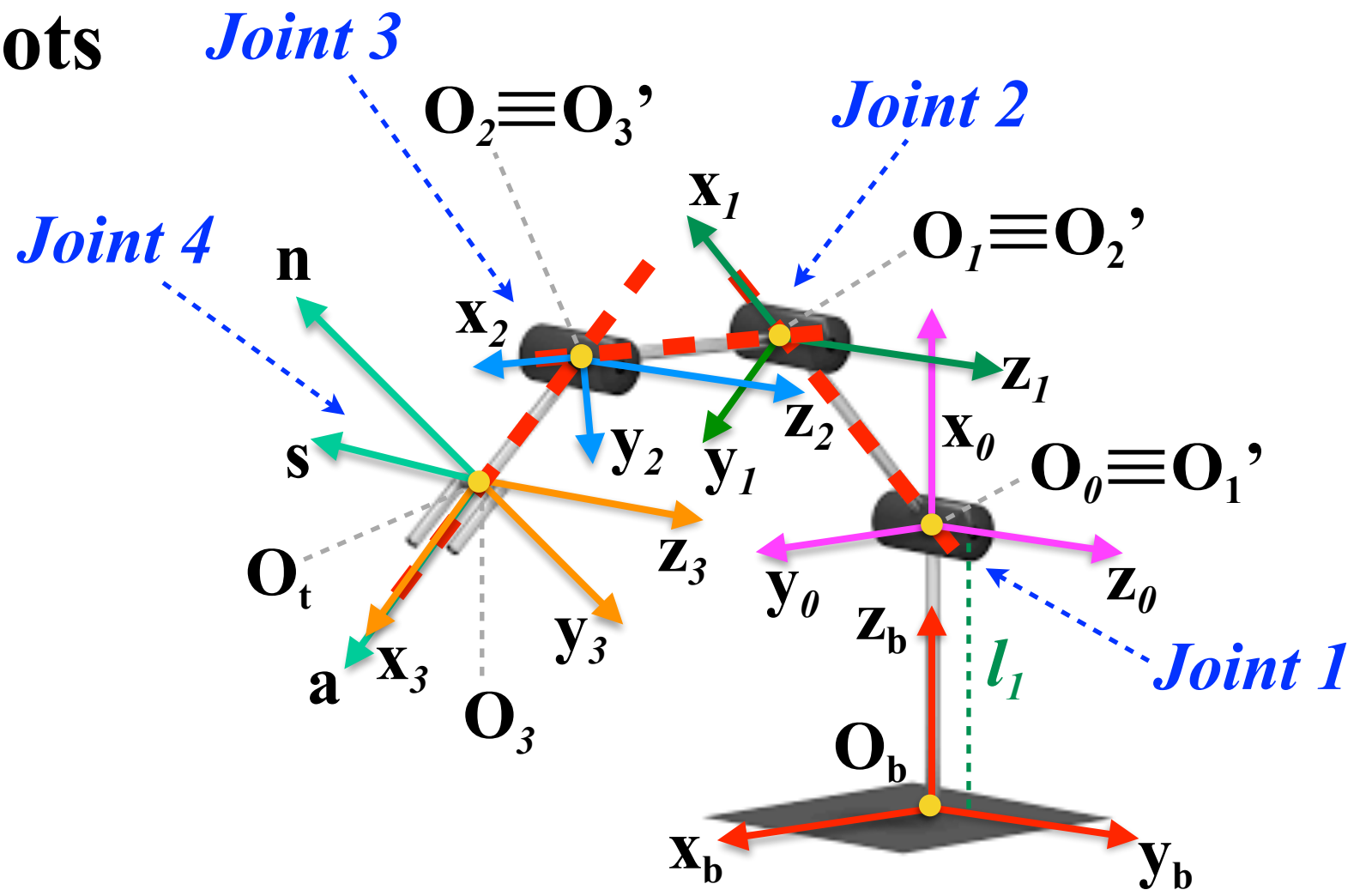


Determine the D-H parameters

How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Anthropomorphic Robots

D-H parameters for reference frame 2

$$\mathbf{a}_2$$
$$\mathbf{d}_2$$
 θ_2 α_2 

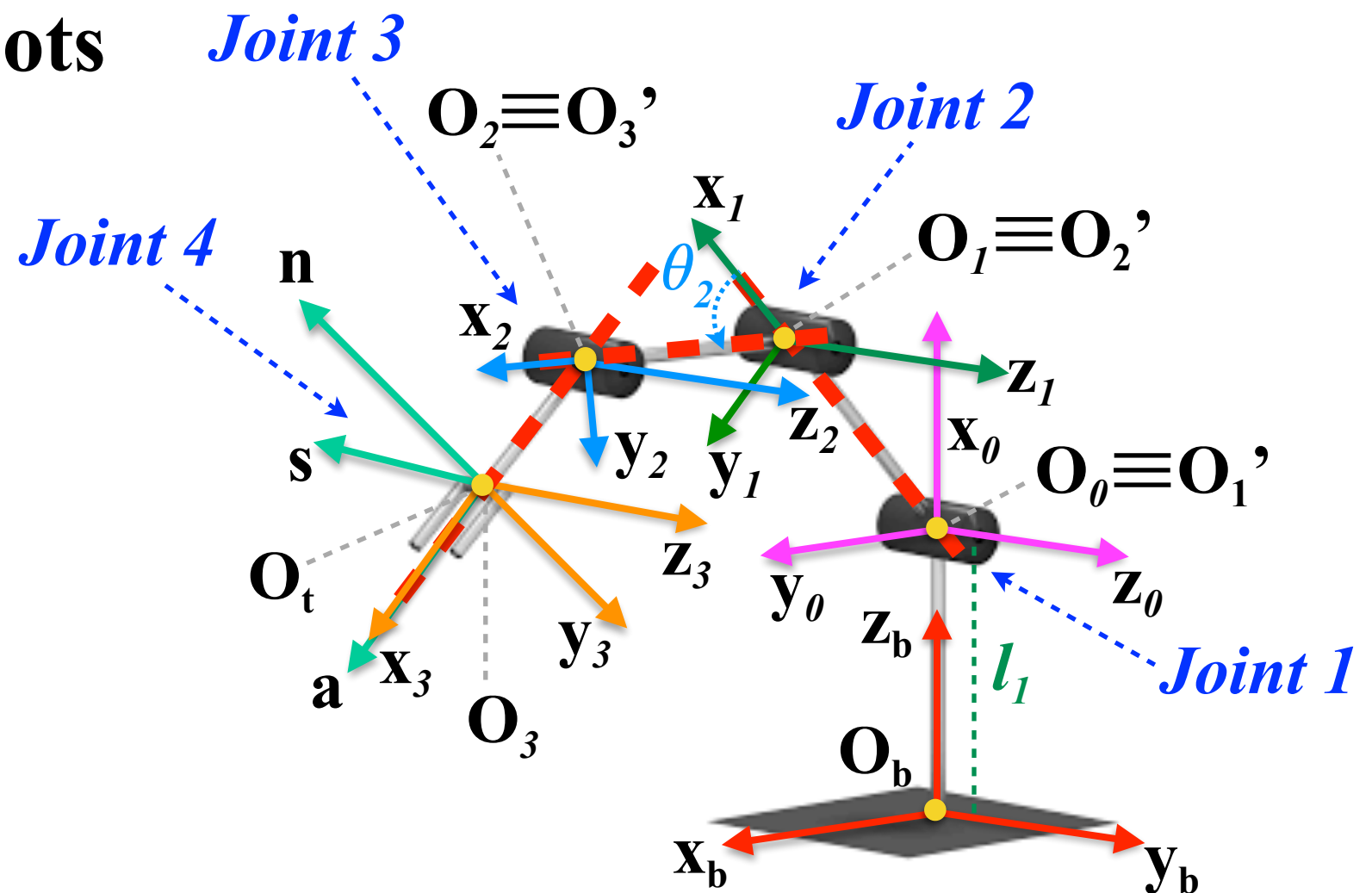
Determine the D-H parameters

How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Anthropomorphic Robots

D-H parameters for reference frame 2

a_2	l_3 [m]
d_2	0 [m]
θ_2	see picture
α_2	0°



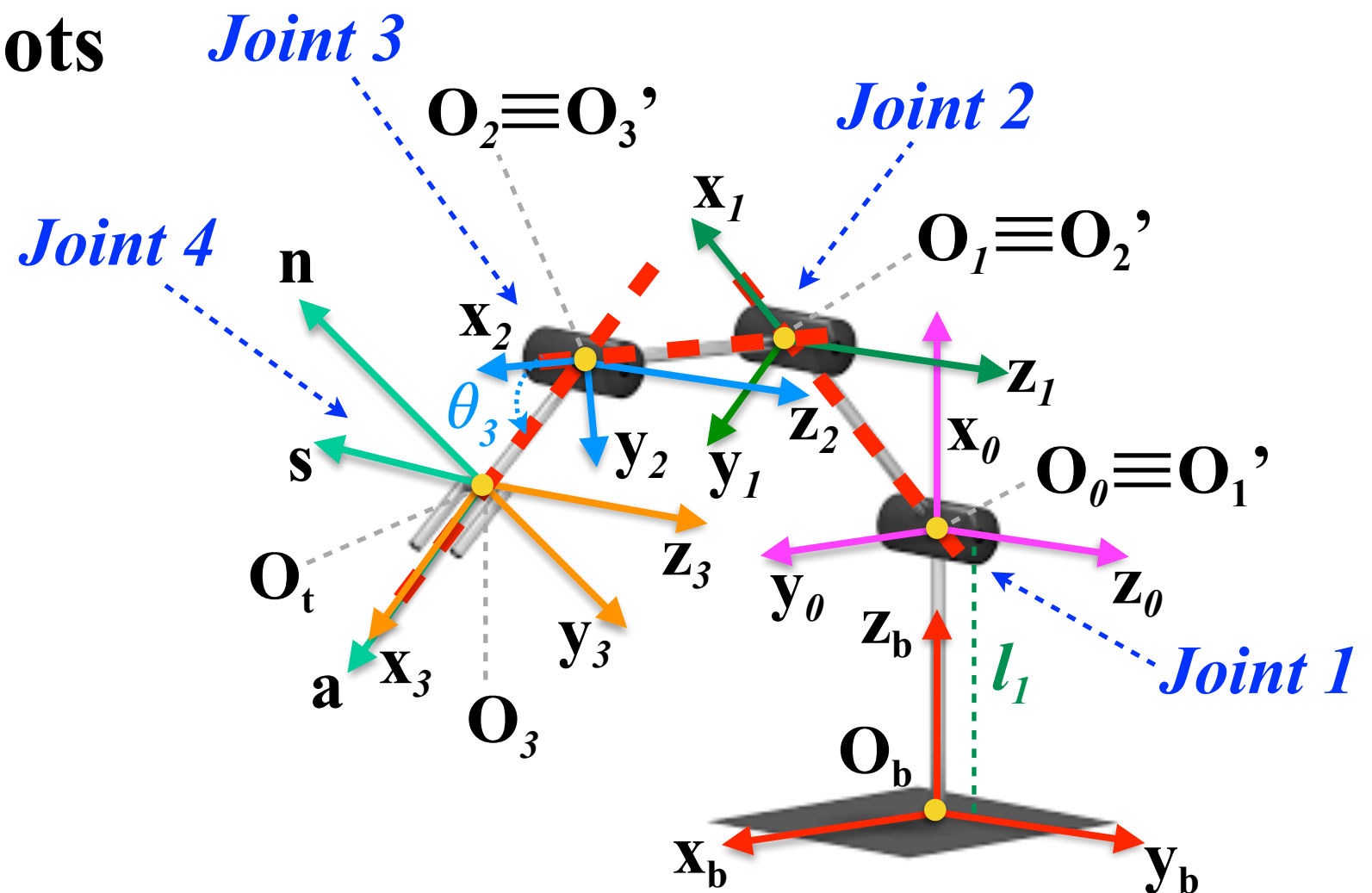
Determine the D-H parameters

How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Anthropomorphic Robots

D-H parameters for reference frame 3

a_3	l_4 [m]
d_3	0 [m]
θ_3	see picture
α_3	0^0



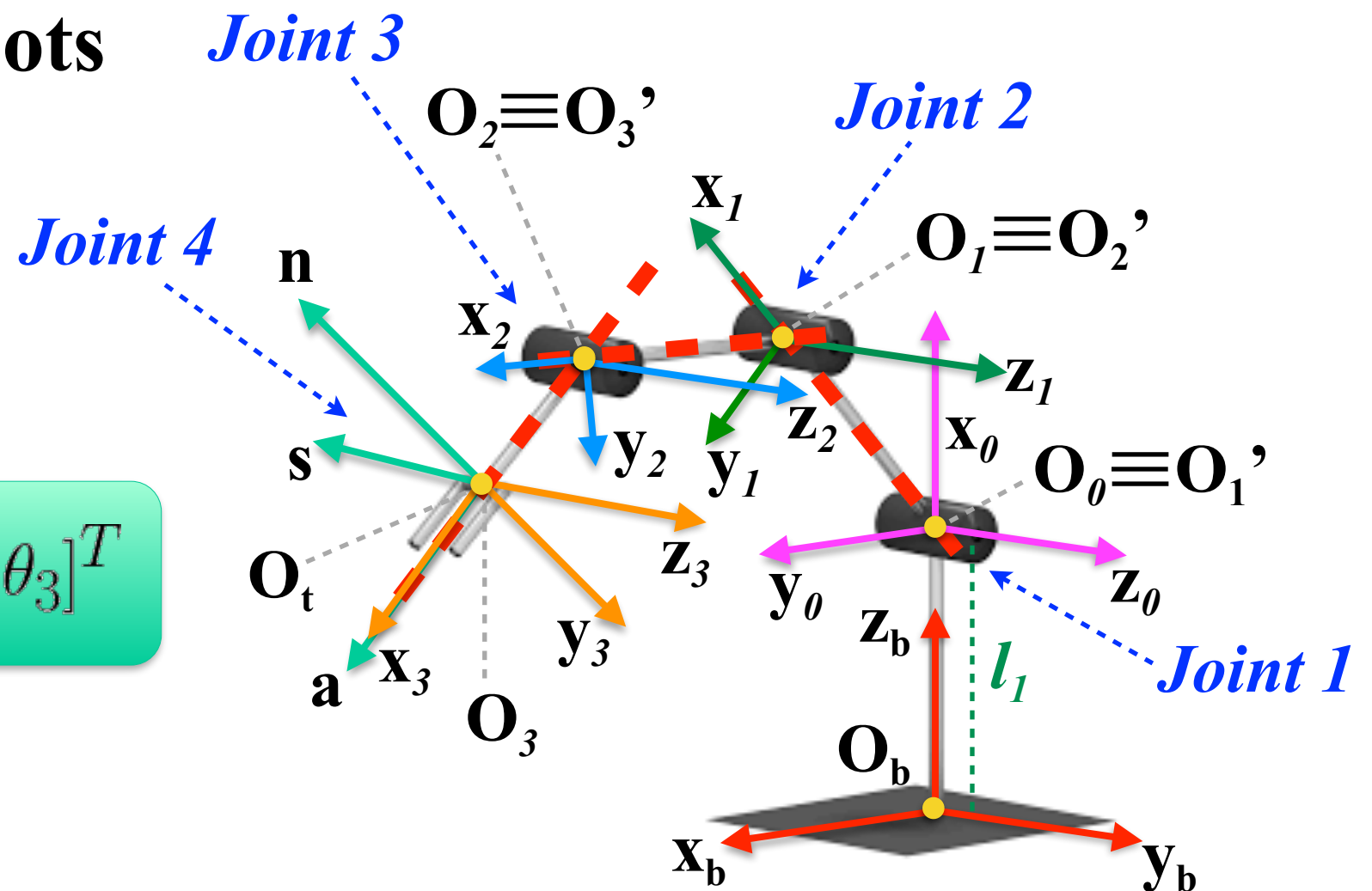
Determine the D-H parameters

How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Anthropomorphic Robots

Define the vector
of the joint variables

$$\underline{q} = [q_1, q_2, q_3]^T = [\theta_1, \theta_2, \theta_3]^T$$



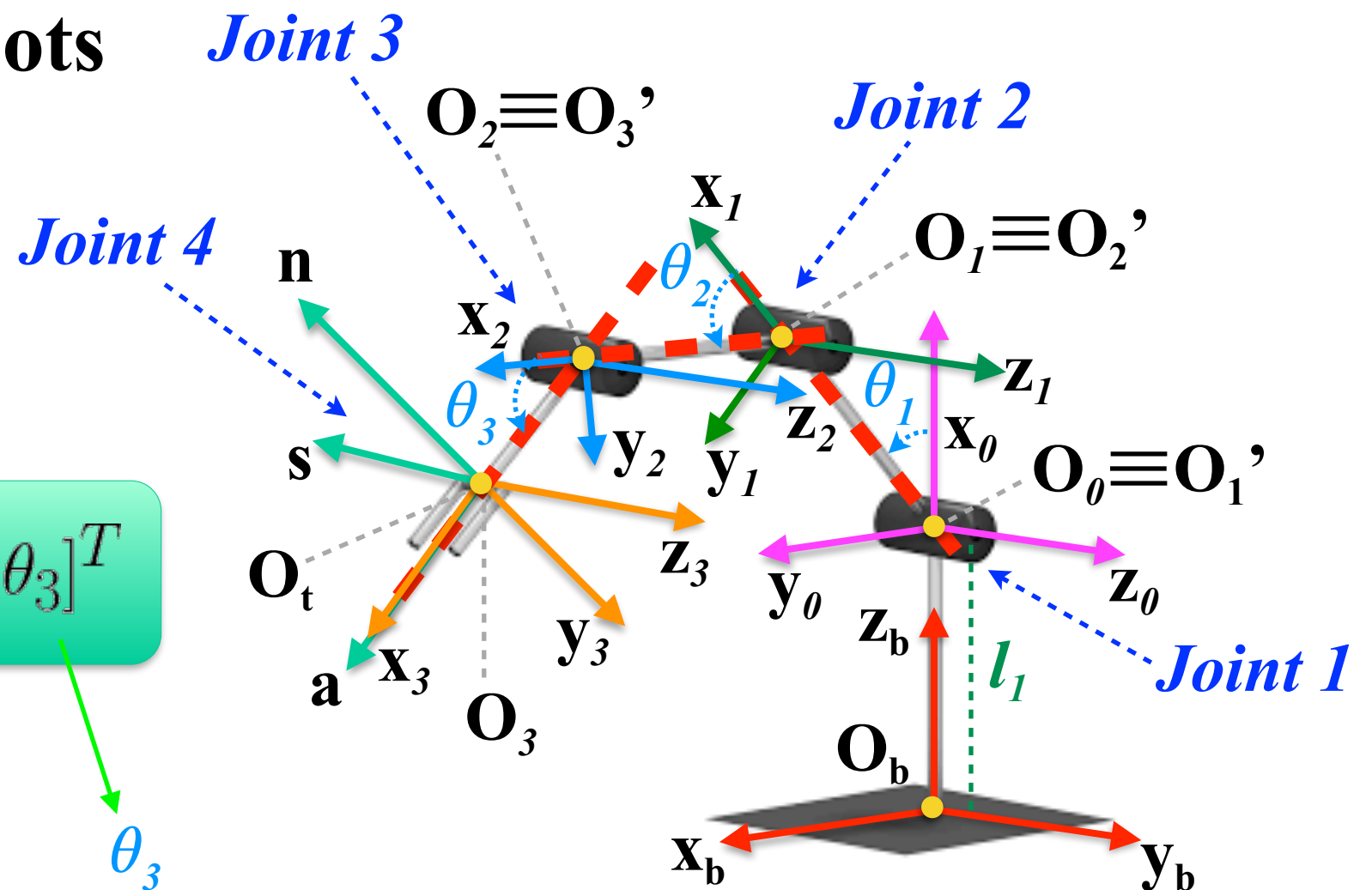
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$$\underline{q} = [q_1, q_2, q_3]^T = [\theta_1, \theta_2, \theta_3]^T$$

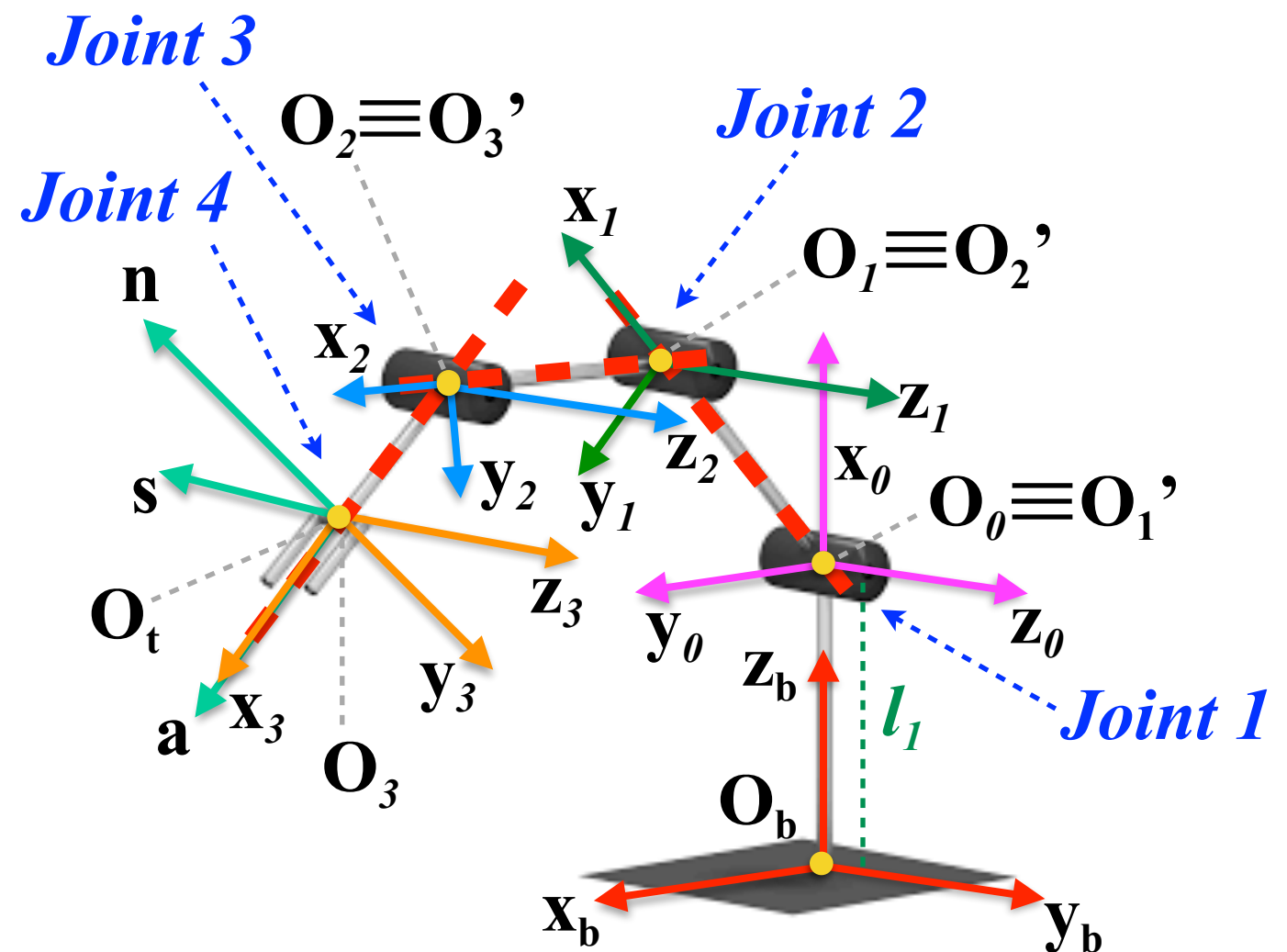
θ_1 θ_2 θ_3



How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Anthropomorphic Robots

Determine the homogeneous transformation matrix

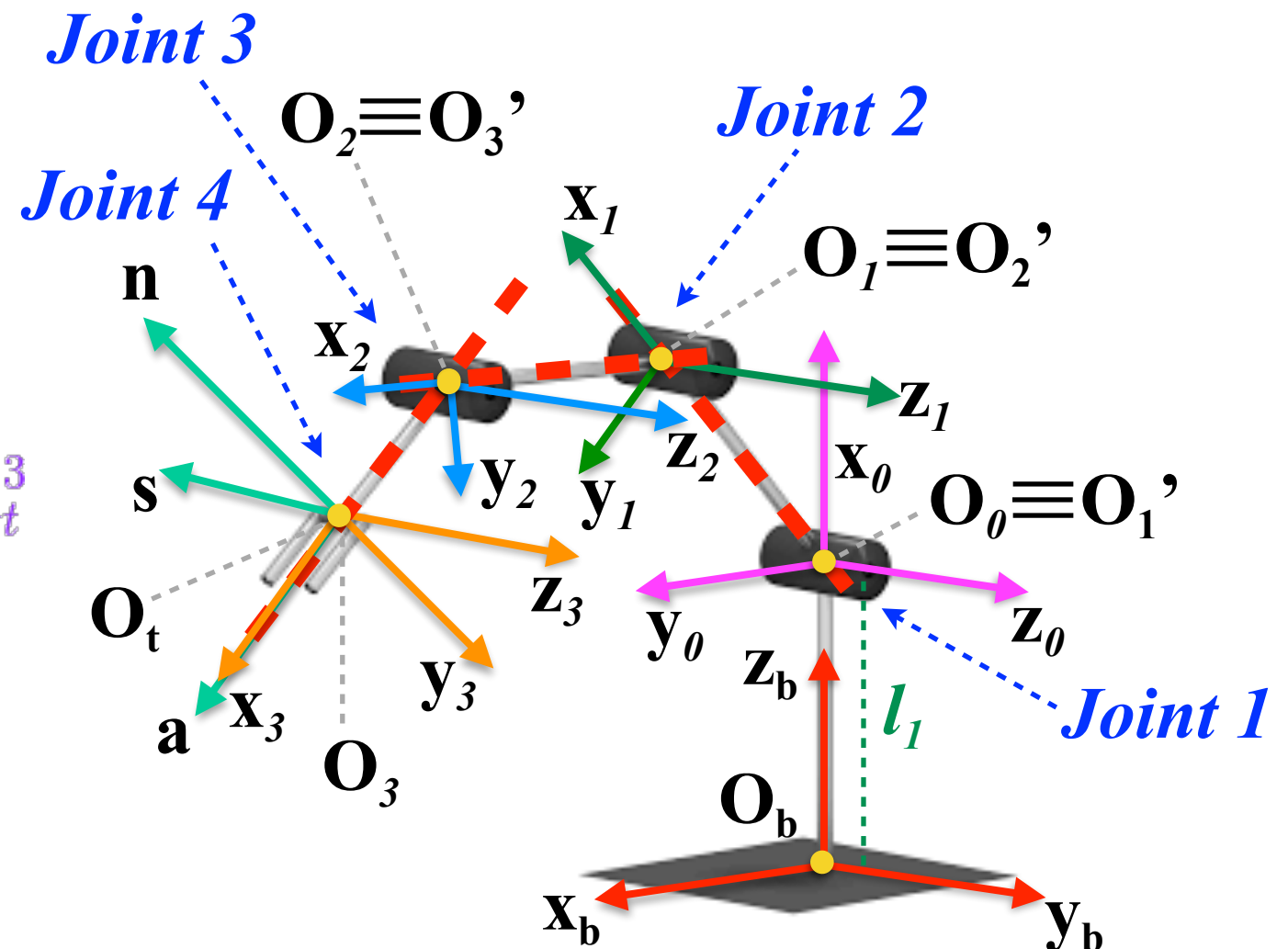


How to Apply the Denavit-Hartenberg Convention Based Approach to Describe the Direct Kinematics of Typical Robots

Anthropomorphic Robots

Determine the homogeneous transformation matrix

$$T_t^b(\underline{q}) = A_0^b \cdot A_1^0(\underline{q}) \cdot A_2^1(\underline{q}) \cdot A_3^2(\underline{q}) \cdot A_t^3$$



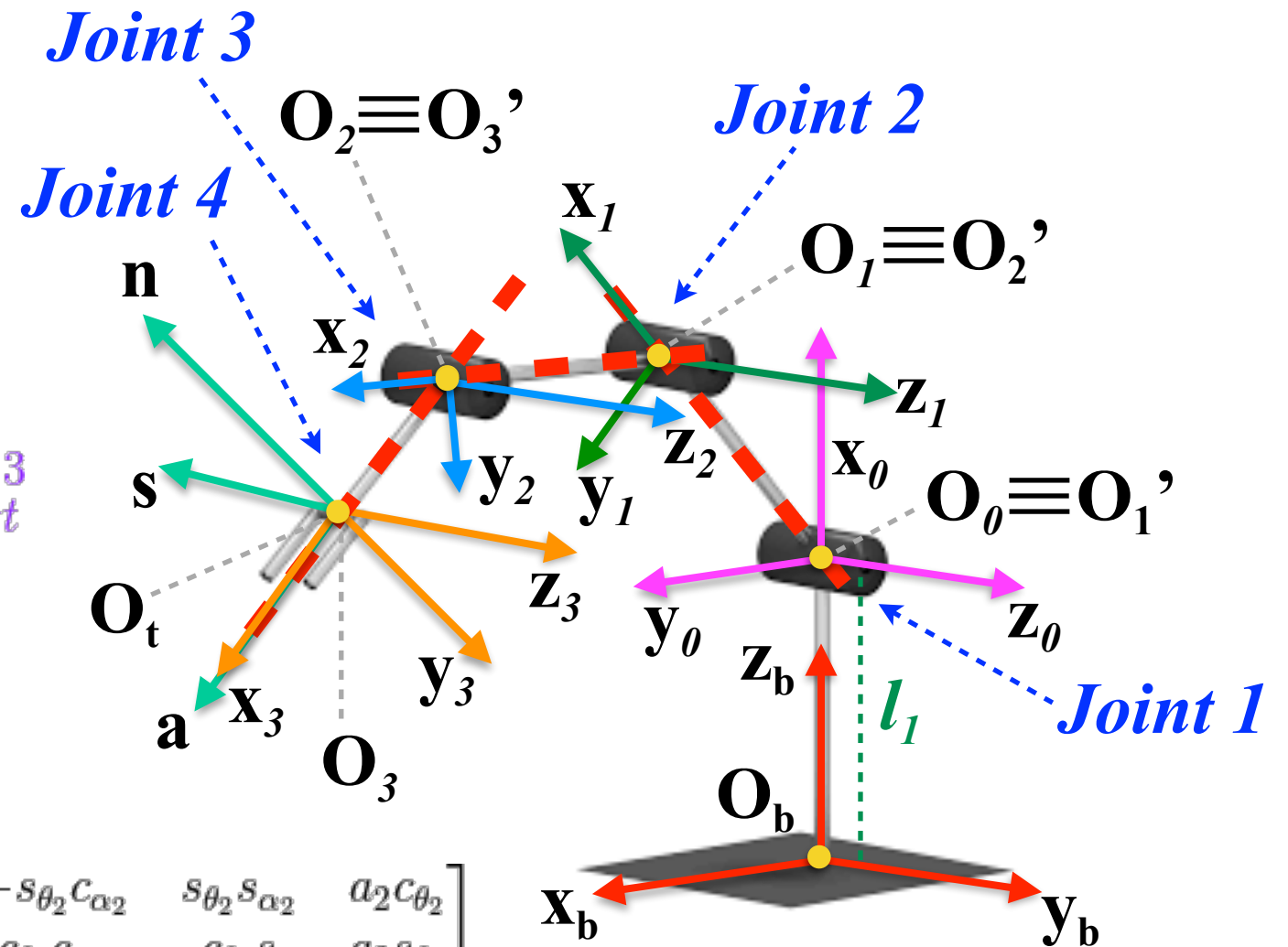
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$$T_t^b(\underline{q}) = A_0^b \cdot A_1^0(\underline{q}) \cdot A_2^1(\underline{q}) \cdot A_3^2(\underline{q}) \cdot A_t^3$$

$$T_t^b(\underline{q}) = A_0^b \cdot \begin{bmatrix} c_{\theta_1} & -s_{\theta_1}c_{\alpha_1} & s_{\theta_1}s_{\alpha_1} & a_1c_{\theta_1} \\ s_{\theta_1} & c_{\theta_1}c_{\alpha_1} & -c_{\theta_1}s_{\alpha_1} & a_1s_{\theta_1} \\ 0 & s_{\alpha_1} & c_{\alpha_1} & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_2} & -s_{\theta_2}c_{\alpha_2} & s_{\theta_2}s_{\alpha_2} & a_2c_{\theta_2} \\ s_{\theta_2} & c_{\theta_2}c_{\alpha_2} & -c_{\theta_2}s_{\alpha_2} & a_2s_{\theta_2} \\ 0 & s_{\alpha_2} & c_{\alpha_2} & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_3} & -s_{\theta_3}c_{\alpha_3} & s_{\theta_3}s_{\alpha_3} & a_3c_{\theta_3} \\ s_{\theta_3} & c_{\theta_3}c_{\alpha_3} & -c_{\theta_3}s_{\alpha_3} & a_3s_{\theta_3} \\ 0 & s_{\alpha_3} & c_{\alpha_3} & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot A_t^3$$



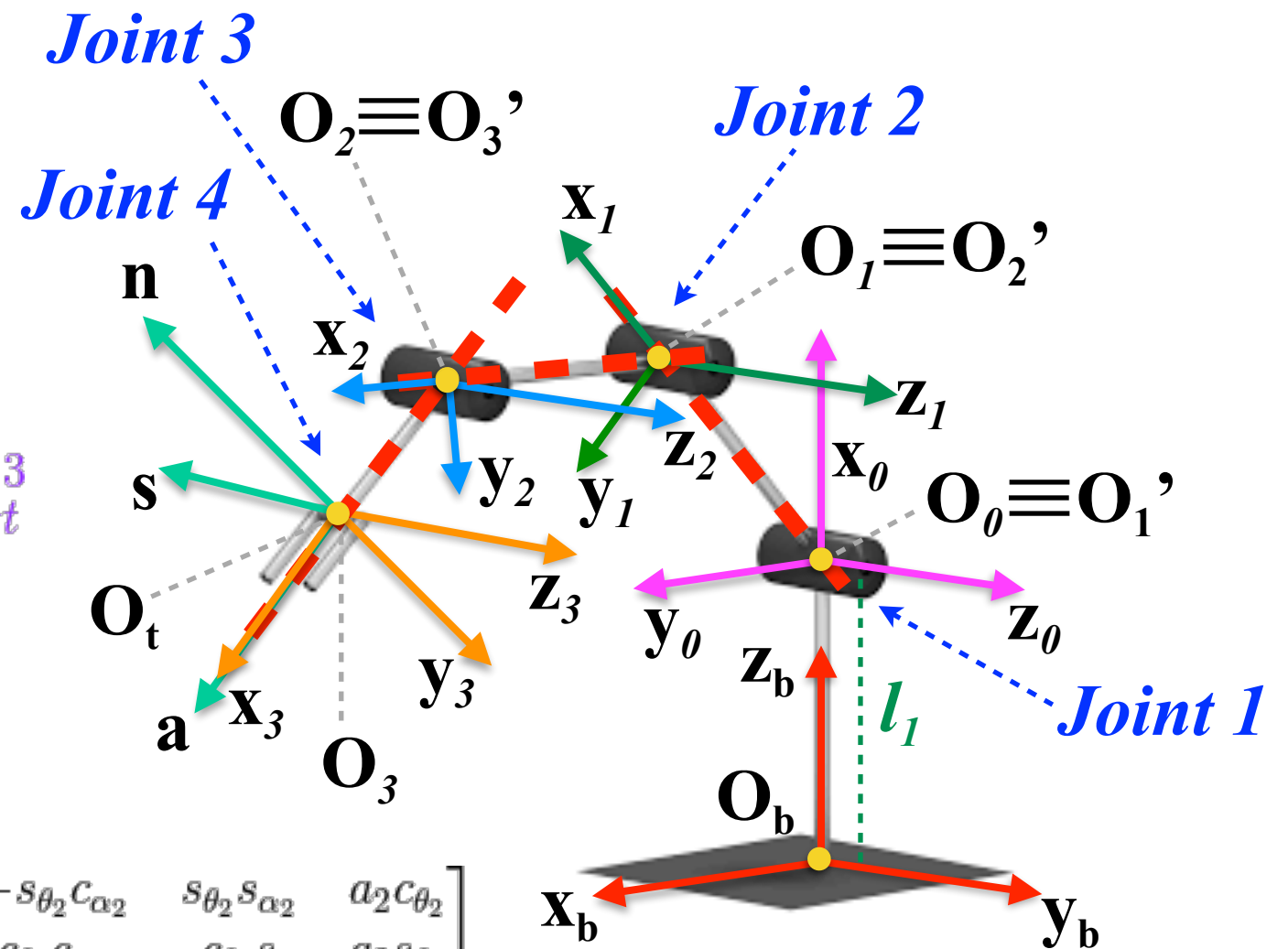
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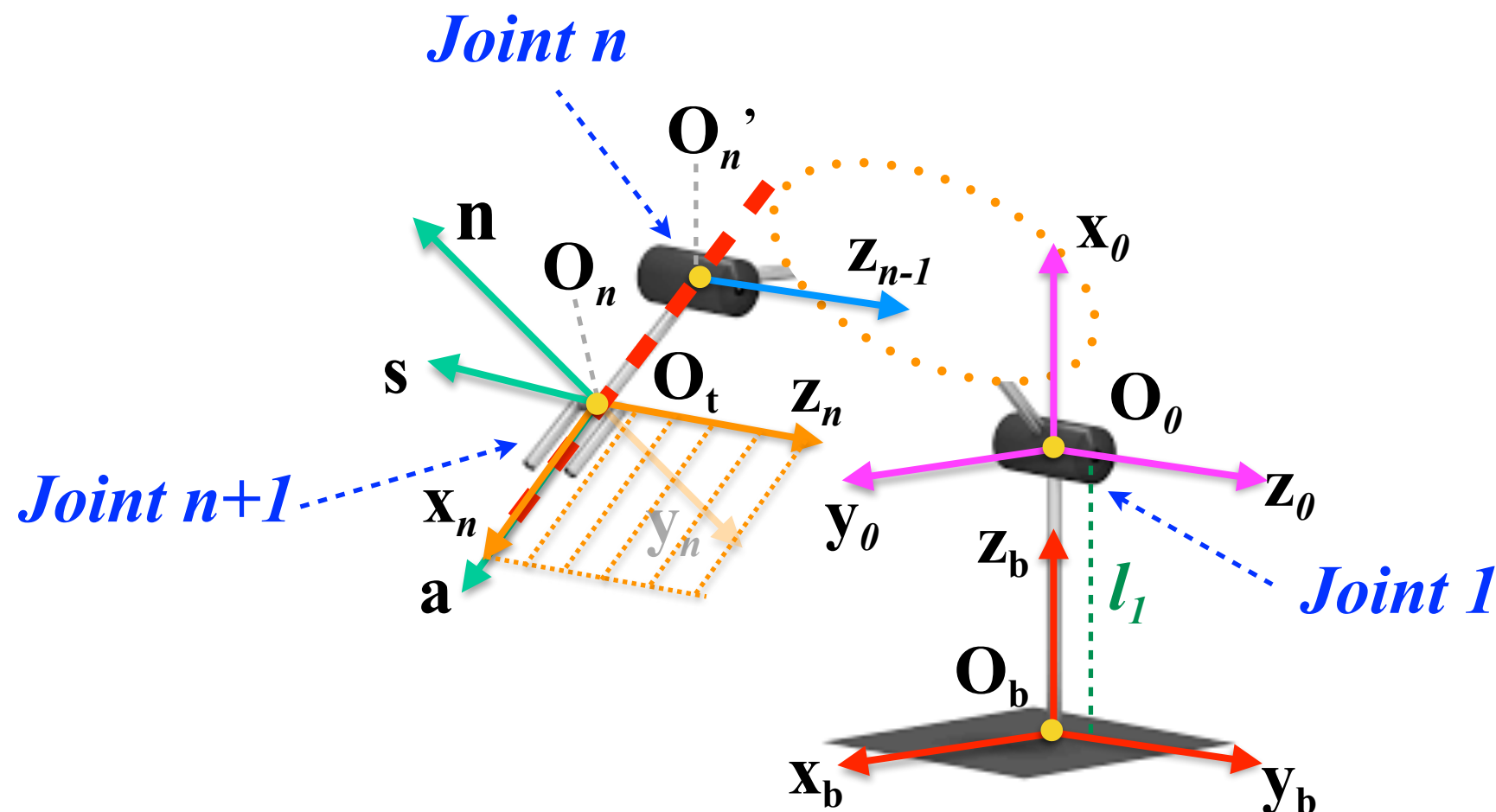
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$$T_t^b(\underline{q}) = A_0^b \cdot \begin{bmatrix} c_{\theta_1} & -s_{\theta_1}c_{\alpha_1} & s_{\theta_1}s_{\alpha_1} & a_1c_{\theta_1} \\ s_{\theta_1} & c_{\theta_1}c_{\alpha_1} & -c_{\theta_1}s_{\alpha_1} & a_1s_{\theta_1} \\ 0 & s_{\alpha_1} & c_{\alpha_1} & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_2} & -s_{\theta_2}c_{\alpha_2} & s_{\theta_2}s_{\alpha_2} & a_2c_{\theta_2} \\ s_{\theta_2} & c_{\theta_2}c_{\alpha_2} & -c_{\theta_2}s_{\alpha_2} & a_2s_{\theta_2} \\ 0 & s_{\alpha_2} & c_{\alpha_2} & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_3} & -s_{\theta_3}c_{\alpha_3} & s_{\theta_3}s_{\alpha_3} & a_3c_{\theta_3} \\ s_{\theta_3} & c_{\theta_3}c_{\alpha_3} & -c_{\theta_3}s_{\alpha_3} & a_3s_{\theta_3} \\ 0 & s_{\alpha_3} & c_{\alpha_3} & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot A_t^3$$



How to compute: A_0^b and A_t^n



$$A_0^b = \begin{bmatrix} R_0^b & \underline{q}_0^b \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

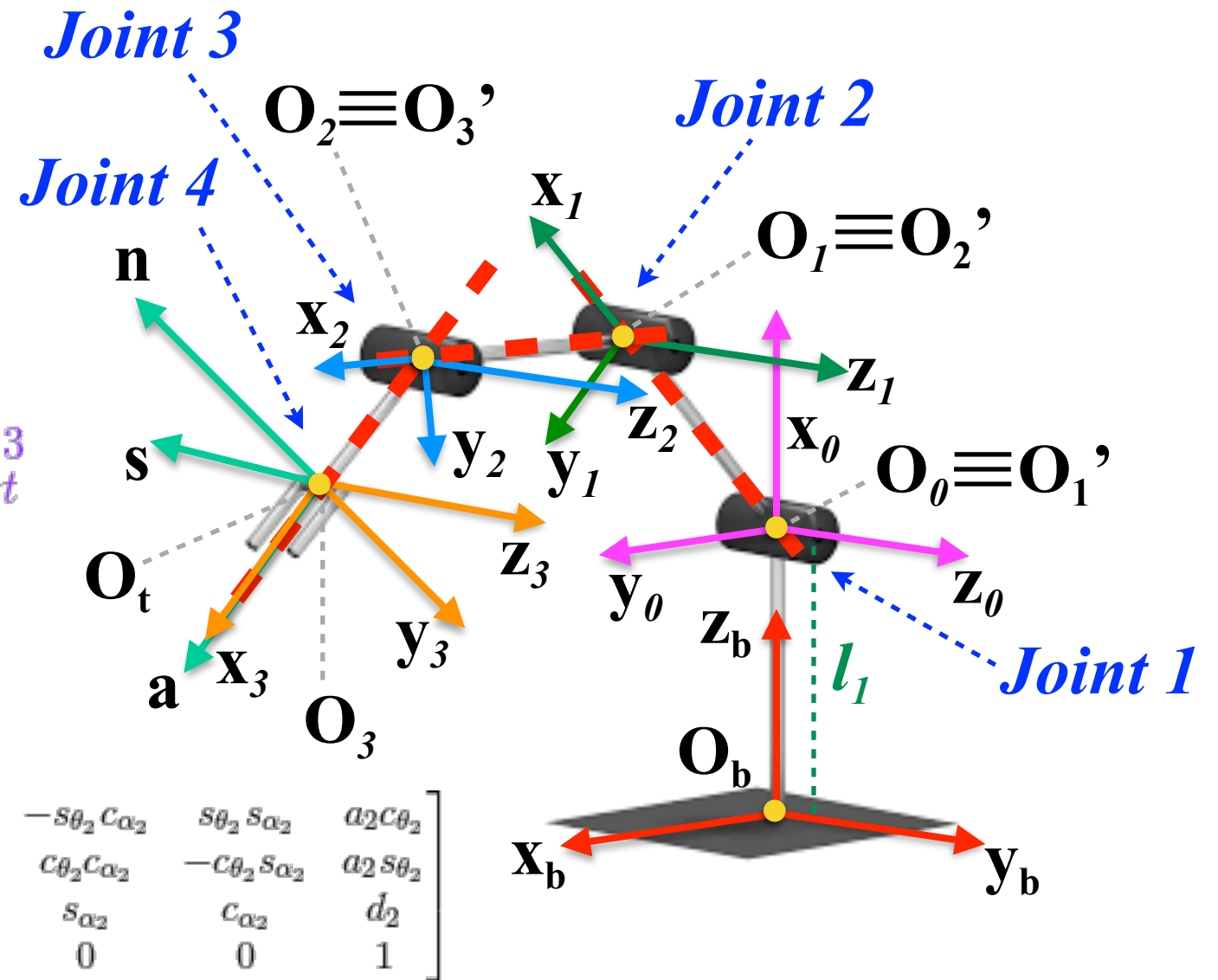
$$A_t^n = \begin{bmatrix} R_t^n & \underline{q}_t^n \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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Determine the homogeneous transformation matrix

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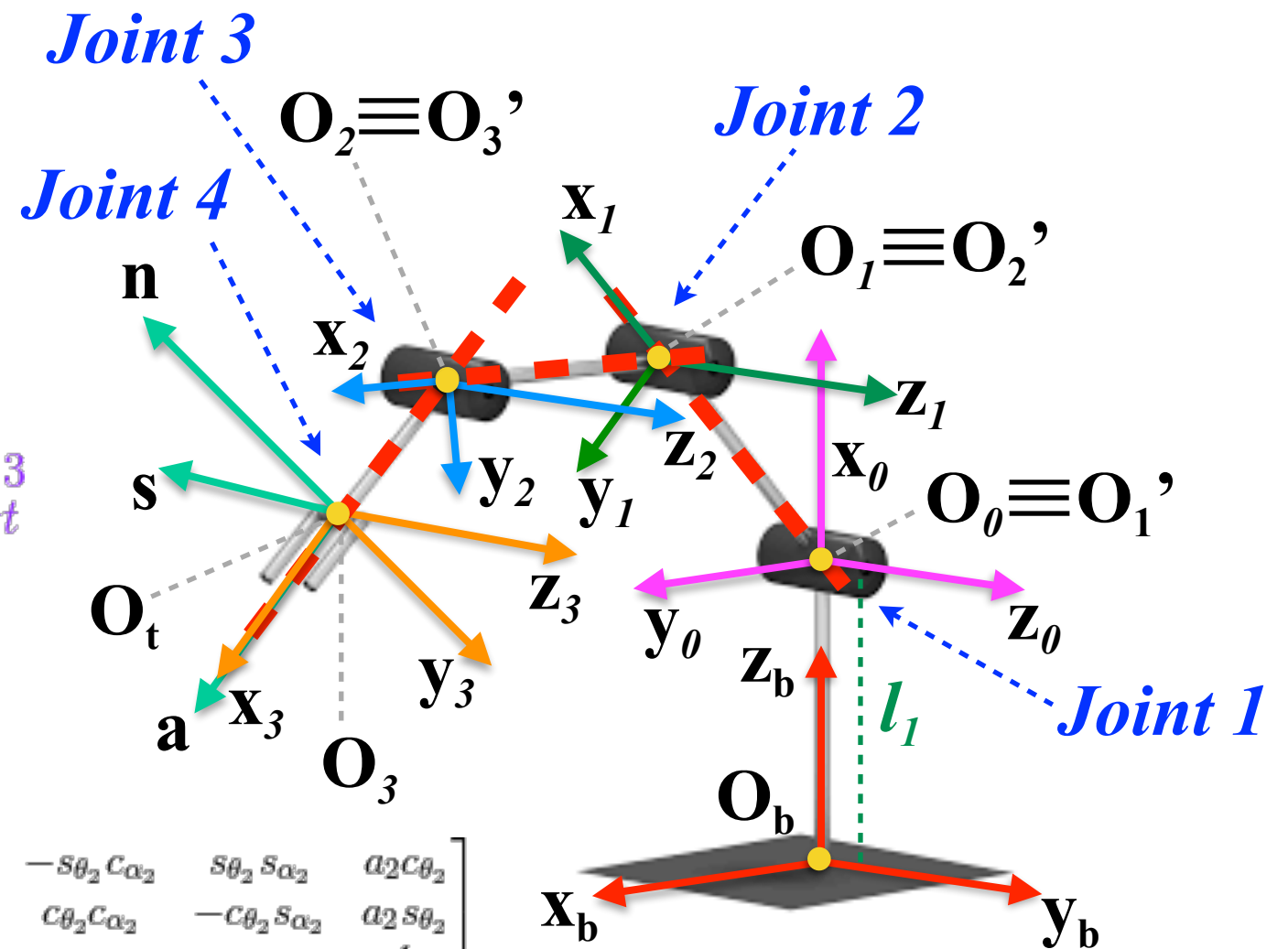
$$T_t^b(\underline{q}) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_1} & -s_{\theta_1} c_{\alpha_1} & s_{\theta_1} s_{\alpha_1} & a_1 c_{\theta_1} \\ s_{\theta_1} & c_{\theta_1} c_{\alpha_1} & -c_{\theta_1} s_{\alpha_1} & a_1 s_{\theta_1} \\ 0 & s_{\alpha_1} & c_{\alpha_1} & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_2} & -s_{\theta_2} c_{\alpha_2} & s_{\theta_2} s_{\alpha_2} & a_2 c_{\theta_2} \\ s_{\theta_2} & c_{\theta_2} c_{\alpha_2} & -c_{\theta_2} s_{\alpha_2} & a_2 s_{\theta_2} \\ 0 & s_{\alpha_2} & c_{\alpha_2} & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_3} & -s_{\theta_3} c_{\alpha_3} & s_{\theta_3} s_{\alpha_3} & a_3 c_{\theta_3} \\ s_{\theta_3} & c_{\theta_3} c_{\alpha_3} & -c_{\theta_3} s_{\alpha_3} & a_3 s_{\theta_3} \\ 0 & s_{\alpha_3} & c_{\alpha_3} & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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$$T_t^b(\underline{q}) = A_0^b \cdot A_1^0(\underline{q}) \cdot A_2^1(\underline{q}) \cdot A_3^2(\underline{q}) \cdot A_t^3$$



Complete the computations as homework

$$T_t^b(\underline{q}) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_1} & -s_{\theta_1} c_{\alpha_1} & s_{\theta_1} s_{\alpha_1} & a_1 c_{\theta_1} \\ s_{\theta_1} & c_{\theta_1} c_{\alpha_1} & -c_{\theta_1} s_{\alpha_1} & a_1 s_{\theta_1} \\ 0 & s_{\alpha_1} & c_{\alpha_1} & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_2} & -s_{\theta_2} c_{\alpha_2} & s_{\theta_2} s_{\alpha_2} & a_2 c_{\theta_2} \\ s_{\theta_2} & c_{\theta_2} c_{\alpha_2} & -c_{\theta_2} s_{\alpha_2} & a_2 s_{\theta_2} \\ 0 & s_{\alpha_2} & c_{\alpha_2} & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_{\theta_3} & -s_{\theta_3} c_{\alpha_3} & s_{\theta_3} s_{\alpha_3} & a_3 c_{\theta_3} \\ s_{\theta_3} & c_{\theta_3} c_{\alpha_3} & -c_{\theta_3} s_{\alpha_3} & a_3 s_{\theta_3} \\ 0 & s_{\alpha_3} & c_{\alpha_3} & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$